

Exam Report

Score: 139 out of 150

1. A 72-year-old man with a 10-day history of fever, nausea and vomiting, and progressive confusion remains hospitalized because his symptoms have not resolved despite treatment with broad-spectrum antibiotics. He has a 6-year history of severe emphysema treated with courses of corticosteroids during the past year. Medications include intravenous ceftriaxone and ampicillin, albuterol inhalers, theophylline, and prednisone. He is confused and stuporous. His temperature is 38.9°C (102°F), pulse is 84/min, and blood pressure is 142/86 mm Hg. Physical examination shows meningismus. An MRI of the brain shows diffuse meningeal enhancement with a basilar predominance. A lumbar puncture is done; cerebrospinal fluid analysis shows a leukocyte count of 322/mm³ (86% lymphocytes), a glucose concentration of 17 mg/dL, and a protein concentration of 212 mg/dL. Which of the following is the most likely diagnosis?

Your Answer: E) Tuberculous meningitis

Correct Answer: E) Tuberculous meningitis

Correct Answer: E

Chronic use of immunosuppressants including glucocorticoids increase the likelihood of tuberculosis reactivation. Tuberculous meningitis typically presents with nuchal rigidity, fever, and altered mental status, similar to the symptoms seen in bacterial meningitis. However, unlike bacterial meningitis which presents acutely, tuberculous meningitis typically shows a subacute presentation. Presence of cranial nerve palsies is also more likely in tuberculous meningitis than bacterial meningitis. Physical examination may show a positive Kernig or Brudzinski sign, indicative of meningeal irritation. Results of laboratory studies will show nonspecific changes including leukocytosis and mild anemia. A tuberculin skin test or interferon-gamma release assay are usually positive, but a negative result is also possible. Cerebrospinal fluid (CSF) evaluation is the initial diagnostic step and will reveal an increased opening pressure, increased white blood cell count with prominent lymphocytes, increased protein concentration, and decreased glucose concentration. Culture of the CSF for acid-fast bacilli may be positive, though the sensitivity of this test for tuberculous meningitis is less than 50%. Radiologic evaluation may show basilar exudates, as in this case, as well as periventricular infarcts and hydrocephalus.

Incorrect Answers: A, B, C, and D.

Herpes simplex encephalitis (Choice A) presents with fever, altered mental status, and neurologic abnormalities, such as seizure, aphasia, hemiparesis, or cranial nerve deficits. Neuroimaging will typically show changes in the temporal lobe. The basilar predominance in this case is more suggestive of a tuberculous meningitis than herpes encephalitis.

Lyme disease (Choice B) caused by disseminated *Borrelia burgdorferi* infection may cause neuropathy and meningitis. However, this would be expected to be accompanied by a rash in the pattern of erythema migrans and history of tick exposure.

Neurosarcoidosis (Choice C) is defined as the presence of neurologic symptoms in a patient with known sarcoidosis. As any portion of the central or peripheral nervous system may be affected, symptoms may stem from cranial nerve palsies, hypothalamic dysfunction, spinal cord involvement, or peripheral neuropathy. This patient does not show other features of sarcoidosis, including pulmonary, eye, or skin involvement.

Neurosyphilis (Choice D) presents with fever, meningismus, ataxia, Argyll Robertson pupil, and tabes dorsalis. Tabes dorsalis refers to infection of the posterior columns and dorsal roots of the spinal cord causing impaired sensation of light touch, vibration, and proprioception along with ataxia. Serologic testing of the CSF for syphilis using a venereal disease research laboratory (VDRL) or rapid plasma reagin (RPR) test is positive.

Educational Objective: Educational Objective: Tuberculous meningitis typically presents in a subacute fashion with nuchal rigidity, fever, and altered mental status. Cerebrospinal fluid (CSF) evaluation is the initial diagnostic step and will disclose an increased opening pressure, increased white blood cell count with prominent lymphocytes, increased protein concentration, decreased glucose concentration, and positive culture for acid-fast bacilli.

2. An 8-month-old girl is brought to the physician by her parents because of increased irritability over the past month. During this period, she has had intermittent low-grade fever and episodes of stiffening of her body. Her father reports that she previously interacted with them and smiled when prompted but has not made eye contact for the past 3 days. She has a history of hypotonia. She cannot yet roll over or sit up. Her parents are third cousins. She is awake and irritable and has poor visual regard. Examination of the skin shows no neurocutaneous lesions. The optic nerves are pale. Neurologic examination shows hyperextension of the head and back, ankle spasticity, and fisting. Deep tendon reflexes are brisk, and Babinski sign is present. An MRI of the brain is most likely to show which of the following?

Your Answer: B) Degeneration of the white matter

Correct Answer: B) Degeneration of the white matter

Correct Answer: B

Correct Answer: B.

This infant's presentation of fever, spasticity, hypotonia, vision loss, optic nerve pallor, and Babinski sign, as well as a family history of consanguinity are consistent with a diagnosis of Krabbe disease, also known as galactosylceramide lipidosis. Krabbe disease is an autosomal recessive lysosomal storage disorder caused by mutations in the GALC gene that encodes galactocerebrosidase, the deficiency of which results in accumulation of galactocerebroside and demyelinating neuronal degeneration. Patients typically present within the first year of life with progressive neurologic degeneration, spasticity, stiffness, seizures, feeding difficulty, and peripheral neuropathy, including optic neuropathy. The diagnosis is confirmed with serum enzyme assays, although MRI of the brain will show white matter degeneration caused by demyelination. There is no effective treatment for Krabbe disease and most affected children die during the first several years of life.

Incorrect Answers: A, C, D, E, F, and G.

Cerebellar hypoplasia (Choice A) and hydrocephalus (Choice D) are both observed in Dandy-Walker malformation. Patients present during the first year of life with hydrocephalus and associated nausea, vomiting, seizures, and downward displacement of the eyes. Dandy-Walker malformation does not typically present with spasticity and stiffness and presents with papilledema rather than optic nerve pallor.

Holoprosencephaly (Choice C) is a congenital malformation in which the prosencephalon does not divide into two hemispheres. Most cases result in death in utero and are also accompanied by craniofacial malformations including cyclopia and the presence of a proboscis. Lysosomal storage disorders, including Krabbe disease, are not frequent causes of holoprosencephaly.

Periventricular calcifications (Choice E) is a frequent finding in several congenital infections, such as in congenital toxoplasmosis or cytomegalovirus infection. While such infections can have profound developmental consequences, they do not typically cause progressive neurodegeneration.

Schizencephaly (Choice F) is a rare congenital malformation characterized by clefts through the cerebral parenchyma between the ventricles and pia mater. Lysosomal storage disorders, including Krabbe disease, are not a frequent cause of schizencephaly.

No abnormalities (Choice G) would not be an expected finding in this infant with profound neurologic degeneration during the first year of life.

Educational Objective: Educational Objective: Krabbe disease is an autosomal recessive lysosomal storage disorder caused by mutations in the GALC gene that encodes galactocerebrosidase, the deficiency of which results in the accumulation of galactocerebroside and secondary demyelinating neuronal degeneration. Patients typically present within the first year of life with progressive neurologic degeneration, spasticity, stiffness, seizures, feeding difficulty, and peripheral neuropathy, including optic neuropathy.

3. A 57-year-old woman is brought to the emergency department 20 minutes after a simple partial seizure. Examination shows mild weakness of the right lower extremity. A CT scan of the head shows a 1.5-cm, uniformly and intensely enhancing parasagittal mass in the left frontal region. Which of the following is the most likely diagnosis?

Your Answer: D) Meningioma

Correct Answer: D) Meningioma

Correct Answer: D

Correct Answer: D.

Meningioma is a common, benign central nervous system (CNS) tumor derived from the meninges that occurs along the surfaces of the brain and spinal cord, often in the parasagittal region. They occur in women more often than men and are associated with neurofibromatosis type 2 and multiple endocrine neoplasia type 1. Patients may be asymptomatic or experience seizures or focal neurologic deficits related to the tumor location. Meningiomas are typically benign but can be malignant. Malignant meningiomas may grow rapidly, leading to symptoms of increased intracranial pressure (eg, postural headache, nausea, papilledema), or may metastasize. Meningiomas appear as well-circumscribed, extra-axial (ie, outside of the brain parenchyma) masses that compress the parenchyma. They show uniform enhancement on imaging. Definitive diagnosis requires histologic examination of tissue, which will show spindle cells arranged in a whorled pattern with psammoma bodies. Treatment may include antiepileptics or surgical resection to alleviate the tumoral mass effect if significant, though often, periodic surveillance is all that is necessary.

Incorrect Answers: A, B, C, and E.

Astrocytoma (Choice A) is a primary CNS neoplasm that can occur as benign pilocytic astrocytoma in children or malignant glioblastoma multiforme (Choice C) in adults. Glioblastoma are highly malignant tumors appearing as intra-axial masses in the cerebral hemispheres, which typically show areas of necrosis and can spread to the opposite hemisphere across the corpus callosum.

Ependymoma (Choice B) is a CNS neoplasm formed from ependymal cells. Location typically involves the fourth ventricle. Histologic characteristics include perivascular pseudorosettes formed by malignant cells arranged around a blood vessel.

Schwannoma (Choice E) are benign tumors that arise from Schwann cells in peripheral nerve sheaths. They commonly affect the facial and vestibulocochlear nerves, and mass effect may result in sensorineural hearing loss or ataxia. They are associated with neurofibromatosis type 2.

Educational Objective: Educational Objective: Meningiomas are benign CNS tumors arising from the meninges that may compress the brain or spinal cord and cause focal neurologic deficits as well as seizures. Neuroimaging shows a spherical and well-circumscribed mass that indents the parenchyma. Lesions show uniform enhancement on imaging.

4. A 67-year-old semiretired corporate executive comes to the physician with his wife because she is concerned about his increasing disorganization at home over the past year. She also notes that he is unable to remember the names of recent acquaintances. He has insisted on working part-time since stepping down from a CEO position. His secretary has told his wife that he has been forgetting scheduled appointments and becomes angry and tearful when he is reminded. He drinks one to two cocktails each evening. Physical examination shows no abnormalities. On mental status examination, he is mildly agitated and has word-finding difficulty and short-term memory deficits. He is oriented to person, place, and time. Which of the following is the most likely diagnosis?

Your Answer: C) Dementia, Alzheimer type

Correct Answer: C) Dementia, Alzheimer type

Correct Answer: C

Correct Answer: C.

Alzheimer dementia (AD) is the most prevalent dementia, presenting with progressive cognitive decline that begins with short-term memory impairment and decline in visuospatial abilities and progresses to apraxia, language abnormalities, and dysexecutive syndrome (resulting in the inability to recall scheduled appointments in this patient). The disease culminates in behavioral and personality changes that prevent the patient from performing basic activities of daily living such as bathing. AD may be treated with drugs that both directly and indirectly increase acetylcholine in synaptic clefts. These include donepezil and cholinesterase inhibitors which non-competitively and reversibly inhibit acetylcholinesterase and thereby increase the amount of synaptic acetylcholine available for neurotransmission. Though these drugs may slow the rate of cognitive decline and modestly improve functionality, they are not curative for AD.

Incorrect Answers: A, B, D, E, F, G, H, I, J, K, L, M, N, O, P, and Q.

Acute stress disorder (Choice A) refers to symptoms of intrusive thoughts, hyperarousal, dissociation, avoidance, and/or depressed mood within 1 month of a traumatic event.

Alcohol-induced amnesic disorder (Choice B) may be seen in alcoholics who are deficient in vitamin B₁ (thiamine). Patients often confabulate to compensate for impaired memory.

Dissociative amnesia (Choice D) is a potentially reversible impairment of memory regarding one's identity, usually in association with a history of psychological trauma.

General paresis (Choice E) is a finding of late stage tertiary syphilis in which dementia is accompanied by general paralysis.

Head trauma (Choice F) may cause a mild traumatic brain injury, also known as concussion, commonly resulting in loss of consciousness, confusion, amnesia, headache, dizziness, nausea, and/or vomiting minutes after the injury.

Hepatolenticular degeneration (Wilson disease) (Choice G) results from impaired metabolism of copper and causes neuropsychologic impairment including parkinsonism, ataxia, dementia, or hallucinations, in addition to liver disease.

HIV encephalitis (Choice H) is characterized by concentration impairment, depressive symptoms, and cognitive and motor retardation seen in patients with poorly controlled HIV.

Huntington disease (Choice I) presents with insidious onset of psychiatric and cognitive symptoms, followed by motor symptoms (chorea, hypotonia, dystonia, loss of voluntary control of movements).

Hypercalcemia (Choice J) may present with anxiety, depression, and/or cognitive dysfunction but not progressive memory loss.

Major depressive disorder (Choice K) may cause cognitive impairment in older patients but would

also disclose depressed mood, anhedonia, or other signs of a depressive disorder. Multi-infarct (vascular) dementia (Choice L) features stepwise cognitive decline following cerebrovascular accidents. Niacin deficiency (Choice M) leads to pellagra, characterized by diarrhea, dementia, and dermatitis. Normal-pressure hydrocephalus (Choice N) presents with symptoms of dementia, urinary incontinence, and ataxia. Parkinson disease (Choice O) is characterized by tremor, rigidity, akinesia, and postural instability. Pick disease (Choice P), or frontotemporal dementia, is a neurodegenerative disorder characterized by atrophy of the frontal and temporal lobes, causing deficits in social behavior or personality along with dementia. Normal aging (Choice Q) includes a slow decline of executive functioning, attention span, and problem-solving after the age of 70, but a successfully aging individual should be able to still function in society.

Educational Objective: Educational Objective: Alzheimer dementia is the most prevalent dementia, presenting with progressive cognitive decline that culminates in the inability to perform basic activities of daily living.

5. A 17-year-old boy is brought to the emergency department by ambulance 2 hours after he dove into a shallow lake and struck the back of his head on the bottom. He immediately had the onset of pain in his neck and a sensation of an electric shock radiating from his neck to his left shoulder. He did not have loss of consciousness. He was able to stand, and two friends helped him out of the water into a supine position on the dock. At the scene, his spine was immobilized with a cervical collar, and his head was taped to a backboard. On arrival, he is alert. Vital signs are within normal limits. Muscle strength is 3/5 on extension of the left wrist. Deep tendon reflexes are normal. Babinski sign is absent. Sensorimotor examination of the right upper extremity and lower extremities shows no abnormalities. Sensation to light touch is decreased over the radial aspect of the left forearm and hand and over the palmar and dorsal aspects of the left thumb. Rectal examination shows normal sphincter tone and sensation. A lateral x-ray of the cervical spine is shown. This patient most likely has which of the following cervical spine injuries?



Your Answer: C) Bilateral dislocation of facets with impingement of the left C6 nerve root

Correct Answer: C) Bilateral dislocation of facets with impingement of the left C6 nerve root

Correct Answer: C

This patient likely has bilateral dislocation of facets with impingement of the left C6 nerve root caused by cervical spinal injury. The C6 dermatome is located on the lateral (radial) aspect of the forearm and extends to the lateral two digits of the hand. Injury to the C6 nerve root will cause decreased sensation, pain, and/or paresthesia in these distributions (or part of this distribution) along with motor dysfunction of elbow flexion and wrist extension. The C6 nerve root carries efferent information for the biceps reflex, so patients with C6 nerve injury may present with a diminished or absent biceps reflex. Dermatomes and myotomes are components of the embryonic somite, a paired segment of paraxial mesoderm. When fully developed, the dermatome is an area of skin whose afferent fibers travel within a single spinal nerve, and myotome is a group of muscles innervated by efferent fibers from a single spinal nerve. Disc herniation and spondylosis are common causes of spinal nerve root compression in older patients, and trauma and consequent vertebral dislocation or fracture is a common cause of root compression in younger patients such as in this case. Sensory deficits, motor dysfunction, and pain typically occur in the dermatomal and myotomal distribution that correlates to the affected nerve root.

Incorrect Answers: A, B, D, and E.

Anterior spinal artery thrombosis (Choice A) typically manifests with bilateral loss of pain and temperature sensation and motor function below the level of the lesion, along with autonomic dysfunction. The anterior spinal artery supplies the corticospinal, spinothalamic, and autonomic tracts.

Bilateral dislocation of facets with impingement of the left C5 nerve root (Choice B) and unilateral dislocation of a facet with impingement of the left C5 nerve (Choice E) would lead to sensory and motor dysfunction in the lateral aspect of the shoulder and upper arm. C5 nerve compression typically results in an absent biceps reflex, as the C5 nerve root carries efferent information for this reflex. However, C5 nerve compression would not cause decreased sensation in the digits.

Brown-Séquard syndrome (Choice D) refers to hemisection of the spinal cord, which damages the anterior horn along with three major spinal cord tracts: the corticospinal tract, dorsal column-medial lemniscus tract, and spinothalamic tract. Because of the decussation of the spinothalamic tract inferior to the lesion, patients typically present with contralateral impairment in pain and temperature sensation and ipsilateral deficits in other sensory domains and motor function.

Educational Objective: Educational Objective: Cervical spinal injury can lead to vertebral dislocation and impingement of the spinal nerve root, which leads to weakness, numbness, paresthesia, and pain in the dermatomal and myotomal distribution that correlates to the affected root. C6 nerve root compression will cause decreased sensation, pain, and/or paresthesia in the lateral forearm and lateral two digits. Additionally, C6 root compression causes motor dysfunction of elbow flexion and wrist extension.

6. A 62-year-old man is brought to the emergency department 2 hours after the sudden onset of headache and left-sided weakness that have become increasingly severe. He has hypertension treated with lisinopril and chronic neck pain treated with 325-mg aspirin. On arrival, he appears mildly lethargic. His pulse is 60/min, respirations are 16/min, and blood pressure is 180/100 mm Hg. Examination shows moderate weakness of the lower left side of the face. Muscle tone is decreased in the left upper and lower extremities. Muscle strength is 4/5 on the left and 5/5 on the right. Deep tendon reflexes are 3+ on the left. Babinski sign is present on the left. A CT scan of the head without contrast is shown. Which of the following is the most likely diagnosis?



Your Answer: B) Hypertensive intracerebral hematoma

Correct Answer: B) Hypertensive intracerebral hematoma

Correct Answer: B

Correct Answer: B.

Hypertensive intracerebral hematomas (ICHs) can occur in patients with chronic hypertension, typically because of hemorrhage from small penetrating arteries in the pons, midbrain, thalamus, putamen, caudate, or cerebellum. Risk factors for hypertensive ICH include uncontrolled hypertension and antiplatelet or anticoagulant therapy, such as with aspirin or warfarin. Patients can present with headache, vomiting, confusion, altered mental status, stupor, or coma. Focal neurologic deficits depend on the site of the ICH and can include hemiparesis, hemisensory symptoms, gaze palsy, dysarthria, aphasia, or if in the cerebellum, ataxia, vomiting, and disequilibrium. Patients may manifest upper motoneuron examination findings such as spasticity, hyperreflexia, and extensor plantar reflex (Babinski sign). Diagnosis is made with noncontrast CT scan of the head which will show hyperdense intraparenchymal products, which is the appearance of acute bleeding in the brain as seen in this patient's right hemisphere.

Incorrect Answers: A, C, D, and E.

Cerebral amyloid angiopathy (Choice A) occurs secondary to amyloid deposition in cerebral blood vessels and leptomeninges. It can cause spontaneous intracerebral hemorrhage in older adults, typically in a lobar location, because of the location of amyloid deposits in cortical vessels. In contrast, hypertensive hemorrhages typically occur in a deep location such as the putamen, thalamus, midbrain, or pons.

Ischemic cerebral infarction (Choice C) can occur as a result of thromboembolic occlusion of cerebral arteries. Patients typically present with acute focal neurologic deficits, especially on one side of the body, and may have speech difficulty, dysarthria, or aphasia. A CT scan would not show hyperdense blood products unless the infarction is complicated by hemorrhagic conversion. Ruptured arteriovenous malformation (Choice D) can lead to intracerebral or subarachnoid hemorrhage. Arteriovenous malformations are uncommon and usually present in individuals younger than 50 years old with intracranial hemorrhage, seizure, neurologic deficit, or headache, or may be an incidental finding. In an older patient with uncontrolled hypertension, a hypertensive intracerebral hematoma is more likely. Ruptured berry aneurysm (Choice E) typically leads to a subarachnoid hemorrhage. Patients typically present with an acute, severe headache (often described as the worst headache of the patient's life), along with neck stiffness, photophobia, and possible stupor or coma in severe cases. CT scan of the head typically shows hyperdense blood products in the cerebral sulci and basal cisterns.

Educational Objective: Educational Objective: Hypertensive intracerebral hematomas can occur in patients with chronic hypertension, typically as a result of hemorrhage from small penetrating arteries in the pons, midbrain, thalamus, putamen, caudate, or cerebellum. Patients can present with headache, vomiting, confusion, altered mental status, stupor, upper motoneuron signs, and focal neurologic deficits such as hemiparesis and hemisensory symptoms.

7. Over the past 6 months, an otherwise healthy 19-year-old cheerleader has had increasingly severe neurologic symptoms. Initially, she had difficulty jumping; 2 weeks later she had a tingling feeling in the toes of her right foot that quickly spread to the left foot. Over the past 2 weeks, she has stumbled and fallen several times. Examination of the upper extremities shows normal findings. Deep tendon reflexes are absent in the lower extremities. She is unable to heel walk and has mild difficulty walking on her toes and rising from a low squatting position. Position sensation is normal, but pinprick feels less sharp distally and occasional light touch stimuli are missed over the toes. Peroneal motor conduction velocity is moderately slowed (22 m/sec; N=42–57). Which of the following is the most likely diagnosis?

Your Answer: A) Chronic inflammatory demyelinating polyradiculoneuropathy

Correct Answer: A) Chronic inflammatory demyelinating polyradiculoneuropathy

Correct Answer: A

Correct Answer: A.

Chronic inflammatory demyelinating polyradiculopathy (CIDP) is an immune-mediated demyelinating condition affecting the spinal nerve roots. The pathogenesis of CIDP involves inflammation of myelin by an unknown mechanism. The ventral nerve roots containing motor efferents are commonly involved, though the dorsal nerve root (controlling sensation) may also be affected. As such, patients typically present with relapsing-remitting or gradually progressive symmetric muscle weakness with depressed or absent deep tendon reflexes (reflecting a lower motoneuron pattern of dysfunction) that affects the proximal and distal lower extremities simultaneously and may also affect the upper extremities. Autonomic dysfunction may occur but is typically mild. Sensory deficits may occur (though these symptoms are typically mild compared to motor dysfunction). A slowing of conduction velocity on nerve conduction studies indicates demyelination and supports the diagnosis. The diagnosis is further supported by increased cerebrospinal fluid protein with normal cell counts on lumbar puncture, termed albuminocytologic dissociation and thought to reflect generalized inflammation. Management includes glucocorticoids, immunomodulators, plasma exchange, or intravenous immunoglobulin therapy.

Incorrect Answers: B, C, D, E, F, G, H, I, J, K, and L.

Diabetic neuropathy (Choice B) commonly results in sensorimotor peripheral neuropathy. This patient has no history of diabetes mellitus.

Epidural spinal cord metastasis (Choice C) and spinal epidural abscess (Choice H) may result in spinal cord compression, which typically leads to sensory, motor, and/or autonomic dysfunction below the level of the lesion in an upper motoneuron pattern of weakness.

Guillain-Barré syndrome (Choice D) is an inflammatory demyelinating polyneuropathy that presents with acute-onset, ascending, bilateral muscle weakness with hyporeflexia. A chronic time course is inconsistent with Guillain-Barré syndrome.

Multiple sclerosis (Choice E) is an inflammatory demyelinating disease that presents with focal neurologic deficits separated in space and time. If present, motor dysfunction would present in an upper motoneuron pattern.

Spinal cord infarction (Choice F) typically occurs in the territory of the anterior spinal artery, which usually manifests with loss of pain and temperature and upper motoneuron dysfunction below the level of the lesion, along with autonomic dysfunction.

Primary spinal cord tumors (Choice G) affecting the cauda equina can lead to unilateral or bilateral lower extremity radicular pain and sensorimotor dysfunction. Pain is generally present.

Spondylotic myelopathy (Choice I) refers to degenerative vertebral disease leading to spinal cord

damage. This otherwise healthy young patient is unlikely to show vertebral degenerative disease. Syringomyelia (Choice J) features a cystic cavity in the central spinal cord and manifests with deficits in pain and temperature sensation and lower motoneuron dysfunction in the bilateral upper extremities.

Vasculitic neuropathy (Choice K) typically presents with painful polyneuropathy. This patient did not endorse pain.

Vitamin B₁₂ (cobalamin) deficiency (Choice L) can lead to subacute combined degeneration, which affects the spinocerebellar tracts, lateral corticospinal tracts, and dorsal columns. Lower motoneuron dysfunction would be atypical.

Educational Objective: Educational Objective: Chronic inflammatory demyelinating polyradiculopathy is an immune-mediated demyelinating condition affecting the spinal nerve roots. Patients typically present with relapsing-remitting or gradually progressive symmetric sensory dysfunction and muscle weakness with depressed or absent deep tendon reflexes that affects the proximal and distal lower extremities simultaneously and may also affect the upper extremities.

8. A 62-year-old man is brought to the emergency department because of progressive weakness of his legs during the past week. He has chronic kidney disease and receives dialysis three times weekly via a percutaneous tunneled catheter. Last week, he had an episode of sweating and chills. His temperature is 38.9°C (102°F), pulse is 100/min and regular, and blood pressure is 150/92 mm Hg. Examination shows severe tenderness to percussion over the lumbar spine. The catheter site appears normal. There is trace edema of the lower extremities. Muscle strength is 3/5 on flexion and extension of the knees and ankles. Deep tendon reflexes are increased at the patellae. Laboratory studies show:

Hemoglobin 11.1 g/dL

Leukocyte count 19,200/mm³

Segmented neutrophils 90%

Bands 5%

Eosinophils 1%

Lymphocytes 4%

Platelet count 290,000/mm³

An MRI of the spine shows an epidural abscess at L4. Blood cultures grow *Staphylococcus aureus*. In addition to antibiotic therapy and drainage of the abscess, which of the following is the most appropriate next step in management?

Your Answer: A) Removal of the catheter

Correct Answer: A) Removal of the catheter

Correct Answer: A

Correct Answer: A.

Epidural abscess is a localized collection of pus in the epidural space between the dura mater and periosteum. It most commonly occurs in the spine but may occur in the brain. Most patients have a history of another preceding or associated (even occult) infection and often have a history of immunocompromise (eg, diabetes mellitus, HIV infection, alcohol use disorder). Epidural abscess is more common in patients with a history of intravenous drug use or indwelling intravenous lines. Infectious organisms usually spread to the epidural space hematogenously, from spinal surgery, or from adjacent osteomyelitis. *Staphylococcus aureus* is a frequent causative organism. Patients present with signs and symptoms of fever, altered mental status, meningismus, localized back pain, and tenderness to palpation, along with weakness, numbness, and paralysis below the affected spinal level as a result of cord compression. Treatment is with broad spectrum antibiotics and surgical drainage of the abscess. It is also crucial to identify and remove potential sources of bacterial colonization, such as indwelling venous catheters.

Incorrect Answers: B, C, D, and E.

Electromyography (Choice B) is useful for the diagnosis of various neuromuscular and inflammatory disorders, such as polymyositis, myasthenia gravis, or amyotrophic lateral sclerosis. This patient's MRI has shown a clear cause of his lower extremity weakness, and further testing for other etiologies is less appropriate than removal of potential sources of bacterial colonization.

Technetium 99m bone scan (Choice C) or bone biopsy (Choice D) may be useful for the diagnosis of osteomyelitis or malignancy of bone. Pursuing further diagnostic testing is less appropriate than removal of potential sources of bacterial colonization.

Lumbar puncture (Choice E) is useful for the diagnosis of meningitis but is not usually necessary for patients with epidural abscess.

Educational Objective: Educational Objective: Epidural abscess may result from hematogenous dissemination of bacterial or fungal infection to the epidural space between the dura mater and periosteum. It presents with fever, altered mental status, meningismus, localized back pain, and tenderness to palpation, along with weakness, numbness, and paralysis below the affected spinal level. Treatment is with broad spectrum antibiotics, surgical drainage of the abscess, and the identification and removal of any potential sources of bacterial colonization, such as indwelling venous catheters.

9. A 67-year-old man is brought to the emergency department by his wife 30 minutes after a 15-minute episode of inability to speak. He has a 15-year history of poorly controlled hypertension. Three weeks ago, he had a transmural myocardial infarction and underwent placement of a stent in the left anterior descending coronary artery. Current medications include clopidogrel, benazepril, and 81-mg aspirin. He has smoked two packs of cigarettes daily for 30 years. On arrival, he is alert and oriented to person, place, and time. His pulse is 60/min, respirations are 15/min, and blood pressure is 160/100 mm Hg. A grade 1/6 systolic murmur is heard best at the second left intercostal space. Neurologic examination shows no focal findings. Carotid duplex ultrasonography shows 40% stenosis of the left carotid artery. Which of the following is the most likely diagnosis?

Your Answer: A) Embolus from a ventricular aneurysm

Correct Answer: A) Embolus from a ventricular aneurysm

Correct Answer: A

Correct Answer: A.

An embolic ischemic stroke involves an event in which a clot breaks off and travels to the brain, occluding blood flow within a cerebral artery. Emboli can originate from proximal vessels, such as the internal carotid artery, or from an atrial or ventricular thrombus. This patient had a recent myocardial infarction, which predisposes him to a ventricular aneurysm. Ventricular aneurysms are more likely to form in the case of large, anterior myocardial infarctions as a result of obstruction of the left anterior descending coronary artery, as was the case in this patient. A small systolic murmur of mitral regurgitation may be heard. Within the aneurysm, as a result of areas of stagnant blood flow and despite being on clopidogrel, a thrombus can form that can embolize and cause an ischemic stroke. If a patent foramen ovale is present, an embolism may arise from either the right or left side of the heart. This patient also has other risk factors for ischemic stroke including hypertension and atherosclerotic disease. Because his symptoms resolved within less than 24 hours, his embolic event is classified as a transient ischemic attack.

Incorrect Answers: B, C, D, and E.

Infarct of the lacunar artery (Choice B) causes ischemia to the deep cerebral white matter, pons, or basal ganglia. A lacunar stroke may present with purely motor symptoms, purely sensory symptoms, ataxic hemiparesis, or dysarthria. Infarcts in these deep penetrating vessels are secondary to hypertension, diabetes mellitus, and hypercholesterolemia. While these risk factors would also predispose to myocardial infarction, this patient's transient ischemic event is most likely caused by an embolism from the left ventricle as opposed to a primary central nervous system infarction. Intraparenchymal cerebral hemorrhage (Choice C) and subarachnoid hemorrhage (Choice E) are both causes of hemorrhagic stroke and are caused by ruptured intracranial or subarachnoid blood vessels in the setting of hypertension. A CT scan of the head without contrast is typically obtained in the acute setting to confirm the presence of blood, which is shown on imaging as hyperdense material. A transient ischemic attack is not caused by hemorrhage.

Spasm of a cerebral artery (Choice D) is a potential complication of subarachnoid hemorrhage which tends to occur within 4 to 10 days after the initial bleed. It leads to subsequent ischemia in the area supplied by the spasmed artery. It is not the cause of this patient's transient ischemic attack.

Educational Objective: Educational Objective: An embolic ischemic stroke is caused by a clot that has broken off from either a proximal vessel or thrombus and has occluded blood flow within a cerebral artery. A ventricular aneurysm after a myocardial infarction can cause a thrombus to form in the left ventricle, increasing the risk for an embolic event.

10. One month after starting isoniazid and rifampin therapy for pulmonary tuberculosis, a 62-year-old woman develops weakness, fatigue, and paresthesias. Examination shows pallor and diminished light touch sensation in the extremities. Laboratory studies show:

Hemoglobin 9 g/dL

Mean corpuscular volume 78 μm^3

Leukocyte count 8000/mm³

Platelet count 210,000/mm³

These symptoms are most likely to respond to treatment with which of the following?

Your Answer: D) Vitamin B₆

Correct Answer: D) Vitamin B₆

Correct Answer: D

Correct Answer: D.

Isoniazid and rifampin are used in the treatment of primary tuberculosis, which results from an initial exposure to *Mycobacterium tuberculosis*. Isoniazid binds vitamin B₆ (pyridoxine) which may result in early excretion and inactivation of pyridoxine, and also inhibits pyridoxine phosphokinase, which is necessary to activate it. Pyridoxine deficiency results, of which primary symptoms are peripheral neuropathy and paresthesia. Pyridoxine deficiency limits synthesis of histamine, hemoglobin, and neurotransmitters including epinephrine, norepinephrine, dopamine, serotonin, and γ -aminobutyric acid (GABA). In addition to peripheral neuropathy, deficiency commonly presents with dermatitis, sideroblastic anemia, glossitis, and seizures. These adverse effects may all be avoided by pyridoxine supplementation in all patients taking isoniazid.

Incorrect Answers: A, B, C, and E.

Folic acid (Choice A) is converted to tetrahydrofolic acid and is used as a coenzyme in the synthesis of nucleotides and nucleosides. Folate is contained in leafy vegetables and absorbed in the jejunum. Folate deficiency is often seen in patients with malnutrition, alcoholism, and patients taking anti-folate medications (eg, phenytoin, methotrexate). It is not caused by isoniazid or rifampin and is not responsible for this patient's symptoms.

Iron (Choice B) deficiency also presents with microcytic anemia. However, it is not induced by isoniazid or rifampin and its supplementation in this case would not address the symptoms of pyridoxine deficiency.

Vitamin B₁ (thiamine) (Choice C) supplementation is not necessary in patients taking isoniazid or rifampin. Thiamine deficiency causes variable manifestations including dry beriberi (peripheral neuropathy with mental status change), wet beriberi (dilated cardiomyopathy leading to heart failure), and Wernicke encephalopathy (confusion, ataxia, and ophthalmoplegia), but not microcytic anemia.

Vitamin B₁₂ (cyanocobalamin) (Choice E) supplementation is necessary in vegans and patients with malabsorption (eg, sprue), pernicious anemia, and Crohn disease. It is not needed when taking isoniazid or rifampin. Vitamin B₁₂ deficiency presents with megaloblastic anemia, not microcytic anemia, and subacute combined degeneration of the spinal cord.

Educational Objective: Educational Objective: Isoniazid therapy for latent tuberculosis without sufficient pyridoxine supplementation can lead to pyridoxine deficiency. This is characterized by

peripheral neuropathy, dermatitis, sideroblastic anemia, glossitis, and seizures.

11. A 22-year-old man who is a collegiate volleyball player comes to the physician because of a 2-month history of progressive pain and weakness in his right shoulder. There is no history of trauma. He is 196 cm (6 ft 5 in) tall and weighs 95 kg (210 lb); BMI is 25 kg/m². There is significant loss of muscle mass in the dorsal scapula region, especially in the area of the scapular spine. Passive abduction and external rotation of the right shoulder and protraction of the scapula reproduce the pain. There is weakness with resisted external rotation and initiation of abduction. Active range of motion shows full flexion and abduction. Sensory examination shows no cutaneous deficits along the right side of the neck or in the right upper extremity. Which of the following is the most likely cause of this patient's symptoms?

Your Answer: C) Brachial plexus palsy

Correct Answer: E) Irritation of the suprascapular nerve

Correct Answer: E

Correct Answer: E.

The patient's symptoms and physical examination findings are most consistent with entrapment of the suprascapular nerve. The suprascapular nerve originates from the superior trunk of the brachial plexus and innervates the supraspinatus and infraspinatus muscles. Nerve compression secondary to traction injury is common among volleyball players. Irritation of the suprascapular nerve results in pain along the course of the nerve and may result in chronic neuropathy with subsequent muscle wasting, atrophy, and degeneration of the infraspinatus and supraspinatus muscles. The supraspinatus is responsible for adduction of the shoulder, and the infraspinatus is responsible for external rotation of the shoulder, which reproduce the pain and disclose weakness in this patient. Initial treatment is conservative with activity modification and physical therapy. Operative intervention may be required.

Incorrect Answers: A, B, C, and D.

Adhesive capsulitis (Choice A) is a common cause of shoulder pain in adults, characterized by severely limited range of motion. Patients typically note a gradual onset of progressive shoulder stiffness and pain. Physical examination would disclose limited range of motion while this patient discloses full range of motion.

Axillary nerve damage (Choice B) results in weakness of the deltoid and teres minor, which abduct and stabilize the shoulder. The axillary nerve gives rise to the superficial lateral cutaneous nerve of the arm, and sensory deficits would be expected if injured.

Brachial plexus palsy (Choice C) describes a range of injury to the nervous system complex that gives rise to the nerves of the upper extremity. The mechanism is usually high-impact trauma or forced abduction of the shoulder. Physical examination typically discloses motor and sensory deficits involving several peripheral nerves.

Herniated cervical disc (Choice D) typically presents with neck pain, occipital headache, and unilateral upper extremity weakness, numbness, and/or pain caused by compression of a cervical nerve root. Like brachial plexus palsy, physical examination typically discloses deficits involving multiple peripheral nerves.

Educational Objective: Educational Objective: Shoulder pain associated with impaired innervation to the supraspinatus and infraspinatus muscles is suggestive of suprascapular neuropathy. Causes

include mass effect as a result of lesions compressing the nerve sheath, trauma, and traction injury (most commonly seen in volleyball players).

12. A 32-year-old woman comes to the physician with her husband because of a 6-month history of fluctuating symptoms including fatigue, loss of balance, blurred vision, and the sensation of pins and needles over her arms and legs. Her husband says that she has also had mood swings with episodes of uncontrollable crying over the past 2 months. Her temperature is 37.3°C (99.2°F), pulse is 80/min, respirations are 16/min, and blood pressure is 128/86 mm Hg. Neurologic examination shows reduced medial gaze of the left eye and numbness to pinprick over the left distal lower extremities. Romberg sign is present. Babinski sign is present on the left. Which of the following is the most appropriate next step in diagnosis?

Your Answer: G) MRI of the brain

Correct Answer: G) MRI of the brain

Correct Answer: G

Correct Answer: G.

Multiple sclerosis (MS), the most prevalent immune-mediated demyelinating disease, may present with episodic or progressive neurologic dysfunction. According to the McDonald criteria for MS diagnosis, patients must have evidence of demyelinating lesions that are separated in time and location. This evidence is typically based on subjective symptoms and radiologic studies. Lesions affecting the left medial longitudinal fasciculus (leading to decreased medial gaze of the left eye), the prefrontal cortex or corticobulbar tracts (leading to pseudobulbar affect, or uncontrollable crying), and the major central nervous system (CNS) tracts innervating the extremities (lesions of the corticospinal tract causing Babinski sign, spinothalamic tract causing decreased pinprick, dorsal column-medial lemniscus causing Romberg sign and paresthesia) are likely separated in space. This patient experienced two discrete periods of symptom onset at 6 months and 2 months, illustrating separation in time. MRI typically shows white matter hyperintensities in the periventricular region, cortex, subcortical region, cerebellum, or spine. Oligoclonal bands of immunoglobulins, indicating immune overactivation, can be found in the cerebrospinal fluid but are not necessary for diagnosis. Treatment includes corticosteroids for symptomatic management of acute flares and long-term disease-modifying treatments such as interferon or monoclonal antibodies.

Incorrect Answer: A, B, C, D, E, F, H, and I.

EEG (Choice A) assists in diagnosing seizures. Hypothalamic seizures can lead to uncontrollable crying. However, seizures typically last seconds to minutes; months of persistent sensory and motor dysfunction would be atypical.

Measurement of serum ammonia concentration (Choice B) supports a diagnosis of hepatic encephalopathy. Hepatic encephalopathy presents acutely with confusion or stupor along with neurologic symptoms such as asterixis, tremor, myoclonus, and frontal release signs.

Measurement of serum calcium concentration (Choice C) would confirm hyper- or hypocalcemia. Hypercalcemia typically presents with confusion, fatigue, polyuria, constipation, and bone pain. Hypocalcemia features muscle spasms, paresthesias, circumoral numbness, and seizures.

Measurement of serum copper concentration (Choice D) diagnoses Wilson disease, in which copper accumulates in the liver, brain, and eyes. Though mood swings and extrapyramidal symptoms (from copper deposits in the basal ganglia) may occur, months of medial gaze palsy along with sensorimotor dysfunction would be atypical.

Measurement of serum vitamin B₁₂ (cobalamin) concentration (Choice E) helps in diagnosing subacute combined degeneration, affecting the spinocerebellar tracts, lateral corticospinal tracts, and dorsal columns.

Measurement of urine porphobilinogen concentration (Choice F) assists in diagnosing acute

intermittent porphyria, which is characterized by episodic abdominal pain, polyneuropathy, psychological symptoms, and port wine-colored urine.

Serum antinuclear antibody assay (Choice H) can assist in diagnosing autoimmune diseases affecting the CNS and are expressed at decreased concentrations in multiple sclerosis. However, characteristic MRI findings are more specific for MS.

Venereal disease research laboratory (VDRL) test in cerebrospinal fluid (Choice I) assists in diagnosing neurosyphilis, which typically presents with progressive dementia or tabes dorsalis.

Educational Objective: Educational Objective: Multiple sclerosis is an autoimmune demyelinating disease that presents with focal neurologic deficits separated in space and time. Though most patients present with episodic deficits, some present with progressive disease. MRI, which confirms the diagnosis, typically shows white matter hyperintensities in the periventricular region, cortex, subcortical region, cerebellum, or spine.

13. A physician makes a house call for an 87-year-old woman who is unable to come to the office because of difficulty with transportation. Her family reports that she has fallen frequently during the past 6 weeks. The patient says that she has had some difficulty feeling where her feet are but that she is otherwise well. She has not had loss of consciousness. She has no history of serious illness and takes no medications. Vital signs are within normal limits. Muscle strength is 4/5 on flexion of the hips and extension of the knees and ankles bilaterally. Proprioception and sensation to vibration are decreased in the lower extremities. Sensation to light touch is intact. She has a slapping gait. Mental status examination shows no abnormalities. The physician counsels the patient about making enhancements to her home to prevent falls. Which of the following is the most appropriate next step in diagnosis?

Your Answer: E) Electromyography

Correct Answer: B) Measurement of serum vitamin B₁₂ (cobalamin) concentration

Correct Answer: B

Correct Answer: B.

Subacute combined degeneration, caused by demyelination of the dorsal columns, lateral corticospinal tracts, and spinocerebellar tracts, is a feature of vitamin B₁₂ (cobalamin) deficiency. Subacute combined degeneration causes a combination of motor and sensory symptoms including ataxic gait, muscle weakness, paresthesia, and loss of proprioception and vibration sense. The effects of these features may include frequent falls, as in this case. Vitamin B₁₂ deficiency is often a result of inadequate dietary intake, seen in veganism, malabsorption, or pernicious anemia. Occasionally, elderly patients with poor nutrition may develop B₁₂ deficiency. Other manifestations include dementia and megaloblastic anemia. In an elderly patient with an acute or subacute change in behavior or activities of daily living, organic causes such as vitamin deficiency should be ruled out.

Incorrect Answers: A, C, D, and E.

Determination of erythrocyte sedimentation rate (Choice A) may be useful in supporting the diagnosis of an inflammatory process, such as polymyositis, as the cause of this patient's lower extremity weakness. However, erythrocyte sedimentation rate is nonspecific and would not suggest a definitive diagnosis. Furthermore, the patient's symptoms are more consistent with vitamin B₁₂ deficiency than an inflammatory process.

X-rays of the lumbosacral spine (Choice C) may be warranted in a patient with recent falls who is suspected to have fractures or other musculoskeletal pathology. However, it is not useful in diagnosing subacute combined degeneration.

EEG (Choice D) is the test of choice in diagnosing a seizure disorder. While seizures could be a cause of recurrent falls, in this case subacute combined degeneration caused by vitamin B₁₂ deficiency is more likely. An EEG is not helpful in making the diagnosis of vitamin B₁₂ deficiency. Electromyography (Choice E) and nerve conduction studies are used to support a diagnosis of inflammatory myopathy, such as polymyositis. The motor weakness seen in this patient is caused by the demyelination of the lateral corticospinal tracts rather than damage to the peripheral muscles, so electromyography will not be useful.

Educational Objective: Educational Objective: Subacute combined degeneration is a feature of vitamin B₁₂ deficiency characterized by a combination of motor and sensory symptoms including ataxic gait, muscle weakness, paresthesia, and loss of proprioception and vibration sense.

14. A 27-year-old nulligravid woman has had café au lait spots and multiple neurofibromas since birth. A diagnosis of neurofibromatosis is made. She is concerned that she might pass this condition on to her children. Her parents and three siblings do not have this disorder. The risk for neurofibromatosis in her children is closest to which of the following?

Your Answer: A) 50% for each child, since neurofibromatosis is autosomal dominant

Correct Answer: A) 50% for each child, since neurofibromatosis is autosomal dominant

Correct Answer: A

Correct Answer: A.

Neurofibromatosis type I is a genetically inherited disorder caused by a defect in NF1, the gene which encodes neurofibromin on chromosome 17. It follows an autosomal dominant pattern of inheritance. It is characterized by a constellation of cutaneous manifestations (eg, neurofibromas, plexiform neurofibromas, café au lait spots, and axillary freckling), a variety of central nervous system tumors including optic nerve glioma and iris hamartoma (ie, Lisch nodule), and bony abnormalities such as long bone dysplasia and sphenoid wing dysplasia. Patients are also at higher risk for developing astrocytomas, brainstem gliomas, and soft tissue sarcomas. Because it follows an autosomal dominant pattern of inheritance, only one allele is needed to develop the condition. Thus, the chance of this patient passing on the dominant allele containing the mutation on chromosome 17 is 50% for each child.

Incorrect Answers: B, C, D, and E.

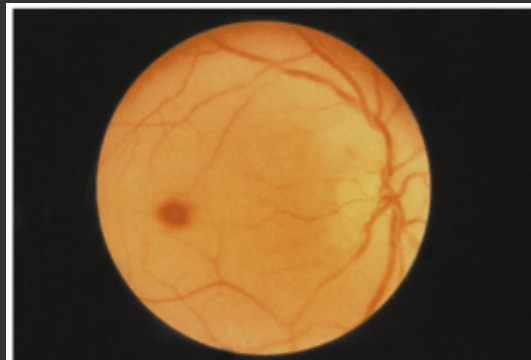
25% for each child (Choice B) would be the chance of a child developing an autosomal recessive disease if both his or her parents were carriers. The defect in NF1 that causes neurofibromatosis type I is inherited in an autosomal dominant pattern.

10% for each child (Choice C) could be the chance of a developing a disease which is controlled by several genes, or for which there is variable penetrance or expressivity, but neurofibromatosis type I is caused by a defect in a single gene, NF1, located on chromosome 17.

No risk for developing neurofibromatosis (Choice D) is incorrect as the disease is a genetic disorder. While the patient likely does represent a new mutation (Choice E) given that her parents are unaffected, this does not change the patient's risk for passing on her mutation to her children.

Educational Objective: Educational Objective: Neurofibromatosis type I is a genetically inherited disorder caused by a defect in NF1, the gene which encodes neurofibromin on chromosome 17. It follows an autosomal dominant pattern of inheritance. The chance of an affected patient having an affected child is 50%.

15. A 57-year-old man comes to the physician 2 hours after the sudden onset of visual loss in his right eye. He has no history of similar episodes. He has a 12-year history of type 2 diabetes mellitus treated with metformin and hypertension treated with lisinopril. He sustained a myocardial infarction 2 years ago. His pulse is 90/min, and blood pressure is 170/96 mm Hg. Findings on right fundusoscopic examination are shown. Which of the following is the most likely diagnosis?



Your Answer: D) Central retinal artery occlusion

Correct Answer: D) Central retinal artery occlusion

Correct Answer: D

Correct Answer: D.

This patient's presenting signs and symptoms of abrupt, painless vision loss and a pale fundus with normal foveal color are consistent with a diagnosis of central retinal artery occlusion (CRAO). CRAO commonly occurs as a result of embolic occlusion of the central retinal artery but may occur secondary to vasculitis, such as in giant cell arteritis (GCA). Patients may have a history of preceding amaurosis fugax. Examination shows profound diminishment of visual acuity, a relative afferent pupillary defect, and a pale fundus with preserved foveal color. CRAO is a medical emergency, as it is an ischemic stroke of the ophthalmic vasculature and indicates a high risk for associated ischemic stroke to the brain. Patients who present with CRAO should be admitted to a stroke center for emergent evaluation, including a search for a possible embolic source with echocardiography and duplex ultrasonography of the carotid arteries. A careful history and laboratory evaluation of erythrocyte sedimentation rate, serum concentration of C-reactive protein, and platelet count should be performed to evaluate for possible GCA. There is no proven medical or surgical therapy for CRAO, and the visual prognosis is poor.

Incorrect Answers: A, B, C, E, F, G, H, I, and J.

Acute angle-closure glaucoma (Choice A) typically presents with pain, conjunctival injection, a fixed, mid-dilated pupil, increased intraocular pressure, and no visible fundus abnormalities. If acute angle-closure glaucoma is not treated promptly it may result in profound, irreversible vision loss. Bacterial conjunctivitis (Choice B) presents with pain, conjunctival injection, and purulent discharge. Cataract (Choice C) is an age-related opacification of the crystalline lens and presents with slow vision loss over years that is more pronounced in dim lighting.

Diabetic retinopathy (Choice E) results from microvascular damage and retinal ischemia and is characterized by microaneurysms, retinal hemorrhage, retinal nerve fiber layer infarctions, retinal edema, exudate deposition, and retinal neovascularization. It may cause abrupt vision loss by causing retinal detachment or vitreous hemorrhage.

Endophthalmitis (Choice F) results from bacterial or fungal intraocular infection and presents with severe pain, conjunctival injection, hypopyon, and vitritis.

Idiopathic intracranial hypertension (Choice G) presents with headache, nausea, vomiting, tinnitus, and transient visual obscuration or blurry vision secondary to increased intracranial pressure.

Orbital melanoma (Choice H) may occur in the conjunctiva or uveal tract (the iris, ciliary body, or choroid) and presents with decreased visual acuity, floaters, and visual field defects.

Sarcoidosis (Choice I) may present with orbital mass lesions, commonly of the lacrimal gland, or with uveitis characterized by keratic precipitates of the corneal endothelium and granulomatous nodules of the iris.

Temporal arteritis (Choice J), also known as giant cell arteritis, is a vasculitis of medium and large vessels in elderly patients and commonly presents with headache, jaw claudication, proximal muscle stiffness, and sudden, irreversible blindness. Patients typically experience amaurosis fugax prior to the onset of abrupt, irreversible vision loss.

Educational Objective: Educational Objective: Central retinal artery occlusion frequently occurs secondary to embolic occlusion of the central retinal artery in carotid or cardiac valvular disease. Patients present with acute, painless vision loss and examination findings including diminished visual acuity, relative afferent pupillary defect, and a pale fundus with preserved foveal color. Patients who present with central retinal artery occlusion should be admitted to a stroke center for emergent evaluation.

16. A 52-year-old woman comes to the physician because of facial numbness for 2 weeks. Her symptoms began as a "funny feeling" over a patch of skin on the left side of her chin and have gradually progressed to a "novocaine" sensation. She has had no facial pain or numbness of the gums or oral mucosa. She has a 3-year history of breast cancer metastatic to the lungs treated with tamoxifen. Examination shows supraclavicular adenopathy. Sensation to pinprick, light touch, and temperature is absent over the left side of the chin. Neurologic examination shows no other abnormalities. Which of the following is the most likely location of the abnormality?

Your Answer: D) Mandible

Correct Answer: D) Mandible

Correct Answer: D

Correct Answer: D.

Numb chin syndrome, or mental neuropathy, is an indication of bony metastasis to the mandible in this patient with a history of breast cancer. The mandibular (V_3) branch of the trigeminal nerve passes through the bony canal of the mandible, branches to form the inferior alveolar nerve, and branches again proximal to the mental foramen to form the mental nerve. The mental nerve branch affected by mandibular bony metastases provides sensory innervation to the ipsilateral skin of the anterior chin (another unaffected mental nerve branch provides sensory innervation to the ipsilateral lower lip and buccal gingiva). Compression of the mental nerve branch by bony metastasis of the mandible leads to anesthesia of the chin. Numb chin syndrome may indicate systemic malignancy, autoimmune disease (eg, multiple sclerosis), or may be related to local infection, trauma, or dental disease. This patient should receive full-body imaging to evaluate for metastatic spread.

Incorrect Answers: A, B, C, and E.

Brachial plexopathy (Choice A) typically presents with weakness, numbness, and/or paresthesia of the shoulder, arm, and/or hand. The chin would be unaffected.

Cavernous sinus thrombosis (Choice B) classically follows an infection or injury involving a structure drained by the cavernous sinus (eg, sinusitis), and results in cranial nerve deficits involving those that traverse the cavernous sinus (generally III, IV, V_1 or V_2 , or VI). Classic symptoms include ipsilateral orbit pain, proptosis, periorbital edema, and impairment in extraocular movement, pupil reactivity, or symmetry.

A mass of the cerebellopontine angle (Choice C) (eg, acoustic neuroma, meningioma, glioma) typically affects the vestibulocochlear nerve and may affect other cranial nerves (eg, trigeminal, abducens, facial nerves). The entirety of the trigeminal nerve would be affected rather than the mental branch specifically.

The parietal lobe (Choice E) is a large brain region that processes sensory information and promotes spatial awareness of the body. Damage to the parietal lobe can result in symptoms such as agraphia, acalculia, finger agnosia, and left-right disorientation (dominant hemisphere, known as Gerstmann syndrome) or hemineglect (non-dominant hemisphere).

Educational Objective: Educational Objective: Numb chin syndrome, or mental neuropathy, may be caused by systemic malignancy, autoimmune disease, or local dental processes. In numb chin

syndrome secondary to malignancy, bony metastasis to the mandible leads to compression of a branch of the mental nerve, resulting in anesthesia of the ipsilateral chin.

17. A 32-year-old woman with AIDS comes to the physician because of a 2-week history of painful burning sensations in her hands and feet. Examination shows no abnormalities. Neurologic examination shows no focal findings. Which of the following is the most appropriate initial pharmacotherapy?

Your Answer: C) Oral nortriptyline

Correct Answer: C) Oral nortriptyline

Correct Answer: C

Correct Answer: C.

Peripheral neuropathy is a frequent finding in patients with HIV and may occur either secondary to direct viral infection of peripheral nerves or secondary to neurotoxicity associated with antiretroviral medications, frequently medications in the nucleoside analog reverse-transcriptase inhibitor (NRTI) class. HIV-associated neuropathy most frequently presents with symmetric, distal polyneuropathy characterized by sensations of burning, tingling, or numbness in a stocking-glove distribution, as well as reduced vibratory or pinprick sensation. As with other forms of distal polyneuropathy, tricyclic antidepressants such as nortriptyline are effective as first-line treatment. Other effective treatments include gabapentin, pregabalin, lamotrigine, and topical capsaicin or lidocaine.

Incorrect Answers: A, B, D, and E.

Oral fluoxetine (Choice A) is a selective serotonin reuptake inhibitor (SSRI) and is not typically used as a first-line agent for peripheral neuropathic pain. While some SSRIs, such as duloxetine, may be effective for the treatment of neuropathic pain, fluoxetine is not effective in treating nerve pain.

Oral lorazepam (Choice B) is a short-acting benzodiazepine and is not recommended for the treatment of neuropathic pain because of its significant adverse effects and risk for dependency.

Oral prednisone (Choice D) is a corticosteroid. While corticosteroids, including prednisone, may be effective for some forms of neuropathic pain, this is not an appropriate initial pharmacotherapy for this patient given the significant adverse effects of corticosteroids, which includes immunosuppression.

Topical lidocaine to the affected areas (Choice E) is not the best initial pharmacotherapy for peripheral neuropathy because they are less effective in comparison with systemic oral therapy such as tricyclic antidepressants.

Educational Objective: Educational Objective: Peripheral neuropathy is a frequent finding in patients with HIV and may occur either secondary to direct viral infection of peripheral nerves or secondary to neurotoxicity associated with antiretroviral medications. Tricyclic antidepressants, such as nortriptyline, are an effective first-line therapy.

18. A 37-year-old man comes to the physician because of a 1-week history of moderate low back pain. He thinks he may have hurt his back while doing yard work. His pain moderately improves with rest; he has been spending a great amount of time in bed. He has not had other symptoms, weight loss, or a recent infection. He has no history of serious illness. His only medication is ibuprofen for the pain. He does not use intravenous drugs or anabolic steroids. He is 178 cm (5 ft 10 in) tall and weighs 102 kg (225 lb); BMI is 32 kg/m². Examination shows mild paraspinal muscle tenderness. Neurologic examination shows no focal findings. Which of the following is the most appropriate next step in management?

Your Answer: D) Continuation of ibuprofen therapy

Correct Answer: D) Continuation of ibuprofen therapy

Correct Answer: D

Correct Answer: D.

Low back pain is a common presenting complaint and it can be difficult to identify an exact underlying cause. Most nonspecific low back pain is attributed to musculoskeletal injury from exertion, overuse, or trauma. The majority of these cases will resolve with conservative management, which includes activity modification, stretching, over-the-counter analgesics such as nonsteroidal anti-inflammatory drugs (NSAIDs), massage, acupuncture, and heat therapy. Certain red flag signs and symptoms should prompt more urgent evaluation for a cause, especially if there is concern for infection, malignancy, or fracture. These include new fevers, intravenous drug use, focal neurologic deficits (such as urinary retention, saddle anesthesia, bowel incontinence which may suggest cauda equina syndrome), progressive lower extremity neurologic symptoms, current or recent cancer history, advanced age, systemic corticosteroid use, and a significant traumatic mechanism. In this young, otherwise healthy patient with a mild inciting event, the next best step in management is continuation of ibuprofen therapy.

Incorrect Answers: A, B, C, and E.

Bed rest for 10 days (Choice A) is not recommended for acute low back pain, as patients tend to have increased pain and a slower return to normal function. Activity modification for symptom relief should be advised, with encouragement to continue normal activities of daily living and non-exacerbating exertion.

X-ray of the lumbar spine (Choice B) and MRI of the lumbar spine (Choice C) should be reserved for patients who present with red flag signs and symptoms to evaluate for evidence of fracture, infection, or malignancy. For otherwise healthy patients with back pain that has persisted for less than 4 weeks, imaging studies do not generally change management.

Oxycodone therapy as needed for pain (Choice E) should be reserved for severe or refractory cases of acute low back pain.

Educational Objective: Educational Objective: Acute low back pain is a common presenting complaint in multiple medical settings. General management for pain persisting less than 4 weeks is conservative with over-the-counter analgesics, activity modification, and nonpharmacologic pain relief options. Initial imaging is not typically recommended in the absence of signs or symptoms suggestive of infection, malignancy, spinal cord compromise, or vertebral fracture.

19. A 37-year-old woman comes to the physician for a follow-up examination 3 weeks after operative repair of a circle of Willis berry aneurysm. Her husband says that her behavior has changed since the operation. She is unmotivated and just sits around all day like a "bump on a log." She has not had depressed mood. Examination shows psychomotor retardation. She is oriented to person, place, and time. She is cooperative and her speech is fluent and grammatical, but there is little spontaneous speech during conversation. Which of the following cerebral cortical regions is most likely affected?

Your Answer: A) Frontal

Correct Answer: A) Frontal

Correct Answer: A

Correct Answer: A.

The patient is exhibiting symptoms of frontal lobe dysfunction. The frontal lobe is the largest lobe of the cerebral cortex, separated from the parietal lobe by the central sulcus. The major areas of the frontal lobe are the primary motor cortex, premotor cortex, prefrontal cortex, Broca area, and the frontal eye fields. Symptoms depend on the location of the lesion. Involvement of the primary motor cortex and premotor cortex result in impaired motor function of the contralateral side of the body and psychomotor retardation. Disruption of Broca area may cause expressive aphasia with retention of language comprehension. Lesions of the frontal eye fields result in ipsilateral deviation of the eyes. Dysfunction of the prefrontal cortex can result in disinhibition, apathy, personality changes, aggressive behavior, and impairments in judgement, concentration, and orientation, as exhibited by this patient. Patients should be evaluated for other medical causes of cognitive impairment, such as vitamin B₁₂ (cobalamin) deficiency, hypothyroidism, syphilis infection, and intoxication. In this patient with a recent invasive procedure, CT scan or MRI of the brain is likely to show the diagnosis.

Incorrect Answers: B, C, D, and E.

Insular (Choice B) cortex lesions may result in impairments in emotional and/or sensory processing, motor control, and speech. Patients may display a progressive expressive dysphasia characterized by non-fluent and grammatically incoherent speech.

Occipital (Choice C) lobe lesions typically present with difficulties in processing visual sensory inputs. Patients may complain of visual field cuts depending on the precise location of the lesion.

Parietal (Choice D) lobe lesions manifest differently depending on the side affected. Dominant parietal lobe lesions, most commonly on the left side, may result in Gerstmann syndrome, characterized by agraphia, aphasia, agnosia, acalculia, and right-left confusion. Nondominant parietal lobe lesions typically manifest with hemineglect of the contralateral side.

Temporal (Choice E) lobe lesions may involve Wernicke area, the limbic system, and/or the amygdala. Wernicke area lesions result in receptive aphasia with loss of language comprehension. Disruption of the limbic system results in dysfunctional emotional processing, and may present with excessive fear, anxiety, and behavioral changes.

Educational Objective: Educational Objective: The frontal lobe contains the primary motor cortex, premotor cortex, Broca area, frontal eye fields, and prefrontal cortex, which is involved in the regulation of behavior, inhibition, and other executive functions. Injury to the frontal lobe may result

in personality or behavior changes, disinhibited behavior, expressive aphasia, and motor dysfunction.

20. A 10-day-old newborn is brought to the emergency department by ambulance because of a seizure that involved clonic jerking of the left arm. He had been well until 2 days ago when he began sleeping somewhat more but continued to feed well. Pregnancy, labor, and delivery were uncomplicated. Vital signs are within normal limits. He is difficult to awaken but moves all extremities when stimulated. He is hypotonic but hyperreflexic. Babinski sign is present. During examination, he has a clonic seizure of the right lower extremity. A CT scan of the head shows no abnormalities. Cerebrospinal fluid analysis shows:

Glucose 55 mg/dL

Protein 110 mg/dL

Leukocyte count 85/mm³

Erythrocyte count 860/mm³

A Gram stain is negative. Which of the following is the most likely diagnosis?

Your Answer: A) Congenital infection with cytomegalovirus

Correct Answer: F) Neonatal herpes simplex encephalitis

Correct Answer: F

Correct Answer: F.

Neonatal herpes simplex virus (HSV) encephalitis is the most likely diagnosis in this 10-day old infant with new onset seizures, a normal head CT scan, and cerebrospinal fluid (CSF) showing increased protein and leukocyte count with a low-normal glucose concentration, which is consistent with aseptic (nonbacterial) meningitis. Further suggestive is the presence of erythrocytes in the CSF which is often seen in herpetic encephalitis. Most cases of HSV encephalitis are caused by HSV-2, although infection with HSV-1 occurs and is treated in the same way. Most patients acquire the virus via transmission from the mother, often during the third trimester, with encephalitis developing usually within a week. Symptoms are nonspecific and include fever, lethargy, poor feeding, and occasionally seizures or focal neurologic deficits. Concomitant oral lesions may occur, but their absence does not rule out the diagnosis. The CSF findings include increased protein, lymphocytic pleocytosis, and a normal or low-normal glucose concentration. In many instances the causative virus can be identified by polymerase chain reaction (PCR) of the CSF, although empiric therapy with intravenous acyclovir is warranted prior to diagnosis. Failure to recognize and treat infection can result in organ dysfunction and death, or long-term neurologic sequelae.

Incorrect Answers: A, B, C, D, and E.

Congenital infection with cytomegalovirus (Choice A) is associated with petechiae, bilateral sensorineural hearing loss, jaundice, microcephaly, hepatosplenomegaly, and hypotonia. The patient lacks these other features. Intracranial calcifications may be present on CT scan.

Congenital infection with syphilis (Choice B) presents typically within five weeks with hepatomegaly, maculopapular rash, rhinitis, and skeletal abnormalities including periostitis or bone demineralization. CSF findings may show pleocytosis and increased protein but would be positive for *Treponema pallidum*.

Congenital infection with toxoplasmosis (Choice C) may present with chorioretinitis, hydrocephalus, and seizures. *Toxoplasma gondii* can be detected in the CSF. Intracranial calcifications may be present on CT scan.

Neonatal *Escherichia coli* meningitis (Choice D) may present with fever, lethargy, and hypotonia, and CSF findings would include a neutrophilic pleocytosis with a decreased glucose concentration. Treatment is with ampicillin if the organism is sensitive, or with a cephalosporin combined with gentamicin.

Neonatal group B streptococcus (GBS) meningitis (Choice E) presents similarly to E. coli meningitis but is more common in patients born to mothers who are colonized with GBS who have not received prophylactic treatment. Colonization is usually detected and treated prior to delivery, and if the status is unknown, prophylactic antibiotics are given.

Educational Objective: Educational Objective: HSV encephalitis should be suspected in neonates who develop fever, lethargy, and/or neurologic symptoms, including focal deficits or seizures shortly after birth. A history of HSV infection in the mother may or may not be present. Evaluation of the CSF typically shows an increased protein concentration with a lymphocytic pleocytosis but normal glucose concentrations. Therapy is with intravenous acyclovir.

21. A 47-year-old woman comes to the physician because of increasingly frequent headaches during the past 3 months. The headaches are throbbing, occur only on the right side, and are associated with nausea, vomiting, and sensitivity to light and sound. She has had one or two headaches yearly since the age of 17 years but now has two or three weekly. She is a schoolteacher and has had no changes in work, lifestyle, or environment. Four months ago, conjugated estrogen therapy was begun for treatment of menopausal symptoms. Her temperature is 37°C (98.6°F), pulse is 76/min, respirations are 14/min, and blood pressure is 135/80 mm Hg. Neurologic examination shows no abnormalities. Which of the following is the most appropriate next step in management?

Your Answer: A) CT scan of the head

Correct Answer: C) Discontinuation of conjugated estrogen therapy

Correct Answer: C

Correct Answer: C.

Migraine is a common condition that presents with recurrent primary headaches. Migraine headaches are typically severe, unilateral, throbbing, and associated with nausea, vomiting, photophobia, and phonophobia. The pain is usually located behind or adjacent to the eye. Some patients additionally experience an aura before or during the headache, which may present with vision loss followed by the perception of scintillating shapes or lines and/or paresthesia. Migraine triggers include stress, menstruation and changes in hormonal concentrations, weather changes, fasting, and alcohol. Sleep may improve migraines. Patients who experience migraines associated with conjugated estrogen therapy may require discontinuation of therapy or a reduction in dose. Migraines are postulated to arise from deficient serotonin inhibition of descending pain pathways. Acute migraine treatment may include triptans and/or nonsteroidal anti-inflammatory drugs (NSAIDs). For patients with disabling or frequent migraines, prophylactic treatment is also recommended and includes β -adrenergic blockers, antidepressants such as tricyclic antidepressants and serotonin-norepinephrine reuptake inhibitors, and anticonvulsants.

Incorrect Answers: A, B, D, and E.

CT scan of the head (Choice A) is unnecessary given a likely cause for the patient's migraines and absence of any neurologic abnormalities.

Butalbital therapy combined with acetaminophen and caffeine (Choice B) may be appropriate in the treatment of a primary migraine, however, it carries risk given the adverse effects and dependency associated with barbiturates. Removal of likely offending agents is a more appropriate first step for the treatment of this patient's likely-triggered migraine.

EEG (Choice D) is useful for the diagnosis of seizure disorders but is rarely useful for the evaluation or treatment of migraine.

Lumbar puncture (Choice E) is useful for the diagnosis of meningitis or idiopathic intracranial hypertension (IIH). While IIH may be associated with hormonal therapy, this patient's symptoms of unilateral, throbbing headache, photophobia and phonophobia are more suggestive of migraine. The headaches associated with IIH are more commonly holocephalic and associated with transient visual obscurations and pulsatile tinnitus.

Educational Objective: Educational Objective: Migraines are typically unilateral, pulsating headaches that are associated with nausea, vomiting, photophobia, and phonophobia, and may present with visual or sensory aura. Changes in hormone concentrations are a common migraine trigger. Patients who experience migraines associated with conjugated estrogen therapy may require discontinuation of therapy or a reduction in dose.

22. A 67-year-old man is brought to the physician by his wife because of a 1-month history of deafness, ringing in both ears, dizziness, and difficulty with balance. He also has had an occasional sensation that the room is spinning. He was discharged from the hospital 21 days ago after prolonged treatment for pneumonia and sepsis with intravenous ceftriaxone and gentamicin. He has chronic obstructive pulmonary disease, type 2 diabetes mellitus, and congestive heart failure. He has a history of tuberculosis 10 years ago. Medications include digoxin, metformin, and losartan. His pulse is 84/min, respirations are 20/min, and blood pressure is 140/85 mm Hg. Examination shows nystagmus on lateral gaze bilaterally. There is severe loss of hearing bilaterally. Air conduction is greater than bone conduction. Results of Weber test are normal. Muscle strength, deep tendon reflexes, and sensory findings are normal. His gait is broad-based, and he is unable to walk in tandem. Which of the following is the most likely diagnosis?

Your Answer: A) Aminoglycoside-induced ototoxicity

Correct Answer: A) Aminoglycoside-induced ototoxicity

Correct Answer: A

Correct Answer: A.

Ototoxicity is a frequent adverse effect of aminoglycosides, such as gentamicin, which exert bactericidal effects by binding the bacterial 30S ribosomal subunit. Aminoglycosides accumulate in the inner ear endolymph and perilymph, and can cause cochlear ototoxicity and vestibulotoxicity, which is typically irreversible. The risk for ototoxicity associated with aminoglycosides may be as high as 10% in some patient populations. While there is an increased risk for ototoxicity with repeat dosing or with high doses, ototoxicity may occur with even the first dose of aminoglycoside. While aminoglycosides are highly effective against a broad range of Gram-negative bacterial infections, such as with *Escherichia coli*, *Klebsiella*, *Proteus*, *Pseudomonas*, and *Serratia* species, their use is limited by their significant adverse effects. In addition to ototoxicity and vestibulotoxicity, aminoglycosides are also associated with nephrotoxicity.

Incorrect Answers: B, C, D, and E.

Cerebellopontine angle tumor (Choice B), such as an acoustic neuroma, meningioma, or glioma, typically affects the vestibulocochlear nerve and other cranial nerves (eg, trigeminal, abducens, facial nerves). Patients may present with hearing loss and vertigo; sensory and motor dysfunction of the face and/or extraocular muscles are also possible.

Meniere disease (Choice C) is a condition of episodic vertigo and sensorineural hearing loss that results from pressure accumulation in the endolymph of the inner ear. This patient's lack of episodic symptoms suggests against this diagnosis.

Pontine infarction (Choice D) may occur as a sequela of basilar artery occlusion and present with truncal ataxia, diplopia, nystagmus, locked-in state, or coma.

Vestibular neuronitis (Choice E), also known as labyrinthitis, is caused by inflammation of the vestibular labyrinths and presents with vertigo and nystagmus over a period of several weeks. Patients typically do not present with hearing loss.

Educational Objective: Educational Objective: Aminoglycosides, such as gentamicin, carry a significant risk for ototoxicity and vestibulotoxicity through accumulation in cochlear endolymph and

perilymph, and may cause toxicity with even the first dose. Ototoxicity and vestibulotoxicity associated with aminoglycosides are typically irreversible.

23. A 72-year-old woman comes to the physician because of a 2-day history of pain in the left thigh and flank and weakness of the left leg. She has a 30-year history of type 2 diabetes mellitus currently controlled with glyburide. Examination shows weakness of the left iliopsoas and quadriceps muscles. Deep tendon reflex of the left quadriceps muscle is decreased. Sensation to pinprick is decreased over the left lateral thigh and medial calf. Which of the following is the most likely location of the lesion?

Your Answer: B) Femoral nerve

Correct Answer: B) Femoral nerve

Correct Answer: B

Correct Answer: B.

Lower extremity neuropathies can be caused by nerve compression, ischemia, inflammation, degeneration, infection, or transection. Diabetic patients may develop spontaneous femoral neuropathies secondary to inflammation and ischemia. The femoral nerve arises from L2 to L4 and exits the lumbosacral plexus by passing under the inguinal ligament. Neuropathy of the femoral nerve or its roots presents with unilateral quadriceps and iliopsoas weakness, decreased sensation over the anteromedial thigh and medial calf (the latter via its saphenous branch), and absent patellar tendon reflex.

Incorrect Answers: A, C, D, E, F, G, H, I, J, K, L, and M.

The axillary nerve (Choice A) arises from C5 and C6. Injury to the axillary nerve presents with weakness of shoulder abduction, as well as numbness over the deltoid region.

The lateral femoral cutaneous nerve (Choice C) provides sensation to the anterolateral aspect of the thigh.

The median nerve (Choice D) arises from C5 to T1. Injury to the median nerve causes weakness of the wrist flexors and muscles of the thenar eminence, and decreased sensation over the thenar eminence and the palmar aspect of the lateral 3½ fingers.

The peroneal nerve (Choice E) is a continuation of the sciatic nerve that supplies innervation to the lateral leg. Injury causes weakness of muscles within the lateral compartment of the leg. A peroneal nerve lesion also causes numbness of the posterolateral side of the leg and the dorsolateral aspect of the foot.

The radial nerve (Choice F) arises from C5 to T1. Injury to the radial nerve causes weakness of the elbow, wrist, and finger extensors, as well as numbness of the dorsal aspect of the thumb and fingers.

C6 Radiculopathy (Choice G) presents with pain of the neck, shoulder, and lateral arm and hand. It is associated with weakness of shoulder abduction and external rotation, elbow flexion, and forearm supination and pronation. Numbness of the lateral forearm, thumb, and index finger, as well as hypoactive biceps and brachioradialis reflexes, are present.

C7 Radiculopathy (Choice H) presents with neck, shoulder, and middle finger pain. It is associated with weakness of the wrist and finger extensors, as well as weakness of finger abduction and adduction. Numbness is primarily present over the medial arm and hand, as well as the medial fingers.

L5 Radiculopathy (Choice I) presents with back pain that radiates down the lateral aspect of the leg and foot. It is associated with weakness of foot dorsiflexion, toe extension, and foot eversion and inversion. Numbness occurs over the lateral leg and dorsum of the foot.

S1 Radiculopathy (Choice J) presents with back pain radiating down the posterior leg and is associated with weakness of foot extension, foot inversion, toe flexion, and plantar flexion.

Numbness occurs over the posterior leg and lateral foot, and loss of the Achilles tendon reflex is common.

Sciatic nerve (Choice K) injuries present with leg pain and weakness of the lower leg and the hamstrings. Sensation will be decreased over the posterolateral leg, and the Achilles tendon reflex is typically absent.

The tibial nerve (Choice L) supplies the posterior aspect of the leg, and injury causes weakness of plantar flexion, toe flexion, and inversion. The tibial nerve also supplies sensation to the posterolateral leg and foot.

The ulnar nerve (Choice M) arises from C8 to T1, and injuries to this nerve lead to weakness of the finger and wrist flexion and finger abduction and adduction, as well as numbness over the surface of the medial 1½ fingers.

Educational Objective: Educational Objective: Femoral neuropathies present with unilateral quadriceps and iliopsoas weakness, decreased sensation over the anteromedial thigh and medial leg, and absent patellar tendon reflexes.

24. A 72-year-old man comes to the physician because of a 2-month history of a progressive tremor of his hands. He and his family have noticed that his movements have become more shaky during this period and that he has been walking more slowly during the past year. He has hypertension. Current medications include hydrochlorothiazide and aspirin. There is no family history of serious illness. He does not smoke or drink alcohol. His pulse is 56/min, and blood pressure is 140/90 mm Hg. Examination shows a fine tremor of the hands at rest; the tremor is exacerbated with movement. There is cogwheel rigidity of the upper extremities. His gait is slow and shuffling. Mental status examination shows no abnormalities. Which of the following is the most appropriate initial pharmacotherapy?

Your Answer: B) Carbidopa-levodopa

Correct Answer: B) Carbidopa-levodopa

Correct Answer: B

Correct Answer: B.

Carbidopa-levodopa is a dopamine agonist used to treat Parkinson disease by increasing the amount of dopamine available in the nigrostriatal pathway. Carbidopa and levodopa are given together in order to increase availability of levodopa to the brain. Levodopa is converted to dopamine by the enzyme dopa decarboxylase in the peripheral circulation and the brain. Dopamine is not able to cross the blood-brain barrier (BBB), so any levodopa which is converted to dopamine peripherally is unable to effect change in the basal ganglia. Carbidopa indirectly increases dopamine availability in the basal ganglia by inhibiting dopa decarboxylase peripherally. In this way, more levodopa is able to cross the BBB and be converted to dopamine in the brain, where it is then able to act on the nigrostriatal pathway. Increased dopamine leads to improvement of Parkinson disease symptoms including the tremor, rigidity, akinesia, and postural instability shown by this patient.

Incorrect Answers: A, C, D, and E.

Atenolol (Choice A), a β -adrenergic blocker, and primidone (Choice E), a barbiturate, are both used in the treatment of essential tremor. This patient's resting tremor accompanied by unsteady gait and cogwheel rigidity is more consistent with Parkinson disease.

Donepezil (Choice C) is an anticholinesterase inhibitor which indirectly increases the concentration of the neurotransmitter acetylcholine. It is used to treat Alzheimer disease. Acetylcholine and dopamine have opposing effects in the basal ganglia related to movement such that decreasing acetylcholine has the same effect as increasing dopamine. Thus, medications which decrease, not increase, acetylcholine are used in Parkinson disease.

Nifedipine (Choice D) is a calcium channel blocker and does not play a role in the treatment of Parkinson disease.

Primidone (Choice E) is a barbiturate used in the treatment of essential tremor. This patient's resting tremor, rigidity, and bradykinesia are consistent with Parkinson disease, not essential tremor.

Educational Objective: Educational Objective: Carbidopa-levodopa is the first-line treatment used in Parkinson disease. Carbidopa cannot cross the blood-brain barrier. It therefore increases the bioavailability of levodopa to the brain without inhibiting the conversion of levodopa to dopamine in the brain.

25. A 52-year-old man comes to the physician because of a 2-hour history of a diffuse, bilateral, dull headache. He has hypertension and has been poorly compliant with his medication. He is alert but confused. His blood pressure is 240/150 mm Hg. Funduscopic examination shows hemorrhages and exudates. Which of the following is the most likely diagnosis?

Your Answer: C) Hypertensive encephalopathy

Correct Answer: C) Hypertensive encephalopathy

Correct Answer: C

Correct Answer: C.

Hypertensive encephalopathy may result from acute and severe increases in blood pressure. It potentially manifests with headache, vomiting, agitation, altered mental status, and seizure, and may progress to obtundation and coma. Visual symptoms may result from cortical dysfunction or from hypertensive retinopathy, including retinal hemorrhages and exudates. Treatment includes carefully lowering the blood pressure while avoiding excessively rapid drops in blood pressure, which can cause end-organ ischemia. In general, a decrease in the mean arterial pressure of 25% within the first hour is appropriate, with gradual correction over the next day or more following. Sodium nitroprusside is a potent vasodilator and is a preferred agent because of its short half-life, allowing it to be titrated via intravenous infusion.

Incorrect Answers: A, B, D, E, F, and G.

Cerebral hemorrhage (Choice A) can be idiopathic or secondary to trauma, aneurysms, arteriovenous malformations, tumors, or hypertension. Patients can present with a severe headache that can be associated with vomiting. In a patient with noncompliance with antihypertensive medications, hypertensive encephalopathy is more likely.

Cerebral infarction (Choice B), or stroke, can occur as a sudden change in mental status as well as focal neurologic deficits and upper motoneuron signs. Patients typically present with acute focal neurologic deficits, especially on one side of the body, and may have speech difficulty, dysarthria, or aphasia.

Idiopathic intracranial hypertension (Choice D) or pseudotumor cerebri is an idiopathic increase in intracranial pressure (ICP) that typically occurs in obese women of childbearing age and presents with headaches that may be postural, visual deficits caused by pressure on the optic nerve, and nausea/vomiting (symptoms of increased ICP).

Migraine (Choice E) is a type of primary headache that usually does not cause neurologic deficits and would be unlikely to cause confusion or altered mental status.

Subarachnoid hemorrhage (Choice F) typically presents with an acute, severe headache (often described as the worst headache of the patient's life), along with neck stiffness, photophobia, and possible stupor or coma in severe cases. Retinal hemorrhage, exudates, and headaches with severely increased blood pressure are more indicative of hypertensive encephalopathy.

A subdural hematoma (Choice G) can occur after blunt head trauma as a result of shearing injury of bridging veins, especially in patients with cerebral atrophy such as elderly patients, or those with severe alcohol use disorder. On imaging, subdural hematoma appears as a crescent-shaped fluid collection abutting the internal surface of the skull.

Educational Objective: Educational Objective: Hypertensive encephalopathy may result from acute and severe increases in blood pressure. It potentially manifests with headache, vomiting, agitation, altered mental status, and seizure, and may progress to obtundation and coma.

26. A 67-year-old woman comes to the physician because of excessive sleepiness and fatigue. She has required more sleep for several years, but the problem has become increasingly worse during the past 7 months. She sometimes falls asleep in the middle of conversations and at dinner. She recently collapsed onto the floor of her apartment; she was aware of her surroundings but unable to move for 5 minutes. She has no history of seizures or other neurologic disorders. Her only medication is celecoxib for treatment of osteoarthritis. She appears well nourished. Physical examination shows limited range of motion in the lower extremities. There is swelling and crepitus of the knees. She has moderate pain on standing. An ECG, EEG, and 72-hour ambulatory ECG monitoring show no abnormalities. Polysomnography is pending. Which of the following is the most appropriate pharmacotherapy?

Your Answer: A) Methylphenidate

Correct Answer: A) Methylphenidate

Correct Answer: A

Correct Answer: A.

Methylphenidate is a non-amphetamine central nervous system stimulant medication that is used in the treatment of narcolepsy, which is the most likely diagnosis in this patient with fatigue and episodes of cataplexy. Methylphenidate works by blocking the reuptake of norepinephrine and dopamine into presynaptic neurons. The pathogenesis of narcolepsy involves the loss of orexin-A and orexin-B, which are neurotransmitters found primarily in the hypothalamus that promote wakefulness and inhibit rapid eye movement (REM) sleep. Loss of these neurotransmitters permits abnormal transition from wakefulness to REM sleep with subsequent development of hallucinations and other features of narcolepsy including cataplexy and sleep paralysis. Diagnosis can be made in patients with daytime somnolence and cataplexy or, in cases where cataplexy is not present, with sleep latency testing. Polysomnography is appropriate to support the diagnosis and to rule out similar sleep-related conditions such as sleep apnea. Occasionally patients undergo lumbar puncture to directly measure concentrations of orexin-A in the cerebrospinal fluid, but this is generally not required for the diagnosis. Pharmacotherapy is directed at promoting wakefulness, and methylphenidate is one relatively safe and effective stimulant used to do this.

Incorrect Answers: B, C, D, and E.

Mirtazapine (Choice B) is an antidepressant that increases presynaptic release of serotonin and norepinephrine. It tends to cause drowsiness and weight gain and is occasionally used to treat patients with depression and sleep disturbances.

Nortriptyline (Choice C) is a tricyclic antidepressant that is used to treat major depressive disorder and is used off label for a number of conditions including chronic pain, postherpetic neuralgia, and diabetic neuropathy. It can cause drowsiness.

Trazodone (Choice D) inhibits the reuptake of serotonin and is also a potent histamine blocker. It is used to treat depression but also causes drowsiness and is used off label to treat insomnia.

Zaleplon (Choice E) binds to γ -aminobutyric acid (GABA) receptors and is used to treat insomnia. It is not useful in the treatment of patients with narcolepsy.

Educational Objective: Educational Objective: Narcolepsy should be suspected in patients with daytime somnolence, hypnagogic hallucinations, or cataplexy. Diagnosis can be made clinically if cataplexy is present but otherwise requires sleep latency testing, which shows a shorter time from wakefulness to the initiation of REM sleep. Treatment is initially with stimulants such as methylphenidate, which promote wakefulness.

27. A 72-year-old right-handed woman is brought to the emergency department 30 minutes after the onset of double vision and right-sided weakness that began while she was eating breakfast. She has type 2 diabetes mellitus and hypertension. Medications include glyburide and atenolol. Her temperature is 37°C (98.6°F), pulse is 80/min and regular, and blood pressure is 170/105 mm Hg. She is alert and oriented. Her speech is fluent but mildly slurred. She can follow complex commands. Cranial nerve examination shows left ptosis. Adduction and elevation of the left eye are impaired. Ocular movements are full on the right. The right pupil is 3 mm, and the left pupil is 5 mm. The right pupil reacts normally, but the left pupil does not constrict either to direct or consensual light stimulation. There is drooping of the right side of the mouth. Muscle strength is normal in the left upper and lower extremities and is 4/5 in the right upper and lower extremities. Deep tendon reflexes are 3+ on the right and 2+ on the left. Babinski sign is present on the right and absent on the left. There is no ataxia. Sensation is intact. Which of the following is the most likely location of this patient's lesion?

Your Answer: D) Left midbrain

Correct Answer: D) Left midbrain

Correct Answer: D

Correct Answer: D.

This patient likely has a lesion of the left midbrain that affects the cerebral peduncle and the oculomotor nerve, which causes contralateral hemiparesis and ipsilateral oculomotor palsy, known as Weber syndrome. The cerebral peduncle refers to the anterior portion of the midbrain and includes the crus cerebri, which contains the corticospinal tract and is supplied by paramedian branches of the posterior cerebral artery. A brainstem lesion of the corticospinal tract leads to contralateral weakness in an upper motoneuron pattern of dysfunction (eg, spastic paralysis, Babinski sign, and hyperreflexia). The corticobulbar tract (located within the cerebral peduncle) may also be affected in Weber syndrome, leading to facial droop and weakness. The oculomotor nerve (cranial nerve III) is located anteriorly between the two cerebral peduncles and may be affected by strokes of the paramedian branches of the posterior cerebral artery, leading to ipsilateral dysfunction of the extraocular muscles and consequent diplopia, ptosis, and primary gaze palsy along with a deficit in pupillary constriction. Cerebral infarctions, or strokes, occur as a result of ischemic or hemorrhagic loss of blood supply to the brain. Risk factors for stroke include smoking, hypertension, diabetes mellitus, carotid or intracranial atherosclerotic disease, history of hypercoagulability, atrial fibrillation, and advanced age. Classically, strokes manifest as a neurologic deficit related to the affected part of the brain.

Incorrect Answers: A, B, C, and E.

Lesions of the left cerebral cortex (Choice A) may cause contralateral hemiparesis of the upper or lower extremities and facial weakness. However, oculomotor palsy would be atypical.

Lesions of the left internal capsule (Choice B) occur in lacunar infarcts. The internal capsule primarily contains the descending corticospinal tract, though it also contains axons of additional pathways, including third-order sensory neurons. Internal capsule lacunar infarcts can present with purely motor deficits that cause contralateral hemiparesis, ataxic hemiparesis, or clumsy hand-dysarthria.

The left medulla (Choice C) contains the nuclei of multiple cranial nerves, the spinal trigeminal nucleus, solitary nucleus, vestibular nuclei, and nucleus ambiguus, as well as tracts including the sympathetic, spinothalamic, and spinocerebellar tracts. Consequently, lesions of the lower medulla may lead to ipsilateral facial numbness, taste loss in the anterior tongue, vertigo with nystagmus,

dysphonia, dysarthria, dysphagia, Horner syndrome, pain and temperature sensation deficits in the contralateral body, and/or ipsilateral ataxia. This constellation of symptoms is known as Wallenberg syndrome.

The left pons (Choice E) contains several cranial nerve nuclei (ie, the vestibular nuclei, facial nucleus, spinal trigeminal nucleus, cochlear nuclei) along with the spinothalamic tract, corticospinal tract, sympathetic fibers, and the middle and inferior cerebellar peduncles. Consequently, lesions of the left pons lead to vertigo, nystagmus, facial paralysis, decreased lacrimation, decreased salivation, taste loss in the anterior tongue, hearing loss, decrease in pain and temperature sensation of the contralateral body, contralateral hemiparesis, ipsilateral limb ataxia, and/or dysmetria. This constellation of symptoms is known as lateral pontine syndrome.

Educational Objective: Educational Objective: In Weber syndrome, a stroke of the paramedian branches of the posterior cerebral artery affects the cerebral peduncle, which contains the corticospinal tract and neighboring oculomotor nerve. Patients typically present with ipsilateral oculomotor nerve palsy (eg, diplopia, ptosis, primary gaze palsy, pupillary constriction deficits) and contralateral hemiparesis with an upper motoneuron-pattern of dysfunction.

28. Over the past 3 months, a 40-year-old woman has had multiple 1- to 2-minute episodes of numbness and tingling that begin in the right hand and then spread to the right arm and the face. Neurologic examination shows normal findings. The most likely cause is a lesion at which of the following sites?

Your Answer: E) Parietal lobe

Correct Answer: E) Parietal lobe

Correct Answer: E

Correct Answer: E.

Focal neurologic deficits that start in one area of the body and spread are suggestive of partial seizure activity. Without an alteration in level of consciousness, this would be defined as a simple partial seizure manifesting with sensory symptoms, also called a simple partial somatosensory seizure. This type of seizure is most commonly associated with lesions of the parietal lobe as a result of the involvement of the primary somatosensory cortex, which processes sensory input from the contralateral side of the body. Patients typically present with numbness, tingling, warmth, and/or pain. Other manifestations of parietal lobe epileptic activity may include receptive aphasia, dyslexia, and other language/comprehension disturbances because of involvement of Wernicke area at the junction of the parietal and temporal lobes.

Incorrect Answers: A, B, C, D, F, G, and H.

Frontal lobe (Choice A) lesions more commonly present with complex partial seizures, with impaired awareness as a result of the disruption of normal frontal lobe cognitive functions. Simple partial seizure activity with motor symptoms may also occur because of involvement of the primary motor cortex.

Brainstem lesions involving the medulla (Choice B), midbrain (Choice C), or pons (Choice F) may result in focal seizure activity characterized by convulsive movements; however, brainstem involvement more typically occurs with generalized tonic-clonic seizures.

Occipital lobe (Choice D) lesions are uncommon causes of focal seizure activity, but typically present with visual manifestations such as visual hallucinations, cortical blindness, and deviated gaze.

Retina (Choice G) lesions are not likely to result in epileptiform activity affecting the upper extremity. The retina is an extension of the central nervous system, and lesions may result in visual disturbances.

Temporal lobe (Choice H) lesions may involve Wernicke area, the limbic system, and/or the amygdala. The temporal lobe is the most common location of focal seizure activity, typically presenting with complex partial seizures that may involve an aura, motor symptoms (especially motor automatisms such as lip-smacking or fidgeting), autonomic dysfunction, and/or confusion.

Educational Objective: Educational Objective: Focal neurologic deficits without loss of awareness are characterized as simple partial seizures. Pure sensory symptoms likely originate from an epileptiform focus or lesion in the primary somatosensory cortex located in the parietal lobe.

29. A 72-year-old man comes to the physician because of a 3-month history of increasingly depressed mood, fatigue, difficulty concentrating, and a decreased appetite resulting in a 2.3-kg (5-lb) weight loss. He was diagnosed with Parkinson disease 2 years ago. He also has hypertension and benign prostatic hyperplasia. His medications are amlodipine, carbidopa-levodopa, and tamsulosin. He has smoked one and one-half packs of cigarettes daily for 47 years. He states that he would like to stop smoking. Vital signs are within normal limits. Physical examination shows mild cogwheel rigidity, resting tremor of the hands, and gait instability. On mental status examination, he has a depressed mood and forlorn affect. Results of laboratory studies, including thyroid function tests, are within the reference ranges. Addition of which of the following is the most appropriate next step in pharmacotherapy?

Your Answer: A) Bupropion

Correct Answer: A) Bupropion

Correct Answer: A

Correct Answer: A.

Bupropion can be utilized in patients with Parkinson disease, major depressive disorder, and nicotine use disorder to treat depression and promote abstinence from nicotine. Depression is a common complication of Parkinson disease. Since this patient's depression began after his Parkinson disease diagnosis, his depression is likely a result of Parkinson disease and should be diagnosed as depressive disorder caused by another medical condition. Many patients with this diagnosis improve once the medical condition resolves. Since Parkinson disease does not remit, the depression should be treated pharmacologically. Selective serotonin reuptake inhibitors are first-line medications for patients with major depressive disorder and are frequently utilized for patients with comorbid Parkinson disease and major depressive disorder because of their effectiveness and low risk for adverse effects. However, serotonergic medications are not effective for smoking cessation. Bupropion is an antidepressant medication that increases the reuptake of dopamine and norepinephrine (which targets depressive symptoms) and antagonizes the nicotinic acetylcholine receptor (which targets tobacco use). Bupropion can increase blood pressure, so this patient with a history of hypertension should have his blood pressure monitored closely.

Incorrect Answers: B, C, D, and E.

Citalopram (Choice B) is a selective serotonin reuptake inhibitor, duloxetine (Choice C) is a selective serotonin and norepinephrine reuptake inhibitor, and mirtazapine (Choice E) is a selective α_2 -receptor antagonist and serotonin modulator. All these medications can be utilized in major depressive disorder or depressive disorder caused by another medical condition (including Parkinson disease). However, none of these medications are effective for smoking cessation. Lorazepam (Choice D) is a positive allosteric modulator of the GABAA receptor and is utilized for acute anxiety, alcohol withdrawal, and convulsions. Lorazepam alone is ineffective for depressive disorders.

Educational Objective: Educational Objective: Bupropion is an antidepressant medication that decreases dopamine and norepinephrine reuptake, and antagonizes the nicotinic acetylcholine receptor. Bupropion is the medication of choice in patients with nicotine use disorder and a comorbid depressive disorder who wish to quit smoking.

30. A 72-year-old woman is brought to the emergency department by her husband because of visual hallucinations during the past 2 weeks. She has a 10-year history of hypertension and a 5-year history of Parkinson disease. Current medications include hydrochlorothiazide, aspirin, metoprolol, and carbidopa-levodopa. Three weeks ago, her carbidopa-levodopa dose was increased because of decreased mobility, and metoprolol was added to her medication regimen because of inadequate blood pressure control. She smoked one pack of cigarettes daily for 40 years but quit 10 years ago. She drinks one glass of wine with dinner every night. Her temperature is 36.8°C (98.2°F), pulse is 58/min, respirations are 18/min, and blood pressure is 148/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. Neurologic examination shows a resting tremor of the hands and cogwheel rigidity. Deep tendon reflexes are absent at the ankles. Her gait is shuffling. Mental status examination shows a constricted affect. She is oriented to person but not to place or time. She can recall one of three objects after 5 minutes. She says she sees ghost-like figures entering the room. Results of laboratory studies are within the reference range. A CT scan of the head shows mild atrophy. Which of the following is the most likely cause of this patient's visual hallucinations?

Your Answer: A) Adverse effect of carbidopa-levodopa

Correct Answer: A) Adverse effect of carbidopa-levodopa

Correct Answer: A

Correct Answer: A.

Adverse effect of carbidopa-levodopa is the most likely cause of this patient's visual hallucinations. Parkinson disease is a progressive neurologic disease that manifests with bradykinesia, rigidity, masked facies, and a resting tremor. It is secondary to the destruction of dopamine-producing neurons in the substantia nigra, so pharmacotherapy focuses on replacing dopamine with carbidopa-levodopa. Frequently, early in the disease course, treatment is effective, but over time, many patients experience an "on-off" phenomenon whereby pharmacotherapy only works for a short time after each dose before the effects dissipate. This often prompts an increase in the dose of carbidopa-levodopa, or a change in the dosing interval, but in elderly patients this can result in multiple off-target adverse effects including hallucinations. This patient has no evidence of another underlying medical illness that may cause hallucinations, so the next step should be to lower the dose of carbidopa-levodopa to resolve her hallucinations. Patients who are intolerant or refractory to pharmacotherapy should be considered for deep brain stimulation.

Incorrect Answers: B, C, D, and E.

Major depressive disorder (Choice B) is diagnosed when patients meet criteria established in the DSM-5 including impaired sleep (either insomnia or hypersomnolence), loss of interest, guilt, loss of energy, poor concentration, poor appetite, psychomotor retardation, or suicidal ideation. This patient does not meet the criteria.

Multi-infarct (vascular) dementia (Choice C) presents frequently with a stepwise decline in cognitive performance over several years and occurs as a result of cumulative damage from many small ischemic strokes. It does not present with hallucinations.

Normal-pressure hydrocephalus (Choice D) classically presents with enlarged ventricles on imaging with normal opening pressure on lumbar puncture, dementia, gait disturbance, and urinary incontinence.

Progression of Parkinson disease (Choice E) is a possibility in this patient as evidenced by her worsening bradykinesia, but this is less likely to be the cause of her hallucinations than the time-correlated effect of increasing the doses of medications with known hallucinogenic adverse effects.

Educational Objective: Educational Objective: Carbidopa-levodopa is the primary treatment for Parkinson disease. As the disease progresses, patients often require increasingly higher doses, which can cause hallucinations. If this occurs, the dose should be lowered. An alternative therapy for Parkinson disease includes deep brain stimulation.

31. A 77-year-old man comes to the physician because of increasingly severe low back pain for 3 months. He has had poor control of bowel and bladder function for 2 weeks. He fell yesterday on his way to the bathroom. He has smoked two packs of cigarettes daily for 60 years. Examination shows marked weakness of the lower extremities, normal upper extremity strength, bilateral brisk deep tendon reflexes in the lower extremities, a sensory level at T9, and a flaccid anal sphincter. There is tenderness over the lower thoracic spine. Babinski sign is present. Which of the following is the most likely diagnosis?

Your Answer: C) Epidural metastatic carcinoma

Correct Answer: C) Epidural metastatic carcinoma

Correct Answer: C

Correct Answer: C.

Epidural metastatic carcinoma resulting in spinal cord impingement is the most likely diagnosis in this patient with bowel and bladder incontinence, bilateral lower extremity weakness with hyperreflexia, and a sensory level at T9 that has developed over the preceding 3 weeks. Vertebral masses may be the first presenting sign of a systemic malignancy, or they may be a late complication of metastatic disease. In either scenario, large masses can result in compression of the spinal cord at any segment, with subsequent sensory and motor deficits below the affected segment. In this instance, the sensory level at T9 likely indicates the presence of a mass at or just above the T9 vertebral body in the epidural space, with compression of the spinal cord at that level. Masses related to cancer must be distinguished from abscesses or hematomas, as the treatment between the entities differs significantly. Depending upon the type of cancer, steroids or radiation may be an initial option to reduce the inflammation associated with malignant compression of the spinal cord. In many cases, surgical decompression is required.

Incorrect Answers: A, B, D, E, F, G, and H.

Carcinomatous meningitis (Choice A) can occur as a result of metastatic spread of a primary malignancy to the meninges around the brain, but it would present with headache and often with cranial nerve deficits. It does not present with neurologic evidence of spinal cord dysfunction.

Cervical spondylosis (Choice B) refers to gradual, age-related degenerative changes of the cervical spine with osteophyte development and vertebral body displacement that can impinge on the spinal cord. Symptoms may include neck pain, radiculopathy, or when severe, myelopathy.

Lumbar spinal stenosis (Choice D) classically causes low back pain and neurogenic claudication whereby patients experience lower extremity pain with prolonged standing or walking. The pain classically improves with bending forward.

Myasthenia gravis (Choice E) is an autoimmune disease caused most often by antibodies against the acetylcholine receptor at the neuromuscular junction. Symptoms include generalized weakness that worsens throughout the day and ptosis.

Syringomyelia (Choice F) refers to a fluid collection that develops in the central spinal cord, most often between C2 and T9 and can cause sensory loss to pain and temperature or weakness in the limbs of the upper and lower extremities, depending upon its location. Vibration and position sense are generally preserved.

Tabes dorsalis (Choice G) is a late finding of neurosyphilis and affects the dorsal columns of the spinal cord. Symptoms include sensory disturbances and lancinating pain that may last for minutes to days. Argyll Robertson pupils are often seen in conjunction with this disorder.

Transverse myelitis (Choice H) refers to an inflammatory syndrome involving a particular spinal cord segment leading to weakness and sensory disturbances below the level of the lesion. It often

follows an infection and symptoms reach maximum intensity within days, not weeks, and gradually improve.

Educational Objective: Educational Objective: Lung, breast, and prostate are the most common malignancies that have a predilection for the spine. Metastatic disease within the epidural space can cause compression of the spinal cord, with consequent development of a distinct sensory level, bowel and bladder incontinence, and weakness below the level of the lesion. Depending upon the type of malignancy, steroids, radiation, and/or surgical decompression may be required to relieve the compression.

32. A 47-year-old man is brought to the emergency department by his friend 30 minutes after being found unconscious in his bed. The patient lives alone in a trailer park, and his friend says he was well the night before. The patient has no history of serious illness and takes no medications. He does not smoke cigarettes or use illicit drugs. He appears well nourished. His temperature is 37°C (98.6°F), pulse is 100/min, respirations are 20/min, and blood pressure is 110/60 mm Hg. The nail beds and oral mucosa are pink. Neurologic examination shows no meningismus. The pupils are 4 mm and sluggishly reactive. The doll's eye (oculocephalic) reflex is intact. There is no verbalization or eye opening in response to painful stimuli. He withdraws all extremities symmetrically to painful stimuli. Deep tendon reflexes are 3+ throughout. Babinski sign is absent bilaterally. Results of laboratory studies are within the reference ranges. Arterial blood gas analysis on room air shows a P_{CO_2} of 30 mm Hg and P_{O_2} of 80 mm Hg. In addition to supportive care, which of the following is the most appropriate therapy for this patient's condition?

Your Answer: A) Hyperbaric oxygen

Correct Answer: A) Hyperbaric oxygen

Correct Answer: A

Correct Answer: A.

Carbon monoxide (CO) poisoning occurs as a consequence of inhaling the gas, often because of its generation in an area of poor ventilation (eg, a garage with door shut and car running, a space heater in a poorly-ventilated room, a malfunctioning basement furnace). CO is a byproduct of combustion. When inhaled, it avidly crosses the pulmonary capillary membrane and tightly binds hemoglobin, inducing a conformational change in the molecule and preventing the binding of oxygen. CO has an affinity for hemoglobin that is hundreds of times greater than that of oxygen, readily displacing oxygen from heme. CO also has affinity with the mitochondrial electron transport chain, thus, in addition to hypoxemia resulting from interference with oxygen carrying capacity, cells may not be able to make use of the oxygen that they do receive via aerobic metabolism. As a result, anaerobic metabolism dominates, and lactic acidosis caused by hypoxia ensues. Symptoms of CO poisoning include headache, loss of consciousness, dizziness, vomiting, and fatigue; in severe cases, seizures, dysrhythmias, coma, and death may occur. Examination may show red skin. Treatment for CO poisoning begins with the administration of 100% oxygen regardless of pulse oximetry readings, and if any neurologic symptoms or signs of end-organ dysfunction are present, as in this case, hyperbaric oxygen therapy may be indicated.

Incorrect Answers: B, C, D, and E.

Methylene blue (Choice B) is an antidote for methemoglobinemia; it serves to reduce iron in methemoglobin to Fe^{2+} , restoring its ability to bind and transport oxygen. Methemoglobinemia may occur secondary to the use of nitrites and anesthetics.

Naloxone (Choice C) is an opioid receptor antagonist used in the treatment of opiate overdose, which presents with bradypnea and miosis. Naloxone has also been used to treat clonidine overdose.

Physostigmine (Choice D) is a reversible acetylcholinesterase inhibitor that is most commonly used in myasthenia gravis but may have a role in the treatment of anticholinergic toxicity as it can cross the blood-brain barrier. Anticholinergic toxicity is characterized by pupillary dilation, dry mouth, tachycardia, and urinary retention.

Vitamin B₁ (thiamine) (Choice E) is used to treat thiamine deficiency such as beriberi or Wernicke encephalopathy. Dry beriberi presents as peripheral neuropathy with mental status change. Wet

beriberi exhibits a dilated cardiomyopathy leading to heart failure. Wernicke encephalopathy classically presents as confusion, ataxia, and ophthalmoplegia.

Educational Objective: Educational Objective: Carbon monoxide (CO) has an affinity for hemoglobin that is hundreds of times greater than that of oxygen, readily displacing oxygen from heme. Treatment for CO poisoning begins with the administration of 100% oxygen and may require hyperbaric oxygen if symptoms are severe.

33. A 5-year-old girl has the insidious onset of a clumsy gait over the past 3 months. Examination shows pes cavus of the left foot. There is generalized diminished girth of the left leg, and a decreased tendon reflex of the left ankle. The tendon reflexes at the knees are brisk. There is a sacral tuft of hair. Which of the following is the most likely diagnosis?

Your Answer: E) Tethered spinal cord

Correct Answer: E) Tethered spinal cord

Correct Answer: E

Correct Answer: E.

Tethered spinal cord, a defect in neural development whereby a caudal portion of the spinal cord becomes tethered to fixed fibrous structures that cause traction on the cord as the patient grows, is the most likely diagnosis. The filum terminale is a band of tissue that stretches between the caudal portion of the spinal cord and the sacrum, and persistent tethering of the spinal cord results in stretching as the child develops and axial growth occurs. Depending upon the severity, symptoms may present at infancy or can develop many years later, but progressive lower extremity neurologic dysfunction is typical as the spinal cord stretches and bends during normal activity. Sensory and gait abnormalities are typical, and bladder or bowel incontinence may also occur. Low back pain that worsens with activity and pain in the perineum can also occur. Clues to diagnosis include a tuft of hair over the sacrum or the presence of hemangiomas or dimples in the same region. Early detection and surgical intervention can improve symptoms and halt the progression of neurologic dysfunction.

Incorrect Answers: A, B, C, and D.

Ataxia-telangiectasia (Choice A) is a genetic disorder that leads to early ataxia with progressively worsening gross and fine motor capabilities as children reach school age. Other findings include oculocutaneous telangiectasias and immunodeficiency.

Friedreich ataxia (Choice B) is caused by a loss of function mutation in the frataxin gene resulting from a trinucleotide repeat. Patients usually present in adolescence with progressive ataxia of all limbs and dysarthria. Distal sensory loss and dysphagia may also be seen.

Juvenile idiopathic arthritis (Choice C) has a diverse presentation but patients usually develop systemic symptoms including fevers and rash in addition to arthritis, which may affect nearly any joint. It does not present with asymmetric gait abnormalities or hyperreflexia.

Spastic diplegia (Choice D) is a subtype of cerebral palsy and presents with symmetric spasticity of the lower extremities with associated sensory loss. In rare cases, the upper extremities may be involved, and intellectual disability can occur.

Educational Objective: Educational Objective: Tethered cord syndrome occurs when a caudal portion of the spinal cord remains attached to fibrous tissue near the sacrum leading to progressive stretching of the spinal cord as the child grows. It presents with either symmetric or asymmetric weakness and sensory changes in the lower extremities, and clues to the diagnosis include the presence of a sacral dimple or tuft of hair.

34. A 42-year-old woman with AIDS comes to the physician because of a 2-week history of progressive right-sided weakness, slurred speech, and headache. She is right-handed. Funduscopic examination shows papilledema. Neurologic examination shows aphasia, right homonymous hemianopia, and right-sided spastic weakness. A CT scan of the head with contrast shows multiple small ring-enhancing lesions with surrounding edema and mass effect in the left frontal, right occipital, and left parietal lobes. Which of the following is the most likely diagnosis?

Your Answer: E) Toxoplasmosis

Correct Answer: E) Toxoplasmosis

Correct Answer: E

Correct Answer: E.

Toxoplasma gondii is an intracellular protozoan parasite. Transmission can occur through the ingestion of oocysts from cat feces or infected meat, direct transmission between a mother and fetus, or from organ transplantation, which leads to invasion followed by dissemination of the parasite where it typically lies dormant. It is a ubiquitous organism and nearly 10% of individuals will be seropositive for *T. gondii*, but the parasite does not usually cause severe disease in immunocompetent individuals. In patients with HIV infection and CD4⁺ T-lymphocyte counts less than 100 cells/mm³, however, reactivation can occur with several distinct clinical syndromes of which cerebral toxoplasmosis results in the most morbidity. Cerebral toxoplasmosis typically presents with multiple ring-enhancing lesions scattered throughout the brain. Seizures, focal neurologic deficits, and systemic symptoms are common, as are signs of increased intracranial pressure. Patients with HIV infection and positive *Toxoplasma* IgG antibodies are typically treated empirically with a sulfonamide and pyrimethamine if cerebral toxoplasmosis is suspected, but confirmatory diagnosis occurs through a brain biopsy, which will show a dense infiltrate of lymphocytes, macrophages, and plasma cells, along with the presence of the *T. gondii* organism. Cerebral toxoplasmosis is an AIDS-defining disease.

Incorrect Answers: A, B, C, and D.

Cryptococcosis (Choice A) is caused by *Cryptococcus neoformans*, an encapsulated fungus and opportunistic pathogen in immunocompromised patients. It can cause cryptococcal meningitis, encephalitis, and pulmonary cryptococcosis. Meningoencephalitis presents subacutely with fever, headache, nuchal rigidity, and photophobia, as well as encephalopathy in severe cases. It is most common in immunocompromised patients. Neuroimaging typically shows no abnormalities, and the diagnosis is made with serum cryptococcal antigen and CSF evaluation.

Cysticercosis (Choice B) is caused by *Taenia solium* and presents with headache, seizures, and increased intracranial pressure. Neuroimaging will show cystic, ring-enhancing, calcified lesions in the brain parenchyma, meninges, ventricles, spinal cord, and/or orbit. It less commonly presents with focal neurologic deficits such as those seen in this patient.

Primary central nervous system lymphoma (Choice C) is a rare form of non-Hodgkin lymphoma that occurs in immunocompromised patients and presents with focal neurologic deficits, vision changes, neuropsychiatric symptoms, and increased intracranial pressure. It appears similar to toxoplasmosis on neuroimaging but is much less common than toxoplasmosis. If this patient failed to improve with pyrimethamine and sulfonamide therapy, central nervous system lymphoma would be the likely diagnosis.

Progressive multifocal leukoencephalopathy (Choice D) is a demyelinating disease of the central nervous system caused by human polyomavirus 2 (formerly JC virus), most commonly in

immunocompromised patients. It presents subacutely with altered mental status, ataxia, vision changes, and hemiparesis. Lesions occur primarily in the white matter and are not enhanced with contrast; rather, they appear as hypodense lesions.

Educational Objective: Educational Objective: Toxoplasmosis is caused by *Toxoplasma gondii*, an intracellular protozoan parasite that is most commonly transmitted via the ingestion of cat feces or infected meat. In immunocompromised patients, particularly AIDS patients with CD4+ T lymphocyte counts less than 100 cells/mm³, it can cause cerebral toxoplasmosis, which presents with seizures, focal neurologic deficits, systemic symptoms, and increased intracranial pressure. Neuroimaging will show multiple ring-enhancing lesions, and treatment is with pyrimethamine and sulfonamide therapy.

35. A 52-year-old man comes to the physician because of a 4-day history of progressive numbness and weakness of his legs. Six years ago, he had non-Hodgkin lymphoma treated successfully with chemotherapy and radiation therapy. He is currently taking no medications. Cranial nerve examination shows no abnormalities. Muscle strength is intact in the upper extremities but 3/5 in the proximal lower extremities and 4/5 in the distal lower extremities. Deep tendon reflexes are 2+ in the upper extremities and 3+ in the lower extremities. Babinski sign is present bilaterally. Sensation to pinprick is decreased below the level of the nipples, and sensation to vibration is decreased over the feet and ankles. Which of the following is the most likely diagnosis?

Your Answer: C) Metastatic spinal cord compression

Correct Answer: C) Metastatic spinal cord compression

Correct Answer: C

Correct Answer: C.

Metastatic spinal cord compression occurs most commonly in prostate, lung, or breast cancer, although it can occur with any malignancy. It is generally caused by vertebral bony metastases that tend to occur in the vertebral body, particularly of the thoracic spine. Its presentation depends on the level of the lesion and the degree of spinal cord compression. Symptoms include back pain, which is often worse at night or with lying down, muscle weakness, sensory abnormalities, bladder or bowel dysfunction, and gait instability. It will generally present with an upper motoneuron pattern of injury in the lower extremities with hyperreflexia and positive Babinski signs, as in this patient, assuming the compression is in the thoracic or upper lumbar region. Ascending numbness or paresthesias are also common. Diagnosis is classically made utilizing MRI of the spine. Patients may be treated with glucocorticoids, radiation therapy, and/or surgical decompression and spine stabilization.

Incorrect Answers: A, B, D, and E.

Chemotherapy-induced polyneuropathy (Choice A) classically presents with sensory loss and paresthesias in the distal extremities that ascend in a "stocking and glove" pattern. Most chemotherapy-induced polyneuropathy will improve over time after chemotherapy is stopped, making this a less likely explanation for this patient's acute onset of symptoms.

Guillain-Barré syndrome (Choice B) is caused by autoimmune-mediated nerve destruction following a systemic infection, such as *Campylobacter jejuni*. It classically presents with ascending paralysis with absent deep tendon reflexes that begins in the lower extremities. It can also ascend to involve the upper extremities, diaphragm, and cranial nerves in some patients.

Myopathy (Choice D) is caused by a disorder affecting the muscle itself, rather than the nerves that innervate it. Generally, it presents with symmetric proximal muscle weakness in response to genetic metabolic abnormalities, toxins (eg, alcohol or medications), endocrine dysfunction (eg, hypothyroidism), or inflammation. It does not present with sensory changes.

Paraneoplastic neuropathy (Choice E) is an immune-mediated destruction of neuronal tissue in the presence of an underlying malignancy. It can present in a variety of ways, such as an acute sensorimotor radiculopathy, chronic sensorimotor neuropathy, or autonomic neuropathy. This patient's symptoms would be most similar to an acute sensorimotor radiculopathy, but this syndrome presents with a predominant ascending motor paralysis.

Educational Objective: Educational Objective: Metastatic spinal cord compression presents in malignancy with back pain worse at night, muscle weakness, sensory abnormalities, bladder or bowel dysfunction, and gait instability. Physical examination may disclose hyperreflexia, Babinski sign, and changes in perception of pinprick or vibratory sensation. Treatment includes glucocorticoids, radiation therapy, and/or surgical decompression and spine stabilization.

36. A previously healthy 62-year-old man has had progressive difficulty with speech and swallowing over the past year. Symptoms were subtle at first, and only he and his family noticed a problem. Recently, two ham radio colleagues have commented that he sounds different on the air. He has also noticed that he has difficulty manipulating objects in the left hand, walking smoothly, and going up and down stairs. He has not had any difficulty with his thought processes. Examination shows slow deliberate speech with mild dysarthria. Cranial nerves are otherwise normal. There is wasting of intrinsic hand muscles and distal weakness in the left hand. Gait is slightly broad-based, but Romberg sign is absent. Deep tendon reflexes are 3+ in the left upper extremity, 2+ in the right upper extremity, and 3+ in the lower extremities. Jaw jerk is hyperactive. His serum potassium concentration is 3.2 mEq/L. An x-ray of the chest shows normal findings. Which of the following is the most likely diagnosis?

Your Answer: B) Amyotrophic lateral sclerosis

Correct Answer: B) Amyotrophic lateral sclerosis

Correct Answer: B

Correct Answer: B.

Amyotrophic lateral sclerosis (ALS) is likely given insidious onset of upper and lower motoneuron deficits. ALS is caused by the progressive destruction of motoneurons. Patients may develop asymmetric weakness of the extremities. Unilateral hand weakness, foot drop, and bulbar symptoms (eg, dysphagia, voice changes) may occur simultaneously. On physical examination, the presence of both upper motoneuron signs such as hyperreflexia and spasticity, combined with lower motoneuron signs such as hyporeflexia suggests the diagnosis. Parkinsonian features such as masked facies, tremor, and shuffling gait may occur, as can autonomic dysfunction. Eventually, patients develop respiratory insufficiency. The clinical progression is relentless and results in death.

Incorrect Answers: A, C, D, E, F, G, H, I, J, K, L, M, N, O, P, and Q.

Alcoholic rhabdomyolysis (Choice A) refers to the breakdown of muscle and the release of creatine kinase in patients who consume large quantities of alcohol. Muscle pain and renal dysfunction may occur.

Becker muscular dystrophy (Choice C) is a myopathy presenting with muscle weakness in the absence of upper motoneuron signs.

Botulism (Choice D) results from the ingestion of botulinum toxin from *Clostridium botulinum*. It presents with progressive, symmetric descending weakness.

Carcinomatous meningitis (Choice E) results from metastatic disease to the meninges and may result in cranial nerve deficits and headaches in patients with history of cancer.

Cervical spondylosis (Choice F) refers to gradual, age-related degenerative changes of the cervical spine; symptoms may include neck pain, radiculopathy, or myelopathy.

Epidural metastatic carcinoma (Choice G) presents with symptoms of spinal cord compression from a tumor including lower extremity weakness, bowel and bladder dysfunction, and a sensory level.

Epstein-Barr virus infection (Choice H) presents with fatigue, sore throat, and atypical lymphocytosis. It does not cause motor deficits.

Guillain-Barré syndrome (Choice I) is typically a self-limited demyelinating polyneuropathy that starts with the distal lower extremities and progresses proximally. Patients usually have normal sensation but profound muscle weakness.

Hypokalemic periodic paralysis (Choice J) occurs as a result of transient hypokalemia caused by a genetic disorder leading to periodic generalized muscle weakness. Patients are often asymptomatic between attacks.

Hypothyroid myopathy (Choice K) results in generalized muscle weakness, occasionally with increased creatine kinase. Symptoms of hypothyroidism would be expected.

Lumbar spinal stenosis (Choice L) classically causes low back pain and neurogenic claudication whereby patients experience lower extremity pain with prolonged standing or walking.

Multiple sclerosis (Choice M) is a demyelinating disease affecting white matter in the brain and spinal cord. This can lead to weakness, hyperreflexia, and spasticity. Optic neuritis is a classic presenting diagnosis.

Myasthenia gravis (Choice N) is an autoimmune disease caused most often by antibodies against the neuromuscular acetylcholine receptor. Symptoms include weakness that worsens with use and ptosis.

Polymyositis (Choice O) is an inflammatory myopathy that causes proximal muscle weakness.

Sensorimotor peripheral neuropathy (Choice P) is a broad term that describes including diabetic, toxic, autoimmune, and/or hereditary neuropathy. Symptoms include paresthesia or pain and lower motoneuron weakness.

Spinal cord compression (Choice Q) can occur as a result of trauma or compression from tumors or infections. Symptoms range from myelopathy and sensory changes to complete paralysis.

Educational Objective: Educational Objective: ALS is a progressive neurodegenerative disorder that results from the gradual destruction of motoneurons in the central and peripheral nervous system. Manifestations include upper and lower motoneuron weakness.

37. A 27-year-old man is admitted to the hospital because of a 4-week history of difficulty with speech and weakness of his right leg. He has a 5-year history of HIV infection treated with antiretroviral therapy. On admission, his temperature is 37°C (98.6°F). Examination shows thrush. There is mild nonfluent aphasia. Muscle strength is 4/5 in the right lower extremity. Deep tendon reflexes are 3+ in the right lower extremity and 2+ elsewhere. Babinski sign is present on the right. An MRI of the brain shows six ring-enhancing lesions in the cerebral hemispheres bilaterally; the largest lesion is in the left frontal lobe. Some of the lesions have surrounding edema. Pyrimethamine and sulfadiazine therapy is started. Two weeks later, there is no improvement in his symptoms. A second MRI of the brain shows no decrease in the size of the lesions, and some have increased in size. Which of the following is the most likely diagnosis?

Your Answer: B) Central nervous system lymphoma

Correct Answer: B) Central nervous system lymphoma

Correct Answer: B

Correct Answer: B.

Central nervous system (CNS) lymphoma is a rare variant of non-Hodgkin lymphoma that occurs most commonly in immunocompromised patients. It can occur in the brain, spinal cord, leptomeninges, or eyes. Presenting symptoms include focal neurologic deficits, vision changes, neuropsychiatric symptoms, and signs of increased intracranial pressure (eg, headache, nausea, emesis). In immunocompetent patients, neuroimaging will most commonly show a solitary, hypodense periventricular mass with well-defined borders and mild surrounding edema. In immunocompromised patients, however, the development of primary CNS lymphoma is strongly associated with Epstein-Barr virus (EBV) infection, with additional risk factors in HIV including a decreased CD4+ T lymphocyte count, increased viral load, and chronic co-infection with hepatitis B or C virus. Imaging in immunocompromised patients more commonly shows multifocal disease, numerous, ring- or heterogeneously enhancing lesions throughout the corpus callosum, periventricular areas, and periependymal regions. The diagnosis is confirmed by biopsy or cerebrospinal fluid (CSF) flow cytometry or cytology, with evidence of EBV DNA in the CSF supporting the diagnosis. Treatment typically consists of high-dose and generally intrathecal chemotherapeutics and antiretroviral therapy.

Incorrect Answers: A, C, D, E, and F.

Candidal abscesses (Choice A) of the brain can occur in immunocompromised patients. These are typically microabscesses that can lead to the development of encephalopathy with altered mental status and are seen on MRI as small, ring-enhancing lesions often with a hemorrhagic component. However, these patients commonly present with fever, nuchal rigidity, and headache, which this patient does not demonstrate.

Cryptococcal meningitis (Choice C) presents subacutely with fever, headache, nuchal rigidity, and photophobia, as well as encephalopathy in severe cases. It is most common in immunocompromised patients. Neuroimaging typically shows no abnormalities, and the diagnosis is made with serum cryptococcal antigen and CSF evaluation.

Listerial meningoencephalitis (Choice D) occurs in patients at the extremes of age (neonates, elderly) and in those that are immunocompromised. It classically presents with fever, headache, nuchal rigidity, altered mental status, cranial nerve palsies, and focal neurologic deficits. While *Listeria monocytogenes* can cause brain abscesses, this is exceedingly rare.

Nocardial abscesses (Choice E) can occur in immunocompromised patients. They classically

present with fever, headache, signs of meningeal irritation, and focal neurologic deficits. It more commonly occurs in a concurrent pulmonary infection, which is associated with fever, weight loss, cough, hemoptysis, shortness of breath, and chest pain.

Toxoplasma encephalitis (Choice F) commonly presents with headache, fever, and neurologic deficits, as well as altered mental status. Toxoplasma lesions appear as ring-enhancing lesions on brain MRI, similar to CNS lymphoma. However, toxoplasma encephalitis would be expected to improve with sulfadiazine and pyrimethamine.

Educational Objective: Educational Objective: CNS lymphoma is a rare form of non-Hodgkin lymphoma that occurs in immunocompromised patients with focal neurologic deficits, vision changes, neuropsychiatric symptoms, and increased intracranial pressure. Neuroimaging will show ring-enhancing lesions that do not improve with sulfadiazine or pyrimethamine therapy.

38. A 67-year-old woman comes for a routine health maintenance examination. She is concerned about developing dementia because she lives alone; she does not want to be a burden to her children. Her father was just diagnosed with dementia, Alzheimer type. She takes no medications other than hydrochlorothiazide for hypertension. Her blood pressure is 124/84 mm Hg. Examination shows no abnormalities. Which of the following measures is most likely to decrease her risk for multi-infarct (vascular) dementia?

Your Answer: A) Continued control of blood pressure

Correct Answer: A) Continued control of blood pressure

Correct Answer: A

Correct Answer: A.

Vascular dementia is the second most common form of dementia after Alzheimer type, characterized by stepwise cognitive decline following recurrent cerebral infarctions (strokes). The rate of decline and specific symptoms depend on the regions of the brain affected, and patients typically develop cognitive impairment, motor dysfunction, and behavioral changes as the disease progresses. Risk factors include age, hypertension, obesity, diabetes mellitus, hyperlipidemia, and prior cerebrovascular accidents. Diagnosis is made clinically with support of neuroimaging demonstrating cerebral ischemic changes. Risk factor reduction is the primary approach to prevention. Continued control of blood pressure is most likely to decrease the risk for this patient developing vascular dementia.

Incorrect Answers: B, C, D, E, and F.

Efforts to reduce overall cardiovascular risk factors should be encouraged, which includes regular aerobic exercise (Choice B). Daily fish oil supplementation (Choice D) and daily vitamin E supplementation (Choice F) have also been shown to have a mild protective effect. However, control of blood pressure has a stronger association with cardiovascular risk reduction.

Daily aspirin prophylaxis (Choice C) is recommended for patients who have existing clinically significant cardiovascular disease (such as history of a myocardial infarction or cerebrovascular event). It is not likely to significantly lower the risk for vascular dementia in this patient.

Daily vitamin D and calcium supplementation (Choice E) is recommended for women who are postmenopausal, patients taking chronic systemic corticosteroids, and patients with advanced renal disease to reduce the risk for osteoporosis and fragility fractures.

Educational Objective: Educational Objective: The risk for developing vascular dementia can be reduced by the management of modifiable risk factors for cardiovascular disease. This includes the control of hypertension, hyperlipidemia, diabetes mellitus, obesity, and antiplatelet agents in the setting of prior cardiovascular disease.

39. A previously healthy 45-year-old woman is brought to the emergency department 1 hour after she lost consciousness while shopping in a department store; on regaining consciousness, she had a severe headache and photophobia. Her temperature is 37°C (98.6°F), pulse is 110/min, respirations are 14/min, and blood pressure is 110/70 mm Hg. Examination shows slight neck stiffness with flexion. There is pain with lateral deviation of the eyes. Neurologic examination shows no other abnormalities. A CT scan of the head shows no lesions or evidence of bleeding. Which of the following is the most appropriate next step in management?

Your Answer: E) Lumbar puncture

Correct Answer: E) Lumbar puncture

Correct Answer: E

Correct Answer: E.

Subarachnoid hemorrhage (SAH) refers to bleeding into the subarachnoid space and can be caused by aneurysmal or non-aneurysmal rupture of an artery in the subarachnoid space. Patients typically present with an acute, severe headache (often described as the worst headache of their life), along with neck stiffness, photophobia, and possible stupor or coma in severe cases. Fever can be associated with SAH and is associated with worse outcomes. CT scan of the head typically shows hyperdense blood products in the cerebral sulci and basal cisterns. However, the sensitivity of CT scan for SAH is lower with small volume bleeding or a delayed presentation. If there is strong suspicion for SAH with a negative CT scan, a lumbar puncture can be performed, which may show red blood cells or xanthochromia, which represents hemoglobin degradation products. As the blood products circulate within the subarachnoid space, complications include the development of communicating and/or obstructive hydrocephalus in the acute stage, along with the potential for re-bleeding and vasospasm in the days following the initial bleed. It is a neurosurgical emergency that may be fatal if progressive or untreated.

Incorrect Answers: A, B, C, and D.

Admission for observation (Choice A) would not be indicated until further workup is performed to evaluate for subarachnoid hemorrhage. First, a subarachnoid hemorrhage should be investigated by lumbar puncture, and if findings are consistent with subarachnoid hemorrhage, vessel imaging should occur to evaluate for aneurysm or arteriovenous malformation.

Discharging the patient with analgesic therapy (Choice B) would be dangerous in the care of this patient, as further evaluation for subarachnoid hemorrhage is indicated. The patient is at risk for hydrocephalus, vasospasm, rebleeding, or severe hemorrhage after a sentinel bleed.

Carotid ultrasonography (Choice C) is frequently used for the evaluation of atherosclerosis at the carotid arteries. Carotid artery atherosclerosis increases risk for ischemic stroke as plaque breakage can travel to and cause blockage within cerebral arteries. This patient does not have focal neurologic deficits or imaging indicating an ischemic stroke, so carotid ultrasonography would not be the next step in management.

MRI of the brain (Choice D) would be indicated for evaluation of ischemic stroke or tumor. The patient does not display any focal neurologic symptoms or aphasia to suggest an ischemic stroke. Tumors can also cause headaches, but typically chronic or subacute headaches, associated with vomiting and papilledema.

Educational Objective: Educational Objective: Patients with SAH typically present with an acute, severe headache (often described as the worst headache of their life), along with neck stiffness, photophobia, and possible stupor or coma in severe cases. If there is a strong suspicion for SAH after a normal CT scan, a lumbar puncture should be performed, which may show red blood cells or xanthochromia.

40. A 72-year-old woman is brought to the emergency department 6 hours after the onset of double vision, loss of balance, and dizziness. She has no history of similar symptoms. She sustained a myocardial infarction 2 months ago. She has hypertension and type 2 diabetes mellitus. Current medications include insulin and clonidine. She appears anxious. Her temperature is 37.1°C (98.8°F), pulse is 100/min and regular, and blood pressure is 160/90 mm Hg in both arms. A left carotid bruit is heard. Pulses are decreased in the lower extremities. Atherosclerotic involvement of which of the following arteries is the most likely cause of this patient's symptoms?

Your Answer: G) Vertebral

Correct Answer: G) Vertebral

Correct Answer: G

Correct Answer: G.

This patient likely has an occlusion of the vertebral artery leading to vertebrobasilar insufficiency. Vertebrobasilar insufficiency can be caused by ischemia in patients with poor sympathetic regulation or thromboembolism commonly originating from an atherosclerotic plaque in or dissection of the vertebral artery (known as a posterior circulation stroke). A thrombus of the vertebral artery commonly forms from atherosclerotic disease but may also form as a result of an intimal defect from trauma or vertebral artery dissection, a known complication of chiropractic manipulation. The brain stem, cerebellum, and occipital lobes are commonly affected. Patients typically present with vestibulocerebellar signs (eg, ataxia, dysmetria, dizziness, imbalance, vertigo, vomiting, nystagmus), ipsilateral cranial nerve dysfunction (eg, dysphagia, dysarthria, vertical or horizontal gaze palsies), contralateral hemiparesis (caused by corticospinal tract damage), contralateral impairment in pain and temperature sensation (caused by spinothalamic tract damage), and/or contralateral homonymous hemianopsia with macular sparing (caused by posterior cerebral artery involvement). Unlike hemispheric strokes arising from the internal carotid arteries, cortical deficits such as cognitive and speech impairment are typically absent. A noncontrast CT scan of the head rules out hemorrhage, and the diagnosis may be confirmed with vessel imaging if the lesion is sufficiently large. If no hemorrhage is seen, acute management of posterior circulation strokes may involve tissue plasminogen activator therapy or thrombectomy. In dissection, antiplatelet and anticoagulant administration is indicated to prevent conversion to a thromboembolic stroke if none has yet occurred.

Incorrect Answers: A, B, C, D, E, and F.

The brachiocephalic artery (Choice A) supplies the arm, head, and neck. Atherosclerotic disease and consequent occlusion of the brachiocephalic artery may lead to neck and upper extremity pain in addition to signs of vertebrobasilar insufficiency that are exacerbated on exercise.

Common carotid artery (Choice B) occlusion from atherosclerotic disease less frequently causes symptoms because of retrograde collateral flow through the external carotid artery. If a stroke were to occur, it would affect the ipsilateral anterior and middle cerebral arterial territory, which would not result in signs of vertebrobasilar insufficiency.

External carotid artery (Choice C) occlusion from atherosclerotic disease typically does not produce symptoms but may lead to an ocular ischemic syndrome.

The internal carotid artery (Choice D) supplies the middle cerebral artery (Choice E), which supplies the lateral frontal and anterolateral parietal lobes. During acute occlusion of the internal carotid artery (typically as a result of a thrombus induced by atherosclerotic disease) or middle cerebral artery (typically related to an embolus from the internal carotid artery), patients may present with

hemiparesis and hemisensory deficit (predominantly of the contralateral face and upper body), along with speech impairment.

The subclavian artery (Choice F) supplies the upper extremity and gives rise to the vertebral artery. Atherosclerotic disease and occlusion of the subclavian artery usually presents with upper extremity fatigability, claudication, and/or rest pain along with symptoms of vertebrobasilar insufficiency that are exacerbated by exercise. This patient lacks upper extremity symptoms or activity-related variability of symptoms.

Educational Objective: Educational Objective: Atherosclerotic disease in the vertebral artery commonly leads to thromboembolic strokes of the posterior circulation, known as vertebrobasilar insufficiency. Patients typically present with vestibulocerebellar signs, ipsilateral cranial nerve dysfunction, contralateral hemiparesis, and/or contralateral homonymous hemianopsia with macular sparing.

41. A 37-year-old woman with alcoholism is hospitalized for detoxification. On the second day after admission, she has an impaired sensorium and a decreased attention span during examination. She appears frightened and disoriented and shrieks, "Make the snakes go away." Her temperature is 38.3°C (101°F), pulse is 110/min, and blood pressure is 190/120 mm Hg. Which of the following is the most appropriate next step in management?

Your Answer: B) Oral chlordiazepoxide therapy

Correct Answer: B) Oral chlordiazepoxide therapy

Correct Answer: B

Correct Answer: B.

This patient has developed delirium tremens, a serious complication of alcohol withdrawal. Alcohol increases the central nervous system activity of gamma-aminobutyric acid (GABA), the main inhibitory neurotransmitter. Abrupt discontinuation of alcohol results in imbalanced and unopposed sympathetic excess. Uncomplicated alcohol withdrawal typically presents with tremors, anxiety, restlessness, headache, nausea, and diaphoresis, which can begin within hours after the cessation of alcohol use. Delirium tremens, the most concerning and life-threatening complication of alcohol withdrawal, involves severe confusion and disorientation, fluctuations in consciousness, agitation, visual hallucinations, and autonomic instability (generally characterized by tachycardia, hypertension, and potentially hyperthermia). As in this patient, delirium tremens can begin 2 to 5 days after the patient's last alcoholic drink. Delirium tremens is managed supportively with benzodiazepines (GABA_A receptor agonists) to address the psychomotor agitation associated with withdrawal and prevent the worsening of withdrawal manifestations. The specific benzodiazepines utilized include chlordiazepoxide, a long-acting benzodiazepine given on a scheduled basis in patients experiencing or at risk for complicated withdrawal (eg, delirium tremens), and/or lorazepam, a short-acting benzodiazepine given as-needed for withdrawal symptoms. Management also includes frequent assessments including vital sign checks, fluid resuscitation, and nutritional supplementation (especially with commonly deficient vitamins and minerals such as thiamine, potassium, and magnesium).

Incorrect Answers: A, C, D, and E.

CT scan of the head (Choice A) and oral thioridazine (Choice C), a first-generation antipsychotic, would be unnecessary in this patient whose acute-onset disorientation and hallucinations are consistent with delirium tremens. Psychosis and agitation related to alcohol withdrawal should be treated with benzodiazepines rather than antipsychotics.

Intravenous antibiotic therapy (Choice D) would be unnecessary in this patient whose hyperthermia and tachycardia are consistent with delirium tremens rather than infection.

Intravenous administration of vitamin B₁ (thiamine) (Choice E) should be administered during this patient's hospitalization to prevent Wernicke-Korsakoff syndrome. However, intravenous thiamine does not emergently address alcohol withdrawal symptoms such as this patient's autonomic dysfunction.

Educational Objective: Educational Objective: Delirium tremens is a life-threatening complication of alcohol withdrawal related to decreased GABA activity that presents with severe confusion, agitation, visual hallucinations, and autonomic instability. Benzodiazepines are GABA_A receptor

agonists utilized to treat agitation during withdrawal and prevent the worsening of withdrawal manifestations.

42. A 52-year-old man is brought to the emergency department 3 hours after the sudden onset of severe low back pain and severe shooting pain down her left buttock and leg. The pain began when she arose quickly from a seated position to open a sliding glass door. Two analgesic tablets leftover from a dental procedure provided no relief. She is an avid cyclist. On arrival, intravenous administration of morphine is begun and provides minimal relief. The patient notes numbness over the perineum when she attempts to void. She is reluctant to move from a flexed, seated position to a stretcher because of the pain. Muscle strength is 3/5 on great toe extension, dorsiflexion, and plantar flexion of the left foot. Knee jerk is present on the left. Achilles tendon reflex is absent on the left. An MRI of the lumbosacral spine shows a large central herniation at L4-5 that compresses the L5 and S1 nerves on the left. Which of the following is the most appropriate next step in management?

Your Answer: E) Lumbar laminectomy

Correct Answer: E) Lumbar laminectomy

Correct Answer: E

Correct Answer: E.

Cauda equina syndrome occurs as a result of compression of the spinal nerve roots comprising the cauda equina. Often, such compression occurs because of malignancy, trauma (eg, compression fractures with fragment retropulsion), disc herniation, spinal stenosis, spinal epidural abscess, or spinal epidural or subdural hematoma. Symptoms include lower extremity numbness, weakness, paralysis, hyporeflexia, perineal anesthesia, fecal incontinence with poor rectal tone, and overflow incontinence marked by inability to void and urinary retention. Emergent MRI of the lumbosacral spine is indicated to investigate the cause of cauda equina syndrome, and to determine the necessity of surgical decompression. Initial treatment for patients with epidural spinal cord compression and neurologic deficits include symptomatic treatment, pain control, and glucocorticoid therapy prior to operative therapy. High-dose glucocorticoid therapy is helpful as a bridge to definitive therapy, can help control pain, and may transiently improve neurologic function secondary to decreasing edema at the site of cord compression. However, cauda equina syndrome is a surgical emergency and requires decompression with lumbar laminectomy as soon as possible.

Incorrect Answers: A, B, C, and D.

A bone scan (Choice A) is used to examine and diagnose abnormalities in bones, such as bony metastases, fractures, or osteomyelitis. It is not used to evaluate neuromuscular weakness.

Physical therapy (Choice B) can be used in the management of other types of back pain, such as lumbosacral strain. However, lumbosacral strain will not cause perineal anesthesia or neuromuscular weakness. Cauda equina is a surgical emergency, making physical therapy an incorrect initial next step.

Bed rest and analgesic therapy (Choice C) is no longer recommended for back pain, as it can lead to prolonged pain, as well as worsened muscle and bone strength. It would certainly not be appropriate in cauda equina syndrome, as delayed surgical decompression can lead to permanent neurologic deficits.

Intravenous administration of muscle relaxants (Choice D) may provide symptomatic relief in lumbosacral strain or other muscular disorders. Patients with cauda equina may similarly gain some symptomatic relief. However, the more pertinent next step in this patient is lumbar laminectomy to prevent permanent neurologic dysfunction.

Educational Objective: Educational Objective: Cauda equina syndrome occurs as a result of compression of the spinal nerve roots of the cauda equina and presents with lower extremity numbness, weakness, hyporeflexia, perineal anesthesia, fecal incontinence, and urinary retention. Diagnosis is made with MRI and definitive treatment includes emergent surgical decompression with lumbar laminectomy.

43. A 32-year-old man comes to the physician 2 hours after the onset of dizziness, nausea, and unsteadiness. He had a whiplash neck injury 2 years ago and is still receiving monthly chiropractic manipulations. His voice is hoarse. There is left ptosis and nystagmus. The left pupil is 2 mm smaller than the right. There is decreased sensation to pain over the left side of the face and right extremities. He has ataxia on the left. Mental status examination shows no abnormalities. Which of the following arteries is most likely to be occluded?

Your Answer: G) Left vertebral

Correct Answer: G) Left vertebral

Correct Answer: G

Correct Answer: G.

This patient likely has an occlusion of the left vertebral artery leading to vertebrobasilar insufficiency. Vertebrobasilar insufficiency can be caused by ischemia in patients with poor sympathetic regulation or thromboembolism commonly originating from an atherosclerotic plaque in or dissection of the vertebral artery (known as a posterior circulation stroke). A thrombus of the vertebral artery commonly forms from atherosclerotic disease but may also form as a result of an intimal defect from trauma or vertebral artery dissection, a known complication of chiropractic manipulation. The brain stem, cerebellum, and occipital lobes are commonly affected. Patients typically present with vestibulocerebellar signs (eg, ataxia, dysmetria, dizziness, imbalance, vertigo, vomiting, nystagmus), ipsilateral cranial nerve dysfunction (eg, dysphagia, dysarthria, vertical or horizontal gaze palsies), contralateral hemiparesis (caused by corticospinal tract damage), contralateral impairment in pain and temperature sensation (caused by spinothalamic tract damage), and/or contralateral homonymous hemianopsia with macular sparing (caused by posterior cerebral artery involvement). In posterior circulation strokes affecting the lateral pons, Horner syndrome may occur, which presents with unilateral ptosis and miosis (pupillary constriction) and ipsilateral facial anhidrosis. If no hemorrhage is shown on noncontrast CT scan of the head, acute management of posterior circulation strokes may involve tissue plasminogen activator therapy or thrombectomy. In dissection, antiplatelet and anticoagulant administration is indicated to prevent conversion to a thromboembolic stroke if none has yet occurred.

Incorrect Answers: A, B, C, D, E, F, H, I, J, K, L, and M.

The basilar artery (Choice A) is supplied by the vertebral arteries. Compared to vertebral artery occlusion, basilar artery occlusion may present with more acute and severe corticospinal and corticobulbar dysfunction, leading to locked-in syndrome. The vertebral arteries are more vulnerable than the basilar artery to chiropractic manipulation-related intimal injury.

The left and right anterior cerebral arteries (ACA) (Choices B and H) supply the medial frontal lobe (primary motor and sensory cortices), which controls the motor and sensory function of the lower extremities. Occlusion of the ACA leads to contralateral hemiparesis and contralateral sensory deficit of the lower extremities.

The left and right internal carotid arteries (ICA) (Choices C and I) supply the left and right anterior and middle cerebral arteries (MCA) (Choices D and J). The MCA supplies the lateral frontal lobe (primary motor and sensory cortices), anterior parietal lobe, and temporal lobe. Occlusion of the ICA or MCA leads to contralateral hemiparesis and contralateral sensory deficit of the face and upper extremities.

The left and right ophthalmic arteries (Choices E and K) supply structures of the orbit. Occlusion may lead to blindness.

The left and right posterior cerebral arteries (PCA) (Choices F and L) supply the posterior parietal

and occipital lobes along with the thalamus and midbrain. Occlusion of the right PCA typically leads to left homonymous hemianopia with macular sparing and spatial disorientation. Right vertebral artery (Choice M) dissection or occlusion would result in ipsilateral Horner syndrome and right-sided cranial nerve deficits with crossed sensory deficit in the contralateral body, which this patient does not demonstrate.

Educational Objective: Educational Objective: Vertebral artery dissection commonly leads to thromboembolic strokes of the posterior circulation, known as vertebrobasilar insufficiency. Patients typically present with vestibulocerebellar signs, ipsilateral cranial nerve dysfunction, contralateral hemiparesis, contralateral decrease in pain and temperature sensation, contralateral homonymous hemianopsia with macular sparing, and/or Horner syndrome.

44. A 37-year-old woman comes to the physician because of a 4-month history of headaches and blurred vision. She has no history of serious illness and takes no medications. Menses occur at irregular 2- to 10-week intervals. She has had an 18-kg (40-lb) weight gain during the past 7 months. She is 163 cm (5 ft 4 in) tall and now weighs 86 kg (190 lb); BMI is 33 kg/m². Her temperature is 37°C (98.6°F). Visual acuity is 20/20 bilaterally. Visual field testing shows no abnormalities. Fundusoscopic examination shows blurred optic discs; there are no spontaneous venous pulsations. The remainder of the examination shows no abnormalities. A CT scan of the head shows small lateral ventricles. On lumbar puncture, the opening pressure is 290 mm H₂O; cerebrospinal fluid (CSF) analysis shows no leukocytes or erythrocytes. Which of the following is the most likely mechanism of these findings?

Your Answer: D) Impaired resorption of CSF

Correct Answer: D) Impaired resorption of CSF

Correct Answer: D

Correct Answer: D.

Impaired resorption of cerebrospinal fluid (CSF) is the primary mechanism behind pseudotumor cerebri, or idiopathic intracranial hypertension (IIH). IIH is characterized by signs of increased intracranial pressure including headache, papilledema, and increased opening pressure on lumbar puncture without any underlying pathology to cause such findings on neuroimaging. It is primarily seen in young women who are overweight. Although there is no malignant cause underlying the increased intracranial pressure, it may cause debilitating headaches and permanent vision loss. Headaches that are secondary to increased intracranial pressure often are worse when lying down and relieved by standing. The most common risk factors for the development of pseudotumor cerebri include female gender and obesity. However, hypervitaminosis A from increased dietary intake is another potential cause of IIH. This can be from excessive dietary intake or from the use of retinoid medications such as isotretinoin or all-trans retinoic acid. Other medications associated with IIH include tetracyclines, danazol, and growth hormone.

Incorrect Answers: A, B, C, and E.

Aqueductal stenosis (Choice A) causes dilation of the lateral and third ventricles and is a cause of hydrocephalus in children. Idiopathic increased intracranial pressure presents with symptoms of increased intracranial pressure (nausea, vomiting, positional headache) without hydrocephalus. Cerebellar astrocytoma (Choice B) is a primary central nervous system tumor seen most often in children and a potential cause of secondary intracranial hypertension and hydrocephalus if it obstructs the flow of CSF. IIH is characterized by increased intracranial pressure but not hydrocephalus.

Choroid plexus papilloma (Choice C), a hamartoma of the choroid plexus and rare intracranial tumor, is a cause of communicating hydrocephalus by increasing the production of CSF. It most commonly presents in a young child with headache secondary to increased intracranial pressure. Lateral venous sinus thrombosis (Choice E) may be induced by an infection of facial skin, orbit, or nasal sinuses which causes a septic thrombus to form. The most common presenting symptom is acute onset of headache, though orbital pain, edema, visual disturbances, fever, and mental status changes are also seen. This young woman with obesity, chronic headache, and signs of increased intracranial pressure suggests IIH instead.

Educational Objective: Educational Objective: Impaired resorption of CSF is the mechanism behind pseudotumor cerebri, or idiopathic intracranial hypertension (IIH), which is characterized by headache, papilledema, and increased opening pressure on lumbar puncture without any underlying pathology to cause such findings on neuroimaging. It is primarily seen in young women who are overweight.

45. A 6-year-old boy is brought to the emergency department 20 minutes after a motor vehicle collision in which he was the unrestrained front seat passenger; he lost consciousness momentarily. Examination shows a 2 x 2-cm area of soft-tissue swelling over the forehead, a small abrasion on the left cheek, and an abrasion on the right elbow. Neurologic findings are normal. Which of the following is the most likely diagnosis?

Your Answer: A) Cerebral concussion

Correct Answer: A) Cerebral concussion

Correct Answer: A

Correct Answer: A.

Mild traumatic brain injury, also known as a concussion, commonly results in the loss of consciousness, confusion, amnesia, headache, dizziness, nausea, and/or vomiting minutes after the injury. A concussion can be caused by direct or indirect blunt head trauma and typically causes self-limited brief neurologic impairment, such as the loss of consciousness, repetitive questioning, or confusion. Symptoms that commonly develop within the following days include mood disturbances, photophobophobia, trouble concentrating, and/or insomnia. Patients do not have focal neurologic symptoms such as limb weakness or vision changes that would indicate traumatic vascular injury or intracranial hemorrhage. All patients with suspected concussion should undergo a thorough neurologic examination, and a subset of at-risk patients should receive neuroimaging. All patients should be monitored closely for neurologic deterioration, which may indicate a developing intracranial hematoma. To prevent cumulative brain injury (eg, from a second concussion), patients with concussion should return to their usual activities in a graded fashion, beginning with the return to school and without return to sports until cleared by a physician.

Incorrect Answers: B, C, D, E, and F.

A cervical spine fracture (Choice B) could present with neck pain, altered mental status, focal neurologic deficits, and decreased neck mobility.

Closed head injury (Choice C) refers to blunt head trauma with no object penetrating the skull and dura mater, and encompasses a variety of injuries, such as hematomas, skull fractures, intracranial hemorrhage, or traumatic brain injury. This patient lost consciousness momentarily and has no other apparent neurologic deficits, making concussion a more specific diagnosis.

Epidural hematoma (Choice D) classically occurs after injury to the middle meningeal artery and can present with an initial lucid interval followed by deterioration of mental status as a result of arterial bleeding and hematoma expansion. Epidural hematomas present on CT scan as lens-shaped, bi-convex hyperdense collections abutting the skull.

Facial bone fracture (Choice E) can occur as a result of direct blunt trauma to the face and can be associated with pain and ecchymosis. Patients can present with a variety of signs or symptoms depending on which facial bone is fractured, including facial deformity, midface instability, malocclusion, nerve entrapment, or diplopia.

A frontal skull fracture (Choice F) is uncommon as a result of the hardness of the bone, but can occur after severe, high-impact blunt head injury and can be associated with intracranial bleeding. It can present with swelling, tenderness, a palpable skull defect or crepitus, and can be associated with altered mental status.

Educational Objective: Educational Objective: A concussion can be caused by direct or indirect blunt head trauma and typically causes self-limited brief neurologic impairment, such as the loss of consciousness, repetitive questioning, or confusion.

46. A previously healthy 6-month-old girl is brought to the emergency department by ambulance because of difficulty breathing. Her mother says that she fell from the couch to a carpeted floor yesterday. She cried immediately and was fine until several hours later when she could not be aroused and seemed to be gasping for breath. Her mother noticed some emesis in the bed. On arrival, she is lethargic. Her temperature is 36.5°C(97.7°F), pulse is 130/min, respirations are 36/min, and blood pressure is 92/60 mm Hg. Examination shows a bulging anterior fontanelle. There are bilateral retinal hemorrhages. Neurologic examination shows no focal findings. A CT scan of the head shows blood along the falx. Which of the following is the most likely diagnosis?

Your Answer: L) Shaken baby syndrome

Correct Answer: L) Shaken baby syndrome

Correct Answer: L

Correct Answer: L.

Severe intracranial injury with a history inconsistent with the severity of the patient's physical findings raises suspicion for abusive head trauma and non-accidental trauma. Patients with abusive head trauma can present with encephalopathy, apnea, seizures, and retinal hemorrhages. The mechanism of abusive head trauma is typically blunt force or shaking, which results in an acceleration-deceleration force on the infant. Infants are vulnerable to deceleration injury because of several factors. The large head and weak neck musculature result in greater movement of the brain in the intracranial space when acted upon by acceleration and deceleration force such as in shaking. In addition, the infant skull is thin, allowing forces to transfer through the skull to the brain. Findings on CT scan associated with non-accidental head trauma include subdural hemorrhage, especially multiple hemorrhages or those in an interhemispheric or posterior fossa location, hypoxic-ischemic injury, and cerebral edema.

Incorrect Answers: A, B, C, D, E, F, G, H, I, J, K, M, and N.

Basilar skull fractures (Choice A) typically occur after high-impact blunt head injury and would be unlikely in this patient. Patients can present with hemotympanum, detected RBC in analysis of cerebrospinal fluid, bloody otorrhea or rhinorrhea, mastoid and periorbital ecchymosis, and cranial nerve palsy.

Caput succedaneum (Choice B) refers to extraperiosteal swelling or ecchymosis on a neonate's scalp with poorly defined margins following vaginal delivery. It is a common and benign condition that resolves spontaneously.

A cephalhematoma (Choice C) occurs typically during prolonged or instrumental delivery because of rupture of blood vessels causing hemorrhage beneath the periosteum, resulting in swelling and ecchymosis with margins defined by suture lines. It can be complicated by jaundice, anemia, and hypotension if bleeding is severe.

A cervical spine fracture (Choice D) could present with neck pain, altered mental status, focal neurologic deficits, and decreased neck mobility.

A depressed skull fracture (Choice E) or frontal skull fracture (Choice I) can occur after severe, high-impact blunt head injury and can be associated with intracranial bleeding. However, interhemispheric and retinal hemorrhages are more specific for shaken baby syndrome.

Diffuse axonal injury (Choice F) can occur with rapid acceleration-deceleration injury but is characterized by microscopic widespread axonal separation that is not typically apparent on a CT scan.

Epidural hematoma (Choice G) classically occurs after injury to the middle meningeal artery such as

from an overlying temporal bone fracture (Choice M) and can present with an initial lucid interval followed by deterioration of mental status as a result of arterial bleeding and hematoma expansion. Epidural hematomas present on CT scan as lens-shaped, bi-convex hyperdense collections abutting the skull.

Facial bone fracture (Choice H) can occur as a result of direct blunt trauma to the face and can be associated with pain and ecchymosis. It would be apparent on CT scan.

Intracerebral hematoma (Choice J) can occur secondary to trauma, aneurysms, arteriovenous malformations, tumors, or hypertension. It would appear on CT scan as an area of intraparenchymal hyperdensity, whereas interhemispheric hemorrhage is more specific for shaken baby syndrome.

An orbital blow-out fracture (Choice K) is characterized by orbital pain, enophthalmos, limited eye movement, and diplopia in direct orbital trauma.

A subdural hematoma (Choice N) can occur after blunt head trauma and would appear as a crescent-shaped fluid collection abutting the internal surface of the skull not bound by suture lines on imaging.

Educational Objective: Educational Objective: Patients with abusive head trauma can present with encephalopathy, apnea, seizures, and retinal hemorrhages. The mechanism of abusive head trauma is typically by blunt force or shaking.

47. A 67-year-old woman comes to the physician because of a 7-month history of progressive bilateral leg pain. She describes bilateral buttock, posterior thigh, and calf pain that begins after standing or walking for 10 minutes. It is relieved in 2 minutes by bending forward at the waist or sitting. She has coronary artery disease and underwent coronary artery bypass grafting 2 years ago. Neurologic examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

Your Answer: D) MRI of the lumbar spine

Correct Answer: D) MRI of the lumbar spine

Correct Answer: D

Correct Answer: D.

The patient is presenting with symptoms suggestive of spinal stenosis with neurogenic claudication. Spinal stenosis describes narrowing of the central canal, neural foramen, or lateral recesses of the spine with subsequent nerve compression and ischemia. Common causes include degenerative arthritis, spondyloarthropathies, degenerative disc disease, hypertrophy of the ligamentum flavum, and osteophyte formation in the facet joints. Neurogenic claudication describes low back pain with radiation to the buttocks and lower extremities that results from increased nerve compression, as may occur with extension of the spine during standing and walking downhill. Reducing narrowing within the spinal column with spinal flexion as during sitting or walking uphill tends to relieve the pain. Diagnostic evaluation involves x-ray imaging followed by MRI of the lumbar spine.

Incorrect Answers: A, B, C, and E.

Carotid duplex ultrasonography (Choice A) is recommended to screen for atherosclerotic plaque in patients with a known cardiovascular disease history or significant risk factors. If a stroke or transient ischemic attack were suspected as the cause of the patient's neurologic symptoms, carotid duplex ultrasonography should be performed to evaluate for intervenable lesions. The exacerbating and alleviating factors in this case are more suggestive of spinal stenosis.

Doppler flow studies of the lower extremities (Choice B) are useful in distinguishing neurogenic claudication from claudication caused by peripheral vascular disease. Vascular claudication is typically relieved by rest as opposed to postural changes.

Electromyography and nerve conduction studies (Choice C) can be used to confirm the presence of radiculopathy caused by nerve root compression or evaluate for other myopathy or neuropathy as the cause of the patient's symptoms. These tests are reasonable if the diagnosis is still in doubt after the initial imaging studies. MRI is the most appropriate next step.

CT myelography (Choice E) is recommended for evaluating the bony structures of the spine in the setting of prior spine surgery, where hardware may cause significant artifact on MRI, or for patients who are unable to obtain an MRI for other reasons.

Educational Objective: Educational Objective: Lower extremity pain that is exacerbated by extension of the spinal column and relieved by flexion is suggestive of neurogenic claudication resulting from spinal stenosis. Diagnostic testing should include x-rays and/or MRI of the spine.

48. A 37-year-old woman has had throbbing right-sided headaches that began gradually 6 weeks ago and have increased in frequency and severity. The headaches typically begin in the morning shortly after arising, last up to 1 hour, and occur several times daily. Acetaminophen has provided partial relief. She is irritable and has been unable to keep up daily activities. She appears upset. Neurologic examination shows a persistent attention deficit and left arm drift. Which of the following is the most likely diagnosis?

Your Answer: A) Brain tumor

Correct Answer: A) Brain tumor

Correct Answer: A

Correct Answer: A.

This patient's examination findings of attention deficit (likely neglect) and left arm drift are suggestive of the presence of a contralateral upper motoneuron lesion, which, in conjunction with her symptoms of new chronic headache, is suggestive of a brain tumor. Hemispatial neglect often occurs as a result of lesions of the parietal cortex and may also occur in stroke. Similarly, left arm drift may be suggestive of an upper motoneuron lesion in a variety of locations, such as the contralateral frontal cortex, cerebellum, or pyramidal tract. Collectively, this patient's findings localize her lesion to the right parietal cortex.

Incorrect Answers: B, C, D, E, F, G, H, I, J, K and L.

Cerebral hemorrhage (Choice B) may present with abrupt and acute onset of headache and focal weakness or numbness.

Cerebral nerve root impingement (Choice C) may cause radiculopathy and present with weakness, numbness, paresthesia, and pain in the dermatomal and myotomal distribution that correlates to the affected root.

Cluster headache (Choice D) is a primary headache that typically presents with short-lived, unilateral, severe pain in the orbital, supraorbital, or temporal region of the head and is accompanied by ipsilateral autonomic symptoms (eg, miosis, ptosis, lacrimation, nasal congestion). Idiopathic intracranial hypertension (Choice E) presents commonly in younger, overweight females taking vitamin A, oral contraceptives, or danazol. It is usually associated with a holocephalic headache and blurry vision.

Increased intracranial pressure (Choice F) may occur as a result of brain tumors, hemorrhage, or may be idiopathic, and presents with headache, nausea, vomiting, and papilledema.

Migraines (Choice G) are a common cause of headache in adult patients. They may present with visual aura of scintillating scotoma or fortification spectra but do not typically cause hemineglect or arm drift. They are generally less than 48 to 72 hours in total duration.

Occipital neuralgia (Choice H) is a rare cause of headache and presents with retro-orbital or occipital pain. It does not typically present with neurologic signs.

Subarachnoid hemorrhage (Choice I) presents with severe headache of acute onset, nausea, vomiting, and meningismus, and may rapidly lead to coma and death. This patient's subacute presentation over 6 weeks is not consistent with a diagnosis of subarachnoid hemorrhage.

Subdural hematoma (Choice J) may result from head trauma and hemorrhage from bridging veins in the subdural space, and presents with subacute onset headache, nausea, vomiting, and may present with focal neurologic deficits or seizures as a result of mass effect.

Temporal arteritis (Choice K) is a vasculitis of medium and large vessels in elderly patients and commonly presents with systemic symptoms, headache, jaw claudication, proximal muscle stiffness, and sudden, irreversible blindness.

Tension-type headache (Choice L) typically presents with bilateral dull, pressure-like or band-like pain across the frontal and temporal areas. Tension-type headache does not present with neurologic signs.

Educational Objective: Educational Objective: Mass lesions of the right parietal cortex may present with left hemineglect and left arm drift. Patients may also present with headache, nausea, and vomiting. Mass lesions typically present with a chronic or subacute course with progressive, non-remitting symptoms over a period of weeks or months.

49. A 77-year-old woman comes to the physician because of a 3-year history of increasingly severe pain, swelling, and numbness of her left arm and shoulder; during the past 8 weeks, she has had difficulty moving her left arm because of these symptoms. Fifteen years ago, she underwent a left radical mastectomy followed by radiation therapy of the chest wall for breast cancer. There has been no evidence of recurrence of breast cancer. She has hypertension treated with atenolol. Her only other medication is ibuprofen as needed for pain. Her temperature is 37°C (98.6°F), pulse is 78/min, respirations are 16/min, and blood pressure is 110/70 mm Hg. Examination shows no neck, axillary, or chest wall masses. There is 2+ pitting edema of the entire left upper extremity. Examination of the right upper extremity shows no abnormalities. Radial and ulnar pulses are 2+ and equal bilaterally. Sensation to pinprick and motor function of the left upper extremity are absent. A photograph of the chest wall and left upper extremity is shown. An MRI of the left upper extremity and shoulder shows fibrosis but no masses. Which of the following is the most likely cause of this patient's symptoms?



Your Answer: B) Radiation-induced brachial plexopathy

Correct Answer: B) Radiation-induced brachial plexopathy

Correct Answer: B

Correct Answer: B.

Radiation-induced brachial plexopathy is a potential complication of radiation to the brachial plexus, most commonly occurring after radiation therapy for breast or lung cancer. Radiation damages the brachial plexus and presents with symptoms of brachial neuritis, including pain, paresthesia, and weakness in the distribution of the affected nerves that may occur months or years after the initial insult. In this scenario, the radiation has likely affected the C5 and C6 nerve roots (called Erb Palsy), leading to dysfunction of the deltoid, supraspinatus, infraspinatus, and biceps brachii muscles, resulting in impaired upper extremity abduction, lateral rotation, flexion, and supination. Consequently, the patient's upper extremity hangs near the body in a medially rotated, extended, and pronated position as shown in the photograph. Lymphedema of the upper extremity can also result from radiation-induced fibrosis of the lymph nodes and is often associated with radiation-induced brachial plexopathy. Neoplastic brachial plexopathy should be ruled out with MRI.

Treatment is supportive and includes brachial decompression surgery, pain management, and physical and occupational therapy.

Incorrect Answers: A, C, D, and E.

Entrapment of a cervical nerve root (Choice A), which commonly results from cervical spinal stenosis (Choice D), leads to weakness, numbness, paresthesia, and pain in the dermatomal and myotomal distribution that correlates to the affected root. This patient shows sensorimotor dysfunction that crosses dermatome and myotome boundaries, indicating that multiple nerve roots are affected.

Reflex sympathetic dystrophy (Choice C), or complex regional pain syndrome, presents with regional pain that is out of the proportion to the insult. Patients may also experience edema or abnormal sensory or motor findings in the region. This patient instead possesses a pattern of findings (Erb palsy and lymphedema) consistent with radiation-induced brachial plexopathy, indicating her symptoms are not out of proportion to the insult.

Syringomyelia (Choice E) features a cystic cavity in the central cervical and/or thoracic spinal cord, which affects midline structures such as the anterior white commissure (decussation of the spinothalamic tracts) and the anterior horns (cell bodies of lower motoneurons). Patients with syringomyelia may present with deficits in pain and temperature sensation and a lower motoneuron-pattern of dysfunction in the bilateral upper extremities.

Educational Objective: Educational Objective: Radiation-induced brachial plexopathy is a potential complication of radiation therapy for breast cancer. Patients typically present with pain, paresthesia, and weakness in the distribution of the affected nerves that may occur months or years after the initial insult. Associated lymphedema is also common.

50. An 82-year-old man is brought to the emergency department 1 hour after he collapsed while walking in his garden. He has hypercholesterolemia. He has smoked two packs of cigarettes daily for 60 years. His temperature is 37°C (98.6°F), pulse is 96/min and irregularly irregular, respirations are 18/min, and blood pressure is 150/90 mm Hg. He is awake but is unable to speak due to facial and tongue paralysis. He is able to blink his eyes and move his eyes vertically on command. He cannot move his eyes laterally. Examination shows complete paralysis of all extremities. Sensation to touch is intact. He undergoes endotracheal intubation for airway protection, and intravenous hydration is begun. An MRI of the brain shows extensive infarction of the base of the pons due to basilar artery occlusion. Conservative treatment is recommended. Although he cannot speak, he can communicate clearly by means of a code using eye blinks. The patient clearly communicates to his physicians repeatedly that he wants no life support or artificial hydration, and wants to die because of his poor prognosis for recovery. His wife and two sons agree with this decision but an estranged daughter does not and is threatening legal action. Which of the following is the most appropriate next step in management?

Your Answer: E) The patient should be made DNR and extubated, and fluids should be withdrawn as he wishes

Correct Answer: E) The patient should be made DNR and extubated, and fluids should be withdrawn as he wishes

Correct Answer: E

Correct Answer: E.

Neurologically impaired patients should not be assumed to lack decisional capacity to make decisions about their care. The medical team should evaluate the decisional capacity of this patient who is making decisions about his end-of-life care. This patient is able to communicate a consistent decision and demonstrate an appreciation of the consequences of his decision, and therefore, he appears to possess decisional capacity to make the decision to withdraw care. The medical team should respect the patient's autonomy and withdraw care. If a patient lacks decisional capacity, the medical team should identify a surrogate decision-maker. Unless the patient has identified a durable power of attorney, the spouse should serve as decision maker (adult children are typically second in line, followed by parents). Further reassuring in this case is that the patient's wife and adult children corroborate his decision. His estranged daughter's wishes may not reflect the patient's wishes, and given demonstrated decisional capacity with support from surrogates, the patient's wishes should be respected.

Incorrect Answers: A, B, C, and D.

A court order (Choice A) or ethics consult (Choice B) are unnecessary. This patient possesses decisional capacity to make the decision to withdraw care. Court orders and ethical consults may be considered when a patient lacks decisional capacity, refuses needed care, and lacks surrogate decision makers.

Leaving the patient on artificial life support (Choice C) or continuing some life sustaining measures (Choice D) counters this patient's autonomy. The medical team should respect the decision of this patient with decisional capacity.

Educational Objective: Educational Objective: Physicians should evaluate all patients' decisional capacity to make decisions about their care. Patients who possess decisional capacity should have the final say in decisions about their care, even if family members disagree with the decision.

1. A 7-year-old girl is brought to the physician because of progressive sleepiness, ataxia, and double vision over the past 2 days. She has a 2-year history of complex partial seizures treated with carbamazepine. One week ago, she developed fever and received erythromycin for pharyngitis. Her brother has attention-deficit/hyperactivity disorder treated with methylphenidate. Her temperature is 37°C (98.6°F). She appears sleepy. Ocular movements appear full. She has no weakness, but is moderately ataxic. Which of the following is the most likely cause of her current symptoms?

Your Answer: A) Carbamazepine-erythromycin interaction

Correct Answer: A) Carbamazepine-erythromycin interaction

Correct Answer: A

Carbamazepine is a common antiepileptic medication. It has been shown to interact with other medications and caution should be exercised when choosing concomitant medications for treating patients. It is a CYP450 inducer, which may increase hepatic clearance of some medications (eg, warfarin, phenytoin, valproic acid), leading to subtherapeutic concentrations in the serum. Other drugs, such as erythromycin, can interfere with the hepatic metabolism of carbamazepine by inhibiting CYP450 enzymes, leading to decreased carbamazepine clearance, increased serum concentration, and toxicity. Carbamazepine toxicity typically presents with altered mental status, drowsiness, ataxia, nystagmus, nausea, and vomiting. Diagnosis can be made by measuring a serum carbamazepine concentration above therapeutic levels, and there is a dose-dependent effect on neurologic depression. Treatment is supportive, with benzodiazepines for seizures or agitation, and involves discontinuation of offending medications.

Incorrect Answer: B, C, D, and E.

Cerebellar astrocytomas (Choice B) are slow-growing and typically benign tumors in children and young adults. Children with astrocytomas can present with headache, nausea, vomiting, irritability, behavioral changes, impaired coordination, and visual impairment, depending on the size and location of the astrocytoma.

Typical allergic reactions to medications, including erythromycin (Choice C), would present with rash, urticaria, and, if severe, signs and symptoms of anaphylaxis such as oropharyngeal, lip, or tongue swelling; voice changes; wheezing; shortness of breath; vomiting; or other gastrointestinal disturbances.

Methylphenidate (Choice D) is a central nervous system stimulant used to treat attention-deficit/hyperactivity disorder by blocking norepinephrine and dopamine reuptake in neuronal synapses. The inappropriate ingestion of methylphenidate can present with agitation, anxiety, palpitations, decreased appetite, tremors, hyperreflexia, delirium, hyperthermia, tachycardia, hypertension, and mydriasis.

Viral encephalitis (Choice E) typically presents with fever, headache, altered mental status, and neurologic abnormalities such as seizure, aphasia, hemiparesis, or cranial nerve deficits. This patient does not demonstrate a fever and a carbamazepine-erythromycin interaction is a more likely cause of this patient's altered mental status.

Educational Objective: Erythromycin can interfere with the hepatic metabolism of carbamazepine by inhibiting CYP450 enzymes, leading to decreased carbamazepine clearance, increased serum concentration, and toxicity. Carbamazepine toxicity typically presents with altered mental status, drowsiness, ataxia, nystagmus, nausea, and vomiting.

2. A 52-year-old man comes to the physician because of progressive bilateral leg pain over the past year. The pain is located on the posterior aspect of the thighs and calves, occurs only when he is standing, and resolves within 30 seconds of sitting or lying down. He has noticed that the pain is less intense if he leans on something as he walks, such as a grocery cart. He can still ride an exercise bicycle without developing any pain. He has hypertension, hyperlipidemia, and a small abdominal aortic aneurysm. He had a left carotid endarterectomy 3 years ago after several transient ischemic attacks in the left hemisphere. Medications include lisinopril, lovastatin, and aspirin. He has smoked one pack of cigarettes daily for 30 years. He is 180 cm (5 ft 11 in) tall and weighs 110 kg (242 lb); BMI is 34 kg/m². His temperature is 37°C (98.6°F), pulse is 70/min, respirations are 12/min, and blood pressure is 150/90 mm Hg. Examination shows a right femoral bruit and diminished pedal pulses. Which of the following is the most likely cause of this patient's lower extremity pain?

Your Answer: B) Lumbar spinal stenosis

Correct Answer: B) Lumbar spinal stenosis

Correct Answer: B

The patient's symptoms are most consistent with neurogenic claudication caused by lumbar spinal stenosis. Spinal stenosis describes narrowing of the central canal, neural foramen, or lateral recesses of the spine with consequent nerve compression. Common causes include degenerative arthritis, spondyloarthropathies, degenerative disc disease, hypertrophy of the ligamentum flavum, and osteophyte formation in the facet joints. Neurogenic claudication describes lower back pain with radiation to the buttocks and lower extremities that results from increased nerve compression, as may occur with extension of the spine during standing and walking downhill. Lengthening the spinal column with spinal flexion as during sitting, walking uphill, or leaning against a grocery cart, tends to relieve the pain. Diagnostic evaluation involves x-rays followed by MRI of the lumbar spine to evaluate the degree of spinal stenosis and whether surgical management is necessary. The absence of ankle reflexes suggests involvement of the S1 nerve root.

Incorrect Answers: A, C, D, and E.

Cerebrovascular disease (Choice A) is a concern in this patient with multiple risk factors and a history of transient ischemic attacks; however, progressive bilateral leg pain that is worsened or relieved by postural changes is more consistent with lumbar spinal stenosis. Persistent or new motor or sensory deficits would be more concerning for cerebrovascular disease.

Lumbosacral plexopathy (Choice C) describes a diverse range of motor and sensory disturbances to the lower extremities, including weakness, paresis, pain, numbness, and tingling, resulting from lesions of the lumbosacral plexus. Patients typically display abnormalities along multiple contiguous nerve root distributions. Diagnostic evaluation includes a thorough neurologic and musculoskeletal examination, imaging of the lumbar spine, electromyography, and nerve conduction studies.

Peripheral vascular disease (Choice D) is also a common cause of claudication, and this patient has multiple risk factors including tobacco use, hypertension, hyperlipidemia, and obesity. His physical examination findings support the presence of peripheral vascular disease with a right femoral bruit and diminished pedal pulses. However, claudication caused by peripheral vascular disease is commonly relieved by rest rather than postural changes. His symptoms are more likely caused by lumbar spinal stenosis.

Trochanteric bursitis (Choice E) classically presents with lateral hip pain that is exacerbated by activity. Physical examination shows point tenderness over the greater trochanter and increased pain with external rotation of the hip as a result of compression of the bursa. Bilateral, posterior leg pain is not consistent with a diagnosis of trochanteric bursitis.

Educational Objective: Lower extremity pain that is exacerbated by extension of the spinal column and relieved by flexion is suggestive of neurogenic claudication resulting from lumbar spinal stenosis.

3. A 57-year-old woman comes to the physician because of a 2-day history of dribbling fluid from her mouth every time she drinks. She has not had difficulty swallowing or any other symptoms. She has no history of serious illness and takes no medications. She has an active lifestyle. Vital signs are within normal limits. Examination shows drooping of the lower lip on the left. The left forehead does not wrinkle on upward gaze. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

Your Answer: B) MRI of the brain

Correct Answer: D) Prednisone therapy

Correct Answer: D

Bell palsy refers to a facial nerve palsy that leads to unilateral facial paralysis. It is most commonly caused by viral-mediated inflammation or immune reaction in the setting of a herpes simplex or herpes zoster infection, although it can also be idiopathic or caused by other viral infections. It presents acutely with unilateral facial paralysis that includes the forehead, consistent with its lower motoneuron pathologic involvement. It can also be associated with loss of taste on the anterior two-thirds of the tongue and decreased lacrimation. Symptoms typically peak around 3 weeks, with slow recovery beginning within 6 months. Diagnosis is most commonly based on clinical symptoms, but electrodiagnostic tests, such as EMG, and imaging of the face and head can be utilized if the diagnosis is unclear. Initial treatment consists of oral glucocorticoids, such as prednisone, and antiviral therapy in severe cases.

Incorrect Answers: A, B, C, and E.

Echocardiography (Choice A) is useful in the evaluation of embolic stroke, since an intracardiac thrombus or patent foramen ovale could explain its cause. However, a stroke that affects the facial musculature would spare the forehead. Additionally, this patient has no risk factors for stroke or symptoms such as extremity weakness or numbness to suggest this as an underlying diagnosis. MRI of the brain (Choice B) is useful in the evaluation of stroke, especially early in the course of the disease, before it may be evident on CT scan. This patient's symptoms are not consistent with stroke.

Doxycycline therapy (Choice C) can be used in the treatment of Lyme disease, which may present with Bell palsy in disseminated Lyme disease. Lyme disease also typically presents with erythema migrans, fatigue, headache, myalgias, arthralgias, and lymphadenopathy, as well as myocarditis, abnormal liver function tests, and kidney injury. This patient has no other signs or symptoms to suggest Lyme disease.

Lumbar puncture (Choice E) is used in the evaluation of central nervous system infections, such as bacterial meningitis and viral meningitis. While these infections can present with focal neurologic deficits, they are commonly associated with fever, nuchal rigidity, headache, and altered mental status.

Educational Objective: Bell palsy is a unilateral, viral-mediated or idiopathic facial nerve palsy that causes facial paralysis that does not spare the forehead. Treatment utilizes glucocorticoids, and occasionally, antiviral therapy.

4. A 72-year-old man comes to the physician because of a 1-week history of imbalance and leg weakness and a 3-day history of difficulty initiating urination. He also has had mild midthoracic back pain for 5 months, but during the past 2 weeks, the pain has worsened and frequently awakens him. He has had a 9-kg (20-lb) weight loss since an episode of bronchitis 6 months ago. He has smoked one pack of cigarettes daily for 50 years. He is cachectic and appears chronically ill. Examination shows scleral icterus. Muscle strength in the lower extremities is decreased in the proximal and distal muscles bilaterally. There is no muscle atrophy, but tone is increased. Deep tendon reflexes are hyperactive, and Babinski sign is present bilaterally. Sensation to pinprick is decreased to the umbilicus. Muscle strength, sensation, and reflexes in the upper extremities are normal. Which of the following is the most likely diagnosis?

Your Answer: A) Epidural spinal cord compression

Correct Answer: A) Epidural spinal cord compression

Correct Answer: A

This patient's presenting signs and symptoms of focal back pain, upper motoneuron signs, and weight loss are suggestive of epidural spinal cord compression caused by metastatic carcinoma. Metastatic disease frequently affects the vertebral column and may cause compression of the spinal cord by extending into the epidural space or by causing bony destruction and fracture of the vertebral bodies. Patients present with back pain, lower extremity weakness, loss of balance, urinary or bowel incontinence, or complete paralysis of the lower extremities. Early detection and treatment are crucial to prevent permanent spinal cord damage and myelopathy. Patients may benefit from short-term treatment with intravenous dexamethasone to decrease spinal cord edema, although definitive treatment often requires a combination of surgical decompression, chemotherapy, and/or radiotherapy.

Incorrect Answers: B, C, D, and E.

Myasthenic (Lambert-Eaton) syndrome (Choice B) is characterized by the formation of autoantibodies against presynaptic calcium channels, resulting in the impairment of acetylcholine release at the neuromuscular junction. It presents with progressive proximal muscle weakness that improves with exercise and does not typically present with upper motoneuron signs.

Spinal cord infarction (Choice C) may result from occlusion of the anterior or posterior spinal arteries and presents with abrupt onset myelopathy. Neuroimaging may be required to distinguish this entity from compressive spinal cord lesions. This patient's history of subacute onset and weight loss is more suggestive of a compressive epidural malignant lesion.

Syringomyelia (Choice D) features a cystic cavity in the central cervical or thoracic spinal cord, which affects midline structures such as the anterior white commissure (decussation of the spinothalamic tracts) and, if large enough, the anterior horns (cell bodies of lower motoneurons). Patients present with deficits in pain and temperature sensation and a potential lower motoneuron pattern of dysfunction in the bilateral upper extremities and torso. The syrinx may progressively expand to affect the lateral corticospinal tract as well, which would result in an upper motoneuron pattern of dysfunction in the lower body.

Transverse myelitis (Choice E) refers to inflammation that extends transversely across the spinal cord, leading to bilateral sensory, motor, and/or autonomic dysfunction, often with a clearly identifiable spinal level below which normal function is decreased or absent. It is associated with multiple sclerosis and neuromyelitis optica, the latter of which is associated with the presence of anti-aquaporin-4 antibodies.

Educational Objective: Patients with epidural spinal cord compression present with back pain, lower

extremity weakness, loss of balance, urinary or bowel incontinence, or complete paralysis of the lower extremities. Metastatic disease is a frequent cause of epidural spinal cord compression.

5. A 67-year-old woman comes to the physician because of a 6-month history of buttock and leg pain. The pain occurs when she walks for more than two blocks or stands for more than 10 minutes; sitting or bending forward provides relief within 1 minute. She has not had weakness or loss of sensation. Bladder function has been normal. She has a 20-year history of lower back pain relieved with rest. She has type 2 diabetes mellitus well controlled with metformin. Muscle strength is full. Deep tendon reflexes are 2+ except for absent ankle reflexes. Sensation is intact. There is decreased lumbar lordosis; lumbar flexion is full and extension is limited. Straight-leg raising is negative. Peripheral pulses are intact. Which of the following is the most likely diagnosis?

Your Answer: B) Lumbar spinal stenosis

Correct Answer: B) Lumbar spinal stenosis

Correct Answer: B

The patient's symptoms are best explained by lumbar spinal stenosis. Spinal stenosis describes narrowing of the central canal, neural foramen, or lateral recesses of the spine with consequent nerve compression. Common causes include degenerative arthritis, spondyloarthropathies, degenerative disc disease, hypertrophy of the ligamentum flavum, and osteophyte formation in the facet joints. Lower back pain with radiation to the buttocks and lower extremities that results from increased nerve compression is called neurogenic claudication. It is exacerbated by extension of the spine during standing and walking downhill. Lengthening the spinal column with spinal flexion as during sitting, walking uphill, or bending forward tends to relieve the pain. The absence of ankle reflexes suggests involvement of the S1 nerve root.

Incorrect Answers: A, C, D, and E.

Femoropopliteal arterial stenosis (Choice A) is a common complication of atherosclerosis and peripheral vascular disease. Patients may present with vascular claudication, which is more commonly provoked by exertion and relieved by rest. Severe stenosis may result in limb ischemia. A bruit may be appreciated over the femoral artery, and peripheral pulses would be diminished in the affected extremity.

Piriformis syndrome (Choice C) results from compression of the sciatic nerve as it exits the pelvis. Patients typically present with neuropathic pain in the posterior gluteal region that radiates down the back of the leg. The pain is reproduced by flexion, adduction, and internal rotation of the hip, which tenses the piriformis muscle. A negative straight-leg test is less suggestive of the diagnosis.

Sacroiliac arthritis (Choice D) is a common cause of low back pain that radiates down the leg, and may result from prior trauma, spondyloarthropathy, osteoarthritis, or scoliosis and other spinal deformities. Pain is typically worse when sitting down or walking uphill. Physical examination would show localized tenderness over the sacroiliac joint.

Trochanteric bursitis (Choice E) classically presents with lateral hip pain that is exacerbated by activity. Physical examination shows point tenderness over the greater trochanter and increased pain with external rotation of the hip as a result of compression of the bursa. Bilateral, posterior leg pain is not consistent with a diagnosis of trochanteric bursitis.

Educational Objective: Lower extremity pain that is exacerbated by extension of the spinal column and relieved by flexion is suggestive of neurogenic claudication resulting from lumbar spinal stenosis.

6. An otherwise healthy 77-year-old woman comes to the physician because of a 1-month history of severe dizziness. Several times every day she has the sudden onset of a spinning sensation and loss of equilibrium that lasts for 20 to 30 seconds. The episodes begin most frequently after she rolls over in bed or turns her head to the right. Her pulse is 76/min, and blood pressure is 140/80 mm Hg. Neurologic examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

Your Answer: A) Positional oculovestibular testing

Correct Answer: A) Positional oculovestibular testing

Correct Answer: A

Benign paroxysmal positional vertigo (BPPV) is a common cause of vertigo and is best diagnosed with positional oculovestibular testing. BPPV occurs when an otolith becomes dislodged from the utricle and enters the semicircular canals. Changes in head posture then cause the malpositioned otolith to move within the semicircular canals, stimulating abnormal movements in endolymph and causing a sensation of vertigo. Patients typically report symptoms of intense, brief episodes of vertigo that occur after changes in position (such as after rolling over or rising from bed), accompanied by visual disturbances (caused by nystagmus), nausea, and vomiting. Positional oculovestibular testing, such as the Dix-Hallpike test, allows the examiner to reproduce the patient's symptoms by repositioning the otolith. Further positional maneuvers, such as the Epley maneuver, can be used to move the malpositioned otolith to its proper position within the utricle. Patients with refractory symptoms may benefit from medical management including scopolamine or meclizine.

Incorrect Answers: B, C, D, E, F, and G.

Tilt table testing (Choice B), or tilt testing, is useful for the diagnosis of orthostasis. While orthostasis may cause position-dependent symptoms of vertigo, presyncope, or syncope, these symptoms are more likely to occur upon suddenly standing, rather than while rolling in bed or turning one's head. Holter monitoring (Choice C), or ambulatory ECG monitoring (24-hour), is useful for the diagnosis of cardiac arrhythmia. Cardiac arrhythmia is unlikely to produce paroxysmal symptoms with a positional component that is dependent on rolling in bed or turning one's head.

Doppler ultrasonography of the carotid arteries (Choice D) is useful for the diagnosis of carotid stenosis, which may produce position-dependent symptoms of presyncope or syncope. However, these symptoms would be more likely to occur upon suddenly standing, rather than while rolling in bed or turning one's head.

EEG (Choice E) is useful for the diagnosis of epilepsy. This patient's lack of postictal symptoms or other suggestive symptoms, such as premonitory aura, generalized muscle spasms, or loss of bladder or bowel control, render this a less likely diagnosis.

Electronystagmography (Choice F) evaluates nystagmus as a method of diagnosing disorders of the vestibular system. While this technique may be indicated for patients with refractory symptoms, it is not necessary for the initial evaluation or treatment of BPPV.

Magnetic resonance angiography (Choice G) is useful for the diagnosis of vertebrobasilar insufficiency, which may cause position-dependent symptoms of vertigo. Similar to blood pressure-dependent causes of presyncope, such as orthostasis or carotid stenosis, these symptoms would be expected to occur upon suddenly standing, rather than while rolling in bed or turning one's head.

Educational Objective: Benign paroxysmal positional vertigo occurs when an otolith becomes dislodged from the utricle and enters the semicircular canals. Patients report symptoms of intense, brief, position-dependent episodes of vertigo accompanied by visual disturbances (caused by nystagmus), nausea, and vomiting. Positional oculovestibular testing, such as the Dix-Hallpike test, allow the examiner to reproduce the patient's symptoms by moving the otolith.

7. A 35-year-old woman has had constant burning pain in her right hand since closing her hand in a car door 1 year ago. Examination immediately following the injury showed extensive ecchymosis and swelling. No fractures were identified, but the hand was placed in a cast for 1 month. She is unable to tolerate light touch or any motion of the wrist or finger joints during the current examination. Which of the following is the most likely explanation for these findings?

Your Answer: A) Abnormal sympathetic neural response

Correct Answer: A) Abnormal sympathetic neural response

Correct Answer: A

Abnormal sympathetic neural response, also known as reflex sympathetic dystrophy or complex regional pain syndrome, can potentially occur after traumatic injuries and has a variety of clinical characteristics. Commonly, patients will develop pain out of proportion to the degree of injury in a distinct anatomic location that does not correspond to a single nerve or dermatome. The exact pathophysiology is unknown but is thought to involve either changes in central pain processing or aberrant stimulation of damaged peripheral nerves that release inflammatory substances such as neuropeptide Y and substance P. Common clinical manifestations include pain out of proportion to the injury with severe allodynia, impaired muscle strength in the location of injury, autonomic changes including increased sweating and color or temperature changes, and occasionally atrophy or fibrosis.

Incorrect Answers: B, C, D, and E.

Increased intracarpal canal pressure (Choice B) leads to carpal tunnel syndrome as a result of compression of the median nerve at the wrist. Symptoms include paresthesias in the palmar aspect of the first two or three digits.

Median nerve fibrosis (Choice C) can sometimes occur following a bone fracture with injury to the nerve or after surgical intervention. Symptoms include sensory and motor disturbances in the distribution of the median nerve and depend upon the location at which the median nerve is affected.

Post-traumatic arthritis (Choice D) describes joint pain that occurs following a traumatic injury to the articular cartilage of a joint. Like most cases of osteoarthritis, pain is localized to the affected joint and worsens with activity but improves with rest. Temperature changes and sensory changes of the skin are not seen unless the traumatic injury also resulted in trauma to local tissues or nerves.

Radial artery thrombosis (Choice E) can occur following trauma or instrumentation, as might occur with the placement of a radial artery catheter. Often, the ulnar artery provides enough collateral blood flow to maintain perfusion to the hand, but in patients where the palmar arch is incomplete, thrombosis can lead to ischemia of the hand.

Educational Objective: Reflex sympathetic dystrophy can occur following traumatic injury to a limb and presents with allodynia and color or temperature changes in the area of injury that do not follow a typical nerve or dermatomal distribution.

8. A 25-year-old woman has had generalized weakness for 2 days. On questioning, she reports numerous hospital admissions for abdominal pain. One episode occurred after treatment of a urinary tract infection with a sulfonamide. Examination shows global weakness and areflexia; Babinski sign is absent. Sensation to pinprick is mildly decreased in a stocking-glove distribution, and sensation to vibration is decreased over the feet. She has tachycardia that does not slow with carotid massage. Which of the following is the most likely diagnosis?

Your Answer: A) Acute intermittent porphyria

Correct Answer: A) Acute intermittent porphyria

Correct Answer: A

Acute intermittent porphyria is the most likely diagnosis in this patient with episodic abdominal pain and peripheral neuropathy in a stocking-glove distribution, which is caused by an autosomal dominant mutation in the porphobilinogen deaminase gene (PBGD). This enzyme catalyzes the third step in heme synthesis, which combines four porphobilinogen (PBG) molecules into hydroxymethylbilane, which is subsequently converted to uroporphyrinogen III. Loss of this enzyme leads to the accumulation of PBG, which is toxic to the nervous system. Additionally, increased activity of the upstream enzyme, delta-aminolevulinic acid synthase (ALAS1) leads to accumulation of ALA, which is also toxic. The mutation affects both erythroid cells and hepatocytes. The disease's manifestations are primarily neurologic and include peripheral neuropathy, bladder dysfunction, and autonomic dysfunction. Abdominal pain is classic and results often from ileus. The penetrance is low, so a family history is not often present, which can delay the diagnosis. Disease flares can occur after exposure to certain triggers such as certain medications, tobacco and alcohol, stress, and starvation.

Incorrect Answers: B, C, D, E, F, and G.

Alcoholic neuropathy (Choice B) presents with a primary sensory neuropathy that starts in the feet and progresses proximally. A history of chronic alcohol intake is required for this diagnosis. Other nutritional deficiencies are often present as well.

Charcot-Marie-Tooth disease (Choice C) is a genetic disorder that presents with slowly progressive peripheral neuropathy often in association with foot deformities such as hammertoes and pes cavus. A family history is typical.

Diabetic radiculopathy (Choice D) occurs most commonly in patients with recently diagnosed or uncontrolled diabetes mellitus, and is most commonly characterized by acute, asymmetric radicular symptoms involving the proximal leg.

Guillain-Barre syndrome (Choice E) manifests as rapidly progressive, ascending weakness and areflexia that often follows a viral or diarrheal illness. It does not affect the arms at the same time as the legs. Patients can develop respiratory insufficiency, which can occasionally require mechanical ventilation.

Leprosy (Choice F) is caused by infection with *Mycobacterium leprae* and is exceedingly rare in the developed world. It presents with characteristic skin findings in conjunction with peripheral sensory neuropathy commonly involving the ulnar, median, radial, or peroneal nerves.

Paraneoplastic neuropathy (Choice G) is most commonly a peripheral sensory neuropathy that is related to an underlying systemic malignancy and is often secondary to the production of autoantibodies. This patient has no evidence of cancer.

Educational Objective: Acute intermittent porphyria is an autosomal dominant disease with low penetrance caused by deficiencies in the PBGD gene leading to the accumulation of hepatically synthesized heme intermediates, which are toxic to the nervous system. Typical findings include

episodic abdominal pain, sensory neuropathy, and autonomic dysfunction that occurs after certain triggers.

9. A 62-year-old woman comes to the physician because of a 6-week history of stiffness and pain of her neck, shoulders, and hips. She has difficulty dressing herself and combing her hair in the morning. She also has had generalized weakness, malaise, and a low-grade fever. She has had a 2.3-kg (5-lb) weight loss during the past month despite no changes in diet. She has no history of serious illness. Her only medication is occasional acetaminophen for pain. Vital signs are within normal limits. Examination shows pitting edema of the hands and the dorsal aspects of the feet. Range of motion of the neck, shoulders, and hips is limited. Her erythrocyte sedimentation rate is 110 mm/h. Additional laboratory findings, including her serum thyroid-stimulating hormone concentration, are within the reference ranges. X-rays of the shoulders and feet show no abnormalities. Which of the following is the most likely diagnosis?

Your Answer: D) Polymyalgia rheumatica

Correct Answer: D) Polymyalgia rheumatica

Correct Answer: D

Polymyalgia rheumatica (PMR) is an inflammatory condition in which aching and stiffness develop in the shoulders, hip, and neck. The most common risk factors are age (older than 50 years) and being female. It is also frequently associated with giant cell, or temporal arteritis, which presents with unilateral headache and vision loss, if untreated. Other inflammatory signs, such as malaise and fever, are often present, though localized muscle weakness is not typically seen. Laboratory evaluation demonstrates increased inflammatory markers, including erythrocyte sedimentation rate (ESR) and C-reactive protein, and thrombocytosis. Polymyalgia rheumatica is responsive to glucocorticoids, which are the first-line therapy for the disease. Low-dose prednisone is the treatment of choice.

Incorrect Answers: A, B, C, E, and F.

Bursitis (Choice A) is inflammation of one or more bursae and is manifested by pain and swelling over a single joint. Some examples include the subacromial bursa which can cause pain in the shoulder, the trochanteric bursa which can cause pain over the lateral aspect of the hip, and the olecranon bursa which can cause pain and swelling posterior to the elbow. When inflamed or injured, fluid can collect inside the potential space of the bursa causing swelling, which can be appreciated on examination as a fluctuant, tender, circumscribed mass. Muscle stiffness is not a typical feature.

Dermatomyositis (Choice B) and polymyositis (Choice E) are autoimmune inflammatory myopathies characterized by muscle weakness, rather than stiffness, of the proximal muscles. Dermatomyositis additionally demonstrates cutaneous findings including a light pink- or heliotrope-colored rash on the eyelids and Gottron papules on the dorsal fingers.

Unlike PMR, fibromyalgia (Choice C) affects middle-aged women at various anatomical sites in conjunction with anxiety, stress, and point tenderness. ESR is normal. This patient's limited involvement to the pelvic and hip girdles along with other inflammatory symptoms including generalized weakness, malaise, and fever suggest a diagnosis of PMR over fibromyalgia.

Rheumatoid arthritis (Choice F) is a common autoimmune disease characterized by joint destruction, pain, and upregulation of immunomodulatory cytokines. Muscle pain and stiffness are not typical features.

Educational Objective: Polymyalgia rheumatica (PMR) is an inflammatory condition in which aching and stiffness develop in the muscles of the hip and shoulder girdles. Low-dose prednisone is the treatment of choice.

10. A 57-year-old woman comes to the physician because of daily headaches for 8 months. The headaches are localized to the right temple and usually worsen throughout the day. At times, the headaches are exacerbated by chewing. She has a history of migraines but has not had one since menopause 6 years ago. She recently divorced and is under a great deal of stress at work. She has smoked one-half pack of cigarettes daily for 25 years. Her temperature is 37°C (98.6°F), pulse is 82/min, and blood pressure is 144/86 mm Hg. Examination shows pain with movement of the jaw and tenderness under the right zygomatic arch. The nasal mucosa is normal. Erythrocyte sedimentation rate is 20 mm/h. Which of the following is the most likely diagnosis?

Your Answer: F) Temporomandibular joint syndrome

Correct Answer: F) Temporomandibular joint syndrome

Correct Answer: F

The temporomandibular joint (TMJ) forms the articulation between the mandibular condyle and the glenoid fossa of the temporal bone. TMJ syndrome occurs because of a combination of structural damage to the TMJ and a variety of neurologic and psychological factors related to pain and stress. Patients usually present with unilateral, dull, occasionally sharp or stabbing pain near the TMJ or anterior to the auricle that is exacerbated by chewing or jaw clenching. Patients may also experience popping or clicking sensations associated with movement of the jaw and a decreased range of jaw motion. Treatment includes physical therapy, cognitive behavior therapy or biofeedback therapy to treat underlying stress, and pharmacologic therapy including nonsteroidal anti-inflammatory drugs or tricyclic antidepressants.

Incorrect Answers: A, B, C, D, E, G, and H.

Bacterial meningitis (Choice A) presents with fever and signs of meningeal irritation (nausea, irritability, confusion, anorexia, headache, and nuchal rigidity). This patient's absence of fever and meningismus renders this diagnosis unlikely.

Cluster headache (Choice B) are primary headaches that typically present with short-lived, unilateral, severe, often lancinating pain in the orbital, supraorbital, or temporal region of the head and are accompanied by ipsilateral autonomic symptoms (eg, lacrimation, mydriasis, perspiration, nasal congestion).

Migraines (Choice C) are a common cause of headache in adult patients. They may present with visual aura of scintillating scotoma or fortification spectra but are not associated with focal tenderness of the jaw or pain with chewing.

Sinusitis (Choice D) may be associated with bilateral, pressure-like, periorbital headaches and nasal congestion. Patients may experience focal tenderness and tenderness to percussion overlying the affected sinus (typically overlying the brow or maxilla) but rarely experience focal jaw tenderness overlying the TMJ.

Temporal arteritis (Choice E), also known as giant cell arteritis, is a vasculitis of medium and large vessels in elderly patients and commonly presents with systemic symptoms, headache, jaw claudication, proximal muscle stiffness, and sudden, irreversible vision loss. This patient's chronic course, lack of associated systemic symptoms, and normal erythrocyte sedimentation rate render this diagnosis less likely.

Tension-type headache (Choice G) typically presents with bilateral dull, pressure-like or band-like pain across the bilateral frontal and temporal areas. It does not tend to present with focal tenderness of the jaw or pain with chewing.

Trigeminal neuralgia (Choice H), also known as tic douloureux, presents with brief, recurrent, sudden episodes of pain in the distribution of the trigeminal nerve. This is typically triggered by

minor stimulation to the affected area, such as light touch, cold air, talking, or brushing one's teeth. Patients often describe the pain as sharp, stabbing, or electric.

Educational Objective: Temporomandibular joint syndrome occurs because of a combination of structural damage to the temporomandibular joint and a variety of neurologic and psychological factors related to pain and stress. Patients present with unilateral preauricular pain exacerbated by chewing or jaw clenching, decreased jaw range of motion, and popping or clicking sensations associated with jaw movement. Treatment includes physical therapy, cognitive behavior therapy or biofeedback therapy to treat underlying stress, and pharmacologic therapy including nonsteroidal anti-inflammatory drugs or tricyclic antidepressants.

11. A 42-year-old man comes to the physician with his wife because of a 6-month history of increasing clumsiness and a 1-month history of irritability. His wife has noticed jerkiness of his limbs when he walks down stairs or walks quickly. During the past month, he has had the same involuntary movements at rest after stressful days at work. He has not had fever, chills, or night sweats. He does not smoke cigarettes, drink alcohol, or use illicit drugs. He goes camping several times yearly. He does not appear to be in distress. Examination shows involuntary movements of the right upper extremity, which the patient is able to suppress when made aware of them. His gait is ataxic. Additional testing is most likely to show which of the following abnormalities?

Your Answer: *A) Atrophy of the caudate nucleus on brain imaging*

Correct Answer: *A) Atrophy of the caudate nucleus on brain imaging*

Correct Answer: A

Atrophy of the caudate nucleus on brain imaging would be expected in this patient who most likely has Huntington disease, a progressive neurodegenerative condition that arises from misfolded proteins as the result of a trinucleotide expansion in the Huntington gene. Patients develop choreiform movements, which are characterized by involuntary flailing motions of the upper extremities. Eventually, these motor disturbances and others, including ataxic gait, greatly interfere with daily life. Concomitant psychiatric disturbances such as emotional lability, anxiety, depression, and psychosis may also occur in the later stages of the disease. Many patients exhibit a family history of similar symptoms, but the disease can also arise de novo. Since this is an autosomal dominant disease, there is a 50% chance that an affected individual will pass it on to offspring. Anticipation, whereby symptoms manifest at progressively earlier ages and/or more severely in subsequent generations, is also typical of this disease.

Incorrect Answers: B, C, D, E, and F.

Decreased serum methylmalonic acid (MMA) concentration (Choice B) is not correct. High concentrations of MMA are seen in vitamin B12 (cobalamin) deficiency, which can lead to subacute combined degeneration of the spine.

Dilated ventricles on brain imaging (Choice C) are seen in patients with hydrocephalus, which can either be communicating or noncommunicating. Patients can present with headaches, nausea, vomiting, or gait instability but would not typically exhibit choreiform movements.

Increased oligoclonal bands in cerebrospinal fluid (Choice D) is characteristic of multiple sclerosis (MS). Patients can develop a variety of sensory and motor symptoms but characteristically present with internuclear ophthalmoplegia. They do not exhibit choreiform movements.

Increased serum acetylcholine esterase inhibitor concentration (Choice E) is seen in patients with organophosphate poisoning. Cholinergic toxicity presents with excess salivation, bradycardia, emesis, and central nervous system depression.

Positive serum IgM Lyme titer (Choice F) can be seen in acute Lyme disease, which presents with erythema migrans following a tick bite. Late neurologic manifestations of Lyme disease include cranial nerve palsies, but choreiform movements would not be expected.

Educational Objective: Huntington disease presents with progressive neurodegeneration leading to choreiform movement, ataxia, and neuropsychiatric decline. On MRI, the caudate nucleus often appears atrophic.

12. A previously healthy 82-year-old man is brought to the emergency department because of a 3-hour history of difficulty communicating. His wife reports that when he awoke this morning, he seemed slightly confused and "did not make sense." He has hyperlipidemia treated with simvastatin. He does not appear to be in distress. He is 170 cm (5 ft 7 in) tall and weighs 100 kg (220 lb); BMI is 35 kg/m². His temperature is 37°C (98.6°F), pulse is 85/min and irregular, respirations are 15/min, and blood pressure is 145/90 mm Hg. Examination shows mild right pronator drift. During the examination, he is attentive but has some difficulty naming objects correctly and following two-step commands. An ECG shows atrial fibrillation. Which of the following is this patient's greatest risk factor for his neurologic condition?

Your Answer: B) Dysrhythmia

Correct Answer: B) Dysrhythmia

Correct Answer: B

The acute onset of focal neurologic deficits, motor weakness, confusion, and dysphasia is concerning for a thromboembolic cerebral infarction, or stroke. Atrial fibrillation is a major independent risk factor for strokes, and stroke prevention is an essential aspect of management. The loss of coordinated atrial contraction in this dysrhythmia results in blood stasis within the atrial compartment and promotion of thrombus formation. Embolism to the cerebral arteries results in ischemic stroke. The presence of right-sided hemiplegia and expressive dysphasia are suggestive of a lesion involving the left middle cerebral artery distribution. The middle cerebral arteries are the most common cerebral arteries involved in acute thromboembolic stroke.

Incorrect Answers: A, C, D, and E.

Advanced age (Choice A), hyperlipidemia (Choice C), increased blood pressure (Choice D), and weight (Choice E) are all additional risk factors for cardiovascular disease and stroke. Other risk factors include tobacco use, diabetes mellitus, presence of mechanical heart valves, hypercoagulable conditions, patent foramen ovale (allowing for passage of paradoxical venous thromboembolism to the systemic circulation), heart failure with severely decreased ejection fraction, and a personal or family history of prior stroke. The presence of atrial fibrillation in this patient is the greatest risk factor for a stroke.

Educational Objective: Atrial fibrillation is a major independent risk factor for ischemic stroke. Preventative anticoagulation should be considered for patients with additional risk factors, such as congestive heart failure, hypertension, advanced age, diabetes mellitus, prior stroke or transient ischemic attack, and atherosclerotic cardiovascular disease.

13. A 77-year-old man is brought to the physician by his daughter because he has become increasingly forgetful during the past 6 months. His daughter states that his short-term memory is quite impaired, and he says that he is frustrated with his memory. He lives alone and is able to perform activities of daily living independently. On questioning, he reports a 1-month history of a mild nonproductive cough, constipation, mild pain in his hips with movement, mild numbness of the toes of both feet, and a few episodes of urinary incontinence. He has hypertension, hyperlipidemia, and well-controlled type 2 diabetes mellitus. Current medications include metformin, lisinopril, atorvastatin, and aspirin. His wife died 6 years ago. He does not smoke or drink alcohol. His temperature is 36.8°C (98.2°F), pulse is 72/min, respirations are 14/min, and blood pressure is 128/72 mm Hg. Muscle strength is normal. Cranial nerves are intact. Deep tendon reflexes are normal. Sensation to vibration is mildly decreased over the distal lower extremities. Proprioception is intact. His gait is wide based, and he has impaired tandem walking. His Mini-Mental State Examination score is 24/30; he recalls one of three objects after 5 minutes. An MRI of the brain shows slightly enlarged ventricles with minimal cortical atrophy. Which of the following is the most appropriate next step in management?

Your Answer: F) Lumbar puncture

Correct Answer: F) Lumbar puncture

Correct Answer: F

The patient is presenting with findings concerning for normal pressure hydrocephalus (NPH), with the characteristic combination of dementia (eg, frontal executive dysfunction), gait instability, and urinary incontinence. NPH refers to a type of communicating (nonobstructive) hydrocephalus in which the ventricles are enlarged, leading to damage of the white matter tracts connecting the frontal lobes and basal ganglia. Signs of increased intracranial pressure (eg, papilledema, postural headaches) are typically absent. Many cases of NPH are idiopathic, though some are secondary to a defect in cerebrospinal fluid drainage from the ventricles as a result of scarring or obstruction of the ventricular lining from prior hemorrhage or meningitis. The diagnosis is primarily clinical, although ventriculomegaly on brain imaging is suggestive. The most appropriate next step in management is a lumbar puncture to assess the opening pressure and clinical response to removing cerebrospinal fluid. Placement of a ventricular shunt may be needed for long-term management.

Incorrect Answers: A, B, C, D, and E.

Recommendation that the patient move to an assisted living community (Choice A) is inappropriate prior to a thorough evaluation for intervenable causes. Recovery of executive function in patients with NPH that has progressed to dementia is uncertain. It may be appropriate to refer to assisted living if the patient continues to decline and is unable to manage activities of daily living.

Discontinuation of atorvastatin therapy (Choice B) and discontinuation of metformin therapy (Choice C) may be considered later in the patient's course depending on the degree of recovery.

Medications that confer long-term benefits are often discontinued in patients with progressive dementia or life-limiting conditions to avoid unnecessary side effects and pill burden.

Addition of donepezil therapy (Choice D) may be considered in the future if the patient's cognitive symptoms do not improve. The next best initial step in management for this patient is to perform a lumbar puncture and assess his clinical response to the removal of cerebrospinal fluid.

Addition of paroxetine therapy (Choice E) would not be indicated currently. Cognitive impairment and memory disturbances may be manifestations of depression, especially in older adults, and patients should be screened for depression as part of the initial evaluation. Initiating an antidepressant would not be appropriate at this time.

Educational Objective: The triad of cognitive impairment, urinary incontinence, and gait instability should raise concern for normal pressure hydrocephalus. Initial evaluation should include neuroimaging and a lumbar puncture to assess the patient's opening pressure and clinical response to the removal of cerebrospinal fluid.

14. A previously healthy 42-year-old man comes to the physician because of a 6-week history of intermittent pain and tingling that radiates from his right shoulder blade to his right thumb and index finger. These symptoms worsen with coughing and sneezing. There is no history of neck, shoulder, or arm injury. His pulse is 72/min, and blood pressure is 120/70 mm Hg. Physical examination shows no abnormalities. The cranial nerves are intact. Muscle strength is normal in all extremities. Deep tendon reflexes are 1+ in the right biceps muscle and 2+ elsewhere. Babinski sign is absent. Sensation to pinprick is decreased over the dorsal aspect of the right thumb and index finger. Which of the following is the most likely mechanism of this patient's symptoms?

Your Answer: E) Cervical nerve root compression

Correct Answer: E) Cervical nerve root compression

Correct Answer: E

Cervical nerve root compression explains this patient's symptoms of cervical radiculopathy. Radiculopathy refers to pathology affecting a specific spinal nerve root that leads to weakness, numbness, paresthesias, and pain in the dermatomal and myotomal distribution that correlates to the affected root. C6 radiculopathy will result in pain that radiates from the neck into the arm along with decreased sensation, pain, and/or paresthesia in the lateral forearm and the lateral two digits. Additionally, C6 radiculopathy causes motor dysfunction of elbow flexion and wrist extension, and a diminished or absent biceps reflex. Radiculopathy symptoms may worsen with activities that increase pressure such as coughing and sneezing. Radiculopathy may be caused by spondylosis (degenerative changes of the spine), disc herniation, infection, tumor, demyelination, or infarction. Dermatomes and myotomes are components of the embryonic somite, a paired segment of paraxial mesoderm. When fully developed, the dermatome is an area of skin whose afferent fibers travel within a single spinal nerve, and myotome is a group of muscles innervated by the efferent fibers from a single spinal nerve.

Incorrect Answers: A, B, C, D, F, G, H, I, J, and K.

Brachial plexus compression (Choice A) typically presents with weakness, numbness, and/or paresthesias of the majority of the upper extremity. Brachial plexus compression typically affects multiple nerve roots and motor dysfunction typically predominates over sensory dysfunction and pain early in the disease process.

Central nervous system (CNS) compression (Choice B), demyelination (Choice C), or ischemia (Choice D) typically affects major central nervous system tracts, leading to sensory, motor, and/or autonomic dysfunction contralateral to the lesion. This patient instead demonstrates deficits that are localized to a single spinal nerve root.

Median nerve compression at the elbow (Choice F) typically causes anterior interosseous syndrome, a pure motor neuropathy leading to a decreased ability to pinch the first two digits and form a fist. Median nerve compression at the wrist (Choice G) initially leads to pain and paresthesias in the first three digits and may progress to hand weakness, known as carpal tunnel syndrome.

Radial nerve compression at the elbow (Choice H) presents primarily with pain of the elbow and dorsal forearm that may radiate to the dorsum of the fingers. Motor and sensory symptoms are less prominent. Radial nerve compression at the spiral groove (Choice I) results in extensor muscle dysfunction of the elbow, wrist, and fingers. Numbness, pain, and/or paresthesias from the triceps to the fingers may occur. The triceps reflex, not the biceps reflex, would be affected.

While ulnar nerve compression at the elbow (Choice J) typically results in numbness and paresthesias of the medial hand and wrist, ulnar nerve compression at the wrist (Choice K) typically

leads to weakness of the intrinsic hand muscles and the consequent loss of hand dexterity. Patients may demonstrate a decreased ability to abduct and adduct the digits.

Educational Objective: Radiculopathy refers to pathology affecting a specific spinal nerve root that leads to weakness, numbness, paresthesias, and pain in the dermatomal and myotomal distribution that correlates to the affected root. C6 radiculopathy will cause decreased sensation, pain, and/or paresthesia in the lateral forearm and the lateral two digits. Additionally, C6 radiculopathy causes motor dysfunction of elbow flexion and wrist extension, along with an absent biceps reflex.

15. A 67-year-old man with hypertension is brought to the emergency department 30 minutes after a stroke. Examination shows Horner syndrome and paralysis of the palate on the right. Sensation is decreased to pain over the right side of the face and left side of the body, and sensation is absent to hot and cold over the right face and left side of the body. There is no paralysis of the extremities. Which of the following is the most likely site of infarction?

Your Answer: B) Right lateral medulla

Correct Answer: B) Right lateral medulla

Correct Answer: B

Lateral medullary (Wallenberg) syndrome is caused by occlusion of the posterior inferior cerebellar artery, leading to an ischemic stroke of the lateral medulla. Five main structures are affected, which include the nucleus ambiguus, vestibular nuclei, lateral spinothalamic tract, spinal trigeminal nucleus, sympathetic fibers, and inferior cerebellar peduncle. Loss of the nucleus ambiguus leads to dysphagia, a decreased gag reflex, and hoarseness. The vestibular nucleus loss causes vomiting, vertigo, and nystagmus. The contralateral body and ipsilateral face will develop decreased pain and temperature sensation from involvement of the lateral spinothalamic tract. Horner syndrome may result if the sympathetic fibers are damaged. Additionally, ataxia and dysmetria from the involvement of the cerebellar peduncles will also be present. However, a patient with lateral medullary syndrome may not necessarily demonstrate all of these findings in equal proportion, as in this case.

Incorrect Answers: A, C, D, and E.

The brachium pontis (Choice A), or middle cerebellar peduncles projecting from the pons, may be affected by ischemic events involving the anterior inferior cerebellar artery. Similar to lateral medullary syndrome, symptoms of an infarct involving the brachium pontis include ataxia and dysmetria, which are absent in this case.

Infarction of the midbrain (Choice C) occurs when the basilar artery becomes occluded. This would impact the oculomotor and trochlear nerves. This patient does not demonstrate any eye movement impairment.

Infarction of the right temporal lobe (Choice D), depending on the side of the patient's dominant hemisphere, may impact regions such as the Wernicke area of speech and Meyer loop of the visual pathway. As well, it may impact the primary auditory cortex, resulting in varying degrees of hearing impairment depending on the extent of infarction.

Infarction of the right thalamus (Choice E) may develop from occlusion of one of four different arteries, the tuberothalamic artery, paramedian artery, inferolateral artery, or posterior choroidal artery. Depending on the artery involved, symptoms may range from cognitive behavioral disturbances and memory deficits, to sensorimotor impairment and visual field defects. When the right thalamus is involved, hemineglect is typically also present.

Educational Objective: Lateral medullary syndrome is caused by occlusion of the posterior inferior cerebellar artery. Symptoms include a combination of dysphagia, decreased gag reflex, hoarseness, vertigo, nystagmus, decreased pain and temperature sensation on the ipsilateral face and contralateral body, Horner syndrome, ataxia, and dysmetria.

16. A 5-year-old girl is brought to the emergency department by her parents because of a 4-hour history of unsteady gait and two episodes of vomiting. Three weeks ago, she had an upper respiratory tract infection that did not require treatment. There is no history of trauma or evidence of toxic ingestion. She takes no medications. Her temperature is 36.7°C (98°F), pulse is 120/min, respirations are 24/min, and blood pressure is 95/60 mm Hg. Examination shows slight horizontal nystagmus. Funduscopic examination shows no abnormalities. The patient is able to sit without support but is unable to walk without support. Deep tendon reflexes are normal, and sensation is intact. Which of the following is the most appropriate next step in management?

Your Answer: C) CT scan of the head

Correct Answer: C) CT scan of the head

Correct Answer: C

The patient is showing signs and symptoms concerning for cerebellar ataxia, including gait disturbance and nystagmus. Other associated findings may include dysarthria, speech abnormalities, impaired coordination, vomiting, and headache. Patients typically do not present with a fever, meningismus, or seizures. Acute cerebellar ataxia usually occurs following an acute febrile illness in children under 6 years of age. In the setting of an acute presentation with neurologic deficits, CT scan of the head is effective to evaluate for acute intracranial hemorrhage, tumor, or hydrocephalus that may require emergent management.

Incorrect Answers: A, B, D, and E.

Re-examination in 24 hours (Choice A) would not be appropriate, as the patient is showing concerning neurologic findings that should be evaluated urgently.

Complete blood count (Choice B) could be helpful in evaluating for infection or anemia, neither of which would be the likely cause of the patient's cerebellar neurologic symptoms.

An EEG (Choice D) would be helpful in evaluating seizures, which are typically characterized by involuntary rhythmic movements, lapse of consciousness, a postictal state, and may be accompanied by tongue biting and/or incontinence. This patient's symptoms are taking place while she is awake, alert, and following commands, making seizures unlikely.

A lumbar puncture (Choice E) would be the test of choice to evaluate for meningitis. Meningitis is characterized by fever, neck rigidity, and headache. While meningitis can cause focal neurologic deficits, they are typically cranial nerve abnormalities. In addition, this patient is not showing associated signs of meningitis.

Educational Objective: Acute cerebellar ataxia typically occurs following an acute febrile illness in children under 6 years of age. In an acute presentation with neurologic deficits, CT scan of the head is effective to evaluate for acute intracranial hemorrhage or hydrocephalus that may require emergent management.

17. A 37-year-old woman with a 10-year history of HIV infection comes to the physician because of headache and fever for 2 days. Her temperature is 38.9°C (102°F). Examination shows confusion and mild weakness of the left upper and lower extremities. Deep tendon reflexes are increased, and Babinski sign is present on the left. Her CD4+ T-lymphocyte count is 4/mm³ (Normal~500), and plasma HIV viral load is 60,000 copies/ml. Which of the following is the most likely finding on MRI of the brain?

Your Answer: D) Mass lesion within the substance of the right frontal cortex with shift to the left of midline structures

Correct Answer: D) Mass lesion within the substance of the right frontal cortex with shift to the left of midline structures

Correct Answer: D

Toxoplasma gondii is an intracellular protozoan parasite. Transmission can occur through the ingestion of oocysts from cat feces or infected meat, direct transmission between a mother and fetus, or from organ transplantation, which leads to invasion followed by dissemination of the parasite where it typically lies dormant. It is a ubiquitous organism and nearly 10% of individuals will be seropositive for T. gondii, but the parasite does not usually cause severe disease in immunocompetent individuals. In patients with HIV infection and CD4+ cell counts less than 100 cells/mm³, however, reactivation can occur with several distinct clinical syndromes of which cerebral toxoplasmosis results in the most morbidity. Cerebral toxoplasmosis typically presents with multiple ring-enhancing lesions within the substance of the cerebral cortex, most commonly the frontal and parietal lobes, although single lesions can occur. It is commonly associated with cerebral edema, which may result in mass effect and shifting of adjacent structures. Seizures, focal neurologic deficits, and systemic symptoms are common, as are signs of increased intracranial pressure. Patients with HIV and positive toxoplasma IgG antibodies are typically treated empirically with a sulfonamide and pyrimethamine if cerebral toxoplasmosis is suspected. If necessary, the confirmatory diagnosis occurs through a brain biopsy, which will demonstrate a dense infiltrate of lymphocytes, macrophages, and plasma cells, along with the presence of the T. gondii organisms. Cerebral toxoplasmosis is an AIDS-defining disease.

Incorrect Answers: A, B, C, and E.

Diffuse leukoaraiosis with periventricular white matter changes (Choice A) describes hyperintense signal alterations on MRI in the white matter surrounding the lateral ventricles. It is associated with chronic vascular pathology, such as hypertension, diabetes, and vascular dementia.

Markedly enlarged sulci with generalized loss of brain matter (Choice B) occurs in cerebral atrophy associated with age, dementia, and alcohol use disorder. This patient with HIV and focal neurologic deficit more likely has an opportunistic infection of the central nervous system.

Markedly enlarged ventricles with minimal changes in sulci (Choice C) is seen in normal pressure hydrocephalus, which classically presents with dementia, urinary incontinence, and ataxia. These symptoms do not describe this patient's clinical presentation.

An opaque mass over convexity of left frontal cortex with compression of brain substance (Choice E) describes a meningioma, which is a benign brain tumor that develops from the meninges.

Symptoms are dependent on the location of the tumor, and patients are often asymptomatic.

Common presenting symptoms include seizures, vision changes, altered mental status, and focal neurologic deficits. However, it does not cause a fever, as seen in this patient.

Educational Objective: Toxoplasmosis is caused by Toxoplasma gondii, an intracellular protozoan parasite that is most commonly transmitted via ingestion of cat feces or infected meat. In immunocompromised patients, particularly AIDS patients with CD4+ T lymphocyte counts less than

100 cells/mm³, it can cause cerebral toxoplasmosis, which presents with seizures, focal neurologic deficits, systemic symptoms, and increased intracranial pressure. Neuroimaging will show one or more ring-enhancing lesions within the cerebral cortex, and empiric treatment is with pyrimethamine and sulfonamide therapy.

18. An 87-year-old man comes to the physician because of headaches over the past 6 weeks; they fluctuate in severity during the day and occasionally awaken him from sleep. He describes the pain as an aching pressure that is worse in the front. His vision is unchanged. He takes aspirin for cervical osteoarthritis. His son died 3 months ago. He is lethargic and yawns during the interview. His temperature is 36.6°C (97.8°F), pulse is 88/min and regular, and blood pressure is 140/90 mm Hg. Funduscopic examination shows cataracts. There is moderate stiffness of the neck with flexion and when the head is turned to the left and right. There is a pronator drift of the right upper extremity. Reflexes are 3+ at the biceps and triceps muscles and knees and 1+ at the ankles. Babinski sign is present bilaterally. He has a slow unsteady gait with short steps. No other abnormalities are noted. Which of the following is the most likely diagnosis?

Your Answer: E) Subdural hematoma

Correct Answer: E) Subdural hematoma

Correct Answer: E

Head trauma, the use of anticoagulants, physical abuse, alcohol use disorder, cerebral atrophy, and malignancy are associated with an increased risk for developing a subdural hematoma (SDH). Elderly patients are at risk for cerebral atrophy, which places traction on dural bridging cortical veins. In this setting, minor trauma such as falls can result in shearing of the veins leading to hemorrhage. In some cases, patients may not report a history of trauma. Patients may present with headaches, vomiting, meningismus, or hemiparesis, or they may manifest upper motoneuron examination findings such as spasticity, hyperreflexia, and extensor plantar reflex (Babinski sign). In some cases, the SDH may be large enough to compress adjacent brain parenchyma; the displacement of the cerebral hemisphere can result in neurologic deficits, altered mental status, seizures, herniation syndromes, and signs of increased intracranial pressure. On imaging, SDH appears as a crescent-shaped fluid collection abutting the internal surface of the skull, not bound by suture lines.

Incorrect Answers: A, B, C, D, and F.

Craniopharyngioma (Choice A) is a rare pituitary mass derived from remnants of the Rathke pouch that may present with bitemporal hemianopsia. It is usually detected in youth.

Hypertensive encephalopathy (Choice B) may result from acute and severe increases in blood pressure and may manifest with headache, vomiting, agitation, altered mental status, and seizure. It may progress to obtundation and coma. This patient's blood pressure is mildly increased but not severe enough to cause hypertensive encephalopathy.

Major depressive disorder (Choice C) may cause cognitive impairment in older patients, which can improve with antidepressant treatment. It would not cause focal neurologic deficits or upper motoneuron signs.

Migraine (Choice D) is a type of primary headache that usually does not typically cause neurologic deficits and would be unlikely to cause altered mental status or upper motoneuron signs.

Temporal arteritis (Choice F) is a common vasculitis of medium and large vessels in elderly patients and commonly presents with systemic symptoms, headache, jaw claudication, proximal muscle stiffness, and sudden, irreversible vision loss. Treatment is with immediate, high-dose corticosteroids.

Educational Objective: Elderly patients are at risk for cerebral atrophy, which causes traction on dural bridging cortical veins such that minor trauma can result in venous injury leading to a subdural hematoma.

19. A 67-year-old woman is brought to the physician because of progressive imbalance and memory difficulties over the past year. Two years ago, she began to have episodes of sudden imbalance with a tendency to fall backwards. She has developed progressive slowness of thought and movement and has had brief episodes of double vision. Examination shows mildly decreased short-term memory and slowness of verbal responses. Downward gaze and upward gaze are markedly impaired. There is marked rigidity of the neck and mild rigidity of the limbs. The gait is broad based with short shuffling steps, and she easily loses her balance, falling backwards. All movements are performed slowly. A CT scan of the head without contrast shows moderate, diffuse atrophy. Which of the following is the most likely diagnosis?

Your Answer: G) Progressive supranuclear palsy

Correct Answer: G) Progressive supranuclear palsy

Correct Answer: G

Progressive supranuclear palsy, an atypical parkinsonian disorder, is a neurodegenerative disease associated with tau-positive neurofibrillary tangles in the basal ganglia, cortex, and brainstem. Patients typically present at an average age of 65 years with gait instability, oculomotor deficits (classically vertical gaze palsy), frontal lobe deficits, bradykinesia, and muscle rigidity. The gait pattern varies in step length and base width, though patients typically walk and pivot too quickly (as a result of impulsivity from frontal lobe damage), leading to backward falls. Definitive diagnosis occurs by neuropathologic examination illustrating a pattern of neurofibrillary tangles in the characteristic distribution. Neuroimaging illustrating midbrain and/or cerebral cortex atrophy supports the diagnosis. Death typically occurs 6 to 9 years after disease onset. Management is supportive and includes physical and occupational therapy along with advanced care planning.

Incorrect Answer: A, B, C, D, E, F, H, and I.

Creutzfeldt-Jakob disease (Choice A) is a human prion disease that presents with rapidly progressing dementia, neuropsychiatric symptoms, cerebellar dysfunction, corticospinal tract dysfunction, basal ganglia dysfunction, and stimulus-induced myoclonus. Vertical gaze palsy would be atypical.

Dementia, Alzheimer type (Choice B) is the most prevalent dementia, presenting with progressive cognitive decline that culminates in the inability to perform basic activities of daily living. Vertical gaze palsy, bradykinesia, and muscle rigidity would be atypical.

Hydrocephalus (Choice C) may present acutely with signs of increased intracranial pressure (eg, positional headache, papilledema, nausea, vomiting). This patient's chronic progressive course and lack of ventriculomegaly on MRI are inconsistent with hydrocephalus.

Hypothyroidism (Choice D) is associated with mood changes but does not lead to focal neurologic deficits.

Multi-infarct (vascular) dementia (Choice E) refers to cognitive impairment caused by cerebrovascular disease. Patients typically demonstrate stepwise, not progressive, focal neurologic deficits and functional decline. MRI of the brain usually demonstrates multiple infarcts.

Parkinson disease with dementia (Choice F) presents with dysfunction of voluntary movements in the form of bradykinesia, resting tremor, rigidity, and postural instability. Dementia may occur late in the disease course. Vertical gaze palsy would be atypical, and the gait pattern is typically narrow based.

Neurosyphilis (Choice H) typically presents with a progressive dementia or tabes dorsalis (disease of the dorsal columns of the spinal cord). Vertical gaze palsy, muscle rigidity, and bradykinesia would be atypical. Patients may demonstrate a tabetic gait (a wide-based, purposeful gait).

Wernicke-Korsakoff syndrome (Choice I), caused by thiamine deficiency, most often in association

with chronic alcohol abuse, is characterized by altered mental status, ophthalmoplegia, ataxia, and retrograde and anterograde amnesia. A narrow-based gait and a purely vertical gaze palsy would be atypical, since ophthalmoplegia typically manifests as nystagmus or horizontal gaze palsy. Further, bradykinesia and muscle rigidity would be atypical. MRI of the brain would likely show focal atrophy (eg, mammillary bodies).

Educational Objective: Progressive supranuclear palsy, an atypical parkinsonian disorder, is a neurodegenerative disease related to tau-positive neurofibrillary tangles in the basal ganglia, cortex, and brainstem. Patients typically present at an average age of 65 years with gait instability, oculomotor deficits (classically vertical gaze, palsy), frontal lobe deficits, bradykinesia, and muscle rigidity.

20. An 80-year-old man briefly loses consciousness and falls to the floor in a grocery store. He awakens within 1 minute and feels dizzy, but after several minutes the dizziness passes. His pulse is 80/min; his blood pressure is 160/90 mm Hg while supine and 150/86 mm Hg while standing. Cardiac examination and ECG show no abnormalities. During the physical examination, when his head is turned to the right, he suddenly loses consciousness and motor tone. While unconscious, his pulse is 40/min. He awakens within 20 seconds and feels slightly dazed, but this feeling passes after 1 minute. Which of the following is the most likely diagnosis?

Your Answer: A) Carotid sinus hypersensitivity

Correct Answer: A) Carotid sinus hypersensitivity

Correct Answer: A

Carotid sinus hypersensitivity syndrome is characterized by an exaggerated response to carotid baroreceptor stimulation, presenting with resultant sinus bradycardia, hypotension, and syncope. Baroreceptors located at the carotid bifurcations and the aortic arch are important in regulating cardiac output and maintaining adequate perfusion pressures to the brain and organs of the upper body. Baroreceptors are sensitive to mechanical pressure; normally, an increase in blood pressure stimulates baroreceptor activity, which activates a reflex arc resulting in parasympathetic innervation of the sinoatrial and atrioventricular nodes, and inhibition of sympathetic vascular tone. This results in a decreased heart rate and peripheral vascular resistance and may result in syncope, especially in older adults. In carotid sinus hypersensitivity syndrome, this response may be triggered by mechanical compression of the carotid sinus from external palpation or with movement of the muscles of the neck. Cardiac examination and diagnostic studies are often unrevealing.

Incorrect Answers: B, C, D, and E.

Complex partial seizure disorder (Choice B), also called focal onset impaired awareness seizures, is characterized by localized motor or sensory symptoms associated with loss of awareness during episodes. Generalized loss of motor tone and bradycardia are more suggestive of carotid sinus hypersensitivity.

Idiopathic orthostatic hypotension (Choice C) is unlikely to explain this patient's symptoms given normal orthostatic vital signs on examination. Orthostatic hypotension is characterized by tachycardia in response to the decreased preload, rather than bradycardia as observed in this patient.

Vasodepressor syncope (Choice D), also called vasovagal syncope, is common and results from an acute decrease in peripheral vascular resistance with an inadequate responsive increase in cardiac output to maintain perfusion pressure to the brain. Common triggers include standing for long periods of time, heat, and intense emotion or pain. In this case, the onset of symptoms when the patient's head is turned to the right raises concern for carotid sinus hypersensitivity.

Vertebrobasilar insufficiency (Choice E) describes inadequate perfusion of the posterior cerebral circulation and may present with vertigo, light-headedness, disequilibrium, ataxia, and nystagmus (caused by transiently decreased blood supply to the vestibular nuclei). It is most commonly caused by atherosclerosis and may be provoked by prolonged neck extension. This patient does not demonstrate vestibular symptoms.

Educational Objective: Baroreceptors in the carotid sinus and aortic arch detect changes in mean arterial pressure to appropriately regulate perfusion pressures to the brain and body organs. Carotid sinus hypersensitivity can result in the abnormal provocation of this reflex arc, causing sinus bradycardia and syncope.

21. An 18-year-old woman is brought to the emergency department 30 minutes after a 5-minute generalized tonic-clonic seizure. She has a 1-year history of sudden jerking movements of her hands and arms; the episodes frequently occur while she is getting ready for work in the morning. On arrival, she is confused but gradually becomes oriented to person, place, and time. Her temperature is 36.9°C (98.4°F), pulse is 80/min, respirations are 20/min, and blood pressure is 114/68 mm Hg. On examination, ocular movements are full. The remainder of the examination shows no abnormalities. Deep tendon reflexes are 2+ throughout. In addition to valproic acid therapy, supplementation with which of the following is the most appropriate next step in management?

Your Answer: A) Folic acid

Correct Answer: A) Folic acid

Correct Answer: A

Supplementation with folic acid should be prescribed in all women of child-bearing age taking anti-epileptic drugs, particularly valproic acid. Valproic acid increases sodium channel inactivation and increases γ -aminobutyric acid (GABA) concentration, leading to anti-epileptic effects. However, it also increases the risk for neural tube defects by twenty-fold when taken during the first trimester of pregnancy. Supplementation of folic acid prior to conception is effective in lowering the risk of neural tube defects. Patients without other risk factors can take the standard dose of folic acid, 0.4 mg daily, while patients on anti-epileptics other than valproic acid should take 1 mg daily, and patients on valproic acid should take 4 mg daily. Valproic acid is also used to treat migraines and bipolar disorder, and the same folic acid supplementation should be recommended in those cases.

Incorrect Answers: B, C, D, E, and F.

Riboflavin (Choice B) deficiency may cause cheilitis and corneal vascularization. It does not have a role in epilepsy management and is not needed in conjunction with valproic acid.

Vitamin B1 (thiamine) (Choice C) deficiency causes variable manifestations including dry beriberi (peripheral neuropathy with mental status change), wet beriberi (dilated cardiomyopathy leading to heart failure), and Wernicke encephalopathy (confusion, ataxia, and ophthalmoplegia). Since the patient does not demonstrate any of these findings, supplementation with thiamine is unnecessary.

Vitamin B6 (pyridoxine) (Choice D) supplementation is needed in patients taking isoniazid. Without it, pyridoxine deficiency develops, which is characterized by peripheral neuropathy, dermatitis, sideroblastic anemia, glossitis, and seizures.

Vitamin D (Choice E) is a steroid hormone involved in calcium and phosphorus homeostasis, and bone mineral density maintenance. Vitamin D supplementation is needed in patients using chronic glucocorticoids, which may lead to decreased bone mineral density.

Vitamin E (Choice F) is an antioxidant that protects cells from free radical damage. Deficiency may present with hemolytic anemia and generalized muscle weakness. Its supplementation is not needed in patients taking valproic acid.

Educational Objective: Valproic acid increases the risk of neural tube defects by twenty-fold when taken during the first trimester of pregnancy. All women of childbearing age on anti-epileptic medications, particularly valproic acid, should take folic acid supplementation.

22. A 67-year-old man with a 30-year history of alcoholism is brought to the emergency department 1 hour after he was found confused and lethargic by his daughter. He appears malnourished and ill. His temperature is 36°C (96.8°F), pulse is 96/min, and blood pressure is 150/80 mm Hg. Scattered rhonchi are heard. Abdominal examination shows no tenderness, a liver span of 18 cm, and a palpable spleen tip. Neurologic examination shows a disconjugate gaze. Which of the following is the most appropriate next step in management?

Your Answer: C) Intravenous vitamin B1 (thiamine) therapy

Correct Answer: C) Intravenous vitamin B1 (thiamine) therapy

Correct Answer: C

Vitamin B1, or thiamine, deficiency causes two potential syndromes: Wernicke encephalopathy and Korsakoff dementia. The classic triad of Wernicke encephalopathy demonstrates encephalopathy, ataxia, and oculomotor dysfunction. If Wernicke syndrome is allowed to progress without treatment, Korsakoff syndrome may develop, which is manifest by retrograde and anterograde amnesia, confabulation, and horizontal nystagmus. Wernicke encephalopathy develops in patients with extreme malnourishment and/or alcohol use disorder. While a serum thiamine concentration may be decreased, this is not useful in managing patients as it cannot be checked quickly. Thus, diagnosis is clinical and should be rapidly treated with intravenous thiamine administration.

Incorrect Answers: A, B, D, and E.

Intensive physical therapy (Choice A) would not address the underlying diagnosis of Wernicke encephalopathy. Delayed diagnosis of this condition potentially leads to coma and death.

Intravenous glucose therapy (Choice B) should not be administered until after thiamine has been given in the setting of Wernicke encephalopathy, since glucose alone can theoretically worsen symptoms. Once thiamine has been replenished, then intravenous glucose may be given.

Intravenous vitamin B12 (cyanocobalamin) therapy (Choice D) may be utilized in the treatment of megaloblastic anemia resulting from vitamin B12 (cobalamin) deficiency. This most commonly occurs in the context of veganism, pernicious anemia, and Crohn disease. It is not necessary to supplement in the case of Wernicke encephalopathy.

Oral lactulose therapy (Choice E) is the treatment of choice for hepatic encephalopathy. This is most commonly seen in patients with cirrhosis whose ability to process ammonia from the gastrointestinal tract is impaired. This patient's liver is enlarged, not cirrhotic on physical examination. As well, disconjugate gaze would be absent in hepatic encephalopathy.

Educational Objective: Vitamin B1, or thiamine, deficiency causes Wernicke encephalopathy, characterized by encephalopathy, ataxia, and oculomotor dysfunction. Diagnosis is clinical and should be rapidly treated with intravenous thiamine administration to avoid progression to Korsakoff syndrome.

23. A 62-year-old woman is brought to the emergency department after an episode of flaccid paralysis and numbness of the left upper and lower extremities that lasted 2 hours. During the episode, she was unable to see objects in her left visual field. Neurologic examination now shows no abnormalities. A CT scan of the head shows no lesions. Which of the following is the most likely diagnosis?

Your Answer: C) Cerebral embolus

Correct Answer: C) Cerebral embolus

Correct Answer: C

Transient ischemic attack (TIA) is defined as a transient episode of neurologic dysfunction lasting less than 24 hours caused by focal brain, spinal cord, or retinal ischemia without evidence of infarction. This patient has developed focal neurologic deficits that resolved after 2 hours, qualifying as a TIA. TIA can be induced by either an embolus arising from a proximal site or from temporarily decreased blood flow to a watershed area in the brain. A hypotensive TIA typically lasts until resolution of the low-flow event while an embolic TIA lasts hours, as in this case. An embolic ischemic stroke involves an event in which a clot breaks off and travels to the brain, occluding blood flow within a cerebral artery. Emboli most commonly originate from proximal vessels, such as the internal carotid artery, or from an atrial thrombus. If the embolus successfully undergoes physiologic thrombolysis, blood flow is restored to the affected area and reverses the clinical symptoms. After a TIA, measures for prevention of an ischemic stroke should be enacted, along with evaluation for the potential source of embolus, such as carotid artery doppler and echocardiography.

Incorrect Answers: A, B, D, E, F, G, and H.

Rupture of an arteriovenous malformation (Choice A) or cerebral artery aneurysm (Choice G) in the brain are causes of intracranial bleeding and hemorrhagic stroke. Hemorrhage is not a typical cause of TIA and is less likely to be transient.

Intracranial neoplasms, including cerebellar glioma (Choice B), frontal lobe meningioma (Choice D), and temporal lobe astrocytoma (Choice H) may cause focal neurologic deficits such as ataxia and dysmetria, social disinhibition, or motor and sensory impairment, respectively, because of mass effect. However, these effects are not transient.

Lacunar infarct (Choice E) causes ischemia to the deep cerebral white matter, pons, or basal ganglia, which is not transient. A lacunar stroke may present with purely motor symptoms, purely sensory symptoms, ataxic hemiparesis, or dysarthria. Occlusion of these deep penetrating vessels is typically secondary to hypertension, diabetes mellitus, and hypercholesterolemia.

Migraine (Choice F) presents with recurrent primary headaches that are typically severe, unilateral, throbbing, and associated with nausea, vomiting, photophobia, and phonophobia. Migraines may present with visual aura symptoms of scintillating scotoma or fortification spectra but do not generally cause loss of visual acuity. A sensory aura is sometimes seen, but flaccid paralysis of the entire left side is more consistent with an ischemic event instead.

Educational Objective: TIA is defined as a transient episode of neurologic dysfunction lasting less than 24 hours caused by focal brain, spinal cord, or retinal ischemia without evidence of infarction. TIA is most commonly induced by a cerebral embolic event.

24. A 25-year-old farm worker is brought to the emergency department because of the sudden onset of headache, dizziness, nausea, and vomiting while picking peaches. His pulse is 52/min, and blood pressure is 90/60 mm Hg. He has generalized weakness, diaphoresis, and pinpoint pupils. Which of the following precautions should medical professionals take while caring for this patient?

Your Answer: B) Wearing rubber gloves

Correct Answer: B) Wearing rubber gloves

Correct Answer: B

Organophosphates are commonly found in insecticides and exposure to them can occur transcutaneously, or via inhalation or ingestion. Organophosphates bind to and are inhibitors of acetylcholinesterase, preventing the breakdown of acetylcholine (ACh), which leads to excess concentrations of ACh in the neuromuscular junction. ACh acts on both nicotinic and muscarinic ACh receptors. Nicotinic ACh receptors are found in autonomic ganglia, adrenal medulla, and skeletal muscle, whereas muscarinic ACh receptors are found in heart, brain, smooth muscle, exocrine glands, and sweat glands. Resultant cholinergic excess leads to bronchospasm, bronchorrhea, bradycardia, diarrhea, lacrimation, and salivation. Because of the potential for organophosphate toxicity via transcutaneous exposure, medical professionals should always wear rubber gloves when treating the patient. Treatment involves decontamination of the patient, along with atropine to antagonize the muscarinic receptors, and 2-pralidoxime to regenerate functional acetylcholinesterase.

Incorrect Answers: A, C, and D.

Organophosphates are not transmitted via aerosolization or droplets, thus wearing protective goggles (Choice A) or wearing a surgical mask (Choice C) would not be as protective as wearing gloves. This patient was likely transcutaneously exposed to organophosphates while picking peaches, so wearing protective rubber gloves is a priority.

Wearing no protective equipment (Choice D) would not be advised, since medical professionals treating the patient could inadvertently be exposed to organophosphates by touching the patient. Educational Objective: Organophosphates are commonly found in insecticides and exposure to them can occur transcutaneously, or via inhalation or ingestion. Because of the potential for organophosphate toxicity via transcutaneous exposure, medical professionals should always wear rubber gloves when treating the patient.

25. A 62-year-old man with amyotrophic lateral sclerosis comes to the physician because of a 1-month history of progressive shortness of breath. The shortness of breath is exacerbated when he is supine. His only medication is an 81-mg aspirin tablet daily. He has smoked two packs of cigarettes daily for 30 years but stopped 10 years ago. His pulse is 76/min and regular, respirations are 18/min while sitting and 26/min while supine, and blood pressure is 112/78 mm Hg. There is no jugular venous distention. Scattered rhonchi are heard on auscultation; they resolve with coughing. An S4 is heard. Abdominal examination shows retractions of the abdominal walls with inspiration. There is no peripheral edema. Which of the following is the most likely cause of his shortness of breath?

Your Answer: B) Diaphragmatic paralysis

Correct Answer: B) Diaphragmatic paralysis

Correct Answer: B

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disorder arising in midlife that affects both lower motoneurons arising from the anterior horn and upper motoneurons of the lateral corticospinal tract within the brain and spinal cord. Neurons degenerate as the result of an unknown cause that may involve RNA processing and/or excitotoxicity. Patient presentations are variable, though typically feature a combination of upper motoneuron (eg, increased muscle tone, hyperreflexia, Babinski sign) and lower motoneuron findings (eg, decreased muscle tone, hyporeflexia, early muscle atrophy) in an asymmetric distribution that may include the bulbar muscles. Some patients may experience cognitive or behavioral changes, though other patients are cognitively normal. The diagnosis is primarily clinical but may be supported by electromyographic findings of acute and chronic denervation. ALS is a progressive disease with death usually occurring within 3 to 5 years of diagnosis, most commonly secondary to neuromuscular respiratory failure. This patient's dyspnea with lying supine is most likely secondary to diaphragmatic paralysis caused by his ALS. Symptoms of diaphragmatic paralysis worsen in the supine position, as gravity no longer assists in contracting and moving the diaphragm downward.

Incorrect Answers: A, C, D, E, and F.

Bronchospasm (Choice A) occurs most commonly in the setting of reactive airway disease, such as asthma or chronic obstructive pulmonary disease. It typically presents with shortness of breath and wheezing on auscultation that is not position-dependent. While this patient may have some degree of reactive airway disease secondary to his significant smoking history, his symptoms are more consistent with diaphragmatic paralysis.

Failure of the intercostal muscles (Choice C) may also occur in the late stages of ALS or in other neuromuscular diseases. Intercostal muscle weakness can lead to dyspnea, given their role in chest expansion, but this shortness of breath would not be dependent on positioning.

Left ventricular failure (Choice D) can cause shortness of breath secondary to pulmonary edema and is commonly associated with lower extremity edema, weight gain, orthopnea, paroxysmal nocturnal dyspnea, rales on lung auscultation, and increased jugular venous pressure. This patient has no signs of heart failure, and his shortness of breath with lying supine is more consistent with diaphragmatic paralysis.

Panacinar emphysema (Choice E) more commonly presents with exertional dyspnea, chronic cough, and sputum production. Dyspnea is not typically exacerbated in the supine position.

Pleural effusions (Choice F) can potentially cause dyspnea with supine positioning and are common in congestive heart failure. However, they would be associated with diminished breath sounds on auscultation, not rhonchi as in this patient.

Educational Objective: ALS is a progressive neurodegenerative disorder that affects both lower and

upper motoneurons. Death typically occurs secondary to neuromuscular respiratory failure, with diaphragmatic paralysis presenting with dyspnea that is worse in the supine position.

26. A 14-month-old infant is brought to the physician because of four episodes of loss of consciousness over the past month. While in the office, the infant has a typical episode. Prior to onset, the infant became fretful when his pacifier was taken away. He cried for 30 seconds, then stopped crying and breathing and became rigid and cyanotic. There are no clonic movements and no response to noxious stimuli. His eyes deviate upward conjugately; the pupils do not respond to direct or consensual light. Faint heart sounds are heard on apical auscultation; a pulse cannot be located. He recovers without intervention and returns to normal behavior within 10 minutes. Which of the following is the most likely diagnosis?

Your Answer: C) Breath-holding episode

Correct Answer: C) Breath-holding episode

Correct Answer: C

Syncope triggered by emotional distress in this infant was likely caused by a breath-holding episode. Breath-holding episodes are common paroxysmal events that occur between ages 6 months and 6 years and are typically triggered by stress-induced autonomic dysfunction in the form of transient profound bradycardia. The typical sequence of events is crying, followed by breath holding in forced expiration and cyanosis, then muscle rigidity and shaking, and finally atonia and loss of consciousness. The eyes may deviate upward, and the pupils may dilate. Episodes may recur several times (weekly on average). Physicians should assess patients for iron deficiency anemia, which is commonly associated with breath-holding episodes, and treat with iron if deficient. Breath-holding episodes typically remit spontaneously.

Incorrect Answers: A, B, D, E, and F.

Absence seizures (Choice A) are generalized seizures that cause transient staring episodes with intact consciousness after the seizure. This patient transiently lost consciousness (did not respond to noxious stimuli), and the characteristic sequence of events for a breath-holding episode would be atypical for an absence seizure.

No medications (Choice B) are known to cause breath-holding episodes.

Carotid artery trauma (Choice D) may lead to carotid artery dissection and ischemic stroke.

However, this patient does not demonstrate persistent neurologic deficits consistent with a stroke. Nor does he present symptoms of carotid sinus syncope which results from pressure against the carotid sinus baroreceptors.

Narcolepsy-cataplexy (Choice E) is a chronic sleep disorder characterized by daytime sleepiness, cataplexy, hypnagogic hallucinations, and sleep paralysis. Cataplexy refers to a sudden loss of muscle tone and usually occurs in response to a positive emotion. Cyanosis would be atypical for cataplexy, and narcolepsy-cataplexy does not typically occur in infants.

Ventricular tachyarrhythmia (Choice F) may rarely lead to syncope in children. Breath-holding episodes are much more common and are consistent with this patient's self-resolving, stress-triggered episode.

Educational Objective: Breath-holding episodes are common paroxysmal events in infants and toddlers typically arising from stress-induced autonomic dysfunction. Patients typically present with breath holding in forced expiration, acute cyanosis, muscle shaking and atonia, and loss of consciousness, which occur after crying.

27. A 22-year-old woman comes to the physician because she has had a black spot that occludes her vision in her right eye for 1 day. She also has pain in the right eye. There is no history of trauma. Examination shows visual acuity of 20/20 in the left eye; the best corrected vision in the right eye is 20/200. The left pupil is normal; the right reacts sluggishly to direct light but consensually. Funduscopic examination shows no abnormalities. The eyes are not injected. Which of the following is the most likely site of the lesion?

Your Answer: B) Cranial nerve II

Correct Answer: B) Cranial nerve II

Correct Answer: B

Optic neuritis results from autoimmune demyelinating inflammation of the optic nerve (cranial nerve II) and presents in young, often female, patients with unilateral vision loss and pain that is exacerbated with extraocular movements. Ocular examination demonstrates decreased visual acuity and color vision, a relative afferent pupillary defect (rAPD), and possibly optic disc edema. Diagnosis is confirmed with MRI demonstrating hyperintensity of the optic nerve. Patients with optic neuritis are at an increased risk for multiple sclerosis (MS), and optic neuritis is a common initial finding in patients with MS. Recently, autoantibodies against myelin oligodendrocyte glycoprotein (MOG) and aquaporin-4 (AQP4) have been identified as a causative factor of some cases of optic neuritis. Patients who develop optic neuritis because of antibodies against MOG and AQP4 often have severe presentations but also have a decreased risk for subsequent MS compared with patients with optic neuritis who lack these autoantibodies. The Optic Neuritis Treatment Trial demonstrated that a short course of treatment with high-dose intravenous methylprednisolone speeds the rate of visual recovery and delays the onset of MS, although it does not impact final visual acuity, and demonstrated that the presence and quantity of periventricular white matter lesions on MRI imaging is the most important prognostic risk factor for the subsequent development of MS.

Incorrect Answers: A, C, D, E, F, G, H, and I.

The cerebellum (Choice A) coordinates ocular saccades, pursuit movements, and gaze holding. Cerebellar lesions do not cause an rAPD.

Cranial nerve III (Choice C), the oculomotor nerve, is responsible for motor innervation of the levator palpebrae superioris; the superior, medial, and inferior rectus muscles; and the inferior oblique muscle. Deficits of the oculomotor nerve present with ptosis, exotropia, and hypotropia.

Cranial nerve IV (Choice D), the trochlear nerve, is responsible for motor innervation of the superior oblique muscle. Deficits of the trochlear nerve present with diplopia, hypertropia, and compensatory head tilt.

Cranial nerve VI (Choice E), the abducens nerve, is responsible for motor innervation of the lateral rectus muscle. Deficits of the abducens nerve present with esotropia and diplopia worst during horizontal gaze.

Extraocular muscles (Choice F) may cause diplopia in the setting of thyroid ophthalmopathy, orbital mass lesions, or entrapment within a fractured orbital bone. Dysfunction of the extraocular muscles does not produce a rAPD.

Lesions of the medial longitudinal fasciculus (MLF) (Choice G) cause internuclear ophthalmoplegia (INO), which presents with an ipsilateral adduction deficit, characterized by an inability of the affected eye to cross midline, and an associated contralateral abduction saccade.

Defects of the neuromuscular junction (Choice H) occur in the setting of myasthenia gravis, which presents with diplopia and ptosis that worsens toward the end of the day.

Defects of the retina (Choice I) may produce visual field deficits and a rAPD but are usually visible

on fundoscopy and are rarely painful.

Educational Objective: Optic neuritis is an autoimmune, inflammatory disorder of the optic nerve (cranial nerve II) that presents with acute onset vision loss and ocular pain that is exacerbated with extraocular movements. Patients typically present with profoundly decreased visual acuity and color vision, along with a relative afferent pupillary defect. Diagnosis is confirmed with MRI and with testing for serum autoantibodies. MRI provides important prognostic information regarding the subsequent risk for developing multiple sclerosis, which commonly occurs in association with optic neuritis. Treatment with high-dose intravenous methylprednisolone speeds visual recovery and delays the onset of multiple sclerosis.

28. A 22-year-old college student comes to the student health center after an episode during which he found himself unable to move for several seconds after awakening. The episode occurred during the college's spring break 2 weeks ago, when he awoke after having fallen asleep on the beach. He has no history of similar symptoms. The patient is not having academic difficulties, although he admits to recent stress caused by the examinations that he took just before the spring break. His medical history is noncontributory. Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate next step in management?

Your Answer: A) Reassurance

Correct Answer: A) Reassurance

Correct Answer: A

Sleep paralysis, a manifestation of the atonia that primarily occurs during REM sleep, happens during the transition from sleep to wakefulness at least once in most people's lives. The precise mechanism of sleep paralysis during the transition to wakefulness is unknown, but stress, psychiatric disorders, sleep deprivation, and substance use may increase the risk. This patient's preceding stress and potential sleep deprivation and substance use during spring break may have contributed to his sleep paralysis episode. Patients are typically distressed during the episode and may experience an associated feeling of impending doom. Other features of REM sleep may also intrude into transitions from sleep to wakefulness or wakefulness to sleep (eg, hypnopompic or hypnagogic hallucinations). Management of an isolated sleep paralysis episode consists of reassurance and education about normal sleep behavior. If sleep paralysis episodes persist, then the patient should be evaluated for narcolepsy.

Incorrect Answers: B, C, D, E, and F.

Polysomnography and multiple sleep latency test (Choice B) would confirm a diagnosis of narcolepsy. Narcolepsy is a chronic sleep disorder that involves deficient orexin secretion, leading to the repeated intrusion of sleep phenomena (eg, sleepiness, cataplexy, hypnagogic hallucinations, sleep paralysis) into times of wakefulness. An isolated episode would not be concerning for narcolepsy.

Psychiatric assessment (Choice C) and toxicology screening (Choice D) are unnecessary. Though this patient experienced stress, possibly contributing to the sleep paralysis, a single episode of sleep paralysis during wakefulness is common and not an indication of a psychiatric or substance use disorder.

Sleep-deprived EEG (Choice E) is utilized to detect epileptiform discharges in patients with a normal standard EEG when the physician has a high clinical suspicion of a seizure disorder. A single episode of sleep paralysis during wakefulness is common and not an indication of a seizure disorder.

MRI of the brain (Choice F) is unnecessary. A single episode of sleep paralysis during wakefulness is common and not an indication of central nervous system damage.

Educational Objective: Sleep paralysis, a manifestation of the atonia that primarily occurs during rapid eye movement (REM) sleep, happens during the transition from sleep to wakefulness at least once in most people's lives. After a single episode of sleep paralysis, reassurance is appropriate, but if sleep paralysis recurs, narcolepsy should be ruled out.

29. An 82-year-old woman is brought to the physician because of a 6-month history of neck pain radiating to her left hand. She has also had neck stiffness and difficulty maintaining her balance while walking. She has osteoarthritis, Crohn disease, gastroesophageal reflux disease, and hyperlipidemia. Current medications include ranitidine, cholestyramine, mesalamine, and ibuprofen. Examination shows moderately decreased range of motion of the neck. There is mild weakness and atrophy of the hands and forearms and increased muscle tone of the lower extremities. The triceps reflexes are absent, the biceps and brachioradialis reflexes are intact, and the knee and ankle jerk reflexes are increased. Babinski sign is present bilaterally. Sensation to pinprick and temperature is decreased over the dorsal forearms and hands. Her gait is broad based and stiff. Which of the following is the most likely diagnosis?

Your Answer: C) Cervical spondylotic myelopathy

Correct Answer: C) Cervical spondylotic myelopathy

Correct Answer: C

Cervical spondylotic myelopathy is a spinal cord syndrome caused by compression of the cervical spinal cord by degenerated vertebral bodies, facet joints, and/or herniated intervertebral discs. The process begins with progressive degeneration of these surrounding bony structures, which then causes impingement on the central spinal canal. Depending on the level of involvement, the exact symptoms will differ. In most cases, compression of the spinal nerves at the level of the degenerative changes will cause lower motoneuron findings including weakness, hyporeflexia, and atrophy, as seen in the upper extremities of this patient. In contrast, compression of the tracts within the spinal cord, including the lateral corticospinal tract, will cause weakness with upper motoneuron findings in the levels inferior to the degenerative changes. As in this patient, hyperreflexia and an upgoing Babinski sign will be present. The spinothalamic tract may also be compressed, leading to impaired pain and temperature sensation.

Incorrect Answers: A, B, D, and E.

Bilateral brachial plexopathy (Choice A) and bilateral median and ulnar mononeuropathy (Choice B) are both pathologies involving peripheral nerves. Impairment of peripheral motor nerves causes lower motoneuron signs including weakness, fasciculations, hyporeflexia, hypotonia, and atrophy. However, neither of these pathologies would explain the patient's upper motoneuron signs in the lower extremities.

Parasagittal meningioma (Choice D) is an intracranial tumor that grows along the falx cerebri and may compress the adjacent cerebral cortex, possibly including the primary motor cortex in the lower extremity portion of the motor homunculus. This has the potential to cause weakness of the lower extremities with findings of upper motoneuron disease. However, this patient also has lower motoneuron findings, which would not be explained by a parasagittal meningioma.

Subacute combined degeneration (Choice E), caused by demyelination of the dorsal columns, lateral corticospinal tracts, and spinocerebellar tracts, is a feature of vitamin B12 (cobalamin) deficiency. Subacute combined degeneration causes a combination of motor and sensory symptoms including ataxic gait, muscle weakness, paresthesias, and loss of proprioception and vibration sense.

Educational Objective: Cervical spondylotic myelopathy is a spinal cord syndrome caused by compression of the cervical spinal cord by degenerated vertebral bodies, facet joints, and/or herniated intervertebral discs. Motor weakness with lower motoneuron findings at the level of degenerative compression are accompanied by upper motoneuron findings below the level of the compression.

30. A 62-year-old man is brought to the emergency department because of a 1-week history of urinary hesitancy, leg weakness, and difficulty walking; he says that he falls occasionally. He has a 1-month history of sharp right-sided abdominal pain that is exacerbated by coughing, sneezing, and lying flat; the pain now radiates posteriorly to the interscapular region. At his last examination 3 days ago, testing for gallbladder disease was negative. He has hypertension treated with atenolol. His temperature is 37.1°C (98.8°F), pulse is 80/min, respirations are 14/min, and blood pressure is 140/70 mm Hg. Examination shows tenderness to percussion over the midthoracic spine. Muscle strength is 4/5 in the iliopsoas, hamstrings, and anterior tibialis muscles bilaterally; it is otherwise full. Deep tendon reflexes are 2+ in the upper extremities and 3+ in the knees and ankles. Babinski sign is present bilaterally. Sensation to pinprick and temperature is decreased below the costal margin, and sensation to vibration is absent below the ankles. Intravenous administration of which of the following is the most appropriate next step in management?

Your Answer: B) Dexamethasone

Correct Answer: B) Dexamethasone

Correct Answer: B

This patient's presenting signs and symptoms of focal back pain and upper motoneuron signs are suggestive of epidural spinal cord compression, which often results from degenerative joint disease or metastatic disease. Metastatic disease frequently affects the vertebral column and may cause compression of the spinal cord by extending into the epidural space or by causing bony destruction and fracture of the vertebral bodies. Patients present with back pain, lower extremity weakness, loss of balance, urinary or bowel incontinence, or complete paralysis of the lower extremities. Early detection and treatment are crucial to prevent permanent spinal cord damage and myelopathy. Patients may benefit from short-term treatment with intravenous dexamethasone to decrease spinal cord edema, although definitive treatment often requires a combination of surgical decompression in the case of a bony structural lesion or chemotherapy, and/or radiotherapy in the case of malignancy.

Incorrect Answers: A, C, D, and E.

Ceftriaxone (Choice A) is a third-generation cephalosporin and is effective for the treatment of osteomyelitis. While osteomyelitis may lead to spinal cord compression as a result of the formation of an epidural abscess, this is usually accompanied by fever.

Furosemide (Choice C), a loop diuretic, is useful for the treatment of hypervolemia such as in congestive heart failure or cirrhosis. Congestive heart failure does not commonly cause symptoms of myelopathy.

Heparin (Choice D) is an anticoagulant and is useful for the treatment of deep venous thrombosis (DVT). While DVTs may cause lower extremity swelling, pain, and discoloration, they are unlikely to cause focal spinal tenderness with upper motoneuron signs.

Mannitol (Choice E) is an osmotic agent that is useful for the treatment of cerebral edema. While cerebral edema may cause upper motoneuron signs, patients usually have other signs and symptoms including altered mental status, headache, nausea, and vomiting. Mannitol is not commonly used to decrease swelling in the setting of epidural spinal cord compression.

Educational Objective: Dexamethasone is useful for decreasing spinal cord edema and inflammation secondary to epidural spinal cord compression prior to definitive treatment. Patients with epidural spinal cord compression present with back pain, lower extremity weakness, loss of balance, urinary or bowel incontinence, or paralysis of the lower extremities.

31. A 67-year-old woman is brought to the emergency department 1 hour after the sudden onset of right facial weakness. She has hypertension treated with lisinopril and hyperlipidemia treated with atorvastatin. She has smoked one pack of cigarettes daily for 35 years. Her pulse is 76/min, and blood pressure is 168/90 mm Hg. Examination shows right facial weakness of the forehead and lower face. There is decreased hearing and tinnitus on the right. Finger-nose testing shows mild ataxia on the right. An MRI of the brain is most likely to show an infarction in which of the following areas?

Your Answer: D) Pons

Correct Answer: D) Pons

Correct Answer: D

Cerebral infarctions, also known as strokes, occur as a result of ischemic or hemorrhagic loss of blood supply to the brain. Approximately 80% to 85% of strokes are ischemic, commonly arising from thromboembolic disease (eg, middle cerebral artery occlusion from a thrombus), whereas 15% to 20% of strokes are hemorrhagic as a result of blood vessel rupture (eg, hypertension-related intraparenchymal hemorrhage from a perforating artery). Risk factors for stroke include smoking, hypertension, diabetes, carotid or intracranial atherosclerotic disease, history of hypercoagulability, atrial fibrillation, and advanced age. Classically, strokes manifest as a neurologic deficit related to the affected part of the brain. Infarction in the distribution of the anterior inferior cerebellar artery affects several cranial nerve nuclei within the pons, which together are referred to as the lateral pontine syndrome. Involvement of the facial nucleus leads to facial droop caused by muscle paralysis, decreased lacrimation, decreased salivation, and decreased taste from the anterior two-thirds of the tongue. Involvement of the vestibular nucleus causes vomiting, vertigo, and nystagmus. Involvement of the middle and inferior cerebellar peduncles will lead to ataxia and dysmetria. Ipsilateral Horner syndrome and contralateral pain and temperature sensation impairment, though not present in this patient, may also result.

Incorrect Answer: A, B, C, and E.

An infarct in the internal capsule (Choice A) will lead to both motor and sensory deficits of the contralateral side of the body. Occlusion of the lenticulostriate artery can cause this constellation of findings.

Infarction of the medulla (Choice B) causes either medial medullary syndrome, if the anterior spinal artery is affected, or lateral medullary syndrome, if the posterior inferior cerebellar artery is affected. In the former, the lateral corticospinal tract, medial lemniscus, and hypoglossal nerve are affected, leading to contralateral muscle weakness, contralateral impaired proprioception, and ipsilateral tongue deviation, respectively. In lateral medullary syndrome, dysphagia, hoarseness, and decreased gag reflex occur because of involvement of the nucleus ambiguus. The vestibular nucleus and lateral spinothalamic tract are also impaired. These findings are not seen in this patient.

Infarction of the midbrain (Choice C) would impact the function of the oculomotor and trochlear nerves. This patient does not demonstrate any eye movement deficits.

Infarction of the temporal lobe (Choice E) may impact regions such as the Wernicke area of speech and Meyer loop of the visual pathway. As well, it may impact the primary auditory cortex, resulting in varying degrees of hearing impairment depending on the extent of infarction.

Educational Objective: Infarction in the pons will affect several cranial nerve nuclei, including that of the facial and vestibular nerves. Facial paralysis is the most common symptom of cerebrovascular hemorrhage or infarction in this distribution.

32. A 52-year-old man is brought to the emergency department from work 30 minutes after the sudden onset of blindness. He is unable to remember all the events leading up to his visual loss, but he states that earlier in the day he saw his boss in a compromising position; he is uncomfortable speaking about this and quickly changes the subject. He has not had any eye pain but says that he has a headache. He has a history of migraines treated with sumatriptan. Ophthalmic examination shows no evidence of trauma, erythema, or infection. He flinches when the physician rapidly raises her hand toward his face. Funduscopic examination shows no evidence of papilledema, optic neuritis, or retinal hemorrhage or detachment. A CT scan of the head shows no abnormalities. On mental status examination, he appears unaffected by his vision loss. His speech is normal, and he does not have any delusional behavior. When his wife arrives at the emergency department, he asks if he can go home and inquires about when he can return to work. Which of the following is the most likely diagnosis?

Your Answer: C) Conversion disorder

Correct Answer: C) Conversion disorder

Correct Answer: C

Conversion disorder, also known as functional neurologic disorder, involves neurologic symptoms such as sensory or motor dysfunction that are not fully explained by objective findings on physical examination or imaging. This patient's flinching in response to a visual stimulus indicates a subconscious ability to see and makes a neurologic injury unlikely. Though the symptoms disrupt daily life, patients with functional neurologic disorder may or may not be preoccupied with their symptoms. Functional neurologic disorder is postulated to represent the subconscious conversion of intolerable emotional distress into physical symptoms as indicated by this patient quickly changing the subject when asked about a stressor. Psychiatric comorbidities such as depression are common but not present in all patients. Physicians should conduct a thorough history and physical examination, and obtain laboratory and imaging studies to evaluate for medical or neurologic disease that may explain or partially explain the symptoms. Treatment of functional neurologic disorder involves education about the diagnosis, physical therapy, and cognitive-behavioral therapy.

Incorrect Answers: A, B, D, E, and F.

Acute stress disorder (Choice A) refers to psychological stress responses (eg, dissociation, intrusive thoughts, depressed mood, avoidance, hyperarousal) occurring within a month of a traumatic event. This patient's response to stress instead manifests as a neurologic symptom.

Brief psychotic disorder (Choice B) is characterized by at least one acute psychotic symptom (eg, delusions, hallucinations, disorganized speech, disorganized behavior) lasting less than 1 month. This patient demonstrates a neurologic symptom, not a psychotic symptom.

Malingering (Choice D) refers to the conscious production of symptoms for a conscious purpose. This conscious purpose is known as secondary gain and includes motives such as obtaining disability payments or missing work. There is no clear secondary gain for this patient who is not focused on his symptoms and wishes to return to work.

Migraine (Choice E) is a primary headache associated with nausea, photophobia, and phonophobia. Ocular migraine is a rare condition producing abrupt, transient monocular vision loss. However, the vision loss would be demonstrated on physical examination and the patient would not flinch in response to a visual stimulus.

An occipital lobe infarct (Choice F) could lead to vision loss presenting as a contralateral homonymous hemianopia with macular sparing. This patient instead endorses loss of the entire visual field, and CT scan of the head was negative for occipital lesions.

Educational Objective: Conversion disorder, also known as functional neurologic disorder, involves

neurologic symptoms such as sensory or motor dysfunction that are not fully explained by objective findings on physical examination or imaging. Functional neurologic disorder is postulated to represent the subconscious conversion of intolerable emotional distress into physical symptoms.

33. A 72-year-old woman is brought to the physician because of a 24-hour history of severe right ear pain, drooping of the right side of her face, and tearing of the right eye. She had a mild upper respiratory tract infection 1 week ago. She has type 2 diabetes mellitus currently treated with glyburide. She is in distress. Her temperature is 37°C (98.6°F). Examination shows rotatory nystagmus on lateral gaze. There is severe palsy of the right peripheral facial nerve with an inability to close the right eye. Examination shows a rash in the external auditory canal and hearing loss on the right. There is no nuchal rigidity. Which of the following is the most likely cause of these findings?

Your Answer: B) Herpes zoster

Correct Answer: B) Herpes zoster

Correct Answer: B

This patient's presenting signs and symptoms of unilateral facial nerve palsy, ear pain and rash, hearing loss, nystagmus, and tearing, are consistent with a diagnosis of Ramsay Hunt syndrome, also known as zoster oticus. Ramsay Hunt syndrome is caused by reactivation of a latent infection with herpes zoster virus within the geniculate ganglion. Viral reactivation within the geniculate ganglion leads to facial nerve palsy, which causes paralysis of the muscles of facial expression (including the forehead) and may cause lagophthalmos and corneal dryness. In addition, the proximity of the vestibulocochlear nerve to the geniculate ganglion increases the risk for its involvement, which causes a characteristic presentation of ear pain and vesicular rash of the external ear canal or auricle, and may also cause hearing loss, vertigo, and nystagmus. The diagnosis is clinical but may be confirmed with PCR amplification testing. Treatment includes oral prednisone and acyclovir.

Incorrect Answers: A, C, D, E, and F.

Borrelia burgdorferi infection (Choice A) causes Lyme disease, which is a common cause of facial nerve palsy. Lyme disease rarely causes palsy of the vestibulocochlear nerve or a rash of the external ear.

Measles (Choice C) is a highly contagious paramyxovirus that presents with fever, cough, coryza, conjunctivitis, and a confluent maculopapular rash that starts at the head or neck and spreads to the trunk, excluding the palms and soles. In severe cases, it may cause subacute sclerosing panencephalitis. It does not typically cause facial or vestibulocochlear nerve palsy.

Mucormycosis (Choice D) is caused by an invasive infection with fungal organisms such as *Mucor* or *Rhizopus*, and results in aggressive infections of the sinuses, nasal cavity, orbits, or lungs, primarily in patients with a history of immunocompromise (eg, diabetes mellitus). Patients present with a necrotic eschar of the affected area and there are fungal hyphae on tissue biopsy. Mucormycosis rarely affects the ear canal.

Pseudomonas aeruginosa infection (Choice E) and Staphylococcal infection (Choice F) are the most common causes of otitis externa, also known as swimmer's ear. Otitis externa presents with erythema, discharge, and edema of the external ear canal and auricle, but it rarely causes facial or vestibulocochlear nerve palsy.

Educational Objective: Ramsay Hunt syndrome, also known as zoster oticus, is caused by reactivation of latent herpes zoster viral infection within the geniculate ganglion with subsequent cranial nerve palsy of the facial and vestibulocochlear nerves. Patients present with facial paralysis, lagophthalmos, exposure keratopathy, hearing loss, vertigo, nystagmus, and a vesicular rash of the external ear canal and auricle. Treatment is with oral prednisone and acyclovir.

34. A 47-year-old woman is brought to the emergency department 1 hour after the onset of a severe headache that began suddenly while she was exercising at the gym. She rates her pain as a 10 on a 10-point scale and says it is present over her entire head. She has type 2 diabetes mellitus controlled with metformin. She is 172 cm (5 ft 7 in) tall and weighs 72 kg (158 lb); BMI is 25 kg/m². She is alert and oriented. Vital signs are within normal limits. On examination, the right pupil is 5 mm and not reactive to light. Meningismus is present. The remainder of the examination, including neurologic examination, shows no abnormalities. The most likely cause of these findings is a lesion in which of the following arteries?

Your Answer: D) Posterior communicating

Correct Answer: D) Posterior communicating

Correct Answer: D

Subarachnoid hemorrhage (SAH) refers to bleeding into the subarachnoid space. It most commonly occurs secondary to trauma, although aneurysmal rupture is the most common cause of nontraumatic SAH. Patients typically present with an acute, severe headache often described as the worst headache of their life, along with neck stiffness, photophobia, and possible stupor or coma in severe cases. Aneurysms can occur at various locations within the intracranial vasculature. This patient presents with oculomotor nerve palsy, raising suspicion for a posterior communicating artery aneurysm. Oculomotor nerve palsy is a common presenting sign of a posterior communicating artery aneurysm because of its location and compression of the oculomotor nerve. The oculomotor nerve carries parasympathetic fibers allowing for pupillary constriction and motor fibers innervating the levator palpebrae superioris and several extraocular muscles. Patients with an oculomotor nerve palsy can present with pupillary dilation, ptosis, and extraocular muscle paresis leading to diplopia.

Incorrect Answers: A, B, C, and E.

The anterior cerebral arteries (Choice A) are connected by the anterior communicating artery, the most common site for aneurysms in the circle of Willis. Because of its location near the optic chiasm, it can cause visual field defects such as bitemporal hemianopsia.

Basilar artery (Choice B) and middle cerebral artery (Choice C) aneurysms can also rupture and cause an acute headache, intracranial hemorrhage, and/or subarachnoid hemorrhage, but oculomotor nerve palsy would be atypical.

Proximal internal carotid artery (Choice E) aneurysms within the neck are much less common than aneurysms of the posterior communicating artery. As well, aneurysmal rupture of the proximal internal carotid artery would not cause cranial nerve palsies, subarachnoid hemorrhage, or findings of meningismus.

Educational Objective: Patients with SAH typically present with an acute, severe headache often described as the worst headache of their life, along with neck stiffness, photophobia, and possible stupor or coma in severe cases. Oculomotor nerve palsy is a common presenting sign of a posterior communicating artery aneurysm because of its location and compression of the oculomotor nerve.

35. A 37-year-old man comes to the physician because of a 6-month history of difficulty walking and a tremor of the right arm. He says that his movements have been slower. He has hypertension, schizophrenia, and gastroesophageal reflux disease. Current medications include hydrochlorothiazide, fluphenazine, benztropine, and ranitidine. His blood pressure is 108/72 mm Hg. Examination shows increased muscle tone in the right upper extremity; there is a resting tremor of the right hand and decreased right arm swing. His gait is short based. Mental status examination shows a restricted affect. Which of the following is most likely to improve this patient's symptoms?

Your Answer: D) Discontinuing fluphenazine

Correct Answer: D) Discontinuing fluphenazine

Correct Answer: D

Fluphenazine, a first-generation antipsychotic medication, is a dopamine-2 (D2) receptor antagonist. By decreasing mesolimbic dopamine signaling, fluphenazine treats the psychotic symptoms of schizophrenia. D2 antagonism in the basal ganglia can cause acute dystonic reactions and parkinsonism, leading to this patient's bradykinesia (manifesting as a decreased right arm swing while walking), resting tremor, increased muscle tone, and postural instability (manifesting as a short, or shuffling, gait). The chronic use of antidopaminergic agents may also upregulate D2 receptors in the basal ganglia, leading to tardive dyskinesia. Management of medication-induced parkinsonism typically involves adding an anticholinergic medication such as benztropine, which counteracts antidopaminergic activity and addresses parkinsonism. In this patient with schizophrenia already on an anticholinergic medication (benztropine), the next step is decreasing or discontinuing the dose of the offending medication and switching to an antipsychotic medication that causes less potent D2 antagonism.

Incorrect Answers: A, B, C, and E.

Donepezil (Choice A) is a cholinesterase inhibitor utilized in dementia that increases cholinergic signaling in the cortex. Benztropine (Choice C) is an anticholinergic medication that antagonizes muscarinic acetylcholine receptors in the central nervous system (CNS), modulating the dopamine-acetylcholine balance in the basal ganglia, and thus treats parkinsonism and acute dystonic reaction. An increase in cholinergic signaling through adding donepezil or discontinuing benztropine would not address this patient's parkinsonism (and in the case of discontinuing benztropine, would likely exacerbate parkinsonism).

Risperidone (Choice B) is a second-generation antipsychotic that antagonizes the D2 receptor and modulates serotonin signaling. Of the second-generation antipsychotics, risperidone demonstrates the most potent D2 antagonism and would likely exacerbate this patient's parkinsonism.

Ranitidine (Choice E) is a histamine-2 (H2) receptor antagonist that decreases gastric acid secretion and is utilized to relieve reflux symptoms or dyspepsia. H2 antagonists may act on the CNS to cause somnolence and confusion but do not lead to motor dysfunction. Therefore, discontinuing ranitidine would not be helpful.

Educational Objective: Fluphenazine, a first-generation antipsychotic medication, is a dopamine-2 (D2) receptor antagonist. D2 antagonism in the basal ganglia can potentially cause acute dystonic reactions and parkinsonism. To address drug-induced parkinsonism in patients with schizophrenia, anticholinergic medications should be added, or the offending medication should be discontinued in favor of an antipsychotic that causes less potent D2 receptor antagonism.

36. A 25-year-old woman comes to the physician because of a 1-year history of depressed mood and episodes of daytime sleepiness despite getting an adequate amount of sleep at night. She takes daily naps that are marked by visions of bloody bodies before she falls asleep or upon awakening. She is frightened by these visions but cannot control her urge to sleep; she wakes up refreshed. She has smoked one pack of cigarettes daily for 8 years and drinks one to two cups of coffee daily. She does not use illicit drugs. She is 180 cm (5 ft 11 in) tall and weighs 75 kg (165 lb); BMI is 23 kg/m². Her temperature is 36°C (96.8°F), pulse is 80/min, and blood pressure is 120/80 mm Hg. Neurologic examination shows no focal findings. Mental status examination shows anxiety and a mildly depressed mood. An ECG shows no abnormalities. Which of the following is the most likely diagnosis?

Your Answer: C) Narcolepsy

Correct Answer: C) Narcolepsy

Correct Answer: C

Narcolepsy is the most likely diagnosis in this patient with hypnagogic hallucinations and daytime somnolence. The pathogenesis of narcolepsy involves the loss of orexin-A and orexin-B, which are neurotransmitters found primarily in the hypothalamus that promote wakefulness and inhibit REM sleep. Loss of these neurotransmitters permits abnormal transition from wakefulness to REM sleep with the consequent development of hallucinations and other features of narcolepsy, including cataplexy and sleep paralysis. Diagnosis can be made in patients with daytime somnolence and cataplexy or, in cases where cataplexy is not present, with sleep latency testing, which shows a shorter time from wakefulness to the initiation of REM sleep. Occasionally patients undergo lumbar puncture to directly measure concentrations of orexin-A in the cerebrospinal fluid. Pharmacotherapy is directed at promoting wakefulness and avoiding cataplexy.

Incorrect Answers: A, B, D, and E.

Brief psychotic disorder (Choice A) is characterized by hallucinations, delusions, disorganized speech or behavior, or catatonia that lasts more than 1 day but less than a month and is not secondary to a mood disorder or substance abuse.

Major depressive disorder (Choice B) is diagnosed when patients meet at least five of the nine criteria, including impaired sleep (either insomnia or hypersomnolence), loss of interest, guilt, loss of energy, poor concentration, poor appetite, psychomotor retardation, or suicidal ideation. This patient does not meet the criteria.

Nightmare disorder (Choice D) describes the presence of recurrent disturbing dreams. This disorder is not characterized by daytime somnolence or cataplexy.

Sleep apnea (Choice E) commonly presents in obese patients with daytime somnolence and morning headaches. A history of snoring is typical, and diagnosis is made by observing periods of apnea during a sleep study. Treatment is with the use of continuous airway pressure devices.

Educational Objective: Narcolepsy should be suspected in patients with daytime somnolence but restorative sleep with the concurrent presence of hypnagogic hallucinations or cataplexy. Diagnosis can be made clinically if cataplexy is present but otherwise requires sleep latency testing, which shows a shorter time from wakefulness to the initiation of REM sleep.

37. A 56-year-old woman is brought to the emergency department 20 minutes after a simple partial seizure. A CT scan of the head shows a 1.5-cm, uniformly and intensely enhancing parasagittal mass in the left frontal region. Further evaluation for a primary lesion is negative. Examination shows mild weakness of the right lower extremity. Which of the following is the most likely diagnosis?

Your Answer: D) Meningioma

Correct Answer: D) Meningioma

Correct Answer: D

Meningioma is a common, benign central nervous system (CNS) tumor derived from the meninges that occurs along the surfaces of the brain and spinal cord, often in the parasagittal region. They occur in women more often than men and are associated with neurofibromatosis 2 and multiple endocrine neoplasia, type 1. Patients may be asymptomatic, or they may experience seizures or focal neurologic deficits related to the tumor location. Meningiomas are typically benign but can be malignant. Malignant meningiomas may grow rapidly, leading to symptoms of increased intracranial pressure (postural headache, nausea, papilledema). Meningiomas appear as well-circumscribed, extra-axial (outside of the brain parenchyma) masses that compress the underlying parenchyma. They demonstrate uniform enhancement on CT imaging. Definitive diagnosis requires histologic examination of the tissue, which will demonstrate spindle cells arranged in a whorled pattern with psammoma bodies. Treatment may include radiation and/or surgical resection to alleviate the tumoral mass effect, if significant.

Incorrect Answers: A, B, C, and E.

Astrocytoma (Choice A) is a primary CNS neoplasm that can occur as a benign pilocytic astrocytoma in children or malignant glioblastoma multiforme (Choice C) in adults. These highly malignant tumors appear as intra-axial masses in the cerebral hemispheres, which typically demonstrate areas of necrosis and can spread to the opposite hemisphere across the corpus callosum.

Ependymoma (Choice B) is a CNS neoplasm formed from ependymal cells. Location typically involves the fourth ventricle. Histologic characteristics include perivascular pseudorosettes formed by malignant cells arranged around a blood vessel.

Schwannoma (Choice E) are benign tumors that arise from Schwann cells in peripheral nerve sheaths. They commonly affect the facial and vestibulocochlear nerves, and mass effect may result in sensorineural hearing loss or ataxia. They are associated with neurofibromatosis 2.

Educational Objective: Meningiomas are benign CNS tumors arising from the meninges that may compress the brain or spinal cord, and cause focal neurological deficits as well as seizures.

Neuroimaging typically demonstrates a spherical and well-circumscribed uniformly enhancing mass that indents the parenchyma.

38. A 72-year-old man with a history of atrial fibrillation comes to the emergency department 2 hours after the sudden onset of visual problems. He states that he was fine until this afternoon when he started walking into objects that were located on his right side. While driving himself to the hospital, he sideswiped a parked car. His only medication is daily aspirin. On arrival, his pulse is 68/min and irregular, and blood pressure is 140/70 mm Hg. Neurologic examination shows a right homonymous hemianopia. Pupils are equal and reactive, and funduscopy examination shows no abnormalities. Muscle strength and sensation are normal. Mental status examination shows no abnormalities. Which of the following arteries is most likely to be occluded?

Your Answer: F) Left posterior cerebral

Correct Answer: F) Left posterior cerebral

Correct Answer: F

Cerebral infarctions, or strokes, occur as a result of ischemic or hemorrhagic loss of blood supply to the brain. Approximately 80% to 85% of strokes are ischemic, commonly arising from thromboembolic disease, whereas 15% to 20% of strokes are hemorrhagic because of blood vessel rupture. Risk factors for stroke include smoking, hypertension, diabetes, carotid or intracranial atherosclerotic disease, history of hypercoagulability, atrial fibrillation, and advanced age. Classically, strokes manifest as a neurologic deficit related to the affected part of the brain. Occlusion of the left posterior cerebral artery (PCA) explains this patient's deficits. The PCA supplies much of the occipital lobes as well as the inferomedial temporal lobe, posteromedial parietal lobe, thalamus, and parts of the brainstem. Occlusion of the left PCA typically leads to right homonymous hemianopia with macular sparing (as a result of damage of the left primary visual cortex). Right homonymous hemianopia likely explains this patient's car accident. Occlusion of the left PCA may also result in right hemisensory deficit (caused by thalamus damage) and right hemiparesis (caused by damage to the cerebral peduncles in the midbrain). As well, damage to the left parietal cortex may occur and lead to agraphia, acalculia, finger agnosia, and left-right disorientation (Gerstmann syndrome).

Incorrect Answers: A, B, C, D, E, G, H, I, J, and K.

The basilar artery (Choice A) supplies the posterior brain stem and feeds the left and right PCA. Consequently, symptoms include cranial nerve deficits such as nystagmus, diplopia, dysarthria, vertigo, dysphagia, nausea, and vomiting; occipital lobe-related vision loss; and occasional long-tract dysfunction leading to sensorimotor deficits. Severe cases lead to quadriplegia and the loss of facial and mouth movements (locked-in syndrome).

The left and right anterior cerebral arteries (Choices B and G) supply the medial frontal lobes (primary motor and sensory cortices), which control the motor and sensory function of the lower extremities. Occlusion of an anterior cerebral artery leads to contralateral hemiparesis and contralateral sensory deficit of the lower extremities.

The left and right internal carotid arteries (Choices C and H) directly supply the anterior and middle cerebral arteries. In the setting of an acute occlusion of the internal carotid arteries, some patients are asymptomatic because of collateral blood supply. Other patients present with hemiparesis, hemisensory deficit, and/or speech impairment.

The left and right middle cerebral arteries (Choices D and I) supply the lateral frontal lobe (primary motor and sensory cortices), anterior parietal lobe, and temporal lobe. Occlusion of the middle cerebral artery leads to contralateral hemiparesis and contralateral sensory deficit of the face and upper extremities. Occlusion of the middle cerebral artery supplying the dominant hemisphere (usually the left hemisphere) leads to aphasia, while occlusion of the middle cerebral artery

supplying the nondominant hemisphere results in hemineglect.

The left and right ophthalmic artery (Choices E and J) supply structures of the orbit. Occlusion may lead to blindness.

The right PCA (Choice K) supplies the right posterior parietal and occipital lobes (including the visual cortex) along with the thalamus and midbrain. Occlusion of the right PCA typically leads to left homonymous hemianopia with macular sparing and the potential for left hemisensory and motor deficits along with hemineglect.

Educational Objective: The PCA supplies the posterior parietal and occipital lobes (including the primary visual cortex) along with the thalamus and midbrain. Occlusion of the left PCA typically leads to right homonymous hemianopia with macular sparing (caused by damage of the right primary visual cortex).

39. A 35-year-old woman comes to the physician because of a 14-day history of progressively severe headaches, mild neck soreness, and photophobia. Twelve weeks ago, she was treated with x-ray radiation therapy and methotrexate, vincristine, procarbazine, and prednisone for stage IVB Hodgkin disease with a large anterior mediastinal mass; the only complication of therapy was a brief episode of fever and neutropenia. Her temperature now is 37.8°C (100°F). She does not appear acutely ill. Examination shows a mildly rigid neck. Neurologic examination shows no papilledema. Lymph nodes, liver, and spleen are not palpable. Cardiopulmonary examination and x-rays of the chest show normal findings. Serum glucose concentration is 70 mg/dL. Results of lumbar puncture show:

Glucose 15 mg/dL

Protein 120 mg/dL

Leukocyte count 500/mm³

Segmented neutrophils 10%

Monocytes 90%

Erythrocyte count 0/mm³

Which of the following is the most appropriate pharmacotherapy?

Your Answer: B) Intravenous amphotericin B

Correct Answer: B) Intravenous amphotericin B

Correct Answer: B

Amphotericin B is the most appropriate therapy for *Cryptococcus neoformans*, which is an encapsulated, nondimorphic yeast and common opportunistic pathogen in immunocompromised patients. Malignancy (especially lymphoma or leukemia), HIV/AIDS, solid organ transplantation, sarcoidosis, cirrhosis, and chronic corticosteroid therapy are risk factors for infection. It can cause cryptococcal meningitis, encephalitis, and pulmonary cryptococcosis. Transmission is through inhalation, with risk for dissemination to the brain and meninges, as well as other distal sites. Pulmonary cryptococcosis typically presents with a low-grade, intermittent fever and nonspecific respiratory symptoms. Central nervous system involvement can cause headache, nuchal rigidity, photophobia, and progressive encephalopathy. Dermatologic spread can occur with pustular, papular, nodular, or ulcerated lesions. Diagnosis is made by analysis of cerebrospinal fluid (CSF), sputum, urine, and/or blood. Histologic examination may show yeast cells with narrow-based budding and a bright red inner capsule on staining with mucicarmine or periodic acid-Schiff. In cryptococcal meningitis, lumbar puncture classically demonstrates an increased opening pressure, increased CSF protein, and a mononuclear cell pleocytosis. Amphotericin B is the treatment of choice for cryptococcal meningitis or severe pulmonary cryptococcosis in an immunocompromised patient.

Incorrect Answers: A, C, D, and E.

Intramuscular meperidine (Choice A) is an opioid used to manage acute pain or postoperative shivering. It does not play a role in the management of cryptococcal meningitis.

Intravenous ceftriaxone (Choice C) is utilized in the treatment of bacterial meningitis, along with vancomycin and, depending on patient age, ampicillin. Bacterial meningitis presents acutely with fever, nuchal rigidity, and altered mental status in a toxic-appearing patient. While immunosuppressed patients are at risk for bacterial meningitis, this patient's subacute history of mild symptoms is more consistent with cryptococcal meningitis. Additionally, CSF analysis in bacterial meningitis would show a neutrophilic predominance.

Intravenous methylprednisolone (Choice D) is used in the treatment of allergic or autoimmune reactions, multiple sclerosis exacerbations, and a variety of other conditions. Multiple sclerosis

presents with various neurologic lesions occurring over varying periods of time, with an acute exacerbation defined as a new neurologic impairment associated with a new brain or spinal cord lesion on MRI. It does not typically present with signs of meningeal irritation. Additionally, CSF in multiple sclerosis consists of a lymphocytic pleocytosis, normal protein concentration, and the presence of oligoclonal bands.

Intravenous penicillin G (Choice E) can be used to treat a variety of bacterial infections, including neurosyphilis, which can present with headache, altered mental status, vision changes, cranial neuropathies, progressive dementia, and possible brain gummas (eg, focal areas of inflammation). Arteritis can also occur, leading to brain ischemia and infarction. CSF evaluation will show a lymphocytic pleocytosis and increased protein concentration.

Educational Objective: Infection with *Cryptococcus neoformans* is common in patients with immunocompromising conditions. Cryptococcal meningitis presents subacutely with headache, nuchal rigidity, photophobia, and progressive encephalopathy. CSF analysis will demonstrate an increased opening pressure, increased protein, and a mononuclear cell pleocytosis. Treatment is with amphotericin B.

40. A 52-year-old woman comes to the physician because of a 2-week history of moderate pain, numbness, and tingling in her right hand. She states that the pain is worse at night and is not relieved with acetaminophen or ibuprofen. She has a 10-year history of chronic lymphocytic (Hashimoto) thyroiditis treated with thyroid hormone therapy. Vital signs are within normal limits. On examination, tapping the palmar surface of the right wrist elicits a tingling sensation in the thumb and index fingers. Her serum thyroid-stimulating hormone concentration is 2 $\mu\text{U/ml}$. Which of the following is the most likely mechanism of this patient's symptoms?

Your Answer: F) Mechanical entrapment of the median nerve

Correct Answer: F) Mechanical entrapment of the median nerve

Correct Answer: F

Mechanical entrapment of the median nerve leading to carpal tunnel syndrome is the most likely mechanism of this patient's symptoms. The carpal tunnel describes a conduit in the wrist that is formed by the flexor retinaculum anteriorly and the carpal bones posteriorly. Both the median nerve and the flexor tendons of the fingers travel through this canal. Increased pressure in this canal can lead to compression of the median nerve, resulting in paresthesias of the first three digits, primarily on the palmar surface. When severe, weakness of the thumb with atrophy of the thenar eminence may occur. The Tinel test involves tapping the volar surface of the wrist in the midline. A positive test, as in this case, leads to tingling in the first two digits and occasionally the third digit, which corresponds with the distribution of the median nerve. This condition occurs commonly in those with jobs that involve repetitive wrist movement such as typing, but it can be idiopathic. It commonly develops in obese patients and is more common in women. Nighttime symptoms are frequently encountered, and the initial treatment requires wrist splints worn at night. Refractory or severe symptoms may be referred for surgical intervention.

Incorrect Answers: A, B, C, D, E, G, and H.

Immunologic damage to the brachial (Choice A), median (Choice B), radial (Choice C), or ulnar (Choice D) nerves may occur in neuralgic amyotrophy, also called Parsonage-Turner syndrome, which is thought to be related to immune-mediated damage of the nerves of the brachial plexus. It most commonly involves the nerves of the upper or middle brachial plexus but can involve more distal, singular nerves. Patients develop debilitating pain and weakness that gradually improve over months.

Mechanical entrapment of the brachial nerve (Choice E), or the brachial plexus, can result in several distinct clinical syndromes depending upon the location of entrapment. Lower trunk entrapment can result in myotomal weakness in the distribution of C8 to T1 while entrapment of the upper trunk causes weakness in muscles innervated by the C5 and C6 nerve roots.

Mechanical entrapment of the radial nerve (Choice G) commonly occurs at the spiral groove where the radial nerve wraps posteriorly around the humerus. It results in wrist drop, impaired finger extension, and loss of sensation over the dorsum of the hand and forearm.

Mechanical entrapment of the ulnar nerve (Choice H) commonly occurs at the medial epicondyle of the elbow and results in sensory loss of the fourth and fifth digits, with occasional weakness of the interosseous muscles in the hand, resulting in impaired grip strength. Compression can occur at the wrist but is less common.

Educational Objective: Carpal tunnel syndrome commonly results from compression of the median nerve within the carpal tunnel, which is formed by the carpal bones and the flexor retinaculum. Symptoms include paresthesias of the first three digits of the affected hand. Treatment is initially with wrist splints, but surgical intervention and carpal tunnel release may be required in some cases.

41. After falling from a horse during a steeplechase, a 42-year-old man has immediate paralysis in all four extremities and has loss of sensation below the clavicles bilaterally. On examination by paramedics, he is alert, dyspneic, and unable to move his extremities or trunk. His pulse is 98/min, respirations are 28/min, and blood pressure is 150/90 mm Hg. After transport to a trauma center, his sensory examination returns to normal, and he regains normal strength of the arms and forearms, although the hands and lower extremities remain weak. Examination shows bladder distention. An MRI of the cervical spine shows contusion with edema of the spinal cord at C3 and C4. Plain x-rays of the cervical spine show a nondisplaced fracture of the vertebral body of C3. Which of the following is the most appropriate next step in management?

Your Answer: F) Surgical exploration of the cervical spine

Correct Answer: D) Intravenous methylprednisolone therapy

Correct Answer: D

Spinal cord compression can occur in the setting of trauma, malignancy with vertebral metastases, and infections (eg, epidural abscess). Its presentation depends on the level of the lesion and the degree of spinal cord compression. Traumatic spinal cord injury is typically caused by fracture or dislocation of the vertebrae. Acute traumatic spinal cord injuries typically cause immediate compression or shearing of the spinal cord, with a secondary injury of the spinal cord occurring after the initial injury because of consequent inflammation, edema, and ischemia. Symptoms include back or neck pain, flaccid paralysis, numbness, bowel and bladder incontinence or retention, and hyporeflexia. Patients may also develop neurogenic shock, depending on the level of injury, with bradycardia and hypotension secondary to injury to the sympathetic outflow tract, as well as respiratory failure caused by diaphragmatic paralysis. Diagnosis is classically made using MRI of the spine, as well as spinal x-rays or CT to visualize skeletal abnormalities. Management includes clinical stabilization, and the use of intravenous glucocorticoid therapy to decrease spinal cord edema. Patients may also require intubation and mechanical ventilation, vasopressor support, cardiac pacing, and/or surgical spine stabilization.

Incorrect Answers: A, B, C, E, and F.

Intravenous ceftriaxone therapy (Choice A) is used in the treatment of bacterial meningitis, along with vancomycin and, depending on patient age, ampicillin. Bacterial meningitis presents acutely with fever, nuchal rigidity, and altered mental status in a toxic-appearing patient, although it can also cause focal neurologic deficits.

Intravenous labetalol therapy (Choice B) is used in hypertensive urgency and emergency, which present with increased blood pressures, typically with the systolic blood pressure greater than 180 mm Hg. End-organ damage is characteristic of hypertensive emergency. This patient's blood pressure is near normal and does not require correction with intravenous medication.

Intravenous mannitol therapy (Choice C) can be used to decrease cerebral edema in the setting of traumatic brain injury, since it is an osmotic diuretic that draws excess fluid out from tissues and excretes it from the body. It plays less of a role in controlling edema of the spinal cord, and its onset of action is longer than that of methylprednisolone.

Oral prednisone therapy (Choice E) is used in rheumatic diseases, allergic reactions, and inflammatory or immune reactions. While it could decrease the spinal cord edema seen in this patient because of its anti-inflammatory properties, its onset of action is longer than methylprednisolone.

Surgical exploration of the cervical spine (Choice F) is typically performed in the setting of an unstable vertebral fracture or dislocation. It may be necessary in this case; however, glucocorticoids can be administered immediately to begin temporizing the effect of the lesion.

Educational Objective: Acute traumatic spinal cord injury occurs in the setting of compression or shearing of the spinal cord; however, subsequent injury can occur because of inflammation, edema, and ischemia of the cord. Initial treatment with intravenous glucocorticoids is critical, and some patients with unstable fractures or dislocation may require surgical stabilization.

42. A 47-year-old man is brought to the emergency department by police after he was found in a state of confusion outside a convenience store. He says he feels anxious. He has not had fever, headache, abdominal pain, nausea, or vomiting. He has a 3-year history of uncontrolled hypertension. He has no allergies and takes no medications. He does not smoke. He usually drinks 12 to 14 beers daily. He says he has never used illicit drugs. He is disheveled and agitated and picks at imaginary things in the air. He is disoriented to person, place, and time. His temperature is 37.5°C (99.5°F), pulse is 110/min, and blood pressure is 160/105 mm Hg. Examination shows diaphoresis and tremulousness. There is seborrhea and rosacea of the face. There are no signs of head trauma. The neck is supple. There is diffuse hyperreflexia. Babinski sign is absent bilaterally. Which of the following is the most likely diagnosis?

Your Answer: E) Delirium tremens

Correct Answer: E) Delirium tremens

Correct Answer: E

This patient has developed delirium tremens, a serious complication of alcohol withdrawal. Alcohol increases the central nervous system activity of γ -aminobutyric acid (GABA), the main inhibitory neurotransmitter. Abrupt discontinuation of alcohol therefore leads to sympathetic overdrive. Uncomplicated alcohol withdrawal typically presents with tremors, anxiety, restlessness, diffuse hyperreflexia, headache, nausea, and diaphoresis, which can occur hours after the cessation of alcohol use. Delirium tremens, the most concerning and life-threatening complication of alcohol withdrawal, involves severe confusion and disorientation, fluctuations in consciousness, agitation, visual hallucinations, and autonomic instability (variable heart rate and blood pressure with hyperthermia). Delirium tremens typically begins 2 to 5 days after the patient's last alcoholic drink. Delirium tremens is managed supportively with benzodiazepines (GABAA receptor agonists) to address the psychomotor agitation associated with withdrawal and prevent the worsening of withdrawal manifestations. Management also includes fluid resuscitation, nutritional supplementation, and frequent assessment including vital sign checks.

Incorrect Answers: A, B, C, D, F, G, and H.

Acute subdural hematoma (Choice A) refers to bleeding between the dura and arachnoid membranes, most commonly caused by head trauma in elderly patients and persons with alcohol use disorder. Patients typically present with a declining neurologic status or coma. This patient does not demonstrate signs of head trauma on physical examination, and sympathetic overdrive is more consistent with alcohol withdrawal.

Alcohol intoxication (Choice B) can cause slurred speech, ataxia, emotional lability, and memory lapses. Sympathetic overdrive is more consistent with alcohol withdrawal.

Brief psychotic disorder (Choice C) is characterized by at least one acute psychotic symptom (eg, delusions, hallucinations, disorganized speech, disorganized behavior) lasting less than 1 month. Sympathetic overdrive and disorientation are more consistent with delirium tremens.

Cocaine intoxication (Choice D) presents with euphoria and increased sympathetic tone, manifesting as tachycardia and hypertension, pupillary dilation, and restlessness. Hallucinations may occur, though disorientation is more consistent with delirium tremens. This patient endorses alcohol but not cocaine use.

Encephalitis (Choice F) refers to inflammation (caused by infection or autoimmune response) of the brain parenchyma and may present with confusion, agitation, and/or hallucinations, along with headache, photophobia, and/or seizures. However, sympathetic overdrive and a history of heavy alcohol use are more consistent with delirium tremens.

Hypoglycemia (Choice G) may manifest with sympathoadrenal symptoms (eg, tremors, palpitations,

diaphoresis, mild hypertension, mild tachycardia) along with cognitive impairment. This patient lacks known risk factors for hypoglycemia and has a history of heavy alcohol use.

Meningitis (Choice H), or inflammation of the meninges, typically presents with headache, neck stiffness, and fever, often caused by a bacterial or viral infection. This patient does not present with symptoms typical for meningitis.

Educational Objective: Delirium tremens is a life-threatening complication of alcohol withdrawal related to decreased GABA activity that presents with severe confusion, agitation, visual hallucinations, and autonomic instability along with other features of sympathetic overdrive (eg, diaphoresis, tremors, diffuse hyperreflexia). Benzodiazepines are GABAA receptor agonists utilized to treat agitation during withdrawal and prevent the worsening of withdrawal manifestations.

43. A 6-month-old boy is admitted to the hospital for evaluation of spasms. The child's father and a paternal aunt are mildly retarded and have seizures. The father has acneiform lesions on his cheeks, but without comedones. Examination of the patient shows five 1- to 2-cm depigmented areas on the chest and flexor surfaces of the upper and lower extremities. A CT scan of the head shows a subependymal calcification near the foramen of Monro on the left. Which of the following is the most likely diagnosis?

Your Answer: E) Tuberous sclerosis

Correct Answer: E) Tuberous sclerosis

Correct Answer: E

Tuberous sclerosis is an autosomal dominant neurocutaneous disorder characterized by benign tumors in several organ systems. Hamartomas can form in the soft tissue (leading to Shagreen patches, or subcutaneous oval nodules) or brain (leading to epilepsy and intellectual disability). Cardiac rhabdomyomas (leading to mitral regurgitation) and renal angiomyolipomas are also common. Other typical dermatologic findings include ash leaf spots (hypomelanotic macules) and angiofibromas, which are acneiform lesions without comedones (clogged hair follicles). Subependymal giant cell tumors are common, which typically demonstrate calcification and may increase the likelihood of developing seizures. Though these tumors are benign, patients are at risk for developing malignancies in the soft tissues, brain, or kidney. Diagnosis is typically clinical but may be confirmed by genetic testing indicating a mutated TSC gene. Management requires anticonvulsant medication for seizures and surgical resection of rapidly growing or symptomatic tumors.

Incorrect Answers: A, B, C, and D.

Idiopathic hypercalcemia (Choice A) is a rare condition that typically presents in infancy with developmental delay, constipation, polyuria, vomiting, feeding problems, and weight loss. Adults can rarely demonstrate an idiopathic increase in vitamin D and consequently become hypercalcemic. Neither variant is hereditary nor associated with dermatologic findings.

Neurofibromatosis (Choice B) is a hereditary neurocutaneous disorder. Type 1 presents with cutaneous neurofibromas (benign neoplasms derived from neural crest cells), cafe au lait spots, pigmented iris hamartomas (Lisch nodules), optic gliomas, and pheochromocytomas. Type 2 presents with cutaneous neurofibromas, bilateral vestibular schwannomas, additional schwannomas of the cranial nerves or in the spinal cord, intracranial meningiomas, and neuropathy. Subependymal brain lesions are atypical, and the dermatologic findings differ between tuberous sclerosis and neurofibromatosis.

Sturge-Weber syndrome (Choice C) is a neurocutaneous disorder that presents with the proliferation of capillary-size blood vessels, seen on examination as nevus flammeus (a benign birthmark also known as a port-wine stain), angiomas of the leptomeninges (leading to seizures, intellectual disability, focal neurologic deficits), and episcleral hemangioma. The causative activating mutation of the GNAQ gene is typically sporadic rather than inherited.

TORCH syndrome (Choice D) refers to fetal infection by any of a group of organisms: Toxoplasma gondii, rubella virus, cytomegalovirus, or Herpes simplex virus, and others. Newborns may demonstrate a variety of consequent symptoms, which often include fever, feeding difficulties, petechiae, hepatosplenomegaly, jaundice, deafness, and/or vision abnormalities. The combination of characteristic dermatologic findings and seizures 6 months after birth (outside of the perinatal period) is more consistent with tuberous sclerosis.

Educational Objective: Tuberous sclerosis is an autosomal dominant neurocutaneous disorder characterized by benign tumors in several organ systems. Patients typically present with

hamartomas of the skin and brain, cardiac rhabdomyomas, and renal angiomyolipomas. Dermatologic findings include hypomelanotic macules (ash leaf spots) and acneiform lesions called angiofibromas, and neurologic findings include subependymal giant cell tumors, which may lead to seizures.

44. A 62-year-old man comes to the physician because of a 4-month history of progressive difficulty climbing stairs, generalized muscle pain, and dry mouth. He has hypertension, type 2 diabetes mellitus, and hyperlipidemia. Current medications include atorvastatin, atenolol, and glyburide. He has smoked two packs of cigarettes daily for 30 years. At his last visit 6 weeks ago, an x-ray of the chest showed a solitary lung nodule. Today, his pulse is 80/min, respirations are 14/min, and blood pressure is 136/82 mm Hg. Examination of the lower extremities shows bilateral proximal weakness and hyporeflexia. Electromyography and nerve conduction studies show an incremental increase in the amplitude of evoked motor response. Which of the following is most likely to have prevented these symptoms?

Your Answer: A) Smoking cessation

Correct Answer: A) Smoking cessation

Correct Answer: A

Lambert-Eaton myasthenic syndrome (LEMS) is an uncommon neurologic condition that affects neuromuscular junction transmission. Normally, action potentials lead to depolarization in the terminal bouton of the axon. Voltage-gated calcium channels consequently open and allow calcium influx, which triggers the exocytosis of neurotransmitter-filled vesicles into the synaptic cleft. Patients with LEMS possess antibodies to voltage-gated calcium channels on presynaptic lower motoneurons, which decreases calcium influx and the subsequent release of acetylcholine. This patient with a smoking history and proximal muscle weakness, with a known lung nodule, likely has paraneoplastic autoantibody production against voltage-gated calcium channels, causing LEMS in the setting of small cell lung cancer. Patients may also have depressed or absent deep tendon reflexes. The muscle weakness typically improves with exercise as small amounts of acetylcholine accumulate in the synaptic cleft over time, increasing the frequency of postsynaptic action potentials in skeletal muscle cells. This can be seen on nerve conduction studies, as well. Autonomic symptoms such as decreased salivation (and consequent dry mouth) may also occur. The diagnosis may be confirmed with nerve conduction studies and electromyography with exercise testing. In this patient, smoking cessation would have prevented the development of lung cancer and consequent LEMS.

Incorrect Answers: B, C, D, and E.

Daily aspirin therapy (Choice B) is used in the prophylaxis and treatment of cardiovascular disease, such as coronary artery disease and stroke. It does not play a role in the prevention of lung cancer development.

Discontinuing atenolol therapy (Choice C) or atorvastatin therapy (Choice D) would not have prevented this patient from developing lung cancer and subsequent LEMS. While β -adrenergic blocker therapy can cause fatigue and statin therapy can cause myopathy, neither causes hyporeflexia.

More aggressive treatment of high blood pressure (Choice E) is an incorrect choice, as it would not have prevented this patient from developing lung cancer. Additionally, this patient's blood pressure is well-controlled.

Educational Objective: Patients with Lambert-Eaton myasthenic syndrome possess antibodies to voltage-gated calcium channels on presynaptic lower motoneurons, which decrease calcium influx and acetylcholine release. This syndrome can be caused by paraneoplastic antibodies, particularly in the setting of small cell lung cancer, which is smoking-related. Patients typically present with progressive proximal muscle weakness that improves with exercise or repeated stimulation.

45. A 30-year-old construction worker comes to the physician because of constant midlower back pain for 1 month. He has no leg pain. Straight-leg raising causes discomfort only in the low back. Sensorimotor examination is normal. Which of the following is the most appropriate next step in management?

Your Answer: A) Restriction of heavy lifting until the pain resolves and instruction in proper body mechanics

Correct Answer: A) Restriction of heavy lifting until the pain resolves and instruction in proper body mechanics

Correct Answer: A

Low back pain is a common condition that is self-limited for many individuals. There are many causes of low back pain, and it rarely can be caused by a serious illness that requires medical treatment or surgical intervention. Risk factors for low back pain include strenuous work, heavy lifting, trauma, obesity, and age. Most patients present with a lumbosacral strain caused by heavy lifting or extended periods of sitting. In patients with a history of trauma or osteoporosis, vertebral compression fractures or other vertebral fractures are a consideration. Signs and symptoms that increase the risk for an urgent medical or surgical cause of low back pain include fever, focal midline spinal tenderness, weight loss, pain increased at night, focal neurologic symptoms, and urinary incontinence or urinary retention. In this patient with no sensorimotor abnormalities, the most likely cause of his low back pain is a lumbosacral strain from heavy lifting and construction work. Most patients with lumbosacral pain will improve within 4 weeks with short-term symptomatic therapy, including pain control, restriction of heavy lifting, physical therapy, and instructions on proper body mechanics, allowing for an early return to activity. Prolonged periods of inactivity and bed rest are to be avoided.

Incorrect Answers: B, C, D, and E.

Bed rest and oral analgesics (Choice B) would not be the most appropriate management. While oral analgesics can help with pain, early movement and physical therapy has been shown to decrease pain and expedite return of function. Bed rest will increase pain and prolong time to recovery.

Vocational rehabilitation and change in occupation (Choice C) would not be appropriate before a trial of physical therapy. The patient is young and will likely recover after a period of restricting heavy lifting and physical therapy to prevent further injuries.

Epidural injection of corticosteroids and nonsteroidal anti-inflammatory drugs (Choice D) would also be inappropriate prior to a period of physical therapy and restriction of heavy lifting. Typically, physical therapy and rehabilitation are sufficient to treat lumbosacral strain in a majority of patients. Intervertebral disc excision and fusion (Choice E) would only be appropriate in the setting of a disc herniation causing severe pain and functional limitation despite a trial of physical therapy. In some cases, disc herniation can cause acute onset neurologic deterioration from spinal cord compression or cauda equina syndrome, which requires emergent surgical management. This patient does not demonstrate any signs of neurologic dysfunction.

Educational Objective: The most common cause of low back pain is a lumbosacral strain. Most patients will improve within 4 weeks with short-term symptomatic therapy, including pain control, restriction of heavy lifting, physical therapy, and instruction on proper body mechanics, allowing for an early return to activity.

46. A 32-year-old man comes to the physician because of a 6-month history of deep aching neck and shoulder pain, progressive weakness of his hands, and poor coordination. Two years ago, he had a whiplash injury that resolved without treatment over a 3-month period. Neurologic examination shows hyporeflexia of the upper extremities. He has atrophy of intrinsic hand muscles bilaterally. There is asymmetric decreased strength of grip and finger abduction and extension. Sensation to pinprick is decreased over the upper torso and shoulders. Which of the following is the most likely cause of these findings?

Your Answer: F) Syringomyelia

Correct Answer: F) Syringomyelia

Correct Answer: F

Syringomyelia refers to a progressively expanding cystic lesion involving midline spinal cord structures, most commonly in the cervical or thoracic spinal cord. Syringomyelia may be associated with congenital malformations such as Chiari malformation type I, spinal cord infections, autoimmune diseases such as multiple sclerosis, neoplasms, or spinal cord trauma. Syringomyelia typically compresses spinothalamic fibers decussating in the anterior white commissure. Consequently, patients present with deficits in pain and temperature sensation in the bilateral upper extremities and/or torso. (Since the ascending spinothalamic tracts in the lateral spinal cord are unaffected, sensation in the lower body is intact.) Neuropathic pain in the upper extremities may occur. The lesion can expand to compress the bilateral anterior horns (the location of the cell bodies of lower motoneurons). Therefore, patients may present with a lower motoneuron pattern of dysfunction in the bilateral upper extremities (eg, decreased muscle tone, hyporeflexia, early muscle atrophy). The cyst may progressively expand to compress the lateral corticospinal tract in severe cases, leading to an upper motoneuron pattern of dysfunction in the lower body.

Incorrect Answers: A, B, C, D, and E.

Amyotrophic lateral sclerosis (Choice A) is a progressive neurodegenerative disorder that affects the anterior horns and the lateral corticospinal tract. Subsequently, patients demonstrate a combination of upper and lower motoneuron findings in an asymmetric distribution. Sensory impairment is atypical, and a pattern of dysfunction that can be localized to a single lesion within the cervical or thoracic spinal cord is more consistent with a syringomyelia.

Anterior spinal cord syndrome (Choice B) results from ischemia of the anterior spinal artery (eg, by thrombosis or trauma), leading to damage to the corticospinal, spinothalamic, and autonomic tracts. The syndrome manifests with loss of pain and temperature sensation and motor function below the level of the lesion, along with autonomic dysfunction. Syringomyelia causes loss of pain and temperature sensation localized to the upper extremities, and autonomic dysfunction is uncommon in syringomyelia because of the lateral location of autonomic cell bodies.

Brown-Sequard syndrome (Choice C), hemisection of the spinal cord, damages the anterior horn along with the corticospinal tract, dorsal column-medial lemniscus tract, and spinothalamic tract. Patients typically present with contralateral impairment in pain and temperature sensation, ipsilateral deficits in other sensory domains, and an ipsilateral upper motoneuron pattern of dysfunction below the level of the lesion. The dorsal column-medial lemniscus (DCML) tract is typically unaffected in syringomyelia.

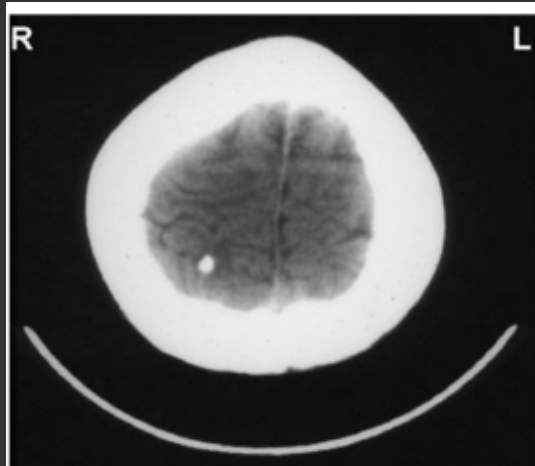
Posterior cord syndrome (Choice D) results from ischemia of the posterior spinal artery, leading to ipsilateral DCML tract damage. Patients present with ipsilateral sensory loss of proprioception, fine touch, and vibration below the level of the lesion. The DCML tract is typically unaffected in syringomyelia.

Subacute combined degeneration (Choice E), resulting from vitamin B12 (cobalamin) deficiency,

occurs as a result of demyelination of the spinocerebellar tracts, lateral corticospinal tracts, and dorsal columns. The DCML tract is typically unaffected in syringomyelia.

Educational Objective: Syringomyelia features a cystic cavity in the central cervical or thoracic spinal cord, which affects midline structures such as the anterior white commissure (decussation of the spinothalamic tracts) and the anterior horns (cell bodies of lower motoneurons). Patients with syringomyelia may present with deficits in pain and temperature sensation and a lower motoneuron pattern of dysfunction in the bilateral upper extremities and/or torso.

47. A previously healthy 13-year-old girl is brought to the emergency department 1 hour after a generalized tonic-clonic seizure that occurred at school. Her teacher noted that the seizure was preceded by jerking of the right arm. She had a similar episode 8 months ago. Examination at that time showed no abnormalities, and no treatment was begun. She was born in Mexico, and she and her parents have made occasional trips to rural Mexico; their last visit was 2 years ago. She was lethargic on arrival to the emergency department, but she is now awake and alert. Her temperature is 37°C (98.6°F). The neck is supple. Neurologic examination shows no abnormalities. Her leukocyte count is 6400/mm³. A CT scan of the head is shown. Which of the following is the most appropriate screening test for family members of this patient?



Your Answer: A) Examination of the stool for ova and parasites

Correct Answer: A) Examination of the stool for ova and parasites

Correct Answer: A

Neurocysticercosis is a central nervous system (CNS) infection caused by the pork tapeworm *Taenia solium* that occurs most commonly in endemic rural regions of Latin America, Asia, and Africa. Symptoms include headache, seizures, signs of increased intracranial pressure, and, rarely, altered vision. Some patients are asymptomatic, while other patients may present with muscle or subcutaneous nodules. Typical manifestations on CT or MRI include cystic, ring-enhancing, calcified lesions in the brain parenchyma, meninges, ventricles, spinal cord, and/or orbit. A scolex (larval head) may also be demonstrated on imaging. Treatment includes symptomatic management of increased intracranial pressure and seizures, antiparasitic therapy, and/or steroids. As this patient has primarily traveled with her family, they are similarly at an increased risk for neurocysticercosis and should be screened for the infection with examination of the stool for ova and parasites.

Incorrect Answers: B, C, D, E, F, and G.

Rapid plasma reagin (Choice B) is used to screen for infection with *Treponema pallidum*, which causes syphilis. Neurosyphilis may present with headache, altered mental status, vision changes, cranial neuropathies, progressive dementia, and possible brain gummas (eg, focal areas of inflammation). Gummas can be seen on neuroimaging as large, enhancing lesions, but they are not typically calcified.

Stool culture for *Coccidioides immitis* (Choice C), a dimorphic fungus present in the Southwestern United States and Central and South America, could be used to evaluate for coccidioidomycosis. Coccidioidal meningitis presents subacutely or chronically with headache, nausea and emesis, vision changes, and altered mental status. Neuroimaging may show hydrocephalus or vasculitis

with associated infarctions, but it rarely demonstrates intracranial, ring-enhancing, or calcified lesions.

Stool culture for *Histoplasma capsulatum* (Choice D) can be used to screen for histoplasmosis. *H. capsulatum*, an endemic mycosis, is primarily found in the Midwestern United States. Disseminated infection most commonly occurs in immunosuppressed patients, whereas isolated infection of the lungs is more common in immunocompetent patients. While disseminated histoplasmosis can present with multiple enhancing brain lesions, it would be unlikely in this otherwise healthy young patient.

Stool culture for *Toxoplasma gondii* (Choice E) can be used to screen for toxoplasmosis. In immunocompetent patients, the infection is generally asymptomatic, although it can present with a flu-like illness. In immunocompromised patients, however, toxoplasmosis can lead to cerebral infection with multiple ring-enhancing brain lesions. This would be unlikely in this otherwise healthy young patient.

Urine culture for cytomegalovirus (CMV) (Choice F) would not be useful in this scenario. While CMV can cause encephalitis in rare circumstances, it more commonly causes mononucleosis with fevers, fatigue, and cervical lymphadenopathy in an immunocompetent host. It does not present with brain lesions on neuroimaging.

CT scan of the head (Choice G) could evaluate for neurocysticercosis in this patient's family members. However, it would unnecessarily expose them to radiation that could be avoided by utilizing the stool examination for ova and parasites to diagnose an infection, making it a less appropriate next step in management.

Educational Objective: Neurocysticercosis is a CNS infection caused by the pork tapeworm *Taenia solium* that occurs in endemic rural regions of Latin America, Asia, and Africa and may infect immunocompetent patients. Typical manifestations on CT or MRI include cystic, ring-enhancing, calcified lesions in the brain parenchyma, meninges, ventricles, spinal cord, and/or orbit. Screening for the infection can be performed by examining the stool for ova and parasites.

48. A 6-year-old girl is brought to the physician because of a 1-week history of progressive weakness and numbness of the legs. This morning, she had urinary incontinence. One year ago, she underwent resection of a medulloblastoma and placement of a ventriculoperitoneal shunt for residual hydrocephalus; her postoperative course had been uncomplicated. Currently, she is drowsy and irritable. Her temperature is 37.5°C (99.5°F), pulse is 82/min, respirations are 24/min, and blood pressure is 126/54 mm Hg. Funduscopic examination shows no abnormalities. Examination of the lower extremities shows weakness and diffuse sensory loss; deep tendon reflexes are decreased. Which of the following is the most likely underlying mechanism for these symptoms?

Your Answer: C) Metastatic spread of the cancer through cerebrospinal fluid pathways

Correct Answer: C) Metastatic spread of the cancer through cerebrospinal fluid pathways

Correct Answer: C

Metastatic spread of cancer through cerebrospinal fluid pathways is the mechanism behind this patient's development of cauda equina syndrome. Cauda equina syndrome occurs as a result of compression of the spinal nerve roots comprising the cauda equina. Often, such compression occurs because of malignancy (as in this case), trauma, disc herniation, spinal stenosis, spinal epidural abscess, or spinal epidural or subdural hematoma. Symptoms include lower extremity numbness, weakness, paralysis, hyporeflexia, perineal anesthesia, fecal incontinence with poor rectal tone, and overflow incontinence marked by an inability to void with associated urinary retention. Emergent MRI of the lumbosacral spine is indicated to investigate the cause of cauda equina syndrome and to determine the necessity of surgical decompression. Initial treatment for patients with epidural spinal cord compression and neurologic deficits include symptomatic treatment, pain control, and glucocorticoid therapy prior to operative therapy. High-dose glucocorticoid therapy is helpful as a bridge to definitive therapy and may transiently improve neurologic function by decreasing edema at the site of cord compression.

Incorrect Answers: A, B, D, E, and F.

Local recurrence of the medulloblastoma (Choice A) would cause neurologic deficits specific to the region of the brain being affected by the tumor. While, depending on the location, this could cause lower extremity weakness, it would cause hyperreflexia rather than hyporeflexia because of the involvement of the upper motoneuron. This patient's lower extremity hyporeflexia indicates a peripheral cause affecting the lower motoneuron.

Metastatic spread of the cancer hematogenously (Choice B) would be more likely to cause foci of malignancy to develop in other organs including the bones, liver, or lungs. This is an uncommon complication of medulloblastoma since extraneural metastases are rare with this particular tumor. Radiation necrosis of the brain stem (Choice D) would cause symptoms of midbrain, pons, and medulla infarcts. The cranial nerve nuclei along with ascending and descending spinal tracts would be affected. Symptoms would include impaired cranial nerve function and decreased sensorimotor function of the ipsilateral face and contralateral body. Lower extremity weakness would be accompanied by hyperreflexia, not hyporeflexia, because of the involvement of the upper motoneurons.

Thrombosis of the anterior spinal artery (Choice E) causes dysfunction of the anterior tracts within the spinal cord including the corticospinal tract (loss of motor function), spinothalamic tract (loss of pain and temperature sensation), and anterior horns.

Thrombosis of the superior sagittal sinus (Choice F) is rare because of the large size of this sinus. However, when it does occur, thrombosis in this area results in severe headaches followed by the potential for altered mental status, bilateral neurologic deficits, and seizures. This patient's findings

are more consistent with cauda equina syndrome secondary to metastatic spread of cancer. Educational Objective: Cauda equina syndrome occurs as a result of compression of the spinal nerve roots comprising the cauda equina. It may occur secondary to metastatic spread of cancer through cerebrospinal fluid pathways. Symptoms include lower extremity numbness, weakness, paralysis, hyporeflexia, perineal anesthesia, fecal incontinence with poor rectal tone, and overflow incontinence.

49. A 45-year-old woman is brought to the emergency department 30 minutes after a generalized tonic-clonic seizure lasting 1 minute. She had a headache 6 hours ago and nausea and vomiting 2 hours ago. She has no history of seizures and has had no medical problems other than hypertension treated with enalapril and hydrochlorothiazide. Her temperature is 37.2°C (99°F), pulse is 100/min and regular, respirations are 16/min, and blood pressure is 230/140 mm Hg. She appears drowsy but is responsive to commands. Examination shows no abnormalities other than the fundoscopic findings shown. Which of the following is the most appropriate initial pharmacotherapy?



Your Answer: D) Nitroprusside

Correct Answer: D) Nitroprusside

Correct Answer: D

This patient's presenting signs and symptoms of hypertension, tachycardia, altered mental status, tonic-clonic seizures, and optic disc edema are consistent with a diagnosis of hypertensive encephalopathy. Hypertensive encephalopathy may result from acute and severe increases in blood pressure and may manifest with headache, vomiting, agitation, altered mental status, and seizures. It may progress to obtundation and coma. Visual symptoms may result from cortical hypertension or from hypertensive retinopathy. Retinal findings may include optic disc edema and nerve fiber layer infarction or hemorrhage. Treatment includes carefully lowering the blood pressure while avoiding excessively rapid drops in blood pressure, which can cause cerebral or cardiac ischemia. In general, a decrease in the mean arterial pressure of 25% within the first hour is appropriate, with gradual correction over the next day or more following. Sodium nitroprusside is a potent vasodilator and is a preferred agent because of its very short half-life, allowing it to be titrated via intravenous infusion.

Incorrect Answers: A, B, D, and E.

Clonidine (Choice A) is an oral or intravenous antihypertensive agent that stimulates α_2 -adrenergic receptors. It is commonly used as an adjunct oral agent for the management of hypertensive emergency but is less appropriate as initial pharmacotherapy.

Furosemide (Choice B) is an oral or intravenous loop diuretic. Diuretics are not typically used in the treatment of hypertensive emergency because of their comparatively long half-life and duration of action.

Lisinopril (Choice C) is an oral angiotensin-converting enzyme inhibitor. Oral agents are not preferred in the initial management of hypertensive emergency.

Propranolol (Choice E) is an oral or intravenous nonselective β -adrenergic blocker. It is not typically used as a primary agent in the treatment of hypertensive emergency because of its comparatively

long half-life and duration of action.

Educational Objective: Hypertensive encephalopathy may result from acute and severe increases in blood pressure, and may manifest with headache, vomiting, agitation, altered mental status, and seizures. It may progress to obtundation and coma. Retinal findings may include optic disc edema and nerve fiber layer infarction or hemorrhage. Sodium nitroprusside is a potent vasodilator and is a preferred agent because of its very short half-life, allowing it to be titrated via intravenous infusion.

50. A 42-year-old woman comes to the physician because of a 4-day history of progressive weakness and numbness in her legs and blurred vision and discomfort in both eyes. She has no history of serious illness and takes no medications. Visual acuity is 20/80 bilaterally. Examination shows bilateral optic disc edema. Muscle strength is 5/5 in the upper extremities and 4/5 in the lower extremities. Deep tendon reflexes are 2+ in the upper extremities and 4+ in the lower extremities. Babinski sign is present bilaterally. A lumbar puncture is performed. Analysis of the cerebrospinal fluid shows:

Glucose 80 mg/dl

Total protein 105 mg/dl

WBC 50/mm³

RBC 5/mm³

Oligoclonal bands 0 bands

An MRI of the brain shows multiple periventricular white matter lesions. An MRI of the thoracic spine shows an abnormal T2 signal and mild enhancement within the thoracic cord spanning T3 to T6. Which of the following abnormalities is most likely to be noted on further testing in this patient?

Your Answer: C) Aquaporin-4 autoantibodies in the cerebrospinal fluid

Correct Answer: C) Aquaporin-4 autoantibodies in the cerebrospinal fluid

Correct Answer: C

Aquaporin-4 autoantibodies in the cerebrospinal fluid (CSF) are diagnostic of neuromyelitis optica (NMO), a disease characterized by intense inflammation and demyelination of both the optic nerves and various spinal cord segments, with involvement of both grey and white matter. Common manifestations include episodes of optic neuritis, which manifests as vision loss and pain, and transverse myelitis, which can affect any level of the spinal cord. Other unique findings include episodes of hiccups with associated nausea and vomiting. The disease course is often described as relapsing and remitting, with some patients experiencing few symptoms between episodes. Treatment is with high-dose intravenous steroids and, occasionally, plasma exchange for acute attacks. Prophylaxis with eculizumab or other biologic agents is warranted at the time of diagnosis.

Incorrect Answers: A, B, D, and E.

Abnormal echocardiography (Choice A) is seen in patients with strokes secondary to septic emboli from endocarditis, but this patient does not have any evidence of endocarditis, and strokes typically present with unilateral symptoms as opposed to bilateral symptoms, which is more consistent with a spinal cord syndrome.

Antinuclear antibodies (ANA) in the serum (Choice B) are a feature of many rheumatologic diseases, including systemic lupus erythematosus. There is also a portion of the population that has positive ANA titers and no identifiable disease; thus, this is a nonspecific finding.

Decreased peripheral nerve conduction velocities (Choice D) are characteristic of peripheral neuropathy, which usually manifests with sensory changes or weakness of the extremities in a stocking-glove distribution. NMO is a disease of the central nervous system (CNS), and this patient's findings of positive Babinski tests and hyperreflexia are consistent with a central pathology. Decreased serum copper concentration (Choice E) is not a feature of NMO. Increased copper concentration is seen in Wilson disease, which can present with psychosis, Kayser-Fleischer rings, and cirrhosis.

Educational Objective: NMO is a disease of the CNS with a characteristically relapsing and remitting course that presents with episodes of optic neuritis, transverse myelitis, and intractable hiccups. The presence of aquaporin-4 autoantibodies in the CSF is diagnostic of this disorder.

1. Over the past 3 weeks, a 64-year-old right-handed man has had five episodes of blindness in the left eye lasting 12 to 20 minutes. His history is otherwise noncontributory. Which of the following is the most appropriate next step in diagnosis?

Your Answer: C) Carotid duplex scan

Correct Answer: C) Carotid duplex scan

Correct Answer: C

Amaurosis fugax refers to transient vision loss lasting from several seconds to several minutes, typically caused by ischemia of the retina or optic nerve. Vision loss is typically unilateral but is occasionally bilateral. The most common cause is ipsilateral carotid artery disease with emboli to the retinal or optic nerve vasculature, although amaurosis fugax may also occur in the setting of vasculitis (eg, temporal arteritis). Amaurosis fugax is a medical emergency because it may herald either impending stroke due to carotid embolization or impending irreversible vision loss due to either thromboembolism or vasculitis. Evaluation should include immediate carotid duplex scan to evaluate for carotid vascular disease. Patients with symptoms suggestive of temporal arteritis, such as fever, optic nerve loss, headache, or jaw claudication, should undergo urgent laboratory evaluation of erythrocyte sedimentation rate, C-reactive protein and platelet count and be considered for immediate, high-dose glucocorticoid therapy. Ophthalmologic examination may show retinal vascular emboli, known as Hollenhorst plaques, although most patients with amaurosis fugax have a normal ocular examination.

Incorrect Answers: A, B, D, and E.

Ophthalmologic evaluation (Choice A) is a crucial component of the evaluation of transient vision loss. However, most patients with amaurosis fugax have a normal ocular examination. While an ophthalmologic evaluation should be performed, obtaining a carotid duplex scan is a more important initial step in diagnosis.

24-Hour ambulatory ECG monitoring (Choice B) is helpful for diagnosis of cardiac arrhythmia but is not usually recommended for patients with amaurosis fugax. While cardiac arrhythmia predisposes to vascular emboli generally, most emboli that are implicated in amaurosis fugax originate from the ipsilateral carotid artery.

CT scan of the head (Choice D) or MRI of the brain (Choice E) may be helpful for identifying prior embolic strokes or cerebral vascular disease but is not as important as carotid duplex scan in establishing the underlying diagnosis.

Educational Objective: Educational Objective: Amaurosis fugax refers to transient vision loss lasting from several seconds to several minutes, typically caused by ischemia of the retina or optic nerve. The most common cause is ipsilateral carotid artery disease with emboli to the retinal or optic nerve vasculature. A careful history and examination are important to exclude the presence of temporal arteritis. Evaluation should include immediate carotid duplex scan to evaluate for carotid vascular disease.

2. A 77-year-old man is brought to the emergency department because of a 2-day history of headache, confusion, and blurred vision. He has not had any recent trauma to the head or eyes. He has chronic hypertension, type 2 diabetes mellitus, irritable bowel syndrome, and bilateral hearing loss. Current medications are hydrochlorothiazide, glipizide, amlodipine, losartan, and chlorthalidone. On arrival, he is holding his head in his hands. He is oriented to name and place but not to date of birth, month, year, or the current president. His temperature is 37°C (98.6°F), pulse is 84/min, respirations are 12/min, and blood pressure is 230/130 mm Hg. Visual acuity is 20/30 in both eyes. Visual fields are full. Funduscopic examination shows blurred optic disc margins and several slit-like hemorrhages in both fundi. A CT scan of the head shows a hypoattenuated area in both occipital lobes and an old lacunar infarct in the basal ganglia and thalami. Which of the following is the most appropriate initial pharmacotherapy?

Your Answer: E) Intravenous labetalol

Correct Answer: E) Intravenous labetalol

Correct Answer: E

Correct Answer: E. Hypertensive encephalopathy results from acute and/or severe increases in blood pressure. It potentially manifests with headache, vomiting, agitation, altered mental status, and seizure. It may progress to obtundation and coma. Visual symptoms may result from cortical dysfunction or from hypertensive retinopathy, including retinal hemorrhages and exudates.

Treatment includes carefully lowering the blood pressure while avoiding excessively rapid drops in blood pressure, which can cause end-organ ischemia. Intravenous agents such as labetalol, esmolol, nicardipine, and nitroprusside are preferred for initial management.

Incorrect Answers: A, B, C, and D.

Oral 81-mg aspirin (Choice A) can be used as secondary prevention of ischemic stroke. While this patient does have evidence of prior strokes on CT scan, his acute encephalopathy is secondary to his increased blood pressure, and primary management should be focused on lowering his blood pressure.

Intravenous alteplase (Choice B) is a fibrinolytic medication used for ischemic strokes that binds to fibrin clots and converts plasminogen to plasmin, which lyses clots. If given early (generally within 3 to 4.5 hours), alteplase may promote neurologic recovery. This patient has evidence of prior strokes on CT scan, likely secondary to his uncontrolled hypertension, but is well outside of the window for receiving alteplase. Additionally, blood pressure greater than 180/110 mm Hg is a contraindication to receiving alteplase as there is increased risk for hemorrhagic conversion.

Intravenous dexamethasone (Choice C) is used in treatment of spinal cord compression or cerebral edema secondary to brain tumors. Corticosteroids are not a part of the management of hypertensive encephalopathy.

Intravenous heparin (Choice D) may be indicated in the event of a thromboembolic ischemic stroke caused by underlying atrial fibrillation. This patient has no evidence of an underlying cardiac arrhythmia, and his strokes are likely caused by his underlying poorly controlled hypertension.

Educational Objective: Educational Objective: Hypertensive encephalopathy results from acute and/or severe increases in blood pressure. It potentially manifests with headache, vomiting, agitation, altered mental status, and seizure, and may progress to obtundation and coma. Treatment

should be focused on carefully lowering the blood pressure. Intravenous agents such as labetalol, esmolol, nicardipine, and nitroprusside are preferred for initial management.

3. A 57-year-old man is brought to the emergency department after he was found walking unsteadily and speaking in a confused and incoherent manner. Neurologic examination shows bilateral abducens nerve palsies, coarse nystagmus in all directions of gaze, and severe truncal ataxia. Which of the following is the most likely diagnosis?

Your Answer: J) Wernicke encephalopathy

Correct Answer: J) Wernicke encephalopathy

Correct Answer: J

Correct Answer: J. Vitamin B₁, or thiamine, plays a complex role in neuronal physiology including in axon conduction, glucose utilization, and neurotransmitter synthesis. Thiamine deficiency results in neuronal damage through a mechanism that likely involves oxidative toxicity. Thiamine deficiency causes two potential syndromes: Wernicke encephalopathy and Korsakoff syndrome. The classic triad of Wernicke encephalopathy demonstrates encephalopathy, ataxia, and oculomotor dysfunction caused by neuronal damage in the cerebellum, midbrain, and pons. Classic oculomotor findings include nystagmus and horizontal gaze palsy as a result of damage to the abducens nuclei. Thiamine deficiency is most common in the setting of alcohol use disorder due to poor nutritional intake, thiamine malabsorption, and impaired cellular utilization of thiamine. Over time, thiamine deficiency can result in damage and atrophy of the mammillary bodies, visible on MRI of the brain, which manifests as Wernicke encephalopathy. The symptoms of Wernicke encephalopathy are reversible but increase the risk for developing Wernicke-Korsakoff syndrome in chronic alcohol use disorder. Korsakoff syndrome is an irreversible condition that features psychosis, anterograde and retrograde amnesia, and the consequent tendency to confabulate, or verbalize false stories or memories to compensate for the inability to remember.

Incorrect Answers: A, B, C, D, E, F, G, and H.

Diabetic polyneuropathy (Choice A) is characterized by sensorimotor peripheral neuropathy. Findings of ataxia and nystagmus would be atypical.

Huntington disease (Choice B) is a hereditary, progressive neurodegenerative disorder associated with atrophy of the caudate nuclei in the basal ganglia. The disease usually presents with an insidious onset of psychiatric and cognitive symptoms, followed by motor symptoms (eg, chorea, hypotonia, dystonia, loss of voluntary control of movements).

Lead encephalopathy (Choice C) presents with cognitive dysfunction, encephalopathy, hearing loss, and peripheral neuropathy. Distal motor neuropathy causes lower motor neuron signs including weakness, atrophy, hypotonia, and hyporeflexia. Non-neurologic manifestations of lead poisoning include renal tubular dysfunction, vomiting, abdominal pain, constipation, and anemia.

Multiple sclerosis (Choice D) is an autoimmune demyelinating disease that presents with focal neurologic deficits separated in space and time. Ataxia and nystagmus are atypical.

Educational Objective: Educational Objective: Vitamin B₁, or thiamine, deficiency causes Wernicke encephalopathy, characterized by encephalopathy, ataxia, and oculomotor dysfunction. Diagnosis is clinical and should be rapidly treated with intravenous thiamine administration to avoid progression to Korsakoff syndrome.

4. A 24-year-old man comes to the physician 1 day after he had a generalized tonic-clonic seizure at work. His symptoms began with a rising sensation in the upper middle abdomen. His coworker told him that he started to stare, licked his lips, and then fell to the floor. He had generalized shaking of his arms and legs. After he awakened, he was confused for several minutes and smelled an odor of burnt rubber that his coworkers did not notice. He has noticed a similar odor three times in the past 2 months. He was hit by a baseball bat at the age of 12 years and had a brief loss of consciousness. He has been otherwise healthy. Physical and neurologic examinations show no abnormalities. Which of the following is the most likely location of the abnormality?

Your Answer: F) Temporal lobe

Correct Answer: F) Temporal lobe

Correct Answer: F

Correct Answer: F. Focal-onset impaired awareness seizures are also called complex partial seizures. As this name suggests, these seizures affect a specific (focal) region of the brain and therefore produce motor or sensory symptoms that are localized to one or a few areas of the body. Patients are unresponsive during the seizure (impaired awareness). Focal seizures can spread to the bilateral hemispheres, evolving into generalized tonic-clonic seizures, as in this patient. Tonic-clonic seizures characterized by clonic movements of the extremities, followed by rhythmic jerking of the extremities, oral automatisms and autonomic disturbances such as an epigastric rising sensation, as in this patient, are suggestive of a temporal lobe lesion, the most common origin of seizures. Manual automatisms and upper limb dystonia can also occur in temporal lobe seizures. Patients may experience visual, auditory, or sensory auras preceding the seizure. Focal-onset impaired awareness seizures are often accompanied by a brief postictal state marked by confusion and fatigue. These seizures will demonstrate localized rhythmic activity with generalized slowing on EEG. Treatment requires antiepileptic medication. For patients with refractory temporal lobe epilepsy, surgical excision of the lesion can be considered.

Incorrect Answers: A, B, C, D, and E.

Brain stem (Choice A) lesions involving the medulla, midbrain, or pons may result in focal seizure activity characterized by convulsive movements; however, brainstem involvement more typically occurs in the setting of generalized tonic-clonic seizures.

Cerebellum (Choice B) lesions can result in focal seizure activity characterized by focal motor or hemifacial seizures, but these seizures are extremely rare. Cerebellar involvement more typically occurs in the setting of generalized tonic-clonic seizures.

Frontal lobe (Choice C) lesions can also present with focal seizures with impaired awareness. Disturbance of the higher cognitive functions of the frontal lobe can lead to features such as speech disturbances, complex automatisms, and strong emotional expressions such as crying or laughing. Focal seizures with intact awareness with motor symptoms may also occur as a result of involvement of the primary motor cortex.

Occipital lobe (Choice D) lesions are uncommon causes of focal seizure activity, but typically present with visual manifestations such as visual hallucinations, cortical blindness, and deviated gaze.

Parietal lobe (Choice E) lesions can result in focal seizure activity involving the primary somatosensory cortex, resulting in sensory disturbances such as numbness, tingling, warmth, or pain. Other manifestations of parietal lobe epileptic activity may include receptive aphasia, dyslexia, and other language/comprehension disturbances due to involvement of Wernicke area at the junction of the parietal and temporal lobes.

Educational Objective: Educational Objective: Focal-onset impaired awareness seizures, or complex partial seizures, produce localized motor or sensory symptoms. Focal seizures can spread to the bilateral hemispheres and evolve into generalized tonic-clonic seizures. Lesions of the temporal lobe, the most common origin site of seizures, can present with oral or manual automatisms, autonomic disturbances such as an epigastric rising sensation, and upper limb dystonia. Patients may experience visual, auditory, or sensory auras preceding the seizure, along with a brief postictal state marked by confusion and fatigue.

5. A 67-year-old man has had four acute episodes of aphasia over the past year. His temperature is 37.1°C (98.8°F), pulse is 84/min, respirations are 12/min, and blood pressure is 142/82 mm Hg. Serum studies show a glucose concentration of 150 mg/dL, total cholesterol concentration of 296 mg/dL, and HDL-cholesterol concentration of 26 mg/dL. Carotid ultrasonography shows 90% stenosis of the left internal carotid artery. Which of the following is the most appropriate therapy to decrease this patient's risk for a stroke?

Your Answer: G) Carotid endarterectomy

Correct Answer: G) Carotid endarterectomy

Correct Answer: G

Correct Answer: G. This patient with four separate episodes of acute aphasia that self-resolved has a presentation concerning for multiple transient ischemic attacks (TIAs). TIA is defined as a transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia without evidence of infarction. The symptoms last less than 24 hours by definition and may last as little as a few minutes. Patients with TIAs are at increased risk for having an ischemic stroke; therefore, management of patients with a TIA should focus on identification and treatment of modifiable risk factors. Such risk factors include smoking, hypertension, hyperlipidemia, and hypercoagulable states. Patients with TIAs should undergo MRI, Doppler ultrasonography of the carotid arteries, echocardiography, and cardiac monitoring. Patients with 70% to 99% stenosis of the carotid arteries should undergo carotid endarterectomy or carotid stenting to decrease their subsequent risk for stroke.

Incorrect Answers: A, B, C, D, E, and F.

Control of blood pressure and control of serum LDL-cholesterol concentration with diet and medication (Choices A and B) and aspirin therapy (Choice E) are also indicated as secondary stroke prevention in this patient. However, carotid endarterectomy should take precedence for this patient with known significant carotid artery atherosclerosis, as this was likely the direct embolic source for his ischemic event and would lead to the greatest reduction in risk for subsequent stroke.

Control of ventricular rate with digoxin (Choice D) would not be appropriate for this patient with a normal heart rate. Digoxin may be used to achieve rate control in patients with atrial fibrillation who do not have an adequate response to more conventional therapies such as beta blockers and calcium channel blockers. While atrial fibrillation is a known risk factor for thromboembolic strokes, controlling the ventricular rate does not decrease that risk. Therefore, patients with atrial fibrillation and increased stroke risk should be on warfarin therapy (Choice F) or another anticoagulant medication such as rivaroxaban or dabigatran.

Educational Objective: Educational Objective: Transient ischemic attack (TIA) is defined as a transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia without evidence of infarction. The symptoms last less than 24 hours by definition and may last as little as a few minutes. Patients with TIAs are at increased risk for having an ischemic stroke, and therefore management of patients with a TIA should focus on identification and treatment of modifiable risk factors. Patients with 70% to 99% stenosis of the carotid arteries should undergo carotid endarterectomy or carotid stenting to decrease their subsequent risk for stroke.

6. A 60-year-old woman is brought to the office by her husband because of a 6-week history of progressive unsteadiness, incoordination, and slurred speech. During the past 6 months, she also has had a nonproductive cough, fatigue, and a 7-kg (15-lb) unintentional weight loss despite no change in appetite. She has not seen a physician in more than 10 years and does not take any medications. She has smoked fifteen cigarettes daily for the past 40 years. She drinks alcoholic beverages only on social occasions. Temperature is 37.0°C (98.6°F), pulse is 70/min, respirations are 13/min, and blood pressure is 130/85 mm Hg. Physical examination shows dysarthria, but no aphasia is noted. Cranial nerves are intact except for bilateral horizontal nystagmus. She has no focal muscle weakness or sensory deficit. Finger-nose testing and heel-shin testing discloses bilateral dysmetria. Deep tendon reflexes are 2+. The patient is unable to tandem walk. MRI of the brain discloses multiple T-2 hyperintensities in both cerebral hemispheres and mild cerebellar atrophy with enhancement of the cerebellar folia. MR angiography of the circle of Willis shows no abnormalities. Carotid Doppler ultrasonography discloses minimal stenosis in the common carotid arteries bilaterally. Which of the following is the most likely cause of this patient's symptoms?

Your Answer: D) Paraneoplastic syndrome

Correct Answer: D) Paraneoplastic syndrome

Correct Answer: D

Correct Answer: D.

A variety of paraneoplastic neurologic syndromes may occur following an abnormal immune response to the presence of an underlying malignancy. The most implicated tumors are small cell lung carcinoma, breast and ovarian carcinoma, and Hodgkin lymphoma. In many cases, the onset of a paraneoplastic disorder may precede detection of the primary malignancy; therefore, patients may present without a known cancer history and a high index of suspicion for an occult tumor must be maintained. Paraneoplastic cerebellar degeneration (PCD) is a rare disorder characterized by dysfunction of the cerebellar cortex with signs and symptoms of ataxia, intention tremor, vertigo, nystagmus, dysarthria, and eventual encephalopathy. PCD is caused by autoantibody production against central nervous system antigens leading to cerebellar Purkinje cells. MRI often shows cerebellar atrophy. Further diagnostic evaluation should be performed to identify the primary malignancy in a patient with suspected PCD.

Incorrect Answers: A, B, C, and E.

Creutzfeldt-Jakob disease (Choice A) is a human prion disease that presents with rapidly progressing dementia, neuropsychiatric symptoms, cerebellar dysfunction, corticospinal tract dysfunction, basal ganglia dysfunction, and stimulus-induced myoclonus. While Creutzfeldt-Jakob disease may cause a rapidly progressive dementia, this patient's history of smoking, cough, and unintentional weight loss are suggestive of malignancy and a paraneoplastic syndrome.

Miller-Fisher syndrome (Choice B) is a subset of Guillain-Barré syndrome, an autoimmune demyelinating disorder. Miller-Fisher syndrome is characterized by the triad of ophthalmoplegia, ataxia, and areflexia. This contrasts with Guillain-Barré syndrome, which presents with acute-onset, ascending, bilateral muscle weakness. This patient has normal deep tendon reflexes, suggesting against the diagnosis.

Multisystem atrophy (Choice C) is a parkinsonian, neurodegenerative disorder characterized by

parkinsonism, autonomic dysfunction, and ataxia. Autonomic symptoms may include orthostatic hypotension, dry mouth, anhidrosis, urinary incontinence, and erectile dysfunction.

Progressive supranuclear palsy (Choice E) is a parkinsonian, neurodegenerative disorder characterized by gait instability, oculomotor deficits (classically vertical gaze palsy), frontal lobe deficits, bradykinesia, and muscle rigidity.

Wernicke encephalopathy (Choice F) results from vitamin B₁ (thiamine) deficiency and presents with the triad of encephalopathy, ataxia, and oculomotor dysfunction (typically nystagmus and horizontal gaze palsy). It most commonly affects patients with a history of alcohol use disorder.

Educational Objective: Educational Objective: Paraneoplastic neurologic syndromes may occur following an abnormal immune response to the presence of an underlying malignancy. Paraneoplastic cerebellar degeneration (PCD) is a rare disorder characterized by dysfunction of the cerebellar cortex with signs and symptoms of ataxia, intention tremor, vertigo, nystagmus, dysarthria, and eventual encephalopathy.

7. A previously healthy 16-day-old male newborn is brought to the emergency department by his parents because of a 6-hour history of temperatures to 38.9°C (102°F), fussiness, and decreased feeding. He was born at 39 weeks' gestation to a 32-year-old woman, gravida 2, para 2, and weighed 3090 g (6 lb 13 oz) at birth. The mother received routine prenatal care. Pregnancy was complicated by group B streptococcal infection at 37 weeks' gestation; no other abnormalities were noted on screening. The mother received three doses of intrapartum ampicillin. The patient was discharged on the second day of life. Well-child examinations on days 3 and 14 of life showed no abnormalities. At 2 weeks of age, he weighed 3232 g (7 lb 2 oz). He is currently breast-feeding. He has not had sick contacts. Today, he weighs 3260 g (7 lb 3 oz). Head circumference is at the 50th percentile. His rectal temperature is 38.8°C (101.8°F), pulse is 142/min and regular, respirations are 48/min, and blood pressure is 58/40 mm Hg. Examination shows no cyanosis or rash. The head is normocephalic with no signs of trauma. The anterior and posterior fontanelles are open and ballotable; the anterior fontanelle is not bulging. There is mild scleral icterus. The liver edge is palpated 1.5 cm below the right costal margin. The spleen is not palpable. The umbilical cord has detached; the site is dry and nonerythematous. A lumbar puncture is performed. Results of urinalysis and blood, urine, and cerebrospinal fluid cultures are pending. Which of the following is the most appropriate next step in management?

Your Answer: E) Intravenous ampicillin and gentamicin therapy

Correct Answer: E) Intravenous ampicillin and gentamicin therapy

Correct Answer: E

Correct Answer: E.

Neonatal sepsis is often categorized on the basis of time from parturition, as the period of time in which symptoms begin may indicate which pathogen is likely involved, thereby guiding therapy. Early-onset neonatal sepsis occurs within 72 hours of birth, typically after exposure to pathogens acquired from the mother, either through hematogenous or transplacental spread or, most commonly, contamination in the birth canal. Late-onset neonatal sepsis presents 72 hours or more after birth and is most likely caused by contact with environmental pathogens or through pathogens acquired through the mother. Organisms commonly associated with late-onset neonatal sepsis include Group B Streptococcus and Escherichia coli. Infants admitted to neonatal intensive care units are also at risk for other Gram-negative bacteria, coagulase-negative Staphylococcus, and Staphylococcus aureus. Syndromes of serious bacterial infection include bacteremia, meningitis, and pneumonia. Workup for suspected neonatal sepsis in this age group should include blood, urine, and cerebrospinal fluid cultures, along with blood cell count, urinalysis, Gram stain, and imaging studies if warranted. The diagnosis is confirmed with positive culture of blood, urine, or cerebrospinal fluid. Until diagnosis is confirmed, empiric treatment with intravenous broad-spectrum antibiotics to cover the common pathogens should be initiated. In neonates, ampicillin and gentamicin provide effective coverage against the most common causative agents of neonatal sepsis.

Incorrect Answers: A, B, C, D, and F.

Discharge after administration of a single dose of ceftriaxone (Choice A) and admission to the hospital for observation only (Choice B) are incorrect, as this septic and hypotensive newborn requires intravenous antibiotics and further workup.

Intramuscular ampicillin and gentamicin therapy (Choice C) could be an appropriate choice if the newborn required only one dose of antibiotics. This patient will require several days of antibiotics,

and therefore intravenous administration is more appropriate as it avoids the need to re-stick the newborn multiple times.

Intravenous ceftriaxone therapy (Choice D) and intravenous ceftriaxone therapy (Choice F) are incorrect. Ceftriaxone should be avoided in newborns at risk for hyperbilirubinemia as it displaces bilirubin from albumin, increasing the free plasma concentrations of unconjugated bilirubin, and thereby increasing the risk for kernicterus. This newborn with scleral icterus likely already has some degree of hyperbilirubinemia.

Educational Objective: Educational Objective: Neonatal sepsis typically occurs after exposure to pathogens acquired from the mother or from the environment. Symptoms include fever, fussiness, decreased feeding, and may progress to shock. Empiric treatment with intravenous broad-spectrum antibiotics should be initiated until diagnosis is confirmed. In neonates, ampicillin and gentamicin provide effective coverage against the most common causative agents of neonatal sepsis.

8. An 18-year-old man comes to the office because of a 3-month history of increasingly severe left knee pain. The pain was originally mild and intermittent but has progressed to be severe and constant. He has noticed fullness in the area above the left knee during the past month; during the past 2 weeks, there has been increased difficulty walking and bearing weight on the affected leg. There is no recent history of trauma to the knee. He says that he "has not been feeling well overall" during the past month and that he has had a 2-kg (4.5-lb) unintentional weight loss because of decreased appetite. He has no history of serious illness, and his only medication is ibuprofen as needed for partial pain relief. Temperature is 37.0°C (98.6°F), pulse is 85/min, respirations are 12/min, and blood pressure is 135/80 mm Hg. Physical examination shows erythema and tenderness of the left distal thigh and above the knee; there is limited flexion of the knee. An x-ray of the knee shows an area of medullary and cortical bone destruction with periosteal new bone formation at the diaphyseal end of the lesion. Which of the following is the most appropriate next step in management?

Your Answer: D) Operative treatment of the knee and chemotherapy

Correct Answer: D) Operative treatment of the knee and chemotherapy

Correct Answer: D

Correct Answer: D.

Osteosarcoma is a rare, primary malignant tumor of bone that affects the metaphysis of long bones. Osteosarcoma has a bimodal distribution, most frequently affecting adolescents and patients over age 60 years. Presenting symptoms include localized pain, pathologic fractures at the areas of involvement, and associated weight loss. Physical examination may show large, tender, palpable soft tissue mass with overlying erythema, as in this patient. Diagnosis is made with radiographic imaging, including x-ray and MRI, and confirmed with bone biopsy. Management includes surgical resection and chemotherapy.

Incorrect Answers: A, B, C, and E.

Intramuscular corticosteroid injection (Choice A) may be used in the treatment of systemic inflammatory conditions. These injections are generally administered in the superior lateral buttocks to avoid injection into the sciatic nerve region. Intramuscular steroid injections are not used in the treatment of osteosarcoma.

Intravenous antibiotic therapy (Choice B) is used in the treatment of septic arthritis and many infectious conditions. The most common causative microorganism in septic arthritis is *Staphylococcus aureus*. Antibiotics targeting *S. aureus* are often used as empiric antibiotic therapy.

Knee brace placement and physical therapy (Choice C) are commonly used in the treatment of acute knee injuries and osteoarthritis. Knee braces provide added stability to the joint, while physical therapy aims to increase muscle strength and flexibility as well as prevent recurrent injury to the affected area.

Oral nonsteroidal anti-inflammatory drug therapy (Choice E) such as naproxen is commonly used in the treatment of acute pain and osteoarthritis. These medications inhibit the COX enzyme. Adverse effects include gastritis, gastrointestinal bleeding, and nephropathy. They may be suitable adjuncts in cases of osteosarcoma but are not the primary treatment.

Oral vitamin D supplementation and bisphosphonate therapy (Choice F) may be used in the treatment of osteoporosis. Osteoporosis is diagnosed by DEXA scan and is most common in postmenopausal female patients.

Educational Objective: Educational Objective: Osteosarcoma is a primary malignant tumor of the bone most frequently affecting adolescent and elderly patients. Patients may present with localized pain, pathologic fracture, and a palpable, tender mass on physical examination. Patients are evaluated with imaging studies and treated with surgical resection and chemotherapy.

9. A 67-year-old man with a 15-year history of emphysema comes to the office because of right-sided shoulder pain for 6 months. He has not had chest pain, palpitations, or headache. He has smoked two packs of cigarettes daily for 40 years. He has worked as a coal miner for 45 years. He appears older than his age. He is not in distress. His temperature is 37.0°C (98.6°F), pulse is 100/min, respirations are 20/min, and blood pressure is 120/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92%. The left pupil is 3 mm, the right pupil is 2 mm; they both react to light. There is ptosis on the right. Examination shows decreased diaphragmatic excursion on the right and a prolonged expiratory phase. There is moderate clubbing of the fingers. Intrinsic muscles of the right hand are atrophic; muscle strength is 4/5 in the right upper extremity. X-ray of the chest shows a 3 x 4-cm right apical mass. Which of the following is the most likely mechanism of this patient's shoulder pain?

Your Answer: C) Infiltration of the brachial plexus with malignant cells

Correct Answer: C) Infiltration of the brachial plexus with malignant cells

Correct Answer: C

Correct Answer: C.

Bronchogenic carcinoma commonly presents with cough, weight loss, hemoptysis, and potential signs of bronchial obstruction such as wheezing or prolonged expiratory phase. If the carcinoma is located within the lung apex, referred to as a Pancoast tumor, it can infiltrate multiple adjacent structures such as the superior vena cava, subclavian vessels, sympathetic chain, recurrent laryngeal nerve, or brachial plexus. This can result in presenting features of Horner syndrome (unilateral ptosis, miosis, and anhidrosis of the ipsilateral shoulder and upper extremity), and hoarseness. If the Pancoast tumor involves the brachial plexus, superior vena cava syndrome may occur, which manifests with unilateral upper extremity and facial swelling, erythema, and vascular engorgement caused by insufficient venous drainage from mass effect or vessel invasion. A Pancoast tumor typically presents as a consolidative, rounded mass in the lung apex on chest x-rays and CT scans.

Incorrect Answers: A, B, D, and E.

Extrinsic compression of the spinal cord (Choice A) can occur with metastatic cancers to the spine or with an infection such as an epidural abscess, which can present with weakness and numbness in a dermatomal or myotomal distribution. It would be unlikely to cause Horner syndrome.

Small cell carcinoma of the lung is a neoplasm of neuroendocrine cells and is associated with numerous paraneoplastic syndromes, including Cushing syndrome caused by increased production of ACTH (Choice B). ACTH produced by small cell carcinoma stimulates the adrenal gland zona fasciculata to increase the production of cortisol. Hypercortisolism, or Cushing syndrome, presents with many manifestations, including proximal muscle weakness, weight gain, central adiposity, hypertension, hyperglycemia, moon facies, acne, striae, and immune suppression. Small cell lung carcinoma is smoking related.

Squamous cell carcinoma is associated with hypercalcemia due to production of parathyroid-like hormone (also known as parathyroid hormone-related peptide) (Choice D). Hypercalcemia can present with nephrolithiasis, chronic bone pain, abdominal discomfort, and psychiatric disturbances.

Jugular vein thrombosis (Choice E) would not compress adjacent structures and would be unlikely to cause Horner syndrome.

Educational Objective: Educational Objective: Pancoast tumor can infiltrate adjacent structures such as the superior vena cava, subclavian vessels, sympathetic chain, recurrent laryngeal nerve, or brachial plexus and can lead to Horner syndrome, ipsilateral shoulder and upper extremity pain, and hoarseness.

10. A 72-year-old man comes to the physician because of a 1-year history of progressive difficulty walking and a 2-month history of urinary incontinence. His wife reports that he is extremely slow to answer questions. Neurologic examination shows bilateral grasp reflexes and increased muscle tone in all extremities. When he tries to walk, he appears "glued" to the floor and cannot take steps without assistance. He is oriented to person, place, and time but speaks very little. Which of the following is most likely to be increased in this patient?

Your Answer: E) Ventricle size

Correct Answer: E) Ventricle size

Correct Answer: E

Correct Answer: E.

This patient presents with findings suggesting normal-pressure hydrocephalus (NPH), with the characteristic combination of dementia, gait instability, and urinary incontinence. NPH refers to a type of communicating (nonobstructive) hydrocephalus in which the ventricles are enlarged, leading to damage of the white matter tracts connecting the frontal lobes and basal ganglia. Signs of increased intracranial pressure (eg, papilledema, postural headaches) are typically absent. Many cases of NPH are idiopathic, though some are secondary to a defect in cerebrospinal fluid drainage from the ventricles as a result of scarring or obstruction of the ventricular lining from prior hemorrhage or meningitis. The diagnosis is primarily clinical, although ventriculomegaly on brain imaging is suggestive. The most appropriate next step in management is a lumbar puncture to assess the opening pressure and clinical response to removing cerebrospinal fluid. Placement of a ventricular shunt may be needed for long-term management.

Incorrect Answers: A, B, C, and D.

Cerebrospinal fluid leukocyte count (Choice A) may be increased in meningitis and encephalitis, which commonly present acutely or subacutely with fever, headache, and altered mental status. This patient does not have a fever or headache, and his symptom onset was chronic and insidious. Cerebrospinal fluid pressure (Choice B) is likely to be normal in this patient as denoted by the name "normal-pressure hydrocephalus." Excess cerebrospinal fluid collects in the ventricles but does not increase opening pressure. Opening pressure is often increased in meningitis and idiopathic intracranial hypertension.

Cortical sulci (Choice C) are increased in patients with neurodegenerative brain diseases such as dementia, Alzheimer type. In NPH, cortical sulci are not significantly increased.

Striatal dopamine (Choice D) is decreased in certain Parkinson disease medications. Dopamine dysregulation is not present in NPH.

Educational Objective: Educational Objective: Normal-pressure hydrocephalus refers to a type of communicating (nonobstructive) hydrocephalus which the ventricles are enlarged, leading to damage of the white matter tracts connecting the frontal lobes and basal ganglia. Patients commonly present with the characteristic combination of dementia, gait instability, and urinary incontinence.

11. A 72-year-old man comes to the physician because of a 2-month history of difficulty walking and burning and tingling sensations in both hands. He has hypertension treated with hydrochlorothiazide. He takes no other medications. Vital signs are within normal limits. Cranial nerve function is intact. There is atrophy of the intrinsic muscles of both hands; grip strength is 4/5. Muscle strength in the lower extremities is 4/5 distally and 5/5 proximally. Muscle tone is increased. Deep tendon reflexes are 1+ in the upper extremities and 4+ in the lower extremities. Babinski sign is present bilaterally. Sensation to pinprick and temperature is decreased over the hands. Sensation to vibration is decreased over all extremities. Which of the following is the most likely diagnosis?

Your Answer: C) Multiple sclerosis

Correct Answer: A) Cervical stenosis

Correct Answer: A

Correct Answer: A.

This patient presents with motor and sensory symptoms concordant with the distribution of the C8 dermatome and myotome. The C8 nerve root contributes a large proportion of nerves to the ulnar nerve, which arises primarily from the C8 and T1 nerve roots of the brachial plexus. The ulnar nerve supplies motor fibers to the intrinsic muscles of the hand and sensation over the anteromedial hand and small finger. Damage to the nerve roots or the nerve along its course (eg, in the cubital tunnel) can result in motor dysfunction of the fingers and numbness or paresthesia involving the medial aspect of the upper extremity. Injury occurring at the C8 root often occurs from cervical stenosis, root impingement from osteophyte, trauma, malignancy, spinal epidural abscess, or spinal hematoma.

Incorrect Answers: B, C, D, and E.

Amyotrophic lateral sclerosis, a motoneuron disease (Choice B) is a progressive neurodegenerative disorder that affects the anterior horns and the lateral corticospinal tract as it descends in the spinal cord. Subsequently, patients demonstrate a combination of upper and lower motoneuron findings in an asymmetric distribution that may include bulbar muscles. Sensory impairment would be atypical. Multiple sclerosis (Choice C) is an autoimmune demyelinating disease that presents with focal neurologic deficits separated in space and time. Patients with multiple sclerosis may present with optic neuritis or transverse myelitis, which may demonstrate a sensory level and a band-like sensation, along with possible autonomic and motor symptoms.

Parasagittal meningiomas (Choice D) are benign tumors that commonly present with headache, seizures, and focal neurologic deficits such as lower extremity weakness and numbness associated with local mass effect on the adjacent cerebral convexity.

A spinal arteriovenous fistula (Choice E) is a rare lesion that can lead to back pain, numbness, weakness, paresthesias, and difficulty walking. If left undiagnosed, bowel and bladder incontinence or retention can occur.

Educational Objective: Educational Objective: Cervical stenosis can lead to impingement of specific nerve roots, leading to motor and sensory symptoms concordant to the distribution of a specific dermatome and myotome.

12. A 38-year-old woman comes to the clinic with her husband because she has had six episodes of severe abdominal pain during the past 4 months. She says when the episodes occur, the pain is continuous and located "all over my belly." During initial episodes, the pain persisted for 3 to 4 hours, but during more recent episodes, the pain has lasted 2 to 3 days. She rates the pain at its worst as an 8 on a 10-point scale. The episodes are accompanied by nausea, bloating, numbness in her hands and feet, severe burning pain in her thighs and calves, and severe constipation; these associated symptoms subside with the resolution of the abdominal pain. She also passed red urine after her most recent episode. Her husband says she often sweats during these episodes and he once measured her pulse rate at 100/min. The patient has no history of serious illness and takes no medications. She has a family history of similar episodes in multiple relatives. Vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following is most likely responsible for the color of this patient's urine?

Your Answer: E) Porphobilinogen deaminase

Correct Answer: E) Porphobilinogen deaminase

Correct Answer: E

Correct Answer: E.

Acute intermittent porphyria is the most likely diagnosis for this patient with episodic abdominal pain and peripheral neuropathy in a "stocking-glove" distribution, which is caused by an autosomal dominant mutation in the porphobilinogen deaminase gene (HMBS). This enzyme catalyzes the third step in heme synthesis, which combines four porphobilinogen (PBG) molecules into hydroxymethylbilane, which is subsequently converted to uroporphyrinogen III. Loss of this enzyme inhibits the accumulation of PBG, which is toxic to the nervous system. Additionally, increased activity of the upstream enzyme, delta-aminolevulinic acid synthase (ALAS1) leads to accumulation of aminolevulinic acid, which is also toxic. The mutation affects both erythroid cells and hepatocytes. The disease's manifestations are primarily neurologic and include peripheral neuropathy, bladder dysfunction, and autonomic dysfunction. Abdominal pain is classic and results often from ileus. The penetrance is low, so a family history is not often present, which can delay the diagnosis. Disease flares can occur after exposure to certain triggers, such as medications, tobacco, alcohol, stress, and starvation. The accumulation of porphyrin precursors, including PBG, can cause urine to turn red or dark after exposure to light.

Incorrect Answers: A, B, C, and D.

Carotene (Choice A) is a naturally occurring pigment that is converted into vitamin A. It is found in many plants, such as carrots, pumpkins, and sweet potatoes, and in many vitamin supplements. It may cause a harmless, orange urinary discoloration when consumed in excess, but does not produce neurologic or gastrointestinal symptoms.

Hemoglobin (Choice B) may impart a reddish discoloration to urine in association with a variety of autoimmune, infectious, inflammatory, and coagulation disorders (eg, paroxysmal nocturnal hemoglobinuria, glucose 6-phosphate dehydrogenase deficiency, thrombotic thrombocytopenic purpura). However, this patient's presentation is most suggestive of acute intermittent porphyria, which does not cause hemoglobinuria.

Hydroxocobalamin (Choice C) is a vitamin used for the treatment of cyanide poisoning. It may impart a harmless, red discoloration to urine following administration.

Myoglobin (Choice D) may impart a red discoloration to urine in the setting of myoglobinuria, which

may occur in the setting of rhabdomyolysis, polymyositis, or a variety of metabolic disorders, such as McArdle disease.

Educational Objective: Educational Objective: Acute intermittent porphyria is an autosomal dominant disease with low penetrance caused by deficiencies in the porphobilinogen deaminase gene HMBS leading to the accumulation of hepatically synthesized heme intermediates such as porphobilinogen (PBG), which are toxic to the nervous system. Typical findings include episodic abdominal pain, sensory neuropathy, and autonomic dysfunction that occurs after certain triggers. The accumulation of porphyrin precursors can cause urine to turn red or dark after exposure to light.

13. A previously healthy 24-year-old man is brought to the physician because of a 3-day history of progressive loss of vision in the right eye and pain behind the eye with movement. He has no history of similar symptoms. Funduscopic examination shows no abnormalities. Visual acuity is 20/100 on the right. Visual field testing shows a right central scotoma. There is a right afferent pupillary defect. Over the next 10 years, this patient is at greatest risk for which of the following conditions?

Your Answer: E) Multiple sclerosis

Correct Answer: E) Multiple sclerosis

Correct Answer: E

Correct Answer: E.

Optic neuritis results from autoimmune demyelinating inflammation of the optic nerve (cranial nerve II) and presents in young, often female, patients with unilateral vision loss and pain that is exacerbated with extraocular movements. Ocular examination demonstrates decreased visual acuity and color vision, a relative afferent pupillary defect (rAPD), and possibly optic disc edema. Diagnosis is confirmed with MRI demonstrating hyperintensity of the optic nerve. Patients with optic neuritis are at an increased risk for multiple sclerosis (MS), and optic neuritis is a common initial finding in patients with MS. Recently, autoantibodies against myelin oligodendrocyte glycoprotein (MOG) and aquaporin-4 (AQP4) have been identified as a causative factor of some cases of optic neuritis. Patients who develop optic neuritis because of antibodies against MOG and AQP4 often have severe presentations but also have a decreased risk for subsequent MS compared with patients with optic neuritis who lack these autoantibodies. The Optic Neuritis Treatment Trial demonstrated that a short course of treatment with high-dose intravenous methylprednisolone speeds the rate of visual recovery and delays the onset of MS, although it does not impact final visual acuity, and demonstrated that the presence and quantity of periventricular white matter lesions on MRI imaging is the most important prognostic risk factor for the subsequent development of MS.

Incorrect Answers: A, B, C, D, F, and G.

Glaucoma (Choice A) occurs as a result of increases in pressure within the anterior chamber of the eye and may present as acute, painful or chronic, painless vision loss. Diabetes mellitus, anticholinergic medications, and Sturge-Weber syndrome have all been associated with increased risk for glaucoma.

Graves ophthalmopathy (Choice B) may present with exophthalmos, eyelid retraction, and lid lag, and is associated with Graves disease.

Herpes encephalitis (Choice C) may present with headache, confusion, fever, and seizures. It is most commonly caused by herpes simplex virus 1.

Macular degeneration (Choice D) may present with insidious blurring of central vision and metamorphopsia. Advanced age is the biggest risk factor associated with developing macular degeneration.

Myotonic dystrophy (Choice F) typically presents with muscle weakness, myotonia, cataracts, frontal balding, cardiac arrhythmias, and hypogonadism. The most frequently affected muscles

include the facial muscles, sternocleidomastoid, distal forearm muscles, intrinsic hand muscles, and ankle dorsiflexors. It is a genetic condition caused by a trinucleotide repeat.

Temporal arteritis (Choice G) is a vasculitis of medium and large vessels in elderly patients that commonly presents with headache, jaw claudication, and proximal muscle stiffness. Patients typically experience amaurosis fugax prior to the onset of abrupt, irreversible vision loss. It has been associated with polymyalgia rheumatica but not optic neuritis.

Educational Objective: Educational Objective: Optic neuritis is an autoimmune, inflammatory disorder of the optic nerve (cranial nerve II) that presents with acute onset vision loss and ocular pain that is exacerbated with extraocular movements. Patients typically present with profoundly decreased visual acuity and color vision, along with a relative afferent pupillary defect. Diagnosis is confirmed with MRI and with testing for serum autoantibodies. MRI provides important prognostic information regarding the subsequent risk for developing multiple sclerosis, which commonly occurs in association with optic neuritis. Treatment with high-dose intravenous methylprednisolone speeds visual recovery and delays the onset of multiple sclerosis.

14. A 56-year-old man is brought to the emergency department by ambulance 1 hour after he was found lying neighbor sitting in the hallway of his apartment building and speaking unintelligibly. On admission, the patient is confused and somnolent; he is unable to provide a medical history. He has alcohol-related cirrhosis complicated by bleeding from esophageal varices, hypokalemia, and metabolic alkalosis. His medications are spironolactone, furosemide, nadolol, and ursodiol. During his last clinic visit 2 weeks ago, the patient stated that he has remained abstinent from alcohol use. His examination at that time showed mild confusion. Today, the patient appears somnolent but is easily aroused. He is 175 cm (5 ft 9 in) tall and weighs 56 kg (123 lb); BMI is 18 kg/m². Temperature is 36.8°C (98.2°F), pulse is 78/min, respirations are 16/min, and blood pressure is 114/72 mm Hg. Physical examination shows palmar erythema and gynecomastia. The abdomen is protuberant with caput medusae radiating from the umbilicus. When asked to outstretch his arms with wrists extended, the patient's hands transiently drop forward in a nonrhythmic manner. Mental status examination shows a reduced attention span. He is oriented to person but not to place, time, or situation. Results of laboratory studies are shown:

Hemoglobin 11.8 g/dL

Hematocrit 34%

Leukocyte count 7800/mm³

Serum

Na⁺ 133 mEq/L

K⁺ 3.2 mEq/L

Urea nitrogen 14 mg/dL

Creatinine 1.2 mg/dL

Ammonia 54 µg/dL

Which of the following is the most appropriate pharmacotherapy at this time?

Your Answer: D) Rifaximin

Correct Answer: D) Rifaximin

Correct Answer: D

Correct Answer: D.

Hepatic encephalopathy is commonly associated with decompensated liver cirrhosis. Signs of hepatic encephalopathy include confusion, stupor, asterixis, tremor, and myoclonus on physical examination. Hepatic encephalopathy results from accumulation of ammonia produced by colonic bacteria. Serum ammonia concentrations in patients with hepatic encephalopathy may be increased, though this is not necessary for the diagnosis. Treatment of hepatic encephalopathy includes rifaximin and lactulose. Rifaximin is a gut-selective, nonabsorbable antibiotic with activity against Gram-positive and Gram-negative bacteria. It acts on the beta subunit of DNA-dependent RNA polymerase to inhibit bacterial synthesis and is bactericidal. Lactulose is degraded by colonic bacteria to create an acidic environment, thereby converting ammonia to ammonium, which cannot be readily absorbed across the gut wall. It is then excreted with feces, lowering total body ammonia.

Incorrect Answers: A, B, C, and E.

Branched-chain amino acid supplementation (Choice A) may be used for supplementation in patients with decompensated liver failure who require protein restriction. Branched-chain amino acids are not used as a treatment for hepatic encephalopathy.

Haloperidol (Choice B) is a first-generation antipsychotic medication that may be utilized to manage acute agitation or psychosis. Benzodiazepines are the preferred medication to treat agitation

associated with hepatic encephalopathy.

Mannitol (Choice C) is an osmotic diuretic used in treatment of cerebral edema. It is not indicated for hepatic encephalopathy.

Sodium polystyrene sulfonate (Choice E) is a resin used in treatment of hyperkalemia. This patient has hypokalemia.

Educational Objective: Educational Objective: Hepatic encephalopathy is caused by the accumulation of ammonia produced by colonic bacteria in patients with decompensated liver cirrhosis. Treatment includes lactulose and rifaximin, a gut-selective, nonabsorbable antibiotic.

15. A 46-year-old woman comes to the emergency department 18 hours after the sudden onset of severe, intermittent, left-sided facial pain. She describes the pain as "an electric jolt that turns on and off every minute." The pain started while she was brushing her teeth the previous evening and is localized to the left premolar area of the upper dental arch. She reports that ibuprofen relieved the pain for 3 hours, but the pain recurred this morning when she rinsed her mouth with cold water. She has hypertension and multiple sclerosis. Her medications are metoprolol, lisinopril, and fingolimod. Her maternal aunt also has multiple sclerosis. The patient appears anxious and occasionally winces in pain. Temperature is 37.1°C (98.8°F), pulse is 64/min, respirations are 18/min, and blood pressure is 146/88 mm Hg. Physical examination shows no abnormalities. Which of the following reflexes should be tested in order to determine if further diagnostic studies are indicated at this time?

Your Answer: D) Pupillary light

Correct Answer: A) Corneal

Correct Answer: A

Correct Answer: A.

Trigeminal neuralgia is a neuropathic pain disorder that presents with severe shooting or burning pain within the innervated regions of the sensory branches of the trigeminal nerve. The trigeminal nerve divides into three sensory branches, V₁, V₂, and V₃. V₁ innervates the forehead (V₁) and the skin overlying the zygomatic arch (V₂). V₂ innervates the skin over the maxilla and upper dental arch (V₂). V₃ innervates the skin over the mandible and lower dental arch (V₃). The pain may be triggered by light touch, chewing, or swallowing; anything that irritates the nerve can trigger an episode of pain. It is associated with multiple sclerosis and may be related to compression of the trigeminal nerve by adjacent blood vessels or a bone spur. Trigeminal neuralgia also mediates the corneal reflex, which involves blinking of the eyelids caused by stimulation of the trigeminal nerve. Trigeminal neuralgia is associated with loss of the corneal reflex. Treatment includes carbamazepine or surgical decompression.

Incorrect Answers: B, C, D, and E.

The gag reflex (Choice B) is mediated by the glossopharyngeal nerve as well as the vagus nerve.

The palmomental reflex (Choice C) is a primitive reflex present in neonates and infants that involves an involuntary contraction of the mentalis muscle of the mandible elicited by stimulation of the thenar eminence.

The pupillary light reflex (Choice D) involves the optic nerve and the oculomotor nerve and causes constriction of the pupils in response to light. Loss of the pupillary light reflex can occur in the setting of an optic nerve injury, brainstem lesion, or oculomotor nerve injury.

The visual threat reflex (Choice E) involves involuntary blinking in response to a visual threat and is associated with the optic nerve.

Educational Objective: Trigeminal neuralgia is a neuropathic pain disorder that presents with severe shooting or burning pain within the innervated regions of the sensory branches of the trigeminal nerve. The trigeminal nerve mediates the corneal reflex, which is lost in trigeminal neuralgia.

16. A 33-year-old woman comes to the physician 24 hours after a fainting episode. Menses have occurred at irregular intervals over the past 6 months. She has had numbness and tingling of her left hand for 2 years. She has had intermittent episodes of sharp, stabbing lower abdominal pain over the past 5 years; she has had no nausea, vomiting, diarrhea, or constipation. Treatment with acetaminophen, ibuprofen, and codeine has been ineffective. Three exploratory laparoscopies did not disclose the cause of the pain, and surgical treatment for uterine adhesions did not provide any significant relief. She has a history of multiple sexual partners but has abstained for the past 3 years because of dyspareunia. Her blood pressure is 140/84 mm Hg. The abdomen is moderately tender to light and deep palpation with no rebound tenderness. Bowel sounds are active. Pelvic examination is difficult to perform because of increased cervical pain. There is bilateral tenderness on palpation of the ovaries. Which of the following is the most likely diagnosis?

Your Answer: E) Somatization disorder

Correct Answer: E) Somatization disorder

Correct Answer: E

Correct Answer: E.

Somatization disorder is characterized by excessive preoccupation with one or more somatic symptoms that result in significant distress or functional impairment. Unlike illness anxiety disorder, in which patients are concerned about developing a serious illness despite absence of symptoms, patients with somatization disorder are preoccupied with existing symptoms that are not fully explained by objective physical examination or laboratory findings. There may be an underlying medical condition that causes some of the symptoms, but the patient's reaction is out of proportion to the severity. Many patients with somatization disorder repeatedly seek medical attention for their symptoms, leading to extensive workups and multiple negative diagnostic studies. The patient's extensive history of abdominal pain, along with multiple negative laparoscopies and treatment for uterine adhesions that did not provide any significant relief are consistent with a diagnosis of somatization disorder.

Incorrect Answers: A, B, C, and D.

Conversion disorder (Choice A), also known as functional neurologic disorder, involves neurologic symptoms such as sensory or motor dysfunction that are not fully explained by objective findings of physical examination or imaging. Functional neurologic symptom disorder is postulated to represent the subconscious conversion of intolerable emotional distress into physical symptoms.

Crohn disease (Choice B) is characterized by transmural inflammation of the gastrointestinal tract and presents with abdominal pain, bloody or watery diarrhea, low-grade fever, and weight loss.

Pelvic pain, dyspareunia, and numbness are not consistent with a diagnosis of Crohn disease.

Endometriosis (Choice C) is caused by ectopic endometrial tissue outside the uterus and presents with chronic dysmenorrhea, dyspareunia, dyschezia, and infertility. Symptoms are typically cyclical. The gold standard for diagnosis is diagnostic laparoscopy, which evaluates for endometriotic implant lesions on the peritoneum. This patient has had three laparoscopies that did not show any endometriotic tissue.

Illness anxiety disorder (Choice D) is characterized by persistent excessive anxiety about having or acquiring a serious illness despite negative diagnostic testing. These patients often experience minimal symptoms and have normal physical examinations. This patient has a preoccupation with her symptoms rather than a specific diagnosis.

Educational Objective: Educational Objective: Somatization disorder is characterized by excessive preoccupation with one or more somatic symptoms that these symptoms disrupt the patient's daily life. Patients persistently devote excessive time and energy to these symptoms or related health concerns (eg, repeatedly going to the doctor).

17. A 57-year-old man comes to the physician because of a 6-month history of progressive hearing loss in his left ear. He initially only had difficulty hearing while talking on the telephone, but now he has difficulty hearing most of the time. During this period, he occasionally has had a 30-second episode of unsteadiness and a spinning sensation. He has been otherwise healthy. Examination shows flattening of the left nasolabial fold and decreased wrinkling of the forehead on the left side. There is rotatory nystagmus on right lateral gaze and profound sensorineural hearing loss on the left. Mental status examination shows no abnormalities. Which of the following is the most likely location of the lesion?

Your Answer: B) Cerebellopontine angle

Correct Answer: B) Cerebellopontine angle

Correct Answer: B

Correct Answer: B.

Acoustic neuromas are tumors derived from Schwann cells involving the vestibulocochlear nerve. The vestibulocochlear nerve originates between the pons and medulla oblongata and exits the brainstem near the cerebellopontine angle. Symptoms associated with the development of an acoustic neuroma include hearing loss and tinnitus associated with the cochlear nerve, and vertigo and gait instability associated with the vestibular nerve. Damage to the facial nerve and trigeminal nerve is also frequently associated with acoustic neuromas because of their proximity to the tumor. Patients with neurofibromatosis 2 may present with bilateral acoustic neuromas. Imaging studies, including CT scan or MRI, can be used to make the diagnosis. Treatment options include surgical resection or radiation therapy.

Incorrect Answers: A, C, D, and E.

Brain stem tegmentum (Choice A) is the portion of the brainstem ventral to the cerebral aqueduct and fourth ventricle. This area would not be affected by an acoustic neuroma.

Cochlea (Choice C) is a component of the inner ear and plays a role in the sensation of hearing and transducing auditory signals to the brain. Patients with damage to the cochlea experience sensorineural hearing loss.

Mastoid bone (Choice D) is located posterior to the ear and contains mastoid air cells. This area is a potential space for infection in patients with fever of unknown origin or following prolonged or spreading otitis media. Mastoiditis, or inflammation of the mastoid bone, would not cause this patient's symptoms.

Temporal lobe (Choice E) contributes to the cerebral cortex and is associated with auditory perception, language recognition, comprehension, and speech. A lesion involving the Wernicke area of the temporal lobe results in Wernicke aphasia where patients can produce speech though it is often garbled and inappropriate.

Educational Objective: Educational Objective: Acoustic neuromas are caused by overgrowth of Schwann cells of the vestibulocochlear nerve. The vestibulocochlear nerve originates between the

pons and medulla oblongata and exits the brainstem near the cerebellopontine angle. Patients with acoustic neuromas may present with hearing loss, tinnitus, vertigo, and gait abnormalities. Bilateral acoustic neuromas are associated with neurofibromatosis 2. Diagnosis is made based on clinical imaging, and patients can be treated with surgery or radiation therapy.

18. An 82-year-old woman is brought to the physician because of a 3-month history of brief memory loss two to three times weekly. She also has had difficulty maintaining her balance. She has glaucoma, hyperlipidemia, and hypertension. Her medications are pilocarpine eyedrops, losartan, and atorvastatin. Her pulse is 92/min, respirations are 16/min, and blood pressure is 132/84 mm Hg. Examination shows arcus senilis and slight restriction of upward gaze. Deep tendon reflexes are decreased at the ankles compared with the knees. On gait testing, she walks flexed at the waist and is unable to tandem walk. On mental status examination, she is fully oriented. She performs serial sevens without difficulty. Her immediate, recent, and remote memory is intact. She displays normal abstract thinking. Which of the following best explains this patient's neurologic findings?

Your Answer: E) Psychomotor slowing

Correct Answer: E) Psychomotor slowing

Correct Answer: E

Correct Answer: E.

This patient with brief episodic memory loss, arcus senilis, decreased Achilles reflexes, and flexed posture is showing signs of psychomotor slowing consistent with normal aging. Normal aging includes a slow decline of executive functioning, attention span, and problem-solving after the age of 70 years. Other findings consistent with aging include decreased or absent Achilles reflex, arcus senilis (grey or white arch visible around the outer edge of the cornea), decreased vision and hearing, decreased immune response, and increased suicide rate. Despite these common changes, an individual experiencing normal aging should still be able to function in society.

Incorrect Answers: A, B, C, and D.

Cerebellar degeneration (Choice A) may present with truncal and limb ataxia, broad-based gait, and intention tremor.

Multiple lacunar infarctions (Choice B) may present with purely motor symptoms, purely sensory symptoms, ataxic hemiparesis, or dysarthria. Infarcts in these deep penetrating vessels are secondary to hypertension, diabetes mellitus, and/or hypercholesterolemia.

Parkinson disease (Choice C) is caused by loss of dopaminergic neurons in the substantia nigra and is characterized by resting tremor, bradykinesia, rigidity, and postural instability.

Peripheral neuropathy (Choice D) has many causes including metabolic, toxic, autoimmune, and/or hereditary neuropathy. Symptoms include paresthesia or pain, and extremity weakness in a "stocking-glove" distribution.

Educational Objective: Educational Objective: Normal aging includes a slow decline of executive functioning, attention span, and problem-solving after the age of 70 years. Other findings consistent with normal aging include decreased or absent Achilles reflex, arcus senilis, decreased vision and hearing, decreased immune response, and increased suicide rate.

19. A previously healthy 37-year-old woman is admitted to the hospital because of a 3-day history of weakness of her arms and legs. Her symptoms began 6 days ago with backache and a prickling sensation in her legs. She is alert and oriented. Vital signs are within normal limits. Cranial nerves are intact. Muscle strength is 3/5 in the distal upper and lower extremities and 4/5 in the proximal upper and lower extremities. Sensation to pinprick is decreased below the knees. Deep tendon reflexes are absent in the Achilles and patellar tendons bilaterally. Babinski sign is absent bilaterally. Laboratory studies show:

Serum glucose 102 mg/dL

Cerebrospinal fluid

Glucose 66 mg/dL

Protein 90 mg/dL

WBC 0-2/mm³

Your Answer: B) Guillain-Barré syndrome

Correct Answer: B) Guillain-Barré syndrome

Correct Answer: B

Correct Answer: B.

This patient most likely has Guillain-Barré syndrome (GBS) leading to an ascending polyneuropathy. In GBS, a preceding infection (eg, *Campylobacter jejuni*, influenza-like illnesses) leads to autoimmune cross-reactivity with the myelin of the nerve roots exiting the spinal cord. The ventral nerve roots, containing motor and sympathetic efferents, are most commonly involved, though the dorsal nerve roots (controlling sensation) may also be affected. As such, patients typically present with acute symmetric muscle weakness with depressed or absent deep tendon reflexes (reflecting a lower motoneuron pattern of dysfunction) that begins in the lower extremities and may rapidly ascend to involve the upper extremities, bulbar muscles, and/or respiratory muscles. Autonomic dysfunction (eg, mood swings, hypotension, cardiac irregularities) is common. Sensory deficits or pain can also occur, though these symptoms are typically mild compared with motor dysfunction. The diagnosis is confirmed by increased cerebrospinal fluid protein with normal cell counts on lumbar puncture, termed albuminocytologic dissociation, thought to reflect generalized inflammation. Management includes respiratory support and, for severe cases, plasma exchange or intravenous immunoglobulin therapy. Intravenous immunoglobulin therapy may work by neutralizing the cross-reactive antibodies that cause GBS.

Incorrect Answers: A, C, D, and E.

Compression of the cervical spinal cord (Choice A), such as with a herniated cervical disc, leads to cervical myelopathy. This can present with varying degrees of sensory and motor neurologic deficits, along with the potential for autonomic dysfunction, including bowel and bladder function. Multiple sclerosis (Choice C) is an autoimmune demyelinating disease that presents with focal neurologic deficits separated in space and time. Patients with multiple sclerosis may present with optic neuritis or transverse myelitis, which may demonstrate a sensory level and a band-like sensation, along with possible autonomic and motor symptoms.

Myasthenia gravis (Choice D) is an autoimmune neuromuscular disorder characterized by the formation of inhibitory anticholinergic receptor autoantibodies. Common symptoms include muscular weakness, fatigue, diplopia, and ptosis that worsens toward the end of the day, dysphagia, and respiratory distress. Patients may present with an associated thymoma.

Syringomyelia (Choice E) features a fluid collection that develops in the central cervical or thoracic spinal cord, most often between C2 and T9 and can cause sensory loss to pain and temperature or

weakness in the limbs of the upper and lower extremities, depending upon its location. Vibration and position sense are generally preserved.

Educational Objective: Educational Objective: Guillain-Barré syndrome (GBS) is typically a self-limited demyelinating polyneuropathy that starts with the distal lower extremities and progresses proximally. Patients usually have normal sensation but profound muscle weakness.

20. A 57-year-old man is brought to the emergency department 1 hour after his wife was unable to arouse him from sleep. He has hypertension but is currently taking no medications. His blood pressure is 190/100 mm Hg. Examination shows both eyes deviated to the left. He does not withdraw his right upper and lower extremities to painful stimuli. Which of the following is the most appropriate next step in diagnosis?

Your Answer: C) CT scan of the head

Correct Answer: C) CT scan of the head

Correct Answer: C

Correct Answer: C.

This patient with uncontrolled hypertension, left eye deviation, right-sided weakness, and depressed mental status has a presentation concerning for intracerebral hemorrhage (ICH). ICH can occur in patients with chronic hypertension, typically due to hemorrhage from small penetrating arteries in the pons, midbrain, thalamus, putamen, caudate, or cerebellum. Risk factors for ICH include uncontrolled hypertension and antiplatelet or anticoagulant therapy, such as with aspirin or warfarin. Patients can present with headache, vomiting, confusion, altered mental status, stupor, or coma. Focal neurologic deficits depend on the site of the ICH and can include hemiparesis, hemisensory symptoms, gaze palsy, dysarthria, aphasia, or, if in the cerebellum, ataxia, vomiting, and disequilibrium. Patients may manifest upper motoneuron examination findings such as spasticity, hyperreflexia, and extensor plantar reflex (Babinski sign). Diagnosis is made with noncontrast CT scan of the head, which will show hyperdense intracerebral blood products.

Incorrect Answers: A, B, D, and E.

Ice water caloric testing (Choice A) is performed to test the oculovestibular reflex. Ice water is injected quickly into the ear canal. Horizontal deviation of both eyes toward the affected ear indicates an intact reflex. This test is often performed as a part of the cranial nerve examination in a patient with suspected brain death.

EEG (Choice B) would be appropriate in the setting of suspected seizure. Patients with seizures often have eye deviation, but this patient lacks other common features consistent with seizures including focal or generalized shaking, incontinence, and tongue biting.

MRI of the spine (Choice D) would be appropriate if spinal cord injury or compression was suspected. Patients with spinal cord compression often present with back or neck pain in addition to weakness or numbness corresponding with the area of spinal involvement.

Lumbar puncture (Choice E) should be performed if the diagnosis of meningitis is suspected. Patients with meningitis often present with fever, headache, neck pain and rigidity, and confusion. Gaze deviation and unilateral upper motor neuron signs are not consistent with meningitis.

Educational Objective: Educational Objective: Hypertensive intracerebral hemorrhage can occur in patients with chronic hypertension, typically due to hemorrhage from small penetrating arteries in the pons, midbrain, thalamus, putamen, caudate, or cerebellum. Patients can present with headache, vomiting, confusion, altered mental status, stupor, upper motoneuron signs, and focal neurologic deficits such as hemiparesis and hemisensory symptoms. Diagnosis is made with noncontrast CT scan of the head, which will show hyperdense intracerebral blood products.

21. A 72-year-old man comes to the physician because of a 3-month history of dizziness on standing. He has angina pectoris, esophageal reflux, degenerative joint disease, and type 2 diabetes mellitus. Medications include cimetidine, aspirin, gemfibrozil, isosorbide dinitrate, and metformin. His pulse is 80/min, and blood pressure is 150/90 mm Hg while supine; his pulse is 68/min, and blood pressure is 118/70 mm Hg while standing. The remainder of the examination shows no abnormalities. His hematocrit is 42%. Discontinuation of which of the following medications is most likely to resolve his orthostasis?

Your Answer: D) Isosorbide dinitrate

Correct Answer: D) Isosorbide dinitrate

Correct Answer: D

Correct Answer: D.

This patient's symptoms of positionally dependent dizziness on standing are consistent with orthostatic hypotension. Orthostatic hypotension is defined as a sustained decrease in systolic blood pressure greater than 20 mm Hg or diastolic blood pressure greater than 10 mm Hg within 3 minutes of standing. The underlying cause is a dysregulation of the sympathetic and parasympathetic autonomic nervous systems resulting in reduced peripheral vascular resistance and decreased cardiac output on standing, which in turn results in hypoperfusion of the brain and peripheral tissues. Orthostatic hypotension can be caused or exacerbated by a variety of medications including antihypertensives such as beta or alpha-1 blockers, narcotics, and antihypertensives such as isosorbide dinitrate, a potent vasodilator used for the treatment of angina pectoris. Patients typically experience presyncope, syncope, or light-headedness that is intermittent and positional. The diagnosis is confirmed with tilt table testing. The patient's heart rate, blood pressure, and symptoms are evaluated while the table is rotated from supine to upright. Patients typically report symptoms with changes in position. Initial treatment involves correction of reversible causes, such as dehydration, and discontinuation of causative medications when possible.

Incorrect Answers: A, B, C, and E.

Aspirin (Choice A) is a nonsteroidal anti-inflammatory drug commonly used in the treatment of coronary artery disease, cerebrovascular disease, and peripheral arterial disease. Adverse effects of aspirin include gastrointestinal bleeding, which may contribute to symptoms of orthostatic hypotension. However, this patient's hematocrit is normal, suggesting that his symptoms are unlikely to be caused by an undetected hemorrhage.

Cimetidine (Choice B) is an H₂-receptor antagonist used for the treatment of peptic ulcer disease. Its adverse effects include dizziness, headache, confusion, and diarrhea. It does not commonly cause symptoms of orthostasis.

Gemfibrozil (Choice C) is a fibrate used for lipid-lowering pharmacotherapy. Adverse effects include dizziness, headache, fatigue, and gastrointestinal upset. It does not commonly cause symptoms of orthostasis.

Metformin (Choice E) is an antihyperglycemic agent used for the treatment of diabetes mellitus. Adverse effects include gastrointestinal upset (eg, nausea, vomiting, diarrhea), abdominal pain, and, rarely, lactic acidosis. It is not a common cause of orthostatic hypotension.

Educational Objective: Educational Objective: Orthostatic hypotension is defined as a sustained decrease in systolic blood pressure greater than 20 mm Hg or diastolic blood pressure greater than 10 mm Hg within 3 minutes of standing. Patients typically experience presyncope, syncope, or light-headedness that is intermittent and positional. Autonomic dysfunction results in a reduced peripheral vascular resistance and decreased cardiac output on standing, which in turn results in hypoperfusion of the brain and peripheral tissues. Nitrates and antihypertensives are common causes of orthostatic hypotension. Initial treatment involves correction of reversible causes, such as dehydration, and discontinuation of causative medications.

22. A 63-year-old woman is brought to the emergency department because of a 3-hour history of double vision. She also has a 1-week history of moderate pain and numbness around her right eye. She has no history of serious illness and takes no medications. Examination shows exophthalmos of the right eye. When asked to look to the right, the patient reports double vision; the right eye does not fully abduct. Sensation to pinprick is decreased over the right forehead and below the right eye. Which of the following is the most likely diagnosis?

Your Answer: A) Cavernous sinus lesion

Correct Answer: A) Cavernous sinus lesion

Correct Answer: A

Correct Answer: A.

Cavernous sinus thrombosis classically follows an infection or injury involving a structure drained by the cavernous sinus and results in cranial nerve deficits involving those that traverse the cavernous sinus (III, IV, V₁, V₂, or VI). Causes include sinusitis, orbital cellulitis, dental infections, pharyngitis, and cellulitis or abscess, especially within the danger triangle of the face (formed by the apex of the nose and the corners of the mouth). Features include fever, orbit pain, exophthalmos, periorbital edema, impaired pupil reactivity, and impaired extraocular movement, along with findings indicative of meningismus, papilledema, and seizures. Oculomotor and abducens nerve deficits and sensory deficits along the branches of the trigeminal nerve (especially V₁ and V₂) are commonly seen. This patient presents with symptoms and findings suggesting a cavernous sinus lesion, including pain and numbness around the right eye, right exophthalmos, right abducens nerve palsy (leading to impaired abduction), and right-sided sensory deficits involving the branches of the trigeminal nerve, as well as double vision. Treatment involves intravenous broad-spectrum antibiotics and/or anticoagulants. Even with prompt recognition and appropriate treatment, the burden of long-term complications, including visual impairment and stroke, remains high.

Incorrect Answers: B, C, D, E, and F.

Graves disease (Choice B) can present with weight loss, insomnia, palpitations, heat intolerance, tremor, diaphoresis, thyromegaly, and exophthalmos. Laboratory studies show increased thyroxine and triiodothyronine, decreased TSH, and the presence of TSH receptor-stimulating antibodies. While this patient has exophthalmos, she lacks other systemic signs of Graves disease.

Mitochondrial myopathy (Choice C) is a hereditary condition that can present with ophthalmoplegia, seizures, myoclonus, peripheral neuropathy, proximal muscle weakness, and fatigue. Diagnosis can be confirmed by ragged red fibers on muscle biopsy and mitochondrial crystalline inclusions on electron microscopy. This patient's acute onset of ophthalmoplegia and lack of a compatible family history are not consistent with this diagnosis.

Paramedian pontine infarction (Choice D) can affect the abducens and facial nerves, the medial longitudinal fasciculus, and the corticospinal tract, leading to impaired abduction of the ipsilateral eye, ipsilateral facial paresis, internuclear ophthalmoplegia (impaired adduction of the ipsilateral eye and contralateral abduction saccade), and contralateral hemiparesis. Exophthalmos and a history of eye pain are not consistent with this diagnosis.

Posterior communicating artery aneurysm (Choice E) may compress the ipsilateral oculomotor cranial nerve, resulting in ptosis, exotropia and hypotropia (lateral and inferior pupil position), diplopia, and mydriasis. The detection of a new oculomotor nerve palsy is a medical emergency, as aneurysmal rupture can lead to life-threatening subarachnoid hemorrhage.

Type 2 diabetes mellitus (Choice F) causes microvascular damage to a variety of tissues, including nerves, and can cause cranial nerve palsy. Microvascular abducens nerve palsy presents with

esotropia, unilateral abduction deficit, and horizontal gaze diplopia. Exophthalmos is not associated with microvascular nerve palsy.

Educational Objective: Educational Objective: Cavernous sinus thrombosis is a potentially life-threatening complication of an infection or injury involving a structure drained by the cavernous sinus. It presents with cranial nerve deficits affecting the oculomotor, trochlear, abducens, and trigeminal nerves, along with exophthalmos, periorbital edema, fever, meningismus, and papilledema. Treatment includes intravenous broad-spectrum antibiotics and anticoagulants.

23. A previously healthy 3-month-old boy is brought to the physician because of a 3-day history of decreased feeding. He also has a 1-day history of cough that worsens when he attempts to feed. His mother reports that he seems hungry but "tires out" while feeding. He was born at 40 weeks' gestation to a 25-year-old primigravid woman following an uncomplicated pregnancy and delivery. He weighed 3600 g (7 lb 15 oz) at birth. Until 1 week ago, he had been exclusively breast-fed; he now receives cow milk-based formula. The patient's father has been excavating the soil around the house to build a foundation for an additional room. On examination, the patient is alert and quiet and has a weak cry. He is at the 50th percentile for length and weight. His temperature is 37.2°C (99°F), pulse is 138/min, respirations are 20/min, and blood pressure is 72/48 mm Hg. Examination of the skin shows no abnormalities. Facial expression is absent. The pupils are 5 mm and sluggishly reactive. The gag reflex is decreased. Transmitted upper airway noises are heard. No murmurs are heard. Abdominal examination shows no abnormalities. Peripheral pulses are 2+ and symmetric. There is head lag and no spontaneous movement of the extremities. Muscle tone is decreased. Deep tendon reflexes are normal. Which of the following is most likely to confirm the diagnosis?

Your Answer: A) Stool toxin assay

Correct Answer: A) Stool toxin assay

Correct Answer: A

Correct Answer: A.

This patient with difficulty feeding, hypotonia, and facial muscle weakness has a presentation concerning for infantile botulism. Botulism is an acute disease of the neuromuscular junction caused by exposure to the *Clostridium botulinum* toxin. *C. botulinum* is a gram-positive, spore-forming, obligate anaerobic bacterium. It produces botulinum toxin, a potent neurotoxin that is absorbed into the bloodstream and transported to the neuromuscular junction where it interferes with the release of acetylcholine, thereby causing weakness and paralysis. Botulism is often encountered as a food-borne illness through contaminated food products. Historically, *C. botulinum* was responsible for outbreaks of "sausage poisoning" through contaminated ham products and acquired its name after the Latin name for sausage, "botulus." In the modern era, it is more often responsible for poisoning via the ingestion of raw honey or canned foods in infants. Infection more commonly occurs through inhalation of spores from contaminated dirt or dust, however, a common mode of exposure is the ingestion of spores in honey, a food that infants should avoid. The incubation period is typically hours to sufficiently mature to produce detectable botulism. Patients typically present with progressive, symmetric hypotonia and hyporeflexia that begins with the cranial nerves, manifesting as ptosis, sluggish pupils, and a diminished gag reflex, and descends to the lower extremities, also causing constipation, poor feeding, and respiratory failure. Diagnosis is confirmed by identification of the botulinum toxin in the stool. Treatment requires hospitalization and administration of immune globulin. As a precautionary measure, the patient's caregivers should be asked about exposure to honey. Infants should not be fed honey.

Incorrect Answers: B, C, D, and E.

Measurement of serum cholinesterase activity (Choice B) can be performed in suspected cases of organophosphate toxicity. Patients classically present with diarrhea, urination, miosis, bronchospasm, bradycardia, vomiting, lacrimation, and salivation.

Measurement of negative inspiratory force (NIF) (Choice C) can be performed to assess respiratory

muscle strength in patients with muscular weakness. A low NIF may indicate that a patient is at risk for respiratory muscle fatigue and decompensation and may necessitate respiratory support. Patients with botulism may develop diaphragmatic weakness and therefore may have a low NIF, but this is a nonspecific finding and is not diagnostic for any one disease. Additionally, measurement of NIF requires patient participation (patient is instructed to fully inhale), which would not be possible for this 3-month-old patient.

Muscle biopsy (Choice D) is a part of the workup for botulism. Biopsy should be normal in patients with botulism, as it is a disease of the neuromuscular junction, not the muscle itself.

Electromyography and nerve conduction studies (Choice E) and repetitive nerve stimulation study (Choice F) can be done to support the diagnosis of botulism, but the gold standard is the stool toxin assay. The diagnosis is supported by a characteristic low amplitude response on repetitive nerve stimulation studies (electromyography) that improves with repeated stimulation.

Educational Objective: Educational Objective: Infantile botulism is caused by the ingestion of *Clostridium botulinum* spores in infants. Patients typically present with progressive, symmetric hypotonia, hyporeflexia, cranial nerve palsies, and respiratory failure. Diagnosis is confirmed by identification of the botulinum toxin in the stool. Treatment requires hospitalization and administration of immune globulin.

24. A 9-year-old girl is brought to the emergency department 30 minutes after being involved in a motor vehicle collision. There is an open fracture of the tibial shaft, and the patient reports severe pain in the area surrounding the fracture. On examination, there is weakness of toe flexion, severe pain with passive extension of the toes, and plantar hyperesthesia. Which of the following is the most likely diagnosis?

Your Answer: B) Complex regional pain syndrome

Correct Answer: A) Compartment syndrome

Correct Answer: A

Correct Answer: A.

Compartment syndrome occurs when there is increased pressure in the intramuscular compartment that compromises circulation and neuromuscular and neurologic function. Compartment syndrome commonly occurs in the setting of trauma, including long bone fractures, and symptoms include pain out of proportion to the injury, paresthesia, and pain with passive stretching of the muscles. Late findings are pallor, pulselessness, and paralysis. Diagnosis is confirmed after clinical suspicion by measuring compartment pressures. Increased compartment pressure or the presence of acute compartment syndrome and correlating blood pressures confirm the diagnosis. Patients with compartment syndrome are treated with emergent fasciotomy to decompress involved compartments; delays may lead to poor outcomes.

Incorrect Answers: B, C, D, and E.

Complex regional pain syndrome (Choice B), or reflex sympathetic dystrophy, presents with regional pain that is out of proportion to the insult. Other symptoms include edema, abnormal sensation, swelling, and erythema. The most common inciting event for complex regional pain syndrome is a soft tissue injury.

Deep venous thrombosis (Choice C) causes extremity swelling, pain, and discoloration. Symptoms may be acute or subacute in nature, and patients are treated with anticoagulants. This patient's clinical presentation is not consistent with deep venous thrombosis.

Ruptured posterior tibial artery (Choice D) may result from posterior knee dislocation, direct trauma, or aneurysmal disease. This injury is usually the result of high-impact trauma, such as a motor vehicle collision. Patients may present with abnormal pulses and pain. Rapid identification of this injury, imaging with angiography, and repair of the injured vessel must be completed promptly to salvage the affected limb.

Transection of the tibial nerve (Choice E) may occur in the case of high-impact trauma or during surgical procedures. Symptoms may differ depending on the level of injury. This patient's symptoms, including pain out of proportion to the injury, are more suggestive of compartment syndrome.

Educational Objective: Educational Objective: Compartment syndrome may arise in the setting of trauma and is associated with severe pain, paresthesia, and pain with passive movement. As vascular compromise worsens, patients may experience pallor, pulselessness, and paralysis. Diagnosis is made by measurement of compartment pressures, and patients are treated with emergent fasciotomy.

25. An 18-year-old woman comes to the office to request genetic testing. She was adopted at birth, and her adoptive parents were told that there was a strong maternal history of psychotic behavior and a movement disorder. The patient has been unable to obtain any further medical information. She has no history of serious illness. She takes no medications. She is planning to start college in the fall and has been an excellent student throughout high school. Physical examination shows no abnormalities. There is no evidence of abnormal movements. Mental status examination shows an anxious mood. She reports no hallucinations, and there is no evidence of delusions. After further counseling, the patient elects to proceed with genetic testing. Testing for which of the following is most appropriate at this time?

Your Answer: A) Fragile X syndrome

Correct Answer: C) Huntington disease

Correct Answer: C

Correct Answer: C.

Based on the reported maternal history of psychotic and movement symptoms, this patient should be offered testing for Huntington disease (HD). HD is a hereditary, progressive neurodegenerative disorder typically arising in the 3rd to 6th decades of life associated with atrophy of the caudate nuclei in the basal ganglia, which can be seen on MRI of the brain. The disease usually presents with an insidious onset of psychiatric (irritability, depression, psychosis) and cognitive symptoms. Motor symptoms (chorea, hypotonia, dystonia, loss of voluntary control of movements) typically appear later in the disease course. Patients with HD typically die 10 to 40 years after disease onset. Chorea, a key HD symptom, can be initially experienced as restlessness. Tremor, muscle spasms, and other involuntary movements may also occur. The key HD gene is located on chromosome 4, which occurs when a series of three nucleotides in repetition are replicated incorrectly and too many repeats are included in the replicated product. It is inherited in an autosomal dominant pattern and also demonstrates anticipation, a genetic inheritance phenomenon in which future generations are affected at a younger age or with more severe symptoms. HD is diagnosed clinically and with genetic testing, which measures the number of trinucleotide repeats in the HD gene.

Incorrect Answers: A, B, D, and E.

Fragile X syndrome (Choice A) is an X-linked dominant cause of inherited intellectual disability that presents due to a CGG-trinucleotide repeat expansion in the FMR1 gene. It presents with clinical features including intellectual disability, an elongated face with a prominent jaw, a high-arched palate, large ears, hyperflexible joints, and postpubertal macroorchidism. These features are less consistent with this patient's reported family history.

Hepatolenticular degeneration (Wilson disease) (Choice B) is caused by impaired metabolism of copper and deposition in the liver and brain. Most patients will have some degree of liver disease at the time of diagnosis. Neurologic and psychiatric manifestations are broad and can mimic other neurologic disorders, especially ones with movement-related symptoms. Patients with neurologic abnormalities will almost always have Kayser-Fleischer rings. This patient lacks features of liver disease. HD disease should be distinguished from Wilson disease from similar disorders. Initial testing involves measurement of serum ceruloplasmin concentrations.

Parkinson disease (Choice D) is characterized by resting tremor, rigidity, bradykinesia, and postural instability. Patients with PD rarely present with psychotic behavior and the disorder is not inherited in an autosomal dominant pattern. While anxiety and depression are common in PD patients, psychosis is rare and is usually a result of medications used in treatment.

Porphyria (Choice E) is a group of inherited metabolic disorders that can cause episodic neurologic

symptoms, abdominal pain, and psychological symptoms. The most common acute porphyria is acute intermittent porphyria, which is caused by defects in enzymes involved in heme synthesis. Porphyria does not typically cause movement disorders.

Educational Objective: Educational Objective: Huntington disease (HD) is a hereditary, progressive neurodegenerative disorder associated with atrophy of the caudate nuclei in the basal ganglia. It is caused by a trinucleotide repeat expansion in the HD gene located on chromosome 4 and is inherited in an autosomal dominant pattern. Patients present with a combination of psychiatric, cognitive, and motor symptoms.

26. A 27-year-old man comes to the physician because of a 1-year history of pain, weakness, and numbness in both arms and hands. Examination shows weakness and atrophy of the muscles of the arms and hands bilaterally. There is loss of sensation to pain and temperature over the neck, shoulders, and upper extremities but preservation of sensation to light touch. Deep tendon reflexes are absent in the upper extremities. He has thoracic kyphoscoliosis. Neurologic examination of the lower extremities and cranial nerves shows no abnormalities. Which of the following is the most likely diagnosis?

Your Answer: **J) Syringomyelia**

Correct Answer: **I) Syringomyelia**

Correct Answer: I

Correct Answer: I.

This patient presenting with a lower motor neuron pattern of dysfunction in the upper extremities with a syringomyelia likely is a result of syringomyelia. Syringomyelia features a cystic cavity in the central cervical or thoracic spinal cord, most commonly in the cervical or thoracic spinal cord. Syringomyelia may be associated with congenital malformations such as Chiari malformation type I, spinal cord infections, autoimmune diseases such as multiple sclerosis, neoplasms, or spinal cord trauma. Syringomyelia typically compresses spinothalamic fibers decussating in the anterior white commissure. Consequently, patients present with deficits in pain and temperature sensation in the bilateral upper extremities and/or torso. (Since the ascending spinothalamic tracts in the lateral spinal cord are unaffected, sensation in the lower body is intact.) Neuropathic pain in the upper extremities may occur. The lesion can expand to compress the bilateral anterior horns (the location of the cell bodies of lower motoneurons). Therefore, patients may present with a lower motoneuron pattern of dysfunction in the bilateral upper extremities (eg, decreased muscle tone, hyporeflexia, early muscle atrophy). The cyst may progressively expand to compress the lateral corticospinal tract in severe cases, leading to an upper motoneuron pattern of dysfunction in the lower body.

Incorrect Answers: A, B, C, D, E, F, G, H, and J.

Amyotrophic lateral sclerosis (Choice A) is a progressive neurodegenerative disorder that affects the anterior horns and the lateral corticospinal tract as it descends in the spinal cord. Subsequently, patients demonstrate a combination of upper and lower motoneuron findings in an asymmetric distribution that may include bulbar muscles. Sensory impairment is atypical, and a pattern of dysfunction that can be localized to a single lesion within the cervical or thoracic spinal cord is more consistent with syringomyelia.

Cerebellar degeneration (Choice B) can occur with chronic alcohol use, nutritional deficiency, or as a paraneoplastic disorder and presents with ataxia, gait instability, dysarthria, and nystagmus, and weakness of all extremities. Gait instability and weakness of the lower extremities with upper motor neuron characteristics are common early symptoms.

Cervical spondylosis (Choice C) refers to gradual, age-related degenerative changes of the cervical spine and presents with neck pain, radiculopathy, or myelopathy.

Friedreich ataxia (Choice D) is a genetic condition that presents with ataxia, spasticity, and impaired speech typically in young children or adolescents. It is associated with hypertrophic cardiomyopathy.

Huntington disease (Choice E) is a hereditary, progressive neurodegenerative disorder that presents with psychiatric, cognitive, and motor symptoms.

Muscular dystrophy (Choice F) refers to a group of inherited myopathies and presents with symmetric proximal muscle weakness in the absence of upper motoneuron signs.

Spinal cord compression (Choice G) can occur as a result of trauma or compression from tumors or

infections. Symptoms range from myelopathy and sensory changes to complete paralysis. Subacute combined degeneration (Choice H) occurs as a result of vitamin B₁₂ deficiency and affects the spinocerebellar tracts, lateral corticospinal tracts, and dorsal columns, causing weakness, paresthesia, and loss of proprioception and vibration sense. Thoracic outlet syndrome (Choice J) is a disorder of the brachial plexus and subclavian vessels that can result in symptoms of pain, numbness, and weakness in the upper extremity.

Educational Objective: Educational Objective: Syringomyelia features a cystic cavity in the central cervical or thoracic spinal cord, which affects midline structures such as the anterior white commissure (decussation of the spinothalamic tracts) and the anterior horns (cell bodies of lower motoneurons). Patients with syringomyelia may present with deficits in pain and temperature sensation and a lower motoneuron pattern of dysfunction in the bilateral upper extremities and/or torso.

27. An 85-year-old man with a 4-year history of amyotrophic lateral sclerosis (ALS) is evaluated in the intensive care unit (ICU) 3 weeks after admission for treatment of respiratory failure. Prior to admission, the patient had experienced progressive weakness since ALS was diagnosed, to the point where he was unable to swallow or speak. After admission to the ICU, he underwent tracheostomy and mechanical ventilation. Since admission, four attempts to wean the patient from the ventilator have failed. During today's evaluation, the patient is awake and remains unable to answer any questions correctly. He has little voluntary movement of the chest wall. Physical examination discloses diffuse atrophy, prominent fasciculations, marked weakness in all extremities, and diffuse hyperreflexia. Prior to admission, he told his family that he did not want extreme measures utilized to keep him alive. The patient does not have a living will, but his daughter has medical durable power of attorney for the patient. Which of the following is the most appropriate recommendation for the physician to give this patient's family?

Your Answer: A) Ask the daughter to make a decision on discontinuation of care

Correct Answer: A) Ask the daughter to make a decision on discontinuation of care

Correct Answer: A

Correct Answer: A.

This patient has expressed a preference regarding end-of-life care prior to his current illness but is currently incapacitated and unable to express a preference regarding his care. Options for directing care in a state of incapacity include durable powers of attorney or advance directives, which enable patients to dictate their preferences for end-of-life care ahead of time or delegate these responsibilities to a designated person. A power of attorney is a legal document by which an individual authorizes another person to make decisions on their behalf regarding medical care. This can be either for medical care or financial matters. In this case, the patient's daughter has medical durable power of attorney (DPOA) for the patient, meaning she is legally authorized to make healthcare decisions on his behalf when he is incapacitated. Therefore, she should make the decision regarding discontinuation of care.

Incorrect Answers: B, C, D, and E.

Consider stem cell therapy for treatment of ALS (Choice B) is incorrect. Consideration of experimental treatments would be inappropriate given this patient's advanced state of disease and current respiratory failure.

Continuing supportive care, administering antibiotic therapy, and attempting to wean the patient from the ventilator (Choice C) is unlikely to be successful given previous failed attempts to wean off the ventilator and given this patient's lack of spontaneous respirations. Discussing discontinuation of care with his designated power of attorney is more appropriate given his overall prognosis. Discontinuing the ventilator today (Choice D) is inappropriate without approval from the patient's designated decision-maker.

Requesting a consultation with the hospital ethics committee (Choice E) may be necessary if there is conflict between a patient's previously stated wishes or advance directive and the decision of a power of attorney or next of kin but would be unnecessary at this time.

Educational Objective: Educational Objective: A power of attorney grants a particular person authority to act on the patient's behalf should the patient become incapacitated and is one option to direct end-of-life decision-making ahead of time should a patient become unable to direct their own care. When a patient is incapacitated, decision-making regarding their care is made by the power of attorney or by an advance directive.

28. A 32-year-old woman comes to the physician because of a 6-month history of daily generalized headaches. She occasionally has blurred vision in the right eye during a headache. Treatment with ibuprofen relieves the headaches. She has had a 23-kg (50-lb) weight gain and menstrual irregularity over the past year. Menses now occur at irregular 20- to 90-day intervals. She is 168 cm (5 ft 6 in) tall and weighs 82 kg (180 lb); BMI is 29 kg/m². Examination shows a visual acuity of 20/20 in both eyes. The visual fields are normal. Optic discs are indistinct, and spontaneous venous pulsations are not observed. The remainder of the neurologic examination shows no abnormalities. Which of the following is the most likely diagnosis?

Your Answer: B) Pituitary tumor

Correct Answer: A) Idiopathic intracranial hypertension

Correct Answer: A

Correct Answer: A.

Idiopathic intracranial hypertension (IIH) is primarily driven by impaired resorption of cerebrospinal fluid. Signs of increased intracranial pressure (ICP) include headache, nausea, vomiting, and transient visual obscurations (blurring), and may be secondary to pressure on the optic nerve or optic chiasm. Evaluation findings include papilledema and increased opening pressure on lumbar puncture without any underlying pathology to cause such findings on neuroimaging. Although there is no malignant cause underlying the increased ICP, it may cause debilitating headaches and permanent vision loss. Headaches that are secondary to increased ICP often are worse when lying down and relieved by standing. The most common risk factors for the development of pseudotumor cerebri include female gender and obesity. However, hypervitaminosis A is another potential cause of IIH. This can be from excessive dietary intake or from the use of retinoid medications such as isotretinoin or all-trans retinoic acid. Other medications associated with IIH include tetracyclines, danazol, and growth hormone.

Incorrect Answers: B, C, D, and E.

Pituitary tumor (Choice B) can present with headache and either hypopituitarism or hyperpituitarism, depending on whether the tumor is nonfunctional or functional in nature. Compression of the adjacent optic chiasm causes bitemporal hemianopsia, not unilateral blurred vision.

Status migrainosus (Choice C) is a refractory migraine attack persisting for over 72 hours.

Treatment includes intravenous nonsteroidal anti-inflammatory drugs, such as ketorolac, as well as dopamine receptor blockers, such as metoclopramide or prochlorperazine.

Subdural hematoma (Choice D) may result from head trauma and hemorrhage from bridging veins in the subdural space. Evidence of increased ICP such as postural headache and papilledema are present, as in this patient, but progressive neurologic decline will also occur. On imaging, it appears as a crescent-shaped fluid collection abutting the internal surface of the skull, not bound by suture lines.

Tension-type headache (Choice E) typically presents with bilateral, dull, pressure-like or band-like pain across the frontal and temporal areas, often arising from stress. It does not present with postural headache, papilledema, visual deficits, or other neurologic signs.

Educational Objective: Educational Objective: Impaired resorption of cerebrospinal fluid is the mechanism behind idiopathic intracranial hypertension, which is characterized by headache,

papilledema, and increased opening pressure on lumbar puncture without any underlying pathology to cause such findings on neuroimaging. It is primarily found in young women who are overweight.

29. A 7-year-old girl is brought to the physician by her mother because of dull, generalized headache for the past 2 days. During the past week, the patient has had difficulty walking and frequently has had loss of balance. She has no history of serious illness, and her growth and development have been appropriate for age. Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 16/min, and blood pressure is 100/75 mm Hg. Physical examination shows tilting of the head to the left. There is marked horizontal nystagmus on lateral gaze bilaterally. Visual field testing shows no abnormalities, and visual acuity is 20/20 in both eyes. Finger-nose testing shows past-pointing bilaterally. She has an ataxic gait, and there is in-toeing of the left foot. Romberg sign is present. Which of the following is the most likely diagnosis?

Your Answer: A) Brain stem glioma

Correct Answer: A) Brain stem glioma

Correct Answer: A

Correct Answer: A.

This patient with headache, ataxia, dysmetria, and nystagmus has a presentation concerning for a brain stem glioma. Gliomas are the most common pediatric brain tumor and are typically low-grade pilocytic astrocytomas. Pediatric gliomas commonly grow in the cerebellum but can occur in the brain stem. Patients typically present with focal neurologic deficits caused by mass effect that may progress over the course of months as a result of the tumor growing larger and affecting additional brain regions. Treatment centers around radiation and chemotherapy for brain stem gliomas.

Incorrect Answers: B, C, D, and E.

Craniopharyngioma (Choice B) is a rare pituitary mass derived from remnants of the Rathke pouch. Due to its proximity to the optic chiasm, patients classically present with bitemporal hemianopsia, but a horizontal nystagmus on lateral gaze as found in this patient would be very unusual.

Great vein of Galen aneurysm (Choice C) is a rare form of arteriovenous malformation that affects one of the intracranial veins. Patients may present with signs and symptoms of heart failure, hydrocephalus, or headache.

Meningitis (Choice D) may present with a headache, neck pain, and fever. This patient's lack of fever in addition to numerous focal neurologic deficits point to an alternative diagnosis.

Tumor of the vestibulocochlear cranial nerve (Choice E), such as an acoustic neuroma, can cause ipsilateral hearing loss and vertigo, and if also compressing the trigeminal nerve, can lead to ipsilateral facial numbness and paresthesia.

Educational Objective: Educational Objective: Gliomas are the most common pediatric brain tumor and are typically low-grade pilocytic astrocytomas that may occur in the cerebellum or brain stem. Gliomas commonly present with focal neurologic deficits caused by mass effect that may progress over the course of months as a result of the tumor growing larger and affecting additional brain regions.

30. A 71-year-old woman comes to the clinic because of a 2-month history of increasingly severe, aching pain in her shoulders, upper arms, and thighs. She says, "I can't roll over in bed any longer and my husband has to assist me in getting up from a chair." She also has had associated muscle stiffness that lasts 1 hour every morning upon awakening; the stiffness sometimes occurs in the afternoon following periods of inactivity. She also reports awakening multiple times during the night, diffuse joint pain, and a 7.7-kg (17-lb) weight loss despite no change in appetite. Menopause occurred at age 56 years. She also has osteoporosis and generalized anxiety disorder. Her medications are estradiol, escitalopram, and vitamin D and calcium supplementation. Her mother and aunt had rheumatoid arthritis, and her brother had sclerosing cholangitis. The patient appears to be in moderate discomfort. She is 174 cm (5 ft 8 in) tall and weighs 64 kg (140 lb); BMI is 21 kg/m³. Temperature is 36.3°C (97.3°F), pulse is 68/min, respirations are 18/min, and blood pressure is 124/76 mm Hg. Muscle strength and tone are normal. She is unable to actively abduct the shoulders bilaterally past 90 degrees because of stiffness. Passive range of motion of the shoulders and hips is decreased. There is no pain on palpation of the temporalis muscles. Results of laboratory studies are shown:

Hemoglobin 8.9 g/dL

Hematocrit 25%

Leukocyte count 8200/mm³

Platelet count 572,000/mm³

Erythrocyte sedimentation rate 82 mm/h

Serum

C-reactive protein 15.3 mg/L

Creatine kinase 112 U/L

Electromyography and nerve conduction studies disclose no abnormalities. Which of the following is the most appropriate next step in management?

Your Answer: F) Prednisone therapy

Correct Answer: F) Prednisone therapy

Correct Answer: F

Correct Answer: F.

Polymyalgia rheumatica (PMR) is an inflammatory condition characterized by proximal muscle and joint (eg, shoulder and hip girdle) pain, stiffness, and increased inflammatory markers, including erythrocyte sedimentation rate and C-reactive protein. Untreated, muscle weakness may result from disuse atrophy. Systemic symptoms such as fatigue and anorexia are associated with PMR. PMR is a diagnosis of exclusion. Symptoms of giant cell arteritis, such as through unilateral headache, vision changes, or jaw claudication, should prompt temporal artery biopsy. The diagnosis of PMR is based on clinical features, and treatment is with prednisone for up to 1 year. It is common for patients to respond to prednisone almost immediately; thus, the absence of response should prompt consideration of a different diagnosis.

Incorrect Answers: A, B, C, D, and E.

Biopsy of the temporal artery (Choice A) is used to diagnose temporal arteritis. Temporal arteritis is a vasculitis of medium and large vessels in elderly patients. Patients commonly present with headache, jaw claudication, and proximal muscle stiffness, and is commonly associated with PMR. This patient does not display any symptoms suggestive of temporal arteritis.

Biopsy of the vastus medialis muscle (Choice B) is used to diagnose inflammatory myopathy, such as polymyositis. Symptoms of polymyositis include proximal muscle weakness, occasionally with

increased creatine kinase, but muscle stiffness is not a typical feature. This patient's clinical presentation is more consistent with PMR than polymyositis.

Dopamine transporter (DAT) scan (Choice C) can be used to support a diagnosis of Parkinson disease in patients who present with tremor, rigidity, bradykinesia, or postural instability. This patient's muscle stiffness, pain, and fatigue are not consistent with Parkinson disease.

Duloxetine therapy (Choice D) is a selective serotonin and norepinephrine reuptake inhibitor that is used in treatment of major depressive disorder, generalized anxiety disorder, fibromyalgia, and chronic pain. This patient has generalized anxiety disorder, but a trial of an SSRI or other antidepressant would not address her muscular pain and stiffness.

Prednisone therapy (Choice E) is indicated in the treatment of polymyalgia rheumatica. Low-dose prednisone is indicated in the treatment of polymyalgia rheumatica.

Educational Objective: Educational Objective: Polymyalgia rheumatica (PMR) is an inflammatory condition in which aching and stiffness develop in the muscles of the hip and shoulder girdles. Low-dose prednisone is the treatment of choice.

31. A 47-year-old man comes to the physician because of severe pain of the left eye and double vision that began after a motor vehicle collision 1 week ago. At the time of the collision, he hit his head but did not lose consciousness. He did not seek medical attention at that time. Examination shows exophthalmos, chemosis, and distention of the orbital and periorbital veins in the left eye. A bruit is heard over the left orbit. There is incomplete abduction of the left eye when looking to the left. Which of the following is the most likely diagnosis?

Your Answer: C) Carotid cavernous fistula

Correct Answer: C) Carotid cavernous fistula

Correct Answer: C

Correct Answer: C.

Carotid cavernous fistula (CCF) results from an inappropriate connection between the carotid artery and cavernous sinus, often caused by head trauma, aneurysmal rupture, inflammation, or vascular or connective tissue disease. Arterial pressure in the cavernous sinus results in cranial nerve deficits involving those that traverse the cavernous sinus (III, IV, V₁, V₂, or VI) as a result of extrinsic compression. Other findings can include vascular congestion of the ipsilateral eye and face, exophthalmos, chemosis, unilateral headache, eye pain, visual loss, and an audible bruit on auscultation of the eye. CCFs can be categorized as low-flow fistulas, which result from communication between the cavernous sinus and cavernous arterial branches, or high-flow fistulas, which result from a direct communication between the cavernous sinus and the internal carotid artery. Carotid angiography is the gold standard diagnostic modality, will show flow through the fistula. CT or MR angiography may also be used. First-line treatment is endovascular intervention with embolization or stenting.

Incorrect Answers: A, B, D, and E.

Acute cerebral infarction (Choice A) can present with a sudden change in mental status, upper motor neuron signs, and acute focal neurologic deficits, especially on one side of the body, and may have speech difficulty, dysarthria, or aphasia, depending on the vascular territory involved.

Exophthalmos and an orbital bruit are not consistent with acute cerebral infarction.

Arteriovenous malformation (Choice B) describes an abnormal tangle of blood vessels located within the brain. A ruptured arteriovenous malformation may result in a subarachnoid or intracerebral hemorrhage (ICH). ICH appears as a hyperdense collection of blood within the brain parenchyma on CT scan. ICH often presents with focal neurologic deficits based on their location with an associated headache and symptoms of increased intracranial pressure (eg, vomiting, altered mental status).

Epidural hematoma (Choice D) is a collection of blood in the epidural space, typically between the dura mater and the skull. It classically occurs after injury to the middle meningeal artery and can present with an initial lucid interval followed by deterioration of mental status caused by arterial bleeding and hematoma expansion, along with headache, vomiting, agitation, seizure, and coma. Epidural hematomas present on CT scan as a lens-shaped, bi-convex hyperdense collection abutting the skull.

Subarachnoid hemorrhage (Choice E) typically presents with an acute, severe headache along with neck stiffness, photophobia, and possible stupor or coma in severe cases. CT scan of the head typically shows hyperdense blood products in the cerebral sulci and basal cisterns. Intracranial aneurysms are the most common cause of non-traumatic subarachnoid hemorrhage, followed by arteriovenous malformations. It is a neurosurgical emergency that may be fatal if progressive or untreated.

Subdural hematoma (Choice F) results from head trauma and hemorrhage from bridging veins in the subdural space. They present with subacute-onset headache, nausea, and vomiting, and may present with focal neurologic deficits or seizures caused by mass effect. Subdural hematomas appear as a crescent-shaped hyperdense collection in parallel to the curvature of the skull and cross the dural sutures.

Educational Objective: Educational Objective: Carotid cavernous fistula (CCF) results from an abnormal communication between the carotid artery and cavernous sinus, often caused by head trauma. It presents with exophthalmos, chemosis, orbital congestion, pain, double vision, and audible bruit on auscultation of the eye.

32. After renovating a new home, a 41-year-old woman has burning pain and tingling in her right index, long, and ring fingers, particularly at night. The thenar eminence becomes atrophied. Which of the following structures is most likely to be damaged?

Your Answer: C) Median nerve

Correct Answer: C) Median nerve

Correct Answer: C

Correct Answer: C.

Carpal tunnel syndrome (CTS) results from compression of the median nerve within the carpal tunnel and presents with signs and symptoms including pain, weakness, and paresthesia of the first three digits and the lateral half of the fourth digit, along with positive provocative test maneuvers such as the Tinel or Phalen tests. The carpal tunnel is located anterior to the carpal bones and is bound by the flexor retinaculum. Flexor tendons of the fingers course through the tunnel, along with the median nerve. The space is rigid and does not generally permit expansion. Thus, any inflammatory, toxic, or space-filling lesion may result in compression of the median nerve, leading to pain and paresthesia in the median nerve distribution, along with weakness and atrophy of the thenar eminence that eventually may atrophy. Symptoms are generally worsened at night and improved with shaking of the hands. CTS is frequently associated with pregnancy, hypothyroidism, rheumatoid arthritis, acromegaly, amyloidosis, ganglion cysts, obesity, and acute trauma to the wrist with impingement, or chronic repetitive stress and scar formation following a wrist injury. Patients with mild symptoms can be treated conservatively with wrist splinting. Wrist splints are typically worn at night but may also be used during the day. Resisting daily activities that involve repetitive wrist motion and limits activities that increase pressure within the carpal tunnel. Patients who have persistent or severe symptoms or muscular weakness may require surgical decompression.

Incorrect Answers: A, B, D, and E.

Injuries to the radial artery (Choice A) and ulnar artery (Choice B) would not result in chronic neurologic dysfunction or muscle atrophy.

The radial nerve (Choice D) courses posterior to the humerus at the midshaft within the radial groove. It is often injured in displaced midshaft humerus fractures, with resulting wrist drop, and hand. The radial nerve supplies motor fibers to the triceps brachii and the extensor muscles within the posterior compartment of the forearm, and sensory fibers to the dorsum of the forearm, wrist, and hand.

The ulnar nerve (Choice E) supplies motor fibers to the intrinsic muscles of the hand, and sensation over the anteromedial ring and small fingers and medial palm including hypothenar eminence.

Damage to the roots or to the nerve along its course (eg, in the cubital tunnel) can result in motor dysfunction of the fingers, and numbness or paresthesia involving the medial aspect of the upper extremity.

Educational Objective: Educational Objective: Carpal tunnel syndrome results from compression of the median nerve within the carpal tunnel and presents with signs and symptoms including pain, weakness, and paresthesia of the first three digits and the lateral half of the fourth digit, along with weakness and atrophy of the thenar eminence.

33. A 21-year-old woman is brought to the emergency department by her mother 1 hour after a 30-minute episode of generalized shaking. One week ago, the patient had a similar episode that began while she was at home watching television with her mother. At that time, the patient fell to the ground and had irregular jerking of her hands and legs. During the episode, her eyes were open and she was occasionally able to respond to her mother's commands. The episode lasted 10 minutes. The patient has had three similar episodes over the past week. She has smoked one pack of cigarettes daily for 6 years. She drinks three to four 12-oz beers weekly. She does not use illicit drugs. She works as a home health aide. She has been under great stress recently following her breakup with her fiancé. Vital signs are within normal limits. Physical, neurologic, and mental status examinations disclose no abnormalities. CT scan of the head discloses no abnormalities. During a video EEG, the patient has a typical seizure; there are no epileptiform EEG changes during the episode. Which of the following is the most appropriate next step in management?

Your Answer: A) Cognitive behavior therapy

Correct Answer: A) Cognitive behavior therapy

Correct Answer: A

Correct Answer: A.

Conversion disorder, also known as functional neurologic symptom disorder, involves neurologic symptoms such as sensory or motor dysfunction that are not fully explained by objective findings on physical examination or imaging. Functional neurologic symptom disorder is often related to psychological factors, such as stress, trauma, or anxiety, and is thought to represent the subconscious conversion of emotional distress into physical symptoms. The shaking episodes described in this patient appear to be non-epileptic based on the lack of epileptiform changes on EEG. The presence of irregular jerking, eye opening, ability to respond to commands during the episode, and lack of postictal confusion are all features that suggest against epileptic seizures. Cognitive behavior therapy (CBT) is a type of psychotherapy that focuses on changing thought and behavior patterns and is an effective treatment for functional neurologic symptom disorder. CBT can help patients identify triggers for their symptoms and develop coping strategies.

Incorrect Answers: B, C, D, and E.

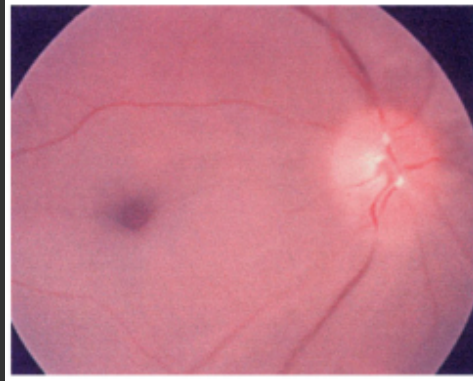
Long-term EEG monitoring (Choice B) can be used to capture a typical seizure and confirm whether it is epileptic or non-epileptic. However, since the patient has already had a typical seizure captured on video EEG, and it was determined to be non-epileptic, further EEG monitoring is unlikely to be helpful.

Measurement of serum prolactin concentration (Choice C) can be used to differentiate between epileptic and non-epileptic seizures. Serum prolactin levels typically rise after an epileptic seizure but not after a non-epileptic seizure. This test could have been useful in the initial workup, but the video EEG has already provided more definitive information.

MRI of the brain (Choice D) would be appropriate if there was concern for a structural lesion causing epileptic seizures or other neurologic symptoms. The CT scan was normal, and the video EEG findings suggest against epileptic seizures, making an MRI less likely to be helpful at this time. Oxcarbazepine therapy (Choice E) is an anticonvulsant medication used to treat epileptic seizures. Since the patient's shaking episodes appear to be non-epileptic, anticonvulsant therapy is not indicated and may cause unnecessary side effects.

Educational Objective: Educational Objective: Functional neurologic symptom disorder, also known as conversion disorder, involves neurologic symptoms that are not explained by objective medical findings and are thought to be related to psychological factors. Non-epileptic seizures are a common presentation of functional neurologic symptom disorder. Treatment often involves psychotherapy, such as cognitive behavior therapy.

34. An 82-year-old man comes to the physician because of a 1-day history of visual loss in the right eye. He has had no eye pain. He has had approximately five episodes of visual loss in the right eye over the past 3 weeks that lasted less than 1 hour. He has hypertension and hyperlipidemia. Medications include captopril and simvastatin. His pulse is 85/min, and blood pressure is 135/90 mm Hg. Corrected visual acuity is 20/40 on the left and limited to hand motion on the right. There is a prominent afferent pupillary defect in the right eye. Fundusoscopic findings of the right eye are shown. Which of the following is the most likely diagnosis?



Your Answer: B) Central retinal artery occlusion

Correct Answer: B) Central retinal artery occlusion

Correct Answer: B

Correct Answer: B.

Central retinal artery occlusion (CRAO) is an ocular emergency characterized by acute, painless vision loss in one eye. It is typically caused by an embolus occluding the central retinal artery, which supplies blood to the inner retina. Risk factors for CRAO include carotid artery disease, cardiac emboli (from atrial fibrillation or valvular disease), hypertension, diabetes, and hyperlipidemia. Fundusoscopic examination typically shows a pale retina with a cherry-red spot in the macula, although the fundus may appear normal in the early stages. A relative afferent pupillary defect (rAPD) is usually present. CRAO is a medical emergency because it is a form of ischemic stroke and indicates a high risk for stroke to the brain. Patients with CRAO should be evaluated immediately for an embolic source with carotid duplex ultrasound, echocardiogram, and potentially cardiac monitoring. Treatment focuses on lowering intraocular pressure to potentially improve blood flow and managing underlying risk factors.

Incorrect Answers: A, C, D, and E.

Age-related macular degeneration (Choice A) is a chronic condition affecting the macula, the central part of the retina responsible for sharp central vision. It typically causes gradual, painless vision loss, often with distortion of straight lines. It is not characterized by acute, episodic vision loss or a pale retina.

Central retinal vein occlusion (Choice C) is caused by a blockage of the central retinal vein, leading to blood backup and hemorrhage in the retina. Fundusoscopic examination shows dilated and tortuous retinal veins, scattered hemorrhages, and cotton wool spots (nerve fiber layer infarcts). Vision loss is typically sudden and painless, but the fundus appearance is distinct from CRAO.

Hypertensive retinopathy (Choice D) refers to changes in the retinal blood vessels caused by chronic high blood pressure. Findings can include narrowed arteries, arteriovenous nicking,

hemorrhages, exudates, and papilledema in severe cases. While this patient has hypertension, the acute, episodic vision loss and pale retina with a rAPD are more consistent with CRAO. Ischemic optic neuropathy (Choice E) is caused by decreased blood flow to the optic nerve and presents with acute, painless vision loss and a rAPD. There are two types: anterior ischemic optic neuropathy (AION), which affects the optic nerve head and causes disc edema, and posterior ischemic optic neuropathy (PION), which affects the optic nerve behind the eye and does not cause disc edema. AION is associated with risk factors like diabetes, hypertension, and sleep apnea. Arteritic AION is caused by giant cell arteritis and is a medical emergency requiring prompt steroid treatment to prevent vision loss in the other eye. While ischemic optic neuropathy can cause sudden vision loss and a rAPD, the fundus appearance of a pale retina with a cherry-red spot is characteristic of CRAO.

Educational Objective: Educational Objective: Central retinal artery occlusion (CRAO) is an ocular emergency caused by embolic occlusion of the central retinal artery. Patients present with acute, painless vision loss and a pale retina with a cherry-red spot in the macula. Evaluation should focus on identifying the embolic source due to the high risk of associated stroke.

35. A 47-year-old woman is brought to the emergency department by her husband 45 minutes after the sudden onset of a severe headache. She has a 25-year history of occasional throbbing headaches. The headaches occur once or twice annually, are associated with nausea and vomiting, and are relieved with sleep. She has hypertension treated with enalapril. On arrival, she can answer questions but is intermittently listless and agitated. Her temperature is 37.7°C (99.8°F), pulse is 120/min, respirations are 24/min, and blood pressure is 154/96 mm Hg. On examination, the left pupil is larger than the right. The left eye deviates slightly downward and outward. The patient resists flexion of the neck and exhibits pain when the neck is flexed. Muscle tone in the right upper and lower extremities is increased. Deep tendon reflexes on the right side are increased. Babinski sign is present on the right. Which of the following is the most likely diagnosis?

Your Answer: D) Ruptured saccular aneurysm

Correct Answer: D) Ruptured saccular aneurysm

Correct Answer: D

Correct Answer: D.

A ruptured intracranial aneurysm is the most common cause of non-traumatic subarachnoid hemorrhage (SAH). SAH occurs when there is bleeding into the subarachnoid space, often from a ruptured aneurysm or arteriovenous malformation. Patients typically present with an acute, severe headache, often described as the "worst headache of my life," along with nausea, vomiting, photophobia, and neck stiffness (meningismus). Aneurysms located at the junction of the posterior communicating artery and the internal carotid artery can compress the oculomotor nerve (cranial nerve III), leading to an ipsilateral oculomotor palsy with a dilated pupil (parasympathetic fibers affected) and impaired eye movements (downward and outward deviation, ptosis). Hemiparesis with upper motoneuron signs on the contralateral side can also occur due to compression of the cerebral peduncle or internal capsule. Diagnosis is typically made with a non-contrast CT scan of the head, which shows hyperdense blood in the basal cisterns and sulci. If the CT scan is negative but suspicion remains high, a lumbar puncture can be performed, which may show red blood cells or xanthochromia (yellow discoloration due to hemoglobin breakdown). SAH is a neurosurgical emergency and requires prompt management to prevent re-bleeding, vasospasm, and hydrocephalus.

Incorrect Answers: A, B, C, and E.

Carotid artery dissection with embolization (Choice A) can cause ischemic stroke symptoms, including headache, neck pain, and focal neurologic deficits (depending on the area of the brain supplied by the affected artery). Oculomotor nerve palsy would be atypical in this setting unless the dissection extends intracranially to compress the nerve.

Hypertensive intracranial hemorrhage (Choice B) typically occurs in patients with chronic hypertension and involves bleeding into the brain parenchyma, commonly in the basal ganglia, thalamus, pons, or cerebellum. While ICH can cause acute headache and focal neurologic deficits, it would be less likely to cause a specific isolated oculomotor nerve palsy with pupillary involvement compared to a posterior communicating artery aneurysm.

Meningoencephalitis (Choice C), inflammation of the brain and meninges, typically presents with fever, headache, altered mental status, and meningismus. Focal neurologic deficits, including

cranial nerve palsies, can occur. However, the acute onset severe headache, dilated pupil, and specific oculomotor palsy pattern are more characteristic of SAH from an aneurysm compressing the nerve.

Vertebrobasilar migraine (Choice E) is a type of migraine with aura symptoms that originate from the vertebrobasilar circulation. Symptoms can include dizziness, vertigo, double vision, speech difficulty, and sometimes headache. While double vision can occur, a dilated pupil and specific oculomotor palsy are not typical features of migraine.

Educational Objective: Educational Objective: Ruptured saccular aneurysms are a common cause of non-traumatic subarachnoid hemorrhage (SAH). Aneurysms arising from the posterior communicating artery can compress the oculomotor nerve (cranial nerve III), leading to an ipsilateral oculomotor palsy characterized by a dilated pupil and impaired eye movements (downward and outward deviation, ptosis). SAH presents with an acute, severe headache, meningismus, and often focal neurologic deficits depending on the location and complications.

36. A 47-year-old man has to quit his job as an automobile mechanic because of hand weakness and unsteadiness. He has bilateral wristdrop and upper limb ataxia. Laboratory studies show:
Hematocrit 34%
Mean corpuscular volume 76 μm^3
Serum creatinine 2 mg/dL
Chronic exposure to which of the following substances is the most likely explanation for these findings?

Your Answer: C) Lead

Correct Answer: C) Lead

Correct Answer: C

Correct Answer: C.

Chronic exposure to lead is the most likely explanation for this patient's symptoms of motor weakness (wristdrop), ataxia, and cognitive dysfunction, given his occupation as an automobile mechanic and laboratory findings of microcytic anemia and renal dysfunction. Lead poisoning affects multiple organ systems. Neurologic manifestations include cognitive dysfunction, encephalopathy, hearing loss, peripheral neuropathy (often predominantly motor, leading to wristdrop or footdrop), and ataxia. Gastrointestinal symptoms can include abdominal pain, constipation, and vomiting. Lead poisoning can also cause renal tubular dysfunction and interstitial nephritis, leading to increased serum creatinine and impaired renal function. Hematologic effects include microcytic, hypochromic anemia due to impaired heme synthesis. Diagnosis is confirmed with measurement of lead concentration in the blood or, for chronic exposure, in bone. Treatment involves chelation therapy with agents such as edetate calcium disodium or dimercaprol. Sources of lead should be investigated and eliminated.

Incorrect Answers: A, B, D, and E.

Arsenic (Choice A) poisoning is uncommon and can cause dermatitis, skin cancer, neuropathy, hepatotoxicity, and, in severe cases, systemic toxicity and death. Acute large exposure to arsenic can cause gastrointestinal distress. QT prolongation that may progress to torsades de pointes, cardiac arrhythmias, shock, and acute respiratory distress syndrome.

Cadmium (Choice B) poisoning may occur in the industrial setting among workers involved in electroplating, metallurgical industries, and battery production. Exposure typically occurs via inhalation and presents with fever, chills, myalgia, chest pain, cough, shortness of breath, pneumonitis, and pulmonary edema.

Mercury (Choice D) poisoning may occur in the industrial setting among workers involved in the manufacture of thermometers, mirrors, fluorescent lighting, and gold. Patients present with headache, ataxia, tremor, hearing and vision loss, insomnia, peripheral neuropathy, and paresthesias.

Thallium (Choice E) poisoning may occur through transcutaneous exposure or via inhalation or ingestion. Thallium was previously widely used as a rodenticide, but thallium poisoning is now rare in the developed world, although it is still used in the manufacture of semiconductors, electronics, and optics. Acute exposure presents with abdominal pain, nausea, and vomiting, while chronic exposure presents with neurologic and psychiatric symptoms of headache, altered mental status, anxiety, psychosis, ataxia, tremor, and painful peripheral neuropathy and distal weakness.

Educational Objective: Educational Objective: Neurologic manifestations of lead poisoning include cognitive dysfunction, encephalopathy, hearing loss, and peripheral neuropathy. Distal motor neuropathy causes wristdrop or footdrop. Lead poisoning also causes renal tubular dysfunction and microcytic anemia.

37. A 27-year-old woman comes to the physician because of a 1-day history of difficulty speaking and moving. Seven days ago she began a course of mitoxantrone and high doses of cytarabine (ara-C) for a recurrence of acute myeloid leukemia. She has been lethargic and mildly confused for the past 24 hours. She is 140 cm (4 ft 7 in) tall and weighs 47 kg (103 lb); BMI is 24 kg/m². Her temperature is 37°C (98.6°F), pulse is 72/min, respirations are 14/min, and blood pressure is 110/70 mm Hg. Examination shows nystagmus and ataxic eye movements. Muscle strength is full, but there is marked ataxia in all extremities. She is unable to maintain her balance sitting or standing unsupported. She is somnolent but can respond appropriately to questions and instructions. Her speech is irregular in rate and volume. Language function and memory are normal. She can calculate well, and her abstract reasoning is intact. Which of the following is the most likely diagnosis?

Your Answer: B) Drug toxicity

Correct Answer: B) Drug toxicity

Correct Answer: B

Correct Answer: B.

Acute-onset neurologic findings in the setting of recent chemotherapy are concerning for drug toxicity. Specifically, nystagmus, ataxia, impaired coordination, postural instability, and dysarthria or other speech abnormalities, in the setting of intact strength and higher cognitive functions are indicative of cerebellar dysfunction. Cytarabine, a pyrimidine analog, can cause loss of Purkinje cells in the cerebellum, leading to acute cerebellar syndrome. Risk factors include underlying neurologic dysfunction and impaired hepatic or renal function. Patients who present with acute cerebellar syndrome should discontinue cytarabine immediately. Some may experience permanent deficits, while others may experience spontaneous resolution.

Incorrect Answers: A, C, D, and E.

Cerebral infarction (Choice A) can present with a sudden change in mental status, upper motor neuron signs, and acute focal neurologic deficits, especially on one side of the body, and may have speech difficulty, dysarthria, or aphasia, depending on the vascular territory involved. However, acute-onset cerebellar signs in the setting of new course of cytarabine are more likely caused by drug toxicity.

Encephalitis (Choice C) presents with fever, altered mental status, and neurologic abnormalities, such as seizure, aphasia, hemiparesis, or cranial nerve deficits. Meningoencephalitis is commonly associated with nuchal rigidity and signs of central nervous system infection, which this patient does not demonstrate, making it a less likely cause of her presentation.

Intracranial hemorrhage (Choice D) appears as a hyperdense collection of blood within the brain parenchyma on CT scan. Intracranial hemorrhage often presents with focal neurologic deficits based on its location with an associated headache and symptoms of increased intracranial pressure (eg, vomiting, altered mental status). This patient does not demonstrate any signs of increased intracranial pressure.

Spread of leukemia to the central nervous system (Choice E), also called leukemic meningitis, is usually asymptomatic. When it does occur, signs and symptoms vary widely depending on the location of spread, and can include headache, signs of increased intracranial pressure, behavioral changes, and focal weakness. Isolated cerebellar signs in the setting of recently started cytarabine is more consistent with drug toxicity.

Educational Objective: Educational Objective: Cytarabine, a pyrimidine analog, can cause loss of Purkinje cells in the cerebellum, leading to acute cerebellar syndrome. Features include nystagmus, ataxia, impaired coordination, postural instability, and dysarthria or other speech abnormalities. Strength, sensation, and higher cognitive functions are intact. Patients who present with acute cerebellar syndrome should discontinue cytarabine immediately. Some may experience permanent deficits, while others may experience spontaneous resolution.

38. A 67-year-old woman with a 15-year history of Parkinson disease comes to the office because of increasing involuntary movements over the past 2 years. She also has had nausea from her medication regimen of carbidopa-levodopa and ropinirole; attempts at medication adjustment have not been successful. She also has a long-standing history of cervical and lumbar stenosis treated with diclofenac and physical therapy. Her pulse is 70/min and blood pressure is 110/60 mm Hg while sitting; pulse is 75/min and blood pressure is 80/50 mm Hg while standing. Her speech is barely audible. Examination shows constant writhing movements of the mouth, neck, and limbs. She walks with her neck flexed and takes small steps with a decreased arm swing. Mental status examination shows no abnormalities. Which of the following is the most appropriate surgical treatment?

Your Answer: E) Pallidotomy

Correct Answer: E) Pallidotomy

Correct Answer: E

Correct Answer: E.

Parkinson disease is a progressive neurodegenerative disorder caused by destruction of dopaminergic neurons in the substantia nigra. Symptoms include bradykinesia, rigidity, resting tremor, and postural instability. As the disease progresses, patients often require increasing doses of dopamine agonists or levodopa, which can cause adverse effects such as nausea and involuntary movements (dyskinesias), as in this patient. Dyskinesias can be troublesome and may not resolve with medication adjustment. For patients with disabling dyskinesias or motor fluctuations despite optimal medical management, surgical options such as deep brain stimulation or lesioning procedures may be considered. Pallidotomy is a lesioning procedure that targets the globus pallidus internus, a component of the basal ganglia involved in motor control. By creating a lesion in this area, pallidotomy can help reduce dyskinesias and motor fluctuations in patients with Parkinson disease.

Incorrect Answers: A, B, C, D, F, and G.

Anterior temporal lobectomy (Choice A) is an effective surgical treatment for intractable temporal lobe epilepsy. It would not help with severe motor symptoms of Parkinson disease.

Carotid endarterectomy (Choice B) is a treatment for atherosclerotic disease of the carotid arteries. It is indicated for patients who have had a stroke or transient ischemic attack or are symptomatic with hemodynamically significant stenosis.

Cervical laminectomy (Choice C) or lumbar laminectomy (Choice D) are spine surgery procedures used to decompress the spinal cord or nerve roots in the setting of spinal stenosis or disc herniation. While this patient has a history of spinal stenosis, these procedures would not address her dyskinesias related to Parkinson disease.

Thymectomy (Choice F) is reserved for patients with tumors of the thymus or with myasthenia gravis. Myasthenia gravis characteristically presents with generalized weakness and ptosis that worsens throughout the day.

Ventriculoperitoneal shunting (Choice G) is used to remove excess cerebrospinal fluid (CSF) in patients with hydrocephalus, either from lesions obstructing CSF outflow or ones that impair CSF reabsorption.

Educational Objective: Educational Objective: For patients with Parkinson disease who experience disabling dyskinesias or motor fluctuations despite optimal medical management, surgical options such as deep brain stimulation or lesioning procedures like pallidotomy may be considered. These procedures target components of the basal ganglia involved in motor control to help reduce symptoms.

39. A previously healthy 37-year-old man is brought to the physician by his wife because of bizarre behavior for 2 days. He is very excitable and jumps when surprised. His wife says that he has been experiencing visual hallucinations. He refuses to drink liquids. For 3 days prior to this behavior, he had headache and sleepiness. Two months ago, he returned from a hunting trip in the mountains. His temperature is 40°C (104°F). Examination shows an inability to accurately answer questions and unusual behaviors such as picking at the air. He seems unable to swallow his secretions, and he is drooling. A CT scan of the brain shows no abnormalities. Cerebrospinal fluid analysis shows:

Glucose 70 mg/dL

Protein 85 mg/dL

Leukocyte count 150/mm³

Erythrocyte count none

Your Answer: H) Rabies

Correct Answer: H) Rabies

Correct Answer: H

Correct Answer: H.

This patient presents with severe, acute-onset neurologic symptoms including bizarre behavior, hallucinations, refusal to drink, inability to swallow secretions (hydrophobia/difficulty swallowing), and hypersalivation, preceded by headache and sleepiness, in the setting of a recent hunting trip and fever, is highly suggestive of rabies. Rabies is a viral encephalitis caused by infection with the rabies virus, a negative-sense, single-stranded RNA virus belonging to the family of Rhabdoviridae. The virus is typically transmitted through direct contact with saliva of an infected animal, most commonly through a bite, although transmission through scratches or mucosal exposure can also occur. In the United States, the most common source of rabies virus infection is wild animals, including bats, raccoons, skunks, and foxes. The incubation period is typically 4 to 12 weeks but can range from days to years. Initial symptoms are nonspecific and include fever, headache, malaise, and fatigue. Neurologic symptoms develop rapidly and can progress to agitation, photophobia, hydrophobia, drooling, and fevers. Once neurologic symptoms develop, rabies is almost universally fatal. Postmortem diagnosis can be made by identifying Negri bodies (intracytoplasmic inclusions) in neuronal tissue. Prevention after exposure includes prompt irrigation and wound care, administration of rabies immune globulin, and administration of rabies vaccine series. Standard care for pets, including vaccination, is essential for preventing rabies transmission. If the species of animal that transmitted rabies is unknown, the patient should be treated as if the animal was infected. Immune globulin should be administered locally into the wound and intramuscularly.

Incorrect Answers: A, B, C, D, E, F, G, and I.

Arbovirus (Choice A) refers to a group of viruses that are transmitted by arthropods (eg, ticks, mosquitoes). West Nile virus is an example, and patients typically present with fever, headache, myalgia, nausea, vomiting, and a non-specific maculopapular rash distributed symmetrically over the extremities and trunk. Meningoencephalitis is associated with acute flaccid paralysis or tremors, and cognitive dysfunction.

Borrelia burgdorferi (Choice B) is transmitted by Ixodes ticks and causes Lyme disease. Neurologic manifestations of Lyme disease include meningitis, cranial nerve palsies (especially facial palsy), radiculopathy, and peripheral neuropathy. Fever, headache, and malaise are common symptoms of acute Lyme disease, and it may present with erythema migrans, a characteristic skin rash. Rabies is

not associated with tick bites.

Cytomegalovirus (Choice C) is a herpesvirus that can cause mononucleosis-like syndrome with fever, fatigue, and cervical lymphadenopathy in immunocompetent individuals. It can cause encephalitis in immunocompromised patients, but this is less common. Rabies is not a herpesvirus. Enterovirus (Choice D) is a common cause of viral meningitis and encephalitis, particularly in children. Symptoms include fever, headache, meningismus, and sometimes rash or focal neurologic deficits. Rabies is not an enterovirus.

Herpes simplex virus (HSV) (Choice E) can cause encephalitis, typically in the temporal lobe. Symptoms include fever, headache, altered mental status, and neurologic abnormalities, such as seizure, aphasia, hemiparesis, or cranial nerve deficits. Rabies is not an HSV.

HIV (Choice F) infection can cause HIV encephalitis, particularly in the late stages of the disease. Symptoms include cognitive dysfunction, motor weakness, and neuropsychiatric symptoms. Rabies is not HIV.

Listeria monocytogenes (Choice G) is a bacterium that can cause meningoencephalitis, particularly in patients at the extremes of age or those who are immunocompromised. It presents with fever, headache, nuchal rigidity, altered mental status, and focal neurologic deficits. Rabies is not caused by bacteria.

Rickettsia rickettsii (Choice I) is a bacterium that is transmitted by ticks and causes Rocky Mountain spotted fever. Symptoms include fever, headache, myalgia, rash, and in severe cases, neurologic complications such as encephalitis or seizures. Rabies is not a rickettsial infection.

Toxoplasma gondii (Choice J) is an intracellular protozoan parasite that can cause cerebral toxoplasmosis in immunocompromised patients. It presents with seizures, focal neurologic deficits, and systemic symptoms. Rabies is not a protozoan.

Educational Objective: Educational Objective: Rabies is a viral encephalitis transmitted by the rabies virus. It presents with acute, rapidly progressive neurologic symptoms including agitation, hallucinations, hydrophobia (difficulty swallowing), drooling, and fevers, preceded by nonspecific symptoms such as headache and sleepiness. Rabies is almost universally fatal once neurologic symptoms develop. Prevention after exposure involves prompt wound care, administration of rabies immune globulin, and rabies vaccine series.

40. A 17-year-old boy with sickle cell disease is brought to the clinic by his father because of a 4-week history of progressive right hip pain. Initially, the pain was intermittent and the patient rated it as a 5 on a 10-point scale; however, during the past 2 weeks it has become constant, and he now rates the pain as a 9 on a 10-point scale. The pain is not worsened by sneezing, coughing, or lifting objects. He was diagnosed with sickle cell disease at age 3 years; his disease course has been complicated by frequent bouts of dactylitis and pneumonia. He has had two blood transfusions done 9 and 6 years ago, respectively. He is taking acetaminophen for the hip pain, but it has provided minimal relief. Family history is remarkable for sickle cell disease in his mother, brother, and two sisters, and for sickle cell trait in his father. The patient appears to be in moderate distress. Temperature is 36.6°C (98.6°F), pulse is 98/min, respirations are 20/min, and blood pressure is 114/68 mm Hg. Physical examination shows a decreased range of motion to hip flexion and abduction. There is no pain on palpation of the lateral aspect of the hip. Muscle strength is 5/5 throughout, and muscle tone is symmetric. Deep tendon reflexes are 1+ in the lower extremities bilaterally. Sensation to pinprick is normal over the lower extremities. Results of laboratory studies are shown:

Hemoglobin 9.8 g/dL

Hematocrit 28.7%

Leukocyte count 7200/mm³

Platelet count 212,000/mm³

Erythrocyte sedimentation rate 13 mm/h

Serum

C-reactive protein 0.2 mg/L

Which of the following is the most likely cause of this patient's pain?

Your Answer: A) Avascular necrosis of the femoral head

Correct Answer: A) Avascular necrosis of the femoral head

Correct Answer: A

Correct Answer: A.

Avascular necrosis, or osteonecrosis, results from impaired vascular supply to the bone with painful degeneration and loss of structural integrity over time. Decreased blood flow may result from direct injury to the blood vessels or from systemic conditions that cause impaired microcirculation, such as sickle cell disease, systemic lupus erythematosus, chronic steroid use, or radiation therapy. Symptoms include localized pain that worsens with activity and weight-bearing, stiffness, and limited range of motion. In sickle cell disease, bone ischemia is a frequent complication, particularly affecting the femoral head and leading to avascular necrosis of the hip. Symptoms may include hip pain, limp, and restricted hip movement. Diagnosis is made with imaging studies, including x-ray, CT scan, or MRI. MRI is the gold standard for diagnosis, showing areas of decreased signal intensity on T1-weighted images and increased signal intensity on T2-weighted images due to bone marrow edema and necrosis. Treatment options depend on the stage of avascular necrosis and can range from conservative management with pain control and limited weight-bearing to surgical intervention, such as core decompression, osteotomy, or total hip replacement.

Incorrect Answers: B, C, D, E, and F.

L2 radiculopathy (Choice B) would lead to pain, numbness, and tingling in the L2 dermatome, which travels from the lateral to medial thigh. While pain is present in this patient, it does not follow a radicular distribution, and the range of motion limitations and hip pain are more suggestive of hip

joint pathology.

Osteoarthritis of the hip (Choice C) is a degenerative joint disease that commonly affects older adults. Patients present with chronic hip pain that worsens with activity and improves with rest, stiffness, and limited range of motion. While osteoarthritis is a possibility in this patient given his age, avascular necrosis is a more common cause of hip pain in patients with sickle cell disease.

Osteomyelitis of the acetabulum (Choice D) is an infection of the bone. It typically presents with pain, fever, and localized swelling and tenderness. While this patient has pain and a history of sickle cell disease (a risk factor for osteomyelitis), he does not have fever or localized swelling, making osteomyelitis less likely.

Osteosarcoma of the femur (Choice E) is a primary malignant tumor of bone that affects adolescents and elderly patients. Patients may present with localized pain, a palpable mass, and weight loss. While this patient has pain and weight loss, the localized hip joint pain and limited range of motion are more suggestive of avascular necrosis. Osteosarcoma would typically appear as a destructive bone lesion on x-ray with periosteal new bone formation.

Trochanteric bursitis (Choice F) classically presents with lateral hip pain that is exacerbated by activity. Physical examination shows point tenderness over the greater trochanter and increased pain with external rotation of the hip as a result of compression of the bursa. The patient's pain is not localized to the lateral aspect of the hip and is more consistent with hip joint pathology.

Educational Objective: Educational Objective: Avascular necrosis, or osteonecrosis, results from impaired vascular supply to the bone. In sickle cell disease, bone ischemia is a frequent complication, particularly affecting the femoral head and leading to avascular necrosis of the hip. Symptoms include localized pain that worsens with activity and weight-bearing, stiffness, and limited range of motion. Diagnosis is made with imaging studies, such as MRI.

41. A 67-year-old woman is brought to the emergency department 24 hours after the onset of double vision, pain behind the right eye, and mild nausea. She describes her double vision as seeing objects side by side. She has hypertension, type 2 diabetes mellitus, peripheral neuropathy, and chronic renal insufficiency. Medications include fosinopril and glyburide. Vital signs are within normal limits. On examination, her double vision is most pronounced when she looks to the right. If she covers either eye, the double vision resolves. The pupils are 3 mm and normally reactive to light. When the patient looks to the right, the right eye does not abduct. Achilles tendon reflexes are absent. Sensation to pinprick is decreased in a stocking distribution over the feet. An MRI of the brain is most likely to show which of the following?

Your Answer: E) Normal findings

Correct Answer: E) Normal findings

Correct Answer: E

Correct Answer: E.

Abducens nerve palsy is characterized by impaired abduction of the ipsilateral eye, leading to horizontal double vision that is most pronounced when looking towards the affected side. Isolated abducens nerve palsy in older adults with vascular risk factors such as diabetes and hypertension is commonly caused by microvascular ischemia of the nerve. Microvascular cranial nerve palsies are typically painless and resolve spontaneously within months. Other causes of abducens nerve palsy include compression from tumors, inflammation, infection, or increased intracranial pressure. In this patient with diabetes and hypertension, a microvascular abducens nerve palsy is the most likely diagnosis. While imaging is often performed to rule out other causes, in patients with isolated abducens nerve palsy and vascular risk factors, normal findings on MRI are common.

Incorrect Answers: A, B, C, and D.

Acute temporal lobe hemorrhage (Choice A) would be more likely to have more pronounced neurologic deficits, such as aphasia or hemiparesis, in addition to headache and potential altered mental status. Diplopia that is centrally mediated is typically bilateral, whereas diplopia that resolves when the patient covers one eye is likely due to an ocular cause.

Enhancing meninges (Choice B) on MRI would be suggestive of an infectious or inflammatory process such as meningitis. While meningitis can present with focal neurologic deficits, patients typically have other signs or symptoms such as fever and nuchal rigidity.

Infarct in the right medulla (Choice C) can lead to lateral medullary syndrome, which can present with ipsilateral ataxia of extremity movements, gait ataxia, Horner syndrome, and can be associated with nystagmus and dysarthria. Isolated abducens nerve palsy is not typical with medullary infarcts. An orbital tumor (Choice D) would be unlikely to result in abducens nerve palsy. It would more likely affect ipsilateral extraocular muscles through local mass effect.

Educational Objective: Educational Objective: Abducens nerve palsy is characterized by impaired abduction of the ipsilateral eye and horizontal double vision. Isolated abducens nerve palsy in older adults with vascular risk factors is commonly caused by microvascular ischemia and often presents with normal findings on MRI.

42. A 32-year-old woman is brought to the emergency department 3 hours after the onset of confusion. She appears agitated. She mistakes the attending physician for her brother and insists that he call their father. She admits to drinking regularly since the age of 22 years. Her last drink was 4 ounces of vodka 72 hours ago. Her pulse is 110/min, and blood pressure is 170/110 mm Hg. Physical examination shows ataxia and a three cycle/sec bilateral hand tremor. On mental status examination, she is oriented to person but not to place or time. She yells, "I saw Dad here; why didn't you tell me he was coming to pay my bail? I could be out by now!" Which of the following is the most likely diagnosis?

Your Answer: B) Delirium tremens

Correct Answer: B) Delirium tremens

Correct Answer: B

Correct Answer: B.

Delirium tremens (DTs) is a severe manifestation of alcohol withdrawal that typically begins 48 to 96 hours after the last drink. It is characterized by an acute onset of disorientation, confusion, agitation, visual or auditory hallucinations (often vivid and unpleasant), fluctuations in consciousness, and autonomic instability (tachycardia, hypertension, fever, diaphoresis). Patients may also experience tremors and seizures. This patient's symptoms of confusion, disorientation, agitation, hallucinations, and autonomic instability (tachycardia and hypertension), occurring 72 hours after her last drink in a patient with a history of regular alcohol use, are consistent with DTs. DTs are a medical emergency and require hospitalization and aggressive management with benzodiazepines to control agitation and prevent seizures and autonomic instability.

Incorrect Answers: A, C, D, and E.

Alcoholic hallucinosis (Choice A) is an alcohol withdrawal syndrome characterized by auditory or visual hallucinations that occur in the setting of alcohol withdrawal, typically within 12 to 48 hours after the last drink. Unlike DTs, alcoholic hallucinosis does not involve significant disorientation, confusion, or autonomic instability. This patient's symptoms of disorientation, confusion, and autonomic instability are more consistent with DTs.

Idiosyncratic alcohol intoxication (Choice C) refers to an uncommon reaction to alcohol characterized by extreme behavioral changes, aggression, or memory loss after consuming small amounts of alcohol. This is not a withdrawal syndrome and does not typically involve hallucinations or autonomic instability.

Schizophrenia (Choice D) is a chronic psychotic disorder characterized by hallucinations, delusions, disorganized thinking, and negative symptoms (eg, flat affect, social withdrawal). Schizophrenia does not typically present with acute onset of confusion, disorientation, or autonomic instability, and the symptoms are not directly related to alcohol withdrawal.

Wernicke-Korsakoff syndrome (Choice E) is a condition caused by thiamine deficiency, often associated with chronic alcoholism. Wernicke encephalopathy, the acute phase, is characterized by encephalopathy (confusion, disorientation, ataxia), oculomotor dysfunction (nystagmus, ophthalmoplegia), and gait ataxia. Korsakoff syndrome, the chronic phase, is characterized by severe anterograde and retrograde amnesia and confabulation. While this patient has confusion and ataxia, the hallucinations, agitation, and autonomic instability are more characteristic of DTs. Thiamine supplementation is important in the management of alcohol withdrawal to prevent Wernicke-Korsakoff syndrome, but it does not acutely treat DTs.

Educational Objective: Educational Objective: Delirium tremens is a severe manifestation of alcohol withdrawal characterized by acute onset of confusion, disorientation, agitation, hallucinations, fluctuations in consciousness, and autonomic instability. It typically begins 48 to 96 hours after the last drink and is a medical emergency requiring aggressive management with benzodiazepines.

43. A 24-year-old woman comes to the office because of a 4-month history of generalized weakness and moderate muscle aches. Initially, she had difficulty standing after she removed clothes from the dryer. Now, she has difficulty walking up stairs and rising from a chair. She has no history of serious illness and takes no medications. Her mother has rheumatoid arthritis. Examination of the patient shows periorbital purplish discoloration bilaterally. Cranial nerves are intact. Palpation of all muscles produces pain. Muscle strength is 4/5 in the deltoids, biceps, hip flexors, and quadriceps bilaterally; this result does not change with repeat testing. Muscle bulk and tone are normal. Bilaterally, deep tendon reflexes are 2+ in the brachioradialis muscles and ankles and 1+ in the biceps and quadriceps muscles. Sensation is intact throughout. Which of the following is the most likely diagnosis?

Your Answer: B) Dermatomyositis

Correct Answer: B) Dermatomyositis

Correct Answer: B

Correct Answer: B.

Dermatomyositis is an inflammatory myopathy that affects both skin and musculoskeletal systems. Patients typically present with symmetric proximal muscle weakness (eg, difficulty rising from a chair, climbing stairs, or lifting objects) and characteristic skin findings, including periorbital purplish discoloration (heliotrope rash) and Gottron papules (erythematous papules over the dorsal aspects of the knuckles, elbows, and knees). Muscle aches and tenderness may also be present. The diagnosis is typically made based on clinical features, elevated muscle enzymes (creatine kinase, aldolase), abnormal findings on electromyography (EMG) and muscle biopsy, and the presence of myositis-specific antibodies. Dermatomyositis is associated with an increased risk of malignancy, particularly in older adults. Treatment involves immunosuppressive therapy, such as with corticosteroids, methotrexate, or intravenous immunoglobulin.

Incorrect Answers: A, C, D, and E.

Charcot-Marie-Tooth disease (Choice A) is a group of inherited peripheral neuropathies that typically present with distal muscle weakness and atrophy, foot deformities (eg, pes cavus), and sensory loss. Proximal weakness is not typical, and there are no skin findings.

Inclusion body myositis (Choice C) is an inflammatory myopathy that typically affects older adults. It presents with asymmetric, slowly progressive weakness and atrophy, often affecting the wrist and finger flexors and quadriceps muscles. It does not typically present with characteristic skin findings or periorbital purplish discoloration.

Myasthenia gravis (Choice D) is an autoimmune disorder of the neuromuscular junction caused by antibodies against the acetylcholine receptor or muscle-specific kinase (MuSK). It presents with fluctuating muscle weakness that worsens with repeated use and improves with rest. Common symptoms include ptosis, diplopia, dysphagia, dysarthria, and proximal limb weakness. There are no skin findings.

Polymyositis (Choice E) is an inflammatory myopathy that causes symmetric proximal muscle weakness. Unlike dermatomyositis, polymyositis does not typically present with characteristic skin findings. Muscle aches and tenderness may be present, and muscle enzymes are usually elevated. Diagnosis is confirmed with EMG and muscle biopsy. Treatment involves immunosuppressive therapy.

Educational Objective: Educational Objective: Dermatomyositis is an inflammatory myopathy that affects both skin and musculoskeletal systems. Patients typically present with symmetric proximal muscle weakness and characteristic skin findings, including periorbital purplish discoloration (heliotrope rash) and Gottron papules. Muscle aches and tenderness may be present.

44. A 19-year-old man is examined in the intensive care unit 3 days after admission for a self-inflicted gunshot wound to the head. Temperature is 37.2°C (99.0°F), pulse is 88/min, ventilatory rate is 16/min, and blood pressure is 100/40 mm Hg. Pupils are nonreactive to light. Corneal reflex is absent. Irrigation of the ear canals with cold water results in eye movement only toward the ear being irrigated. On application of noxious stimuli, the patient has only spinal reflex movement. He is not initiating breaths spontaneously above the ventilator. Which of the following findings prevents a diagnosis of brain death in this patient?

Your Answer: A) Caloric response

Correct Answer: A) Caloric response

Correct Answer: A

Correct Answer: A.

Brain death is determined by the presence of three factors: absence of brainstem reflexes, persistent coma, and lack of spontaneous respirations. Commonly tested brainstem reflexes include the pupillary light response (which tests the optic and oculomotor cranial nerves), the corneal reflex (which tests the trigeminal and facial cranial nerves by lightly touching the cornea with a strand of cotton or saline while observing for blinking or tearing), the pharyngeal and tracheal reflexes (which test the glossopharyngeal and vagus cranial nerves by observing gag or cough), oculocephalic (doll's eyes) reflex (which tests the oculomotor and vestibular cranial nerves by rapidly turning the head while observing for eye movement), and oculovestibular (caloric) reflex (which tests the oculomotor and vestibular cranial nerves by irrigating the ear canal with cold water while observing for eye movement). A normal response to eye movement away from the direction of the head turn, whereas an abnormal response is characterized by eye movement toward the ear being irrigated, or no eye movement (absent reflex). The patient's response to ice water caloric testing (eye movement only toward the ear being irrigated) indicates a preserved brainstem reflex (oculovestibular reflex), which prevents a diagnosis of brain death. All other findings listed in this patient's examination (nonreactive pupils, absent corneal reflex, lack of respiratory effort, and only spinal motor reflexes) are consistent with brain death.

Incorrect Answers: B, C, D, and E.

The corneal reflex (Choice B) tests the trigeminal and facial cranial nerves by lightly touching the cornea with a strand of cotton while observing for blinking or tearing. This patient's corneal reflex is absent, which could be consistent with a diagnosis of brain death.

The pupillary light reflex (Choice C) tests the optic and oculomotor cranial nerves by assessing for pupillary constriction in response to light. This patient lacks a pupillary light response, which could be consistent with a diagnosis of brain death.

Motor response (Choice D) testing assesses for a motor response to painful or noxious stimuli. The presence of spinal reflex movement in this patient does not indicate an intact motor response.

Respiratory effort (Choice E) is assessed with apnea testing by withdrawing the patient from ventilator support. This is typically the final step in diagnosis of brain death. While this patient does not yet have a positive apnea test, lack of spontaneous breaths above the ventilator is consistent with a diagnosis of brain death.

Educational Objective: Educational Objective: Brain death is determined by the presence of three factors: absence of brainstem reflexes, persistent coma, and lack of spontaneous respirations. Commonly tested brainstem reflexes include the pupillary light response, corneal reflex, pharyngeal and tracheal reflexes, oculocephalic reflex, and oculovestibular (caloric) reflex. The presence of any brainstem reflex prevents a diagnosis of brain death.

45. A 4-year-old boy is brought to the physician because he walks predominantly on his toes. He was born at 29 weeks' gestation and has a history of hyaline membrane disease and a grade I intraventricular hemorrhage. He takes albuterol for asthma, and his immunizations are up-to-date. He sat up at the age of 9 months and walked at 18 months; he currently uses full sentences when talking. He is at the 50th percentile for both height and weight. Examination shows generalized hyperreflexia, a persistent Babinski sign bilaterally, heel cord contractures, and decreased abduction and external rotation of the hips. Which of the following is the most likely diagnosis?

Your Answer: A) Cerebral palsy

Correct Answer: A) Cerebral palsy

Correct Answer: A

Correct Answer: A.

Cerebral palsy (CP) is a group of non-progressive neurologic disorders resulting from injury to the developing brain during the perinatal period. CP is characterized by impaired motor control and posture, often accompanied by intellectual disability, seizures, and visual or hearing deficits. Risk factors for CP include premature birth, low birth weight, intrauterine infections, perinatal stroke, and intraventricular hemorrhage. Symptoms depend on the location and extent of brain injury and may include spasticity, hypotonia, dystonia, ataxia, or choreoathetosis. Spastic diplegia, a subtype of CP, is characterized by symmetric spasticity of the lower extremities, leading to toe walking, scissoring gait, and heel cord contractures, as in this patient. Physical examination typically shows hyperreflexia and a persistent Babinski sign in affected limbs. Diagnosis is made based on clinical features and supported by imaging studies, such as MRI, which may show areas of brain injury.

Incorrect Answers: B, C, D, and E.

Multiple sclerosis (Choice B) is a progressive demyelinating disease that affects white matter in the brain and spinal cord. It is typically diagnosed in young adults and is characterized by relapsing-remitting or progressive neurologic deficits. Symptoms depend on the location of lesions and can include weakness, spasticity, sensory changes, and vision loss. It does not typically present in early childhood with the pattern seen in this patient.

Muscular dystrophy (Choice C) refers to a group of inherited myopathies that cause progressive muscle weakness and atrophy. Symptoms depend on the type of muscular dystrophy and the affected muscles, but typically involve symmetric proximal or distal weakness. Muscle tone and reflexes are usually decreased or absent. CP is a disorder of the central nervous system, whereas muscular dystrophy is a disorder of the muscle itself.

Myasthenia gravis (Choice D) is an autoimmune disorder of the neuromuscular junction that causes fluctuating muscle weakness that worsens with activity and improves with rest. Common symptoms include ptosis, diplopia, dysphagia, dysarthria, and limb weakness. It does not typically present in early childhood with the pattern seen in this patient.

Neurofibromatosis (Choice E) is a hereditary neurocutaneous disorder that presents with cutaneous findings (eg, café au lait spots, neurofibromas) and an increased risk of tumors in the nervous system and other organs. It does not typically present with the motor and gait abnormalities seen in this patient.

Educational Objective: Educational Objective: Cerebral palsy (CP) is a group of non-progressive neurologic disorders resulting from injury to the developing brain during the perinatal period. Spastic diplegia, a subtype of CP, is characterized by symmetric spasticity of the lower extremities, leading to toe walking, scissoring gait, heel cord contractures, hyperreflexia, and a persistent Babinski sign.

46. A 42-year-old woman comes to the office because of a 2-month history of generalized weakness. During this period, she has had difficulty climbing one flight of stairs and combing her hair. She also reports double vision that occurs when she reads. She has no history of serious illness and takes no medications. Vital signs are within normal limits. Examination shows bilateral ptosis. Muscle strength is 4/5 in all extremities. Deep tendon reflexes are 2+. Which of the following is the most appropriate next step in management?

Your Answer: E) Repetitive nerve stimulation study

Correct Answer: E) Repetitive nerve stimulation study

Correct Answer: E

Correct Answer: E.

This patient's presenting findings of intermittent ptosis, diplopia, and proximal muscle weakness are suggestive of myasthenia gravis. Myasthenia gravis (MG) is a B-cell driven autoimmune disorder characterized by the formation of inhibitory anticholinergic receptor autoantibodies. Common symptoms include fluctuating muscle weakness that worsens with repeated use and improves with rest. Symptoms commonly affect the oculomotor muscles (leading to ptosis and diplopia), bulbar muscles (dysphagia, dysarthria), and proximal limb muscles. Severe weakness can lead to respiratory failure. Diagnosis is made based on clinical features and supported by laboratory studies (positive for acetylcholine receptor or muscle-specific kinase [MuSK] autoantibodies), electromyography (EMG) findings (decreased amplitude on repetitive nerve stimulation), and pharmacologic testing (edrophonium test). Treatment includes pyridostigmine (an acetylcholinesterase inhibitor), immunosuppressive therapy (eg, corticosteroids, azathioprine, mycophenolate mofetil), and, in severe cases, plasmapheresis or intravenous immunoglobulin. Thymectomy is also a treatment option for patients with MG, particularly those with a thymoma.

Incorrect Answers: A, B, C, and D.

Measurement of serum creatine kinase (CK) activity (Choice A) and muscle biopsy (Choice D) may be appropriate in the setting of a suspected myopathy, such as dermatomyositis or drug-induced myopathy. Since myasthenia gravis is a disorder of the neuromuscular junction, CK activity and muscle biopsy would be expected to be normal.

Plasmapheresis (Choice B) is a treatment used for patients with myasthenic crisis. Patients with myasthenic crisis may develop weakness of the muscles of the throat, diaphragm, and chest wall leading to dysphagia and acute respiratory failure requiring mechanical ventilation. This patient is not exhibiting signs or symptoms of a myasthenic crisis.

MRI of the brain (Choice C) could be useful if this patient's symptoms were caused by a stroke, but proximal weakness in all extremities and bilateral ptosis do not fit with this clinical picture. MRI of the brain in a patient with myasthenia gravis would be expected to be normal.

Educational Objective: Educational Objective: Myasthenia gravis is an autoimmune disorder of the neuromuscular junction that causes fluctuating muscle weakness that worsens with repeated use and improves with rest. Common symptoms include ptosis, diplopia, dysphagia, dysarthria, and proximal limb weakness. Diagnosis is confirmed by laboratory studies (autoantibodies), electrophysiology (repetitive nerve stimulation), and pharmacologic testing.

47. A 37-year-old man comes to the physician because of a 3-week history of joint pain. The symptoms initially began with increasing pain in his left knee and have progressed to include his right knee and left hip. There is no history of trauma. Six weeks ago, he had fever, malaise, and a 5-cm rash with central clearing on his chest that resolved spontaneously after 7 days. He is sexually active with multiple partners and uses condoms inconsistently. His temperature is 37.1°C (98.8°F), and blood pressure is 148/90 mm Hg. Examination of the left knee shows no warmth or erythema. Range of motion of the left knee is full and does not produce pain. Range of motion of the right knee is limited by pain; there is moderate joint effusion. Abduction and adduction of the left hip against resistance produce pain. Which of the following is the most likely diagnosis?

Your Answer: C) Lyme arthritis

Correct Answer: C) Lyme arthritis

Correct Answer: C

Correct Answer: C.

Lyme arthritis, one of several manifestations of Lyme disease, is caused by disseminated *Borrelia burgdorferi* infection transmitted by Ixodes ticks, most commonly from Ixodes ticks. Initial symptoms of Lyme disease include the erythema chronicum migrans rash, which is an expanding rash with a central ring-like clearing, and targetoid appearance. It generally appears within 1 to 2 weeks of the tick bite. Untreated, a subset of patients will develop disseminated disease that may result in Bell palsy, meningitis, myocarditis, or arthritis, as in this patient. Lyme arthritis typically presents as intermittent or persistent joint pain and swelling, often affecting large joints such as the knee. Diagnosis is confirmed with serologic testing for antibodies to *B. burgdorferi*. Although a positive result is indicative of exposure, false-positive results can occur, so clinical context is important. Lyme arthritis typically responds to treatment with doxycycline or amoxicillin. Patients who do not respond to initial treatment may require intravenous antibiotics.

Incorrect Answers: A, B, D, and E.

Gonococcal arthritis (Choice A) arises from disseminated *Neisseria gonorrhoeae* infection and may present with fever, malaise, and migratory polyarthritis or oligoarthritis. Patients will have purulent arthritis of one or more extremity joints (eg, knees, elbows, or ankles), and synovial fluid analysis will show greater than 50,000 leukocytes/mm³ with a predominance of neutrophils, and may grow the organism on culture. Patients with gonococcal arthritis are treated with intravenous antibiotics such as ceftriaxone, along with possible irrigation of the affected joints.

Gouty arthritis (Choice B) is an inflammatory arthropathy induced by hyperuricemia, with deposition of uric acid crystals inside the joint and surrounding tissues. This type of arthritis presents following consumption of a purine-rich meal or alcohol (which competes with renal excretion of uric acid). Joint aspirate will show negatively birefringent needle-shaped crystals and a leukocyte count of typically less than 50,000/mm³.

Osteoarthritis (Choice D) is caused by repetitive wear-and-tear injury to a joint with subsequent cartilage degeneration, bony sclerosis, and joint pain. Risk factors include age, obesity, and prior intra-articular injury. Treatment options include physical therapy, braces, weight loss, acetaminophen, intra-articular corticosteroid injections, and, if severe, joint replacement.

Rheumatoid arthritis (Choice E) is a chronic, progressive, autoimmune disease that primarily leads to destruction of synovial joints. The disease typically presents with symmetric patterns of arthritis, particularly in the wrists and hands, and is treated with disease-modifying anti-rheumatic drugs and/or tumor necrosis factor- α blockers, such as adalimumab and infliximab.

Educational Objective: Educational Objective: Lyme arthritis is a manifestation of Lyme disease caused by *Borrelia burgdorferi* infection. It typically presents as intermittent or persistent joint pain and swelling, often affecting large joints such as the knee. Diagnosis is confirmed with serologic testing and treatment involves doxycycline or amoxicillin.

48. A 57-year-old man is brought to the emergency department because of a 24-hour history of progressive agitation and confusion. He has used a wheelchair since an accidental gunshot wound transacted his spine at the T9 level 20 years ago. On arrival, he is too confused to provide his medical history. His medical record indicates that his medications are pregabalin for neuropathic pain and diazepam for spasms as needed. His family reports that he does not drink alcohol or use illicit drugs. He appears well nourished. He is uncooperative and hits anyone who touches him. His temperature is 38.6°C (101.4°F), pulse is 106/min, and blood pressure is 110/76 mm Hg. Physical examination shows paraplegia. He is oriented to person but believes he is at home. He appears to be responding to internal stimuli. Laboratory studies show:

Leukocyte count 22,000/mm³

Urine Nitrites positive

Leukocyte esterase positive

Bacteria >10,000 colonies/mL

A CT scan of the head shows no abnormalities. In addition to administration of antibiotics, which of the following is the most appropriate pharmacotherapy?

Your Answer: B) Haloperidol

Correct Answer: B) Haloperidol

Correct Answer: B

Correct Answer: B.

Delirium is defined as an acute-onset alteration of consciousness and cognition, with symptoms that wax and wane throughout the day. In elderly patients, especially those with underlying dementia, systemic infection can rapidly lead to the development of delirium. Other common causes of delirium include intoxication and adverse effects from medications such as anticholinergics, benzodiazepines, or opiates. All patients with delirium should be evaluated for evidence of systemic infection with urinalysis and culture, chest x-ray, and blood cultures, and urine analysis, depending on the symptoms. In this case, the patient's laboratory findings and physical examination are consistent with a urinary tract infection (UTI). Treatment for delirium should focus on treatment of the underlying infection if one is detected, frequent reorientation, maintenance of normal sleep-wake cycles, and placement of the patient in a calm and soothing environment. For patients with refractory symptoms, anti-psychotic therapy may be used, including haloperidol.

Incorrect Answers: A, C, D, and E.

Diazepam (Choice A) and Midazolam (Choice D) are benzodiazepines commonly used for the treatment of seizure disorders, sedation, anxiety, and muscle spasms. Benzodiazepines bind to γ -aminobutyric acid (GABA) receptors and increase the frequency of their opening. GABA is an inhibitory neurotransmitter that counterbalances neuronal excitation, making benzodiazepines an effective treatment for seizure. Benzodiazepines are a known trigger of delirium in elderly patients and are not recommended for the treatment of delirium unless triggered by ethanol withdrawal. Labetalol (Choice C) is a non-specific alpha- and beta-blocker that may be used in the treatment of hypertension disorders and aortic dissection. Labetalol is not commonly used in the treatment of delirium unless triggered by hypertensive encephalopathy.

Vitamin B₁ (thiamine) (Choice E) is used to treat Wernicke encephalopathy, characterized by encephalopathy, ataxia, and oculomotor dysfunction. Diagnosis is clinical, and Wernicke encephalopathy should be rapidly treated with intravenous thiamine administration to avoid progression to Korsakoff syndrome.

Educational Objective: Educational Objective: Delirium is an acute-onset alteration of consciousness and cognition, with symptoms that wax and wane throughout the day. It is a common complication of systemic infection in elderly patients. Treatment focuses on treating the underlying cause and providing supportive care. For refractory symptoms, anti-psychotic therapy may be used.

49. A 27-year-old woman comes to the physician because of a 3-week history of headaches that are dull, constant, and localized to the back of her head. The pain is not relieved by ibuprofen. She has not had fever, nausea, vomiting, or weakness. She has a 5-year history of AIDS currently treated with three-drug antiretroviral therapy. Her temperature is 37.8°C (100°F), pulse is 90/min, respirations are 14/min, and blood pressure is 130/90 mm Hg. Examination shows multiple, 1-cm, cervical lymph nodes that are mildly tender. There is no papilledema. Examination of the oropharynx shows thrush. Neurologic examination shows no focal findings. Her CD4+ T-lymphocyte count is 25/mm³ (Normal ≥500), and plasma HIV viral load is 5000 copies/mL. A CT scan of the head with contrast shows five 1- to 2-cm, ring-enhancing lesions. Which of the following is the most likely diagnosis?

Your Answer: D) Toxoplasmosis

Correct Answer: D) Toxoplasmosis

Correct Answer: D

Correct Answer: D.

This patient has a history of AIDS with low CD4+ count and CT scan showing ring-enhancing lesions, which is suggestive of cerebral toxoplasmosis. Cerebral toxoplasmosis is an AIDS-defining disease caused by *Toxoplasma gondii* and is the most common cause of multiple ring-enhancing lesions on CT scan of the head in patients with AIDS. Transmission can occur through the ingestion of oocysts from cat feces or infected meat, direct transmission between a mother and fetus, or from organ transplantation, which leads to invasion followed by dissemination of the parasite where it typically lies dormant. It is a ubiquitous organism, and nearly 10% of individuals will be seropositive for *T. gondii*, but the parasite does not usually cause severe disease in immunocompetent individuals. In patients with HIV infection and CD4+ counts less than 100 cells/mm³, however, reactivation can occur with several distinct clinical syndromes of which cerebral toxoplasmosis results in the most morbidity. Cerebral toxoplasmosis typically presents with multiple ring-enhancing lesions scattered throughout the brain. Seizures, focal neurologic deficits, and systemic symptoms are common, as are signs of increased intracranial pressure. Patients with HIV and positive *Toxoplasma* IgG antibodies are typically treated empirically with a sulfonamide and pyrimethamine if cerebral toxoplasmosis is suspected, but confirmatory diagnosis occurs through a brain biopsy, which will show a dense infiltrate of lymphocytes, macrophages, and plasma cells, along with the presence of the *T. gondii* organism. Cerebral toxoplasmosis is an AIDS-defining disease.

Incorrect Answers: A, B, C, and E.

Cryptococcus neoformans infection (Choice A) of the central nervous system (CNS) presents subacutely with headache, nuchal rigidity, photophobia, and progressive encephalopathy. Cryptococcal fluid (CSF) analysis will show an increased opening pressure, increased protein, and a mononuclear cell pleocytosis. Ring-enhancing lesions on CT scan of the head are not a feature of cryptococcal meningitis.

Cytomegalovirus (CMV) infection (Choice B) can cause a mononucleosis-like syndrome in immunocompetent patients but can cause a variety of other diseases in patients who are immunodeficient, including esophagitis, retinitis, pneumonitis, and colitis. Congenital CMV may present with CNS findings and calcifications on CT scan of the head, but it is not known to commonly cause CNS findings in adults.

Lymphoma (Choice C) of the CNS is a rare form of non-Hodgkin lymphoma that occurs in immunocompromised patients. Patients may present with focal neurologic deficits, vision changes, neuropsychiatric symptoms, and increased intracranial pressure. Neuroimaging will show ring-

enhancing lesions similar to those in toxoplasmosis. Toxoplasmosis is more common than CNS lymphoma, and empiric therapy for toxoplasmosis is the initial diagnostic step. If symptoms do not improve with sulfadiazine or pyrimethamine therapy, CNS lymphoma should be considered. Tuberculosis (Choice E) meningitis typically presents in a subacute fashion with nuchal rigidity, fever, and altered mental status. CSF evaluation is the initial diagnostic step and will show an increased opening pressure, increased white blood cell count with prominent lymphocytes, increased protein concentration, decreased glucose concentration, and positive culture for acid-fast bacilli. While CNS tuberculosis can cause tuberculomas that appear as ring-enhancing lesions on imaging, this is less common than meningitis in AIDS patients.

Educational Objective: Educational Objective: Toxoplasmosis is caused by *Toxoplasma gondii* and is a common cause of multiple ring-enhancing lesions on CT scan of the head in patients with AIDS with low CD4+ count. Patients typically present with seizures, focal neurologic deficits, and systemic symptoms. Empiric treatment with sulfadiazine and pyrimethamine is the initial diagnostic step.

50. A 32-year-old man comes to the emergency department because of a 2-day history of low back pain that began 4 hours after he lifted a sofa. His symptoms have gradually become worse, and he now has severe pain with any movement of his back. The pain radiates to the right posterior thigh. He has had no changes in bowel or bladder function. His temperature is 37°C (98.6°F), pulse is 88/min, respirations are 18/min, and blood pressure is 120/80 mm Hg. On examination, flexion and extension of the waist and twisting movements of the torso increase the pain. There is tenderness over the lumbar paraspinal musculature bilaterally that is more prominent on the right. Straight-leg raising on the right causes pain in the right sacral area at 45 degrees. Neurologic examination shows no other abnormalities. Sensation is intact. Which of the following is the most appropriate initial step in management?

Your Answer: D) Ibuprofen therapy

Correct Answer: D) Ibuprofen therapy

Correct Answer: D

Correct Answer: D.

This patient's presentation is suggestive of acute onset of low back pain, which lasts up to 4 weeks and may be attributed to musculoskeletal injury from exertion, overuse, or trauma. Symptoms include pain with movement and paraspinal tenderness in the absence of neurologic symptoms or bony tenderness. Most cases resolve with conservative management, which includes activity modification, stretching and physical therapy, over-the-counter analgesics such as nonsteroidal anti-inflammatory drugs, massage, and heat therapy. Certain red flag signs and symptoms should prompt more urgent evaluation for an alternative cause, especially if there is concern for infection, malignancy, spinal cord compromise, or vertebral fracture. These include new fevers, intravenous drug use, focal neurologic deficits (such as urinary retention, saddle anesthesia, bowel incontinence which may suggest cauda equina syndrome), progressive lower extremity neurologic symptoms, current or recent cancer history, advanced age, systemic corticosteroid use, and a significant traumatic mechanism. In this young, otherwise healthy patient with a mild inciting event, the most appropriate initial step in management is ibuprofen therapy.

Incorrect Answers: A, B, C, and E.

Bed rest for 1 week (Choice A) is not indicated in patients with acute low back pain, as these patients may experience increased pain and slower return to normal function. Activity modification for symptom relief should be advised, with encouragement to continue normal activities of daily living and non-exacerbating exertion.

X-rays of the lumbosacral spine (Choice B) and MRI of the lumbosacral spine (Choice C) should be reserved for patients who present with red flag signs and symptoms to evaluate for evidence of infection, malignancy, spinal cord compromise, or vertebral fracture. For otherwise healthy patients with back pain that has persisted for a subacute time course, imaging studies do not generally change management.

Ibuprofen therapy (Choice D) is an appropriate initial step in management for this patient's back pain. Ibuprofen is a nonsteroidal anti-inflammatory drug (NSAID) commonly used for pain relief in musculoskeletal injuries.

Nerve conduction studies of the lower extremities (Choice E), including electromyography, should be reserved for patients who experience neurologic symptoms including weakness or altered sensation. This patient's presentation does not warrant additional nerve conduction studies.

Educational Objective: Educational Objective: Acute low back pain is a common presenting finding in multiple medical settings. General management for pain persisting less than 4 weeks is conservative with over-the-counter analgesics, physical therapy, and nonpharmacologic pain relief options. Initial imaging is not typically necessary in the absence of red flag signs or symptoms suggestive of infection, malignancy, spinal cord compromise, or vertebral fracture.